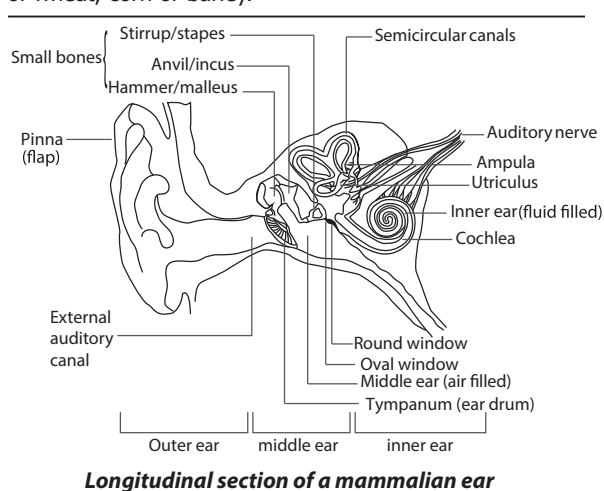


bonds, numbers are used to designate the bonds, for example, (2E, 4Z)-2,4-hexadienoic acid). The system is less ambiguous than the cis/trans system of describing isomers. E and Z are the first letters, of the German words (*entgegen* and *zusammen*), which mean 'opposite' and 'together' respectively.

EAGLE: A common name for more than sixty species of the order *Falconiformes*, in the bird family *Accipitridae* of several genera. Most of the species are found in Eurasia and Africa; and a few others are found in North America, Central America, South America and Australia. Eagles have strong hooked beak, large wings and raptorial feet. They also have exceptional eyesight to enable them to spot potential prey from a very long distance. Eagles are generally divided into four groups based on their physical characteristics and behaviour: **Sea or Fish Eagles**, some of whose members include the Bald eagle (*Haliaeetus leucocephalus*) and the Vulturine fish eagle (*Gypohierax angolensis*); **Snake or Serpent Eagles**, which includes the Banded snake eagle (*Circoetus cinerascens*) and Brown snake eagle (*Circaetus cinereus*); **Booted or True Eagles** some of whose members include Spanish imperial eagle (*Aquila adalberti*), Golden eagle (*Aquila chrysaetos*) and Wedge-tailed eagle (*Aquila audax*); and **Harpy or Buteonine Eagles**, whose members include Harpy eagle (*Harpia harpyja*); Crowned eagle (*Harpyhaliaetus caranatus*) and Solitary eagle (*Harpyhaliaetus solitarius*). The size of an eagle is dependent on the species. The Stellar's Sea Eagle (*Haliaeetus pelagicus*), one of the world's biggest eagles, for example, has a length of 85-105 cm, a wingspan of 220-245 cm and a weight of 6.8-9 kg. Some eagles can reach a height of over 1 m. Eagles are dominant predators, known as birds of prey and feed on smaller birds and mammals. Many species of the eagle are listed as endangered and vulnerable on the World Conservation Union's (IUCN) Red List of Threatened Species.

EAR: 1. Part of the auditory system, which acts as a receiver of sound and also plays a major role in the sense of balance and body position. It responds to the vibrations that constitute sound and these are translated into nerve signals and passed to the brain. The typical vertebrate ear consists of a sound-collecting outer ear separated by the tympanic membrane from a sound-transmitting middle ear that in turn is separated from a sensory inner ear by membranous fenestrae. 2. Any of various organs, as of a bird or fish, capable of detecting vibratory motion. 3. The spike of spike-like inflorescence, such as the ear of wheat, corn or barley.



Longitudinal section of a mammalian ear

EAR BONES (AUDITORY OSSICLES; OSSICULA AUDITUS; OSSICULAR CHAIN): The three small bones of the middle ear, consisting of the malleus, the incus and the stapes, which are articulated to form a chain for the transmission of sound from the tympanic membrane to the oval window.

EAR CANAL (AUDITORY CANAL; EXTERNAL AUDITORY MEATUS): The slightly curved tube leading from the outer ear to the eardrum, which separates it from the middle ear. In humans, the ear canal is about 2.5 cm long. It is lined with skin that extends to

cover the tympanic membrane. The structure of the external auditory canal is the same in all mammals. The outside third of the canal wall consists of cartilage and the inner two-thirds of the wall are made of bone. The ear canal has hairs near the outer end, which keep foreign objects like dust or flies out. The ear canal also has special oil glands that produce ear wax or cerumen, a yellowish waxy substance that protects the skin of the human ear canal, assists in cleaning and lubrication and also provides some protection from bacteria, fungi, insects and water.

EARDRUM (TYMPANIC MEMBRANE): A thin membrane separating the outer and middle ear in tetrapod vertebrates. It vibrates in response to sound waves and transmits these vibrations via the ear ossicles of the middle ear to the inner ear.

EAR LOBE: The fleshy soft lower part of the outer ear or pinna. It does not contain cartilage but it is made up of tough areolar and adipose connective. It lacks the firmness and elasticity of the rest of the pinna but has a large blood supply, which may help to warm the ears and maintain balance. Generally, however, earlobes are not considered to have any major biological function.

EAR NOTCH: A system of permanently marking animals by cutting notches from their ears.

EAR OSSICLES (AUDITORY OSSICLES): The three small bones, namely the malleus, incus and stapes in the mammalian middle ear, which form a link between the tympanum and the fenestra ovalis. Their function is to transmit and amplify vibrations of the tympanum across the middle ear to the fenestra ovalis, which transfers them to the inner ear through the fluid-filled labyrinth or cochlea. The absence of the ossicles leads to a moderate-to-severe hearing loss.

EARDRUM (TYMPANIC MEMBRANE): A thin membrane that separates the outer ear from the middle ear. It closes externally the cavity of the middle ear and functions in the mechanical reception of sound waves and in their transmission to the site of sensory reception by vibrating in response to sound waves which it transmits via the ear ossicles of the middle ear to the site of hearing.

EARED (AURICULATE): Having ears or bearing earlike projections.

EARLY UNIVERSE: The study of cosmology relating to the time soon after the big bang, which occurred around 13.3-13.9 billion years ago. The term *Big Bang* was used to refer to the idea that the universe was originally extremely hot and dense at some given time in the past and has since cooled by expanding to the present diluted state - and continues to expand today. Theories of the early universe, particularly the grand unified theories, have led to a mutually beneficial interaction between cosmology and the theory of elementary particles. It was proposed that there were very high temperatures in the early universe and many of the broken symmetries became unbroken at these temperatures in gauge theories. As the universe cools after the big bang, it is thought to be a sequence of transition to broken symmetry state.

EARLY WARNING (EW) SATELLITES: Military spacecraft used to detect the launch of missiles and rockets from Earth's surface. Information from these satellites would provide first alert of the start of a major missile attack and is used to track the long-term patterns of foreign nations' space programmes.

EARNshaw'S THEOREM: It states that a collection of point charges cannot be maintained in a stable static equilibrium configuration solely by the electrostatic interaction of the charges. This theorem was first proved by Samuel Earnshaw in 1842 and it is a fundamental result concerning bodies that interact according to inverse square laws, such as Coulomb's law of electrostatics and Newton's law of gravitation. It applies to the classical law forces (electric and gravitational) and also to the magnetic forces of permanent magnets and paramagnetic materials but not diamagnetic materials.

EARTH: 1. The third planet in distance from the Sun and the fifth largest in the solar system. It is almost spherical, flattened slightly at the poles and bulging at the equator. It is composed of three concentric layers: the core, the mantle and the crust. The Earth has an equatorial diameter

of 12,760 km, polar diameter of 12,713.56 km, circumference at the equator 40,000 km, total surface area 510,072,000 km², land surface area 148,940,000 km² (29.2 %), water surface area 361,132,000 km² (70.8 %), volume 1.08321×10¹² km³, mass 5.9736×10²⁴ kg, maximum surface temperature 57.8°C (331 K) and minimum surface temperature -89.2 °C (184 K). The Earth is surrounded by an envelope of gases called the atmosphere, of which the greater part is nitrogen and oxygen. It is the only planet on which life is known to exist. Earth interacts with other objects in space, especially the Sun and the Moon. Earth orbits the Sun once every 366.26 in a leap year. It rotates about its own axis, which is equal to 365.26 solar days or one sidereal year. The Earth's axis of rotation is tilted 23.4° away from the perpendicular of its orbital plane, producing seasonal variations on the planet's surface with a period of one tropical year (365.24 solar days). Earth's only known natural satellite is the Moon. 2. Metallic oxide not readily reducible by chemical means. These include alkaline earths, rare earths and alumina. The term is also applied to certain absorbent clays such as fuller's earth and compounds such as carbonates, silicates and hydroxides. 3. Also known as **ground**, it is an electrical connection between an appliance and the ground so that it shares any charge it has with the earth. As the earth is so large this means in effect that an earthed object is of zero potential and cannot keep a charge hence making it harmless. In the event of a fault in the electrical appliance involving a connection between the live part of the circuit and the outer casing, for example, the current flows to the earth without causing any harm to the user. The earth in an electrical circuit is the reference point from which other voltages are measured.

EARTH OBSERVATION SATELLITE: A satellite designed to observe Earth's surface and atmosphere from orbit. Included in this category are Earth resources satellites, weather satellites, geodesy satellites and, for military and intelligence applications, reconnaissance satellites.

EARTH PROBES: A NASA programme that complements Earth Observing System (EOS). Earth Probes are specific, highly-focused missions that investigate processes requiring special orbits or short development cycles of one to three years. Within this programme are the Earth System Science Pathfinder (ESSP) missions, the University Earth Systems Science (UnESS) programme and various specialised Earth-observing projects such as Total Ozone Mapping Spectrometer (TOMS), Triana and Shuttle Radar Topography Mission (SRTM).

EARTH RECEIVED TIME (ERT): The time at which an event that has occurred on a spacecraft will be known on Earth. It corresponds to spacecraft event time plus one-way light time and does not include any delay in signal processing once the signal has reached the ground.

EARTH RESOURCES SATELLITES: Unmanned orbiting spacecraft that collect information about Earth's atmosphere, land and oceans for scientific research and resource management. They fall into the more general category of Earth observation satellite.

EARTH SCIENCE (GEOSCIENCE): The scientific study of the planet earth as a whole, putting together several traditional subjects such as geology, meteorology, oceanography, geophysics, geochemistry and palaeontology. Typically, however, Earth scientists use tools from physics, chemistry, biology, chronology and mathematics to build a quantitative understanding of how the Earth system works and how it evolved to its current state. Since Earth is the only known life-bearing planet, it is regarded as a special case in planetary science. There are both reductionist and holistic approaches to Earth sciences.

EARTH STATION: Equipment on Earth that can transmit or receive satellite communications, though generally, the term refers to receive-only stations. An Earth station usually contains a combination of antenna, low noise amplifier, down-converter and receiver electronics.

EARTH SUMMIT (UNITED NATIONS CONFERENCE ON ENVIRONMENT AND DEVELOPMENT - UNCED): An international meeting in Rio de Janeiro, Brazil, in June 1992, which drew up measures towards world environmental protection. The UN sought

to help Governments rethink economic development and find ways to halt the destruction of irreplaceable natural resources and pollution of the planet. Hundreds of thousands of people from all walks of life were drawn into the Rio process.

EARTH WIRE (GROUND WIRE): A safety wire connected to the metal case of an electrical appliance and to the ground or earth, which is zero potential. If a fault causes the live to touch the case, the larger current flows through the earth wire and the fuse blows or causes a circuit breaker to trip. Without the earth wire, a person could receive an electrical shock from the case in such events. The earth in an electrical circuit is the reference point from which other voltages are measured. In Europe and other parts of the world areas using a 50 Hz system, the colours of earthwire are green and yellow stripes. In the U.S.A., ground wire of a circuit may be bare or insulated with green colour or green stripes.

EARTH-CLASS PLANET: A planet having a mass of 0.5 to 10 times that of the Earth (M_e) and a radius 0.8 to 2.2 times that of the Earth. Planets more massive than 10 M_e have enough gravity to attract a large hydrogen-helium atmosphere and so become gas giants. Those less massive than 0.5 M_e are likely to lose any potentially life-supporting atmosphere because of low gravity and lack of plate tectonics needed to recycle carbon dioxide back into the atmosphere.

EARTH'S ATMOSPHERE: The layer of gases that surrounds the Earth together with the clouds, which protects life on Earth by absorbing ultraviolet solar, warming the surface through heat retention (greenhouse effect) and reducing temperature extremes between day and night. It has a mass of about 5×10^{18} kg, three quarters of which is within about 11 km of the surface. It becomes thinner with increasing altitude. There is no definite boundary between the atmosphere and outer space, however, at an altitude of 120 km atmospheric effects become noticeable during atmospheric re-entry of spacecraft. The Kármán line, which lies at 100 km, is often regarded as the boundary between the atmosphere and outer space.

EARTH LEAKAGE CIRCUIT BREAKER (ELCB): A former safety device used in electrical installations with high earth impedance to prevent shock. It has now been replaced by residual current circuit breakers which detect leakage current directly and opens a circuit automatically, when an abnormally high current flows through the circuit.

EARTH TWIN: A new discovery on July 23, 2015 of an Earth-like planet by NASA (National Aeronautics and Space Administration), named Kepler-452b, which it is claimed is about one-and-a-half times the diameter of Earth (about 60 percent larger in diameter than that of Earth), circling a star very much like the Earth's Sun and has the right temperature within the habitable zone (the area around a star where liquid water could pool on the surface of an orbiting planet). Although other Earth-like planets have been discovered, Kepler-452b, is the closest found so far to match our Earth-sun system and is dubbed Earth's "older cousin". Some of the Earth-like planets discovered so far are Kepler-22b, Kepler-69c, Kepler-62f and Kepler-186f.

EARTHERNWARE: Pottery such as bowls, pots, mugs or other vessels made of slightly porous clay and fired at low heat.

EARTH'S MAGNETIC FIELD (GEOMAGNETIC FIELD): A region of space surrounding the Earth in which the resulting magnetic force of the Earth can be detected. It operates in a region above the Earth's surface known as the magnetosphere which effectively extends several tens of thousands of kilometres into space. It weakens with distance from its source. Studies of Australian red dacite and pillow basalt estimated the magnetic field to be at least 3.5 billion years old.

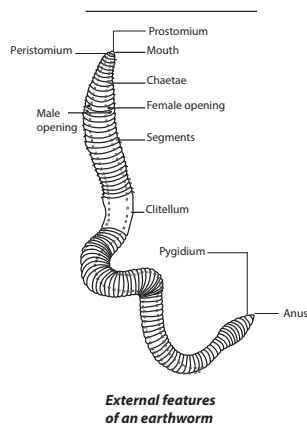
EARTHQUAKE (QUAKE; TREMOR; TEBLOR; SEISMIC ACTIVITY): The sudden and violent shaking of the Earth as a result of the sudden release of stresses or energy built up in the Earth's crust, which creates seismic waves, when the lithospheric plates that compose the surface of the Earth shift in relation to one another. It is measured and recorded by a seismograph and seismometer. Richter

scale rates earth tremors on a scale from 1 to 9, with 9 being the most powerful and each number representing an increase of ten times the energy over the previous number. According to this scale, any quake that is higher than 4.5 can cause damage to stone buildings; quakes rated a magnitude of 7 and above are considered very severe. A less-known scale, the Mercalli scale, was devised by Italian seismologist Giuseppe Mercalli to measure the severity of an earthquake in terms of its impact on a particular area and its inhabitants and buildings.

EARTHSHINE: Sunlight reflected from the surface of the Earth. An observer in space may see nearby objects dimly illuminated by earthshine, as things on Earth may be illuminated by moonlight.

EARTHSIDE: The movement of a layer of dry soil or part of an earth embankment, down a slope. The movement may be sufficient to break up blocks and crush or grind the material in such a way that it moves in a somewhat fluid manner.

EARTHWORM: A common long thin worm with segmented body, which is tapered at both ends. It exists in all soils of the world that have sufficient moisture and organic content. There are more than 1,800 species of these terrestrial worms, belonging to the class Oligochaeta of the phylum Annelida (segmented worms). It produces slime that binds particles and improves soil structure. Plant remains carried down its burrows enrich the soil. Earthworms eat decaying organisms and, in the process, ingest soil, sand and pebbles, which aerate the soil, promoting drainage and improving the soil's nutrient content for plants. Earthworms are eaten by many animals.



EBCDIC: In computing, it is the acronym for **Extended Binary Coded Decimal Interchange Code**, a binary (8-bit) code used for storing and communicating alphabetic and numeric characters, developed by IBM for its larger operating systems and is capable of holding 256 different characters, although only 85 of these are defined in the standard version.

EBERLEIN-SMULIAN THEOREM: The theorem, named after William Frederick Eberlein and Witold Lwowski Schmulian that asserts the equivalence of compactness (Heine-Borel compactness), sequential compactness and limit point compactness in the weak topology for a Banach space.

EBOLA: A virus that causes a haemorrhagic fever, which is believed to have been long present in animals. It was first identified in humans in Zaire in 1976. Ebola is acquired through contact with contaminated body fluids. There is no specific treatment and the death rate can be as high as 90%. Symptoms include severe headache, muscle pains and high fever.

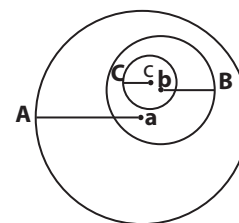
EBONY: A common name for various dark-coloured tropical trees and also woods of the genus *Diospyros* and family *Ebenaceae*. The wood is very heavy, hard, almost black and fine-textured. Due to its colour, durability, hardness and ability to take high polish, it is used for cabinet work and inlaying, piano keys, knife and umbrella handles and turned articles.

EBRACTEATE: A flower that has no bracts.

EBULLIOSCOPE CONSTANT (MOLAL ELEVATION OF THE BOILING POINT): The ratio of the elevation of the boiling point of a solvent caused by dissolving a solute to the molality of the solution, taken at extremely low concentrations and relates molality to boiling point elevation. Its symbol is K_b .

EBULLIOSCOPY: The use of boiling point elevation to determine relative molecular masses or the determination of the concentration of materials in a solution based on boiling point deviations.

ECCENTRIC: 1. Of geometric figures, such as two or more circles, not having a common centre; having different centres. 2. Of an ellipse or ellipsoid, having widely separated foci, thus an eccentricity close to unity. 3. In technology, it is a disk or circular frame, fixed on a revolving shaft or axis which does not pass through the centre of the disk. It is used for taking an alternating rectilinear motion from a revolving shaft.



For circle A, the centre is a, for circle B, the centre is b, and for circle C, the centre is c

Eccentric circles

ECCENTRICITY: 1. Also called **linear eccentricity** and often denoted e , it is a property that identifies a conic section (parabola, hyperbola or ellipse), which is equal to the distance from the curve to a fixed point (focus) divided by the distance to the fixed line (directrix) or the ratio of the distance of any point on a conic section from a focus to its distance from the corresponding directrix. 2. Also known as **numerical eccentricity** and denoted by ϵ , it is the ratio of the linear eccentricity to half the length of the major axis of a conic, that is constant for a family of similar curves. In other words, it is a measure of how closely a conic section resembles a circle. This ratio is constant for any particular conic section. For an ellipse the eccentricity is less than 1, for a hyperbola it is more than 1 and for a parabola it is unity (1).

ECCRINE GLANDS (MEROCRINE GLANDS): One of the two types of sweat glands, the other one is apocrine gland. Eccrine glands are linked to the sympathetic nervous system and are widely distributed over the body surface. They are composed of an intra epidermal spiral duct (the acrosyringium), a straight dermal portion and a coiled acinar (grape like) portion in the dermis or hypodermis. They produce a clear, odourless substance, consisting primarily of water and sodium chloride, with the odour being produced by bacterial activity on the secretions of the apocrine glands. The sodium chloride is reabsorbed in the duct to reduce salt loss. Apocrine glands develop in areas abundant in hair follicles, such as on the scalp, armpits and groin and open into the hair follicle just before it opens onto the skin surface.

ECDYSIS (MOULTING): The moulting of the cuticula in arthropods and related groups (ecdysozoa) or the periodic shedding of the outer cuticle or exoskeleton by insects and other arthropods to allow for growth or the periodic loss of the outer layer of the epidermis of reptiles (except crocodiles) to allow growth to occur. Since growth is constrained by the rigidity of the exoskeleton, the animal grows during this short period. In about one or two hours, the cuticle hardens and darkens following a tanning process similar to that of the tanning of leather. Growth of the limbs and other parts normally covered by hard exoskeleton is achieved by transfer of body fluids from soft parts before the new skin hardens. The old, empty exoskeleton is called an **exuvia** or **exuvium**.

ECDYSIS-TRIGGERING HORMONE: A peptide hormone, produced by neuro secretory cells in insects, that works in conjunction with other hormones such as ecdysone and eclosion hormone, to trigger moulting of the cuticle.

ECDYSONE: A steroid hormone produced by insects and crustaceans that stimulates moulting and metamorphosis or any of several insect hormones that triggers the expression of critical genes during larval development. It is a prohormone of the major insect moulting hormone, 20-hydroxyecdysone. Insect moulting hormones are generally called ecdysteroids and they not only act as moulting hormones of arthropods but also occur in other invertebrates where they play different roles. Ecdysteroids also appear in many plants mostly as protection agents (toxins or antifeedants) against herbivore insects. These hormones are being researched as safer alternatives to pesticides.

ECDYSOZOA: A group consisting of a common ancestor and all its descendants (clade) of protostome organisms, characterised by their habit of periodic moulting (ecdysis). These include the arthropods (insects, chelicerata, crustaceans and myriapods), nematodes, onychophorans and tardigrades.

ECG: Acronym for **electrocardiogram**, a graphical recording of the electrical changes in the heart muscle, as detected by electrodes placed on the chest or a transthoracic interpretation of the electrical activity of the heart over time as captured and externally recorded by skin electrodes.

ECHO SOUNDER (SONAR DEVICE): A device or type of sonar that detects objects under water by using reflected sound waves. It is used basically for determining the depth of water by means of an acoustic echo and made up of a transducer which acts as a transmitter and receiver of acoustic signal, an oscillator, a receiver and an amplifier which generate and receive the electrical impulses to and from the transducer and a recorder or other indicator, calibrated in terms of the depth of water. A pulse of sound sent from the ship is reflected from the sea bottom back to the ship, with the interval of time between transmission and reception being proportional to the depth of the water. Most boats are equipped with echo sounders to measure the water depth beneath them.

ECHELLE GRATING (ÉCHELLE GRATING): A diffraction grating in which the grooves are quite widely spaced and have a zigzag or step-like cross-sectional profile. The light to be dispersed is made to fall on the grating at right angles to the faces of the grooves. This has the effect of producing a series of overlapping spectra with a high degree of resolution. A second, low-dispersion grating or a prism, arranged perpendicular to the échelle, serves to separate out the overlapping spectra.

ECHELLE SPECTROGRAPH (ÉCHELLE SPECTROGRAPH): A spectrograph that uses an échelle grating to achieve high spectral resolution and wide wavelength coverage. The overlapping spectra produced by the grating are separated by a prism or a grism before reaching the detector. The sensitivity brought to modern échelle spectrographs by CCD detectors has made them indispensable in such fields as astrophysics and extrasolar planet searches.

ECHELON: A series of glass plates of equal thickness arranged like steps, used in spectroscopy for studying the fine structure of spectral lines.

ECHELON MATRIX (ECHELON FORM MATRIX): A matrix derived from another by a Gaussian elimination to perform elementary row operations to modify the matrix until the bottom left-hand corner of the matrix becomes zeros. A **row-reduced echelon** or **row echelon form** matrix has the following properties: (i) if there are any rows containing only zero elements, they are grouped together at the bottom of the matrix; (ii) the first non-zero element in each row, called the leading entry is 1; and (iii) the leading entry of each non-zero row after the first occurs to the right of the leading entry of the previous row. In a **column echelon form**, the reverse of the properties of the row echelon form apply. An example of a reduced

echelon form matrix is
$$\begin{bmatrix} 1 & 5 & 8 \\ 0 & 1 & 6 \\ 0 & 0 & 0 \end{bmatrix}.$$

ECHINATE: Covered with prickles or bristles as found in some plant parts.

ECHINODERM (ECHINODERMATA): Marine invertebrates such as starfish of the phylum Echinodermata (spiny skinned), characterised by pentaradial symmetry (body parts are arranged along five rays of symmetry) found in all oceans. They are biologically important because few other groupings are so abundant in the biotic desert of the deep sea, as well as the shallower oceans. On the other hand, they

are geologically important because their ossified skeletons are major contributors to many limestone formations, thus providing valuable clues to the geological environment. It is also believed that their radiation was responsible for the Mesozoic revolution of marine life.

ECHINULATE: Having very small prickles or spines.

ECHO: The reflected sound wave from a hard surface or wall. It is the repetitions of a sound wave, radar or sonar signal at the sound source by reflections from the surface or the reflection of a wave by a surface or object so that a weaker version of it is heard or detected shortly after the original. The reflection of the sound wave back to its source occurs in sufficient strength and with sufficient time lag to be separately distinguished. Since the velocity of sound is 344 m/sec at a normal room temperature of about 20°C (293 K), a reflecting surface must be more than 17.2 m from the sound source at this temperature for an echo to be heard by a person at the source. In this case the sound requires $\frac{1}{20}$ sec to reach the reflecting surface and the same time to return. When a surface reflects sound it partially absorbs and partially reflects the energy and as the process is repeated the sound becomes weaker and weaker and eventually ceases. A sound wave that is reflected again and again from different surfaces, as by parallel walls in a tunnel, is called **reverberation**. The application of echo may be shown by the following examples: (i) Radar sets broadcast radio waves, usually between 100 and 10,000 megahertz, pick up the portion reflected back by objects and electronically determine the distance and direction of the objects. (ii) Sonar and depth sounders work by analysing electronically the echo time lag of sound waves, generally between 10 and 50 kilohertz, produced by underwater transducers. (iii) Bats navigate by listening for the echo of their high-frequency cry.

ECHOCARDIOGRAM: A diagnostic test that uses ultrasound waves to create an image of the heart muscle. The picture is much more detailed than X-ray image and involves no radiation exposure. An echocardiogram provides information about the size and shape of the heart and how well the heart's chambers and valves are working. The test also can identify areas of heart muscle that are not contracting normally due to poor blood flow or injury from previous heart attack(s). In addition, a type of echocardiography called Doppler ultrasound shows how well blood flows through the chambers and valves of the heart. Echocardiography can detect possible blood clots inside the heart, fluid build-up in the sac around the heart (pericardium) and problems with the aorta (the main artery that carries oxygen-rich blood out of the heart).

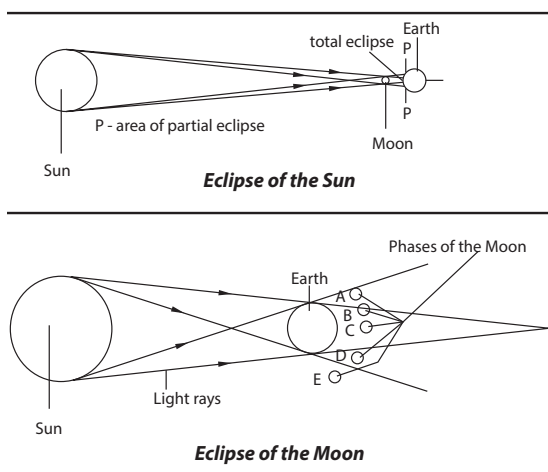
ECHOLICS: Agents or substances containing properties that initiate, hasten or accelerate child birth. They include herbs such as fenugreek, black cohosh, and raspberry leaf.

ECHOLOCATION: A method used by some animals such as bats, dolphins, porpoises, toothed whales and certain birds to detect objects in the dark. The animal emits a series of high pitched sounds that echo back from the object and are detected by the ear or some other sensory receptor. From the direction of the echo and the time between emission and reception of the echo the object is located, very often, accurately. Many scientists believe that these animals use sound to "view" objects in the same or greater detail than humans can see with reflected light.

ECLAMPSIA: The occurrence of convulsions not associated with any other disease, such as epilepsy, in a pregnant woman with abnormally high blood pressure and fluid retention (pre-eclampsia, toxemia of pregnancy). The condition can threaten the lives of mother and baby and must be treated immediately or coma and death will follow.

ECLIPSE: The total or partial hiding of light from a celestial or an astronomical body as it passes behind or through the shadow of another body. The term is usually employed for solar and lunar eclipses which may be partial or total. A solar eclipse occurs when the Moon is between the Sun and the Earth, while a lunar eclipse occurs when the Earth is between the Sun and the Moon. The obscuration of a planet

or star by the Moon or a planet is called an **occultation**, while the eclipse of a small portion of the Sun by Mercury or Venus is called a **transit of the planet**.



ECLIPSE SEASON: A period in a given year during which the Earth-Sun line falls near the line of nodes of the Moon's orbit, so that lunar and solar eclipses can take place. Eclipse seasons recur every 173.31 days and last about 24 days for lunar eclipses and about 37 days for solar eclipses. Two eclipse seasons make one eclipse year.

ECLIPSED (ECLIPSING): Two atoms or groups attached to adjacent atoms are said to be eclipsed if the torsion angle between the three bonds is zero or approximately so.

ECLIPSING BINARY: A double or binary star in which the two stars periodically pass in front of each other as seen from Earth.

ECLIPSING STRAIN (BOND OPPOSITION STRAIN; PITZER STRAIN; TORSIONAL STRAIN): Intramolecular strain present due to nonbonded interactions between two eclipsed atoms or groups. It is, for example, one of the interactions responsible for the existence of a rotational barrier and hence restricted rotation about the C-C bond in ethane.

ECLIPTIC: The great circle in which the plane of the Earth's orbit round the Sun intersects the celestial sphere or the intersection of the celestial sphere with the ecliptic plane, which is the geometric plane containing the mean orbit of the Earth around the Sun. It is thus, the Sun's apparent annual path across the sky or path against the background of stars that the Sun appears to follow each year as the Earth orbits the Sun. It appears to move westwards on an imaginary spherical surface (the celestial sphere) relative to the almost fixed stars.

ECLIPTIC COORDINATES: A system that uses the plane of the ecliptic as a reference plane. Ecliptic coordinates of objects with respect to the centre of Earth, known as geocentric ecliptic coordinates, are given in terms of celestial latitude (the perpendicular distance of the orbit from the ecliptic in angular measure) and celestial longitude (the angular distance along the ecliptic between the plane through the object and the vernal equinox). Ecliptic coordinates referring to the centre of the Sun, known as heliocentric ecliptic coordinates, are given in terms of heliocentric latitude and heliocentric longitude.

ECLOGITE: A coarse to medium grain metamorphic rock consisting mainly of green pyroxene and red garnet. A very dense rock, it is formed deep in the Earth's crust under conditions of very high temperature and pressure. It occurs in schists in Norway and in diamond-bearing pipes of kimberlite (South Africa).

ECLOSION HORMONE: A peptide hormone secreted by cells of an insect's nervous system, especially the brain that triggers the sequence of events leading to emergence of the adult from the pupa.

EOCLINE: A cline that has been shown to be due to a specific environmental factor. For example, a gradual increase in

heavy-metal tolerance can be correlated with increasing metal concentration in the soil.

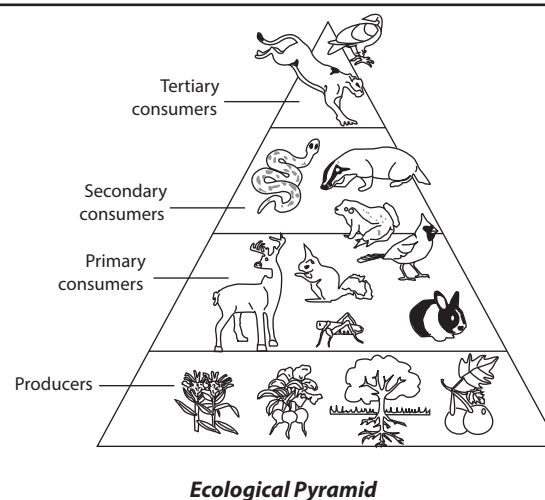
ECOLOGICAL BALANCE (FOOD BALANCE): The balance between food supply and the demand for food from a population. It is a state of dynamic equilibrium within a community of organisms in which genetic, species and ecosystem diversity remain relatively stable, subject to gradual changes through natural succession. This balance may be disturbed due to the introduction of new species, the sudden death of some species, natural hazards or man-made causes.

ECOLOGICAL EQUIVALENCE: Unrelated organisms that occupy similar habitat and resemble each other.

ECOLOGICAL ISOLATION (HABITAT ISOLATION): A form of reproductive isolation in which closely related species live in the same territory, but live in different habitats making them to reproduce in isolation of the others in order to prevent interbreeding between species.

ECOLOGICAL NICHE: The status, role or relational position to each other, of an organism in its environment or echo system. This is defined by the types of food it consumes, its predators, temperature, tolerances, etc. Two species cannot exist stably if they occupy identical niches. It, therefore, describes how an organism or population responds to the distribution of resources and competitors, for example, by growing when resources are abundant and when predators, parasites and pathogens are scarce and how it in turn alters those same factors such as limiting access to resources by other organisms, acting as a food source for predators and a consumer of prey.

ECOLOGICAL PYRAMID: A graphic or diagrammatic representation of the quantitative relationships of numbers of organisms, biomass or energy flow between the trophic levels of an ecosystem. Because large amounts of energy and biomass are dissipated at every trophic level, these diagrams nearly always take the form of pyramids. Ecological pyramids are often represented in a way that the producers are at the bottom level and then proceeds through the various trophic levels in which the highest is on top. There are three different types; pyramid of numbers, pyramid of biomass and pyramid of energy.



ECOLOGICAL PYRAMID OF BIOMASS (BIOMASS PYRAMID): A graphical representation to show the relative masses of consumers and food are represented at the same time. There are two types of biomass pyramids: the upright and the inverted. The upright type occurs in most ecosystems and results when the combined weight of producers is larger than the combined weight of consumers. An inverted type results when the combined weight of producers is smaller than the combined weight of consumers.

ECOLOGICAL PYRAMID OF ENERGY (ENERGY PYRAMID; PYRAMID OF ENERGY): A type of ecological pyramid graphical that shows how the energy flows through a food chain, how the amount of energy is decreasing and becoming less available for organisms

as it enters each trophic level and how much of the energy in the ecosystem is lost to the atmosphere as heat. Since there is always a loss of energy at each trophic level the energy pyramid is always upright.

ECOLOGICAL RELEASE: A population increase in which a species expands its living habitat as well as the resources it utilises into areas that have a lower density of species in terms of diversity.

ECOLOGICAL STATISTICS: Procedures for studying the dynamics of natural communities and their relation to the environmental variables.

ECOLOGICAL SUCCESSION (SUCCESSION): In ecology, it is the sequence of ecological communities that develop in an area from the initial stages of colonisation until a stable mature climax community is reached or the series of changes that create a full-fledged plant and animal community, such as from the colonisation of a bare rock to the establishment of a forest. There are two types of ecological succession: **primary succession**, which is the ecological succession that occurs following an opening of uninhabited, barren habitat or that occurs on an environment that is devoid of vegetation and usually lacking topsoil; and **secondary succession**, which is the ecological succession that occurs on a pre-existing soil after the primary succession has been disrupted or destroyed due to a disturbance that reduced the population of the initial inhabitants. Numerous factors which may be brought about or by the colonisers affect the nature of the succession. For example, shrubs produce deep soil, which in turn produce trees, which may attract animals seeking to use them as shelter, whose droppings act as manure which results in the growth of other plants, etc.

ECOLOGICAL SYSTEM (ECOSYSTEM): A self-sustaining unit comprising a community of living organisms and their environment. There is a continuous flow of energy and matter through the system. Ecosystems may be small or large and simple or complex in structure. They range from small freshwater ponds or pools to the earth itself.

ECOLOGY (ENVIRONMENTAL BIOLOGY): A sub discipline of biology that studies the ways in which communities of plants and animals interact with one another and with their non-living environment. It is concerned with the patterns of distribution and abundance in space and time, seeking to explain the factors that determine the range of environments that organisms occupy and that determine how abundant they are within those ranges. It also emphasises functional interactions between co-occurring organisms. It is both a synthetic and an integrative science, drawing information and concepts from other sciences such as physiology and meteorology, to explain the complex organisation of nature.

ECOMORPHOLOGY (ECOLOGICAL MORPHOLOGY): The study of the relation between an individual's ecological role and its form and structural adaptations.

ECONOMETRICS: It is the application of statistical methods to economics data. Economists use statistics to test their theories or to make forecasts. Since economics data are not experimental in nature and often contain some randomness, econometrics uses stochastic models rather than deterministic models. Examples are regression model and time series model.

ECONOMIC POTENTIAL: In climatology, it is the portion of technological potential for greenhouse gas emissions reductions or energy efficiency improvements that could be achieved cost-effectively through the creation of markets, reduction of market failures, or increased financial and technological transfers.

ECONOMICAL NUMBER: A number that has no more digits than there are digits in its prime factorisation (including powers). If a number has fewer digits than are in its prime factorisation it is known as a **frugal number**. The smallest frugal number is 125, which has three digits, but can be written as 5^3 , which has only two. The next few frugals are 128 (2^7), 243 (3^5), 256 (2^8), 343 (7^3), 512 (2^9), 625 (5^4) and 729 (3^6). An **equidigital number** is an economical number that has the same number of digits that make up its prime factorisation. The smallest

equidigitals are 1, 2, 3, 5, 7 and $10 = 2 \times 5$. All prime numbers are equidigital. An **extravagant number** is one that has fewer digits than are in its prime factorisation. The smallest extravagant number is $4 = 2^2$, followed by 6, 8 and 9. There are infinitely many of each of these kinds of numbers.

ECONOMY: A mathematical model of an economic system in which, commonly, there are m producers, each with a production set P_i and n consumers, each with a consumption set C_i and an associated ordering $\leq_i$. These sets lie on an Euclidean space, whose dimension corresponds to the number of goods in the economy. There is a total resource level w . The producers each produce a vector y_i in P_i and the consumers have a demand x_i in C_i . These $m + n$ elements define a state of the economy. The excess demand is $\sum x_i - \sum y_i - w$ and a market equilibrium occurs if excess demand is zero. Individual consumers will try to maximize their preference satisfaction; this leads to the study of **equilibria** that occur as a market equilibrium together with a pricing of resources that every producer maximizes profit and every consumer optimizes satisfaction.

ECOSTATE: A leaf without a mid-rib or nerves.

ECOSYSTEM: A unit made up of a group or community of organisms consisting of living things such as plants and animals (biotic components) interacting with each other and with their non-living or physical environment (abiotic components) such as air, soil, water and sunlight. Ecosystems can be permanent or temporary and they usually form a number of food webs.

ECOSYSTEM APPROACH: A method of resource management for the integrated management of land, water and living resources that promotes conservation and sustainable use in an equitable way. It acknowledges that the different components of an ecosystem including structure, function, and species composition are interlinked, and this factor must be taken into consideration while restoring and protecting the ecosystem's natural balance.

ECOSYSTEM MANAGEMENT: The use of good management practices to conserve the environment such as taking appropriate measures to reduce pollution, controlling the use of fertilisers and pesticides and adopting sensible methods of agricultural practices such as crop rotation, etc. Its goals are more proactive than reactive and aim at protecting a habitat as best as possible. It also protects the connections between different ecosystems.

ECOSYSTEM SERVICES: A multitude of resources and processes that are supplied by natural ecosystems, from which humans receive benefits. They are divided into four broad categories namely, provisioning, regulating, supporting and cultural. Provisioning ecosystem sources include products such as food, fibres, fuel, drinking water and medicinal products. Regulating ecosystem services include such processes as control of the climate, purification of air and water, amelioration of erosion, reduction of pests and diseases and provision of pollination mechanisms. Cultural ecosystem services comprise the non-material benefits that accrue from the landscape, such as recreational pursuits, spiritual renewal and aesthetic purposes. Supporting ecosystem services include primary productive nutrient cycling, water cycling, decomposition of wastes and soil formation.

ECTOMESENCHYME: A mass of tissue consisting of neurocrest cells present in the early formation of an embryo. It eventually forms the hard and soft tissues of the neck and cranium.

ECOTONE: A transition zone or region separating two biomes. For example, in central Asia an area of wooded steppe, the grove belt, separates the coniferous forest from the temperate grassland.

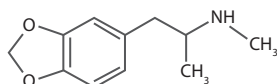
ECOTOURISM: Responsible visit to sites that are of ecological interest that conserves the environment and sustains the well-being of the local people such as bringing about employment and income. The main attractions in the sites visited are the flora, fauna and cultural heritage. Ecotourism minimises physical, social, behavioural and psychological impacts and does not do damage to the environment as much as tourism in which thousands of people go to the same place, often at the same time of year.

ECOTOXICITY: The degree to which a chemical released into an environment by human activities affects the organisms that live or grow there.

ECOTYPE: A subspecies or breed, that has adapted to its local environment and as a result is distinctive from other members of the species, but can still successfully interbreed with other members of the species. In some cases individuals belonging to different ecotypes cannot interbreed, for example, where accumulated genetic differences are too great.

ECSTASY (METHYLENEDIOXYMETHAMPHETAMINE; MDMA):

A drug, with the chemical formula $C_{11}H_{15}NO_2$, initially intended as an appetite suppressant, which produces a feeling of euphoria and hallucinogen. It is derived from the amphetamine family and a relative of the stimulant methamphetamine. Apart from increasing the amount of the neurotransmitter dopamine, it also increases the production of the neurotransmitter serotonin and blocks its re-absorption in the brain. It stimulates the central nervous system and gives users a feeling of increased energy and lower social inhibition. It is now an illegal drug used as a stimulant and relaxer of inhibitions.



Chemical Structure
of Ecstasy

ECTOCARP (EXOCARP): The outermost layer of the pericarp of an angiosperm fruit that is outside of the mesocarp. It is usually thin.

ECTODERM: The external layer of cells or tissue of the gastrula in a developing animal embryo which develops into the epidermis, the outermost parts of the nervous system in the adult and various other outer and external organs, structures and systems, depending on the organism. It occurs in all animals except sponges and placozoans.

ECTOPARASITE: A parasite that lives on the surface of its host's body, for example, the skin or attached to hair follicles. Ectoparasites include those that stay permanently on their hosts' bodies such as lice, ticks and fleas; and those such as tsetseflies, *Glossina palpalis* that stay on their hosts' bodies only, when feeding. Ectoparasites do not contribute to the benefit or survival of the host. The species of flea that most commonly affects animals and humans is *Ctenocephalides felis*. Lice can live on farm animals like cattle, sheep, goats and dogs; and also in the hair of humans. The human head lice is *Pediculus humanus capitis*. Ticks may be seen on the legs and ventral surface of the body, ears and neck of mammals, birds and reptiles feeding on blood. Tsetseflies feed on the blood of humans and animal. Ectoparasites such as lice, ticks and fleas are transmitted by host to host or by fomites to host. A **fomite** is any object or substance such as a dish, toy, book, doorknob or clothing that is capable of carrying infectious organisms, such as germs or parasites. Some common examples of ectoparasites are lice, tick and tsetsefly.

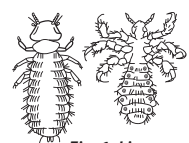


Fig. I: Lice

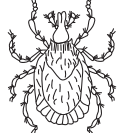


Fig. II: Tsetsefly



Fig. III: Tsetsefly

Examples of Ectoparasite

ECTOPIC PREGNANCY: Pregnancy that occurs in the Fallopian tube or elsewhere outside the uterus or womb. The embryo cannot develop normally and spontaneous abortion often occurs. If not, urgent surgery is necessary to save the mother from serious, possible fatal haemorrhage.

ECTOPLASM: The outer, non-granulated part of a cell's cytoplasm that surrounds the nucleus.

ECTOTHERM: Cold-blooded animals that rely on external warmth either by absorbing radiation or through conduction to raise their body temperature so that they can become active. To cool their body, they have to seek out a cooler environment. Ectotherms include the fishes, amphibians, reptiles and invertebrates. **Ectothermy** is the process an organism adopts to maintain its body temperature by availing itself of heat from the environment.

ECTOTROPHIC MYCORRHIZA (ECTOPHYTIC MYCORRHIZA):

A form of mycorrhizal association in which there is a well developed mycelium forming a mantle on the outside of the root. Such associations are found on most trees and often a tree will not grow properly unless the appropriate fungus is present.

ECZEMA: A general term for any type of inflammatory skin condition characterised by dryness, rashes, itching, formation of blister and the exudation of fluid that typically affects the insides of the elbows, backs of the knees, the face and sometimes covering most of the body. The commonest is atopic dermatitis, often called eczema. Other types of eczema include **contact dermatitis** caused by contact with irritants and characterised by burning, itching and redness; **dyshidrotic dermatitis** affects mostly affects women causing blistering to the fingers, palms and soles of the feet. fingers, palms of the hand and soles of the feet, characterised by itchy, scaly patches of skin that flake constantly or become red, cracked and painful; **nummular dermatitis** usually affects the leg causing dry, round patches of skin. It is more common in men. **Seborrheic dermatitis** causes itchy, red, scaly rashes, particularly on the scalp, on the eyebrows, on the eyelids, on the sides of the nose and behind the ears.

EDAPHIC CLIMAX: A stable ecological community that results from the content or properties of the soil rather than the climate.

EDAPHIC FACTORS: Abiotic factors relating to the physical or chemical composition of the soil found in a particular area or environment that affects the lifestyle of an organism living in that habitat. They include the composition and structure of the soil. For example, a very alkaline soil may be an edaphic factor limiting the variety of plants growing in a region.

EDAPHOLOGY: The science that deals with the influence of soils on living things, particularly plants, including man's use of land for plant growth. Its subfields include agricultural soil science or agrology and environmental soil science. It is one of two main divisions of soil science, the other is **pedology**, which is the scientific study of soils and their weathering profiles.

EDDINGTON ECLIPSE MEASUREMENTS: Tests conducted by British astronomer and physicist, Arthur Stanley Eddington in 1919 during the eclipse of the Sun to measure the bending of light at a total solar eclipse of the apparent change in position of a star as predicted by Albert Einstein that the bending of light by the gravity of the Sun, one of the components of his general theory of relativity, can be tested during a solar eclipse, when stars with apparent position near the Sun become visible. He first measured the "true" positions of the stars in January and February 1919. Then in May he measured the stars' positions during the eclipse on the island of Príncipe in the Gulf of Guinea off the west coast of Africa, as viewed through the Sun's gravitational lens. The group also took measurements from Sobral, Brazil, in case the eclipse was blocked by clouds over Príncipe. Both locations had clear skies, and the astronomers took several pictures during the six minutes of total eclipse. These measurements were one of the earliest confirmations of general relativity theory.

EDDINGTON LIMIT (EDDINGTON LUMINOSITY): A theoretical limit for the maximum value of the brightness of a star of a given

mass, named after British astronomer and physicist, Arthur Stanley Eddington and given as $L = \frac{4\pi GMm_p c}{\sigma_T}$, where G is the gravitational pressure, M is the mass of the luminous object, m_p is the mass of a proton, c is the speed of light and σ_T is the effective area of an electron when it is illuminated by radiation. This limit exists because the radiation pressure caused by the nuclear fission reactions powering the star has to counter, but not exceed the gravitational force that would cause gravitational collapse of the star. A star emitting radiation at greater than the Eddington limit would break up. In a star it is defined as the point where the gravitational force inwards equals the continuum radiation force outwards, assuming hydrostatic equilibrium and spherical symmetry. When exceeding the Eddington luminosity, a star would initiate a very intense continuum driven stellar wind from its outer layers. Since most massive stars have luminosities far below the Eddington limit, however, their winds are mostly driven by the less intense line absorption.

EDDY: In turbulent fluid motion, a blob of the fluid that has some definitive character and moves in some way differently from the main flow.

EDDY COVARIANCE TECHNIQUE: A micrometeorological technique for simultaneously measuring fluxes of carbon dioxide, water vapour and energy (heat) above a plant canopy and hence the heat exchange of carbon between an ecosystem and the atmosphere (by measuring the covariance between fluctuations in vertical wind velocity and carbon dioxide mixing ratio). This technique has a wide application in ecological studies, such as for determining the amounts of carbon dioxide that are absorbed or emitted by different types of vegetation cover and whether particular sites are sinks or sources of carbon. This method is most accurate when the atmospheric conditions such as wind, temperature, humidity, carbon dioxide are steady, the underlying vegetation is homogeneous and it is situated on flat terrain for an extended distance upwind.

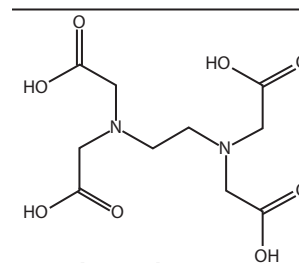
EDDY CURRENT (FOUCAULT CURRENT): An electric current induced in a conductor located in a changing magnetic field or when that conductor moves through a non-uniform magnetic field. The conductor could be a piece of metallic sheet such as the supporting frame of an electrical machinery like transformers and motors. The effect is most intense in solid metallic conductors, although they can be induced in any electrical conductor. They can cause much wasted energy in the cores of transformers and other electrical machines; however, they can be useful in the speedometers of cars, in induction heating and in damping out oscillations in various devices. Such currents can be reduced by laminating a conductor with many thin sheets that are insulated from each other rather than making it a single solid piece. The laminations must be properly oriented to cut across the paths of the eddy currents and by so doing, confine the currents largely to single laminae, where the paths are long, making higher resistance.

EDDY DISPERSION (DIFFUSION): The process by which substances are mixed in the atmosphere or in any fluid system due to eddy motion.

EDENTATE: A member of one of the four families of mammals making up the order Edentata or Xenarthra. They are the anteaters, armadillos, two-toed sloths and three-toed sloths. Currently, the armadillos are classified in the order Cingulata, while the sloths and anteaters belong to the order Pilosa. The anteaters are completely toothless but the sloths and armadillos have primitive molars. Extinct edentates include the giant ground sloth (Megatherium and the two enormous armadillos, Doedicurus and Glyptodon).

EDETIC ACID (ETHYLENEDIAMINETETRAACETIC ACID; EDTA; ETHYLENEDINITRILO-TETRAACETIC ACID):

A colourless, water-soluble polyamino carboxylic acid, with the chemical formula $C_{10}H_{16}N_2O_8$. It is a hexadentate ligand and chelating agent of metal ions such as Fe^{3+} , Ca^{2+} and Mg^{2+} . It is produced as several salts, including disodium EDTA and calcium disodium EDTA. It is widely used to dissolve scale.



Chemical Structure of Edetic Acid

The sodium salt is used as an antidote for metal poisoning, an anticoagulant and an ingredient in a variety of industrial reagents.

EDGE: 1. A line segment joining two vertices or corners of a two-dimensional shape or two faces of a three-dimensional object. A cube, for example, has 12 edges. 2. A directed or undirected line segment joining two nodes or vertices of a graph. 3. Acronym for **Enhanced Data rates for Global Evolution** or **Enhanced Data rates for GSM Evolution**, a GSM mobile cellular phone technology that provides improved data transmission rates nearly three times faster than the GPRS system. Although it meets the requirements for a 3G network, it is usually classified as 2.75G. It can provide data rates of 384 kbps to 473.6 kbp. To use this technology, one needs to have a GSM EDGE handset that is EDGE compatible.

EDGE CONNECTOR: In computing, the broad metallic tracks on a printed circuit board that plugs directly into a matching socket to provide the electrical connection.

EDGE DEVICE: 1. Any server or other networking device that is located closer to the client machines rather than being in the backbone of the network. For example, a cache server is an edge device that sits inside the firewall and holds frequently requested pages. 2. A network device used to convert LAN frames (Ethernet, etc.) to ATM cells and vice versa. It is typically a switching device with one ATM port and multiple LAN ports.

EDGE EFFECT: 1. An increase in growth and yield in crop plants growing at the edge of a plot or field. 2. The effect of juxtaposing or placing side by side of contrasting environments on an ecosystem. The term is commonly used in conjunction with the boundary between natural habitats, especially forests and disturbed or developed land. It is especially pronounced in small habitat fragments, where they may extend throughout the patch. Increasing edge effects allow more habitat structure to increase biodiversity within the area.

EDGE-WEIGHTED GRAPH (WEIGHTED GRAPH): A graph having a weight or number, associated with each edge. Some algorithms require all weights to be nonnegative, integral, positive, etc.

EDGEWORTH BOX: A diagrammatic representation in general equilibrium analysis that uses the indifference curve analysis to analyse the interaction of two individuals trading two different commodities. It is named after its inventor, Irish economist, Francis Ysidro Edgeworth (1845- 1926). The satisfaction of the first trader increases from bottom left to top right corner and that of the second correspondingly decreasing.

EDGEWORTH-PARETO POINT (EFFICIENT POINT; PARETO EFFICIENCY; PARETO OPTIMALITY): A state of allocation of resources in which it is impossible to make any one individual better off without making at least one individual worse off.

EDI: In computing, it is the acronym for **electronic data interchange**, a set of standards for exchanging business data that helps specify how electronic data such as orders, confirmations, tracking shipments and electronic invoice should be handled or formatted between organisations.

EDIFACT: Acronym for **Electronic Data Interchange For Administration, Commerce and Trade**, a set of internationally agreed standards, directories and guidelines for conducting **electronic data interchange (EDI)** transactions.

EDISON CELL (EDISON ACCUMULATOR; NICKEL-IRON BATTERY; NiFe BATTERY): A storage battery developed by the American inventor, Thomas Edison in the 1900s. It has a nickel(III) oxide-hydroxide anode and an iron cathode, with an electrolyte of potassium hydroxide held in nickel-plated steel tubes or perforated pockets. It is a very robust battery which can withstand overcharge, over discharge and short-circuiting and has a very long life, more than 20 years. Its disadvantages are low specific energy, poor charge retention, of giving off hydrogen gas during charging and high cost of production. This battery is used principally in heavy industry applications. They are currently gaining popularity for off-the-grid applications, where daily charging makes them an appropriate technology.

EDISON EFFECT: The flow of electric current when an electrode sealed inside the bulb of an incandescent lamp is connected to the positive terminal of the lamp. The current flows only when the heated wire is more negative, because it is carried by free electrons released from the wire by its heat. The Edison effect, named after its discoverer, Thomas Edison, was the principle behind "vacuum tubes" used in radio and television receivers and transmitters before the invention of the transistor.

EDMAN DEGRADATION: A systematic method of sequencing proteins or amino acids in a peptide, proceeding by stepwise removal of single amino acids from the amino terminal of a polypeptide chain. It was developed by Pehr Edman and involves labelling and cleaving the amino-terminal residue from the peptide without disrupting the peptide bonds between other amino acid residues.

EDO RAM: Acronym for **extended data out random-access memory**, a faster type of RAM introduced in the mid-1990s that improved the performance of fast page mode (FPM).

EDP: In computing, it is the abbreviation for **electronic data processing**, the use of computers in recording, classifying, manipulating and summarising data.

EDUCATIONAL PSYCHOLOGY: The branch of psychology concerned with the scientific study of human learning from both cognitive and behavioural perspectives, including development, learning, motivation, instruction, assessment and related issues that influence the interaction of teaching and learning.

EDUTAINMENT (EDUCATIONAL ENTERTAINMENT): Any entertainment program such as a computer-aided form of learning that is designed to educate as well as to entertain. Most computer games designed for children entertain and also educate them.

EEG: Abbreviation for **electroencephalogram**, which in medicine is the graphic measuring and recording of the electrical activity in the brain by placing between 16-20 electrodes on the scalp which are connected to an electroencephalograph and electronic instrument. It is helpful in the diagnosis of some brain disorders, such as epilepsy, tumours or brain damage as well as in establishing clinical death.

EEPROM: Acronym for **Electrically Erasable Programmable Read-Only Memory**, a non-volatile rewritable memory chip that maintains its data without electrical power. Its contents can be erased and reprogrammed. EEPROMs are typically used on circuit boards to store small amounts of instructions and data.

EFFECTIVE: Of a procedure, able to be carried out mechanistically by following a finite number of steps of an algorithm; in particular, a value may be effectively computable.

EFFECTIVE APERTURE: 1. In telecommunications, it is also known as **antenna effective area** and is the functionally equivalent area from which an antenna directed toward the source of the received signal gathers or absorbs the energy of an incident electromagnetic wave. It is usually expressed in square metres. It determines how much

an antenna is receiving radio waves. 2. In optics, it is the diameter of the image of the aperture stop of an optical system, as viewed from the object; especially the apparent diameter of the diaphragm opening in a camera lens as seen through the front of the lens.

EFFECTIVE BANDWIDTH: 1. The width of the band in wavelength units at half-peak height. 2. The bandwidth of an assumed rectangular band-pass having the same transfer ratio at a reference frequency as a given actual band-pass filter and passing the same mean-square value of a hypothetical current having even distribution of energy throughout that bandwidth. The concept is used for wavelength selectors or monochromators.

EFFECTIVE CADMIUM CUT-OFF ENERGY: In a given experimental configuration, the energy value determined by the condition that the detector response would be unchanged if the cadmium cover surrounding the detector was replaced by a fictitious cover opaque to neutrons with energy below this value and transparent to neutrons with energy above this value.

EFFECTIVE CHARGE: Change in effective charge is a quantity obtained by comparison of the polar effect of substituents on the free energies of rate or equilibrium processes with that on a standard ionisation equilibrium. Provided the effective charges on the states in the standard equilibrium are defined, then it is possible to measure effective charges for states in the reaction or equilibrium under consideration.

EFFECTIVE CHIMNEY HEIGHT (STACK HEIGHT): A height used for the purposes of calculating the dispersion of emitted gases from a chimney and which differs from the real chimney height by an amount which depends on such factors as the exit velocity, buoyancy effects and wind speed. It may also be affected by the local topography. It denotes the maximum height of the centre of a plume path above the level of the ground. The effective height may be above or below the actual chimney height, although the former case is most common.

EFFECTIVE COLLISION: A collision in which the particles meet with sufficient energy and an orientation that allows them to react.

EFFECTIVE DOSE (ED): 1. In pharmacology, it is the dose or amount of drug that produces a therapeutic response or desired effect in some fraction of the subjects taking it. 2. In radiation, it is the measure of the stochastic effect that a radiation dose to part of the body will have on the whole organism. It is obtained by taking the equivalent dose (dose equivalent) and multiplying by a tissue weighting factor which relates to the organs/tissues under consideration. Effective dose can, therefore, also be considered a doubly weighted absorbed dose since it takes into account the type of radiation (radiation weighting factor) and the target organ/tissue. The quantity can be used to express detriment to the whole body as a summation of several different doses of radiation with varying radiation weighting factors (radiation type) and targets.

EFFECTIVE HALF-LIFE: The time required for the amount of a radioactive substance deposited in a living organism to be reduced by 50% as a result of the combined action of radioactive decay and biologic elimination.

EFFECTIVE MOLALITY: The sum of the concentration of all solute particles in a solution.

EFFECTIVE NUCLEAR CHARGE: The apparent nuclear or net positive charge which an outermost electron actually experiences through attraction in a multi-electron atom as a result of the shielding effect by other electrons. Its symbol is Z_{eff} . Since the shielding effect of negatively charged electrons prevents higher orbital electrons from experiencing the full nuclear charge by the repelling effect of inner-layered electrons, the term 'effective' is used to qualify the nuclear charge. The strength of the nuclear charge can be determined by looking at the oxidation number of the atom. Where there is only one electron in an atom, that electron experiences the full charge of the positive nucleus, in which case, the effective nuclear charge can be calculated from Coulomb's law. However, in a multi-electron atom the outer electrons are simultaneously attracted to the positive nucleus

and repelled by the negatively charged electrons. The effective nuclear charge on such an electron is given by the expression: $Z_{\text{eff}} = Z - S$, where, Z is the number of protons in the nucleus (atomic number) and S is the average number of electrons between the nucleus and the electron in question (the number of nonvalence electrons).

EFFECTIVE POPULATION SIZE: In ecology, it is the average size of a population expressed in terms of individuals assumed to contribute offspring equal to the next generation.

EFFECTIVE PERMEABILITY: In geology, it is the measure of the ability of a particular fluid to flow through a rock when other immiscible fluids are also present in the pore spaces.

EFFECTIVE PRECIPITATION: The portion of the total precipitation that is actually added and stored in the soil and becomes available for plant growth. If less than 5mm of daily rainfall, it would not be considered effective, as this amount of precipitation would likely evaporate from the surface before soaking into the ground.

Ineffective precipitation is that portion which is lost by surface run-off, unnecessary deep percolation losses, the moisture remaining in the soil after the harvest of the crop and which is not useful for next season's crop.

EFFECTIVE RADIUS: The distance from the centre of a galaxy from within which half the galaxy's luminosity is emitted.

EFFECTIVE TEMPERATURE: 1. The effective temperature of a body such as a star or planet is the temperature of a black body that would emit the same total amount of electromagnetic radiation, often used as an estimate of a body's temperature when the body's emissivity curve (as a function of wavelength) is not known. 2. In meteorology, it is the temperature at which motionless, saturated air would induce, in a sedentary worker wearing ordinary indoor clothing, the same sensation of comfort as that induced by the actual conditions of temperature, humidity and air movement. 3. In statistical mechanics, it is that temperature which can be inserted in the Boltzmann distribution formula to describe the relative populations of two energy levels which may or may not be in thermal equilibrium.

EFFECTIVE VALUE (ROOT MEAN SQUARE; RMS): A statistical measure of the magnitude of a varying or alternating current or voltage, which is equal to the product of 0.707 times the peak amplitude. It is useful in sinusoids. RMS is used in various fields, including electrical engineering. One of the more prominent uses of rms is in the field of signal amplifiers.

EFFECTOR: 1. A specialised animal tissue, cell or organ that produces a physiological response to stimulus from the environment, when stimulated by a nerve impulse or motor neurones. There are two types of effectors, activators and inhibitors. Examples include a molecule affecting enzyme activity or a nerve ending activating a muscle. 2. A molecule, including small and regulatory ones, that binds to a protein and thereby changes the activity of that protein. A modulator molecule binds to a regulatory site during allosteric modulation and modulates the shape of the protein allosterically (interaction inducing altered activity at another site).

EFFECTOR NEURONE: A nerve cell, such as a motor neurone, that transmits impulses from the central nervous system to an effector in order to bring about a physiological response to changes in the environment.

EFFERENT: Carrying blood, nerve impulses, etc, away from the centre of a body or organ towards peripheral regions. The term is usually applied to describe nerve fibres that carry impulses away from the brain or spinal cord or blood vessels that carry blood away from an organ. For example, an efferent artery carries blood away from the kidney.

EFFERVESCENCE: Gas bubbles forming in a liquid as a result of a chemical reaction or the escape of gas from an aqueous solution and the foaming or fizzing that results from a release of the gas; or the bubbling of a solution of an element or chemical compound as a result of the emission of gas, without the application of heat. For example, if a few pieces of marble or an antacid tablet are put in hydrochloric acid in a test tube fitted with a cork, effervescence of carbon dioxide can be observed. It is also seen if hydrochloric acid is added to a

block of limestone. The escape of carbon dioxide from carbonated water is another example. This process is generally represented by the following reaction, where a pressurised dilute solution of carbonic acid in water releases gaseous carbon dioxide at decompression: $\text{H}_2\text{CO}_3 \rightarrow \text{H}_2\text{O} + \text{CO}_2$.

EFFICIENCY: 1. A measure of the performance or how well a machine or other energy changing device changes energy into useful work. It is the measure of the percentage ratio of work output to work input.

For a simple machine, it is given by: $E = \frac{\text{Energy Output}}{\text{Energy input}} \times 100\%$. For

example, a gear transmission is 97% efficient, when the useful energy output is 97% of the input, with the other 3% being lost as heat due to friction. Automobile engines have low efficiency (below 30%) because out of the total energy content of fuel converted to heat, only a small portion provides motive power, while a substantial amount is lost in the radiator and car exhaust. 2. In chemistry, it is the ratio between the useful energy delivered or bound and the energy supplied: energy output/energy input. It is also used in the sense of a quantitative measure of the relative rate of a given step involving a species with respect to the sum of the rates of all of the parallel steps which depopulate that species. Its symbol is η .

EFFICIENCY OF ATOMISATION/ATOMIZATION: The ratio of the amount of analyte that passes through the flame cross-section at the observation height, as free neutral (or ionised) atoms, to the amount of analyte aspirated.

EFFICIENCY SPECTRUM: A plot of the efficiency of a step (η) against wavelength or photon energy.

EFFICIENT CODE: In information theory, an error-correcting code containing a high ratio of information to word-length.

EFFICIENT POINT (EDGEWORTH-PARETO POINT; PARETO EFFICIENCY; PARETO OPTIMALITY): A state of allocation of resources in which it is impossible to make any one individual better off without making at least one individual worse off.

EFFLORESCENCE: The process in which hydrated crystalline solids lose part of their water of crystallisation as vapour spontaneously to the atmosphere when exposed to air, resulting in a dry powdery surface. The thermodynamic condition for efflorescence is that the partial pressure of water vapour at the surface of the solid (its dissociation pressure) must exceed the partial pressure of water vapour in the air. The spontaneous loss of water normally requires that the crystal structure be rearranged and consequently, efflorescent salts usually become microcrystalline powders when they lose their water of hydration. An example of an efflorescent substance is hydrated sodium carbonate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$).

EFFLUENT: 1. Liquid waste discharged from a factory, sewage system, nuclear power station or other industrial plant. 2. A stream or river that flows out of a larger body of water such as a larger stream or a lake. 3. In chromatography, it is the mobile phase, which leaves the column.

EFFORT: The force given to a machine to move a load and expressed as: $\text{Effort} = \text{Load (Force)}/\text{Distance}$.

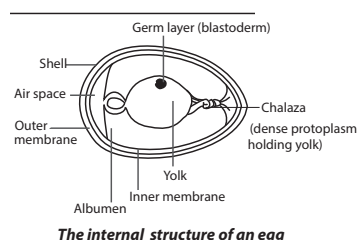
EFFORT ARM (IN-LEVER ARM): The part of a lever that applies force to the resistance arm.

EFFUSION: 1. The process in which individual molecules flow through an aperture without collisions between molecules, occurring when a gas flows through an aperture of a diameter, which is considerably smaller than the mean free path of the molecules. 2. The escape of a fluid from anatomical vessels by rupture or exudation.

EFFUSIVE ERUPTION: An eruption dominated by the outpouring of lava onto the ground. Lava flows generated by effusive eruptions vary in shape, thickness, length and width depending on the type of lava erupted, discharge, slope of the ground over which the lava travels and duration of eruption. An effusive eruption is distinct from the violent fragmentation of magma by an explosive eruption.

EGESTION: Discharging or excreting from the body. The term generally refers to the removal of undigested or waste food materials or faeces from the intestine or body.

EGG: A reproductive system in female egg-laying animals which is an organic vessel that facilitates the embryonic development of a zygote (the union of sperm (male sex cell) and an ovum (female sex cell) until the eggs mature to be laid and hatched. For an ovum to become an egg, the transfer of the genetic material should take place. The matured egg itself is spherical or ellipsoidal in shape.



The internal structure of an egg

E

EGG BINDING (EGGBOUND): An unsuccessful attempt by a hen to lay egg. In farming, aviculture or bird rearing and animal husbandry, the term refers to a medical condition in birds where the female produces egg and yet is unable to pass out the egg that has formed. The egg may be stuck near the cloaca or further inside. It is a common and often dangerous condition that can lead to infection or damage to internal tissue. The bound egg may be gently massaged out or broken *in situ* and removed in parts. The oviduct should be cleaned of shell fragments and egg residue to avoid damage or infection, if the egg is broken.

EGG CELL (OVUM): A mature female reproductive cell in animals and plants or more specifically, a fertilised ovum in egg-laying animals such as birds and insects, especially a female sex cell when it is ready for fertilisation. The mature ova (plural of ovum) are generally spheroidal and large and the number of ova produced at one time varies from millions in many marine animals that spawn into the surrounding sea water to about a dozen or less in mammals in which adaptations for internal nourishment of the developing embryo and care of the young are highly developed. In lower plants and algae, the ovum is also called oosphere. The internal structure of an egg of an egg-laying animal is shown in the diagram.

EGG DUMPING: A method adopted by a brood parasite to get another bird (the host) to hatch and raise its bird by laying its eggs in the nest of the host.

EGG MEMBRANE (CHORION): The layer of material that covers an animal egg cell. It is the outermost membrane around the embryo in mammals, birds and reptiles and it develops from an outer fold on the surface of the yolk sac. In insects, it is the outer shell of the insect egg, which lies outside the vitelline envelope, developed by the follicle cells, while the egg is in the ovary.

EGG PLANT (GARDEN EGG PLANT): A tropical plant, *Solanum melongena* of various sizes and colours, belonging to the solanaceae or nightshade family. Members produce edible fruits such as *aubergine* or *melongene*, which varies in size and may be black, purple, white or striped used as vegetables. Eggplants are classified in the division Magnoliophyta, class Magnoliopsida order Polemoniales, family Solanaceae.

EGLANDULAR: Having no glands.

EGO: In psychology, it is the processes concerned with the self and a person's conception of himself or herself, encompassing values and attitudes. In Freudian psychology, ego is the psychological component of our personality that is represented by our conscious decision making process. According to Sigmund Freud's psychoanalytic theory there are three parts to the personality: ego, id (the instinctual element) and superego (the social component of our personality and your conscience). The ego consists of conscious decision making and prevents us from acting on our basic urges created by the id, but also works to achieve a balance with our moral and idealistic standards created by the superego.

EGOROFF'S THEOREM: Named after the Russian analyst, D. F. Egoroff (1869-1931), the result that if a sequence of almost everywhere finite and measurable functions converges almost everywhere to a finite limit, on a set of finite measure, E , then there is, for each positive ϵ , a measurable subset F with a measure at most ϵ such that uniform convergence occurs on $E \setminus F$.

EGRESS: The act of or the mechanism for exit from an enclosure. In a spacecraft this can describe the act of a crew member exiting from the vehicle or the exit chamber, pressure lock and hatchways.

EGRET (ENERGETIC GAMMA RAY EXPERIMENT TELESCOPE): One of the four main instruments installed on board National Aeronautics and Space Administration's (NASA) Compton Gamma-ray Observatory to detect high energy gamma ray from 30 MeV to 30 GeV. The Compton Gamma-ray Observatory operated from April 1991 to June 2000.

EGYPTIAN FRACTION: A number of the form $\frac{1}{x}$, where x is an

integer or the sum of distinct unit fractions such as $\frac{1}{2}, \frac{1}{3}, \frac{1}{2}$, which sums up to $\frac{43}{48}$. Each fraction in the expression has a numerator equal to 1 and a denominator that is a positive integer and all the denominators differ from one another. The sum of an expression of this type is a positive rational, $\frac{a}{b}$. Every positive rational number

can be represented by an Egyptian fraction. Sums of this type and similar sums such as $\frac{2}{3}$ and $\frac{3}{4}$ as summands, were used as a serious notation for rational numbers by the ancient Egyptians and other civilisations. At present they have been replaced by vulgar fractions and decimal notations. They are, however, studied in modern number theory, recreational mathematics and modern historical studies of ancient mathematics.

EGYPTIAN MULTIPLICATION: A multiplication algorithm used by ancient Egyptians based on repeated doubling of one factor. They used addition to get to the answer of a multiplication problem. They only had to memorise the 2 times table multiplication table. A number is either multiplied by 2 or added to itself. This is done repeatedly until the problem is solved.

EHRENFEST CLASSIFICATION: A classification of phase transitions in terms of their thermodynamic properties. Under this scheme, phase transitions were labelled by the lowest derivative of the free energy that is discontinuous at the transition. There are two phases: A first order transition such as melting and boiling, where there is nonzero change in enthalpy, entropy and volume at the transition temperature and the second-order transition, where there is no jump in the value of the enthalpy, entropy and volume at the transition temperature such as the transition in the ferromagnetism and order-disorder transitions in alloys. There could in principle be third, fourth and higher-order phase transitions under this classification scheme. Though useful, the scheme is an inaccurate method of classifying phase transitions, because it does not consider the situation where a derivative of free energy diverges, which is only possible in the thermodynamic limit. For example, in the ferromagnetic transition, the heat capacity diverges to infinity.

EHRENFEST THEOREM: The theorem that a quantum-mechanical wave packet obeys the equations of motion of the corresponding classical particle when the position, momentum and force acting on the particle are replaced by the expectation values of these quantities.

EICOSANOIDS (ICOSANOIDS): Any of a group of unsaturated 20-carbon (C_{20}) fatty acids or their related compounds having one or more double bonds. They include signalling molecules and various biologically important molecules made by oxidation of 20-carbon essential fatty acids. Unlike hormones, they are not produced in a central location and then transported to the target organ. They act locally in the cells in which they are created and disappear quickly due to their rapid inactivation and extremely short half-lives. In general, eicosanoids bind to receptors on target cell plasma membranes in various tissues and stimulate or inhibit the synthesis of other messengers such as cyclic AMP. There are many different types of eicosanoids and they have a wide variety of functions. Three types: the leukotrienes, the thromboxanes and particularly the prostaglandins have been the focus of the majority of eicosanoid research.

EIDETIC IMAGE: An exceptionally vivid mental image which an individual recalls or "sees" either projected onto a suitable background such as a wall or with eyes closed in a darkened room. The image may be a memory (eidetic memory) or a fantasy and may be either voluntary or spontaneous. An eidetic person claims to continue to "see" an object that is no longer objectively present; and behave as if they are actually seeing an item, either with their eyes closed or while looking at some surface that serves as a convenient background for the image. Furthermore, eidetic persons describe the image as if it is still present and not as if they are recalling a past event. Eidetic imagery is most common among children.

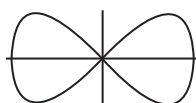
EIGEN FUNCTION: An allowed wave function of a system in quantum mechanics, which is also the associated solution of an eigenvalue equation. The importance of eigenfunctions and eigenvalues in applied mathematics results from the widespread applicability of linear equations as exact or approximate descriptions of physical systems. However, the most fundamental application of these concepts is in quantum mechanics, where they enter into the definition and physical interpretation of the theory.

EIGENVALUE (CHARACTERISTIC VALUE; CHARACTERISTIC NUMBER/ROOT; LATENT ROOT): In a characteristic polynomial, for a square matrix A , in which x is a nonzero vector called the eigenvector that does not change its direction under the associated linear transformation and satisfies the equation $Ax = \lambda x$, the scalar, λ is the eigenvalue. In other words, it is a root, λ , of the characteristic equation $\det(A - \lambda I)$. For a linear operator A , it is a solution, λ , of the equation $AX = \lambda X (X \neq 0)$.

EIGENVALUE EQUATION: A parameter-dependent equation that possesses non-vanishing solutions only for particular values (eigenvalues) of the parameter. Eigenvalue equations appear in many contexts, including the solution of systems of linear algebraic equations (matrix equations), differential or partial differential equations and integral equations.

EIGENVECTOR (CHARACTERISTIC VECTOR; LATENT VECTOR): In a square matrix, it is a nonzero vector, x that does not change its direction under the associated linear transformation and satisfies the equation $Ax = \lambda x$, where A is a square matrix and λ is a scalar called an eigenvalue.

EIGHT CURVE (LEMNISCATE OF GERON): A curve that has the Cartesian equation $x^4 = a^2(x^2 - y^2)$ and the appearance of a figure eight such as the symbol for infinity, lying on its side.



Eight Curve

EIGHTEEN-ELECTRON RULE: An electron-counting rule to which an overwhelming majority of stable diamagnetic transition metal complexes adhere. The number of nonbonded electrons at the metal plus the number of electrons in the metal-ligand bonds should be 18. The 18 electron rule in transition metal chemistry is a full analogue of the 'Lewis octet rule'.

EINSTEIN COEFFICIENTS: Coefficients used in the quantum theory of radiation, which are related to the probability of a transition occurring between the ground state and an excited state in the process of induced emission and spontaneous emission. The coefficients were developed by Albert Einstein in his analysis of the quantum theory of radiation.

EINSTEIN RING: A gravitational lens effect in which the image of a remote background object, such as a quasar, is distorted into a ring by a foreground galaxy. A perfect ring will only result if the source, the lensing object and the observer are exactly lined up and, in addition, the mass of the lensing object is evenly distributed. An Einstein ring image can be up to 30 times brighter than the image of the distant galaxy would be in the absence of the lensing effect.

EINSTEIN SHIFT (EINSTEIN DISPLACEMENT): A shift toward longer wavelengths of spectral lines emitted by atoms in strong gravitational fields.

EINSTEIN-SMOLUCHOWSKI EQUATION: An unexpected connection revealed independently by Albert Einstein in 1905 and

Marian Smoluchowski in 1906 in their papers on Brownian motion, which is a relationship between the diffusion coefficient D and the distance λ that a particle can jump in a time τ . The equation is given as $D = \lambda^2/2\tau$ and it gives a connection between the microscopic details of particle diffusion and the macroscopic quantities associated with the diffusion. This equation is an early example of a fluctuation-dissipation relation and it is frequently used in the electro diffusion phenomena.

EINSTEIN'S CLOSED ELEVATOR THOUGHT EXPERIMENT: An experiment which shows that an object dropped inside a closed elevator will accelerate toward the floor. This could be explained in one of two ways: either the elevator is far from any planet but accelerating upward or the elevator is at rest on the surface of planet.

EINSTEIN, ALBERT (1879-1955): German-born American physicist and Nobel laureate, who is best known as the creator of the special and general theories of relativity and for his hypothesis concerning the particle nature of light. In 1905, he published three papers, each of which had a profound effect on the development of physics. In one paper, he proposed the theory of special relativity, which provides a correct description for particles travelling at high speeds. The two postulates of the special theory of relativity were that the speed of light in a vacuum is constant and that the laws of physics are the same for all inertial reference frames. He is best known for his mass-energy equivalence formula $E = mc^2$. He received the 1921 Nobel Prize in Physics "for his services to theoretical physics and especially for his discovery of the law of the photoelectric effect". He is probably the most well known scientist of the 20th century.

EINSTEIN'S EQUATION: 1. It refers to Einstein's mass-energy relationship, which is expressed as $E = mc^2$, where E is a quantity of energy, m (the mass) and c (the speed of light). Alternatively, it is defined as the relationship $E_{\max} = hf - W$, where $E_{\max}$ is the maximum kinetic energy of electrons emitted in the photo-emissive effect; h , the Planck's constant; f , the frequency of the incident radiation and W the work function of the emitter. 2. Einstein's Field Equations (EFE) or Einstein's equations are a set of ten equations in Albert Einstein's theory of general relativity which describe the fundamental interaction of gravitation as a result of space-time being curved by matter and energy. Apart from obeying local energy-momentum conservation, the EFE reduces to Newton's law of gravitation, where the gravitational field is weak and velocities much less than the speed of light. In much the same way that electromagnetic fields are determined (using charges and currents via Maxwell's equations) the EFE is used to determine the space-time geometry resulting from the presence of mass-energy and linear momentum relationship between the metric tensor and the Einstein tensor and it allows the EFE to be written as a set of non-linear partial differential equations when used in this way. The solutions of the EFE are the components of the metric tensor. These equations are also used to study phenomena such as gravitational waves. Also, special classes of exact solutions are most often studied because they model many gravitational phenomena, such as rotating black holes and the expanding universe.

EINSTEINIUM: Named in honour of Albert Einstein (1879-1955), it is a silver-coloured, highly radioactive metallic transuranic element belonging to the actinoids, with the chemical symbol Es and atomic number 99. It has eleven known isotopes and they are all highly toxic. The mass number of the most stable isotope is 254 with a half life of 270 days. Its position on the Periodic Table indicates that its chemical and physical properties are similar to other metals. According to tracer studies conducted using isotope ^{253}Es , this element has chemical properties typical of a heavy trivalent, actinide element. It is of no commercial importance.

EISENSTEIN'S CRITERION: A condition that is sufficient for a polynomial with integer coefficients to be irreducible. It states that it suffices to find a prime p that does not divide the leading coefficient of the polynomial and that does divide all other coefficients, but is such that p^2 does not divide the constant coefficient. For example, the polynomial $x^n - 3$ is irreducible for all n , but $x^n - 9$ need not be. The corresponding result holds in a principal ideal domain. The criterion

is named after German mathematician, Ferdinand Gotthold Max Eisenstein (1823-1852).

EJACULATION: The discharge of semen out of the erect penis due to powerful rhythmic contractions of the urethra. An ejaculation coincides with the peak of sexual excitement or orgasm and is accompanied by various physiological effects in the body, such as increased respiration rate and heart rate. It is usually the final stage and natural objective of sexual stimulation and an essential component of natural conception. In rare cases ejaculation occurs because of prostatic disease. It may also occur spontaneously during sleep and called **nocturnal emission**.

EJACULATORY DUCT: Any of the short paired ducts that connects the vas deferens from each testis to the urethra. They pass through the prostate and empty into the urethra at the *Colliculus seminalis*. During ejaculation, semen passes through the duct and exits the body via the penis.

EJECT: 1. To escape from an aircraft in an emergency by means of an ejection seat or special capsule. 2. The removal of small particles such as electrons, atoms or even molecules from a lattice as a result of the bombardment of the material with a fast moving particle.

EJECTA: Fractured and/or molten rocky debris thrown out of a crater during an impact event or, alternatively, material, including volcanic ash, pyroclasts and bombs, discharged by a volcano.

EJECTA BLANKET: A generally symmetrical apron of ejecta that surrounds an impact crater. It is layered thickly at the crater's rim and thin to discontinuous at the blanket's outer edge. The extent of an ejecta blanket is determined by a number of factors including the size and mass of the impacting body (meteorite, asteroid or comet), the surface gravity and the atmospheric pressure. Where there is no atmosphere and the surface gravity is low, the ejecta blanket may extend to several times the radius of the central crater. On the other hand, in a world where both the atmospheric pressure and surface gravity are relatively high, the ejecta blanket will not reach far beyond the crater rim.

EJECTION CAPSULE: 1. A detachable compartment in an aircraft or manned spacecraft which serves as a cockpit or cabin and may be ejected as a unit and parachuted to the ground. 2. A boxlike unit in an artificial satellite, probe or unmanned spacecraft, which usually contains recording instruments or records of observed data and may be ejected and returned to Earth by parachute or some other deceleration device.

EKPYROTIC UNIVERSE (EKPYROTIC SCENARIO): A model of the universe originating from researches done by Neil Turok and Paul Steinhardt, which postulates that the big bang occurred because of the collision of two parallel branes of a type predicted to occur in string theory.

EL NIÑO-SOUTHERN OSCILLATION (ENSO): An irregularly occurring periodic variation in winds and sea surface temperatures over the tropical eastern Pacific Ocean, affecting the climates of much of the tropics and subtropics. It has two phases: **El Niño**, its warm phase and **La Niña**, its cooler phase. El Niño occurs every few years (2 to 7 years, usually lasting nine to 12 months), typically in late December bringing warm surface ocean temperatures that affect the equatorial Pacific region and beyond, including the coasts of Ecuador and Peru, and sometimes causing catastrophic weather conditions. It is characterised by high air pressure in the western Pacific and low air pressure in the eastern Pacific. It brings reversal of wind patterns across the Pacific, drought in Australasia, and unseasonal heavy rains in South America. El Niño can have serious effects on weather patterns around the world. **La Niña** is the cooler version of El Niño, occurring at irregular intervals. It is characterised by cooler surface ocean temperatures in the equatorial central and eastern Pacific. It brings widespread changes in weather patterns across the globe,

which can lead to intense storms in some places and droughts in others; but generally its effects are not as extensive and damaging as those of El Niño.

ELAIOPLAST: Types of leucoplasts which are specialised for the storage of lipids in plants. They keep oil body deposits as rounded plastoglobuli, which are essentially fat droplets. Being a variety of leucoplast, they are non-pigmented and fall into the much broader organelle category of plantplastids. Another example of a storage-specialised leucoplast is the amyloplast.

ELAMINATE: A leaf with no blades.

ELASTANCE: The reciprocal of capacitance (ability to store electric charge). It is measured in farad⁻¹.

ELASTIC: Of an object or material, having the ability to resist being permanently distorted or deformed and to resume its normal shape spontaneously after being stretched or compressed.

ELASTIC COLLISION: A collision between two or more bodies in which the total kinetic energy is conserved (remains constant) or a collision in which the total kinetic energy of the colliding bodies after collision is equal to their total kinetic energy before collision. Elastic collision occurs only if there is no conversion of kinetic energy into other forms. This type of collision occurs between individual atoms as well as between a spacecraft and a gravitating body. An example of elastic collisions in the collision of atoms is Rutherford backscattering.

ELASTIC LIMIT: The maximum load (force), which a body can hold and still regain its original size and shape once the load (force) has been removed. On the other hand, the **yield strength** or **yield point** of a material is the stress at which a material begins to deform plastically and prior to the yield point, the material will deform elastically and will return to its original shape when the applied stress is removed.

ELASTIC MODULES (ELASTIC MODULUS; MODULUS OF ELASTICITY): The ratio of the stress applied to a body to the strain produced or the slope of its stress-strain curve in the elastic deformation region. It is the mathematical description of an object's tendency to be deformed elastically, when a force is applied to it.

It may be expressed as $\lambda = \frac{\text{stress}}{\text{strain}}$, where λ (lambda) is the elastic

modulus, stress the force causing the deformation divided by the area to which the force is applied and strain, the ratio of the change caused by the stress to the original state of the object. If stress is measured in pascals, since strain is a unitless ratio, then the units of λ are also pascals. Due to the fact that the denominator of the above equation becomes unity if length is doubled, the elastic modulus becomes the stress needed to cause a sample of the material to double in length. Since this endpoint is not realistic because most materials will fail before reaching it, it is practical, due to the fact that small fractions of the defining load will operate in the same ratio.

ELASTIC RATIO: The ratio of the elastic limit to the ultimate strength of a solid.

ELASTIC TISSUE: A yellow connective tissue composed mainly of elastic fibres produced by fibroblasts that can stretch up to 1.5 times their length and snap back to their original length when relaxed. These tissues are found in many places in the body such as the walls of arteries and veins, dermis of the skin, lung tissues, bronchial tubes, and certain ligaments and tendons. Its function is to enable organs and parts of organs to stretch and contract.

ELASTICITY: 1. In physics, the ability of a body to recover its original shape after the deforming forces acting on it are removed or the ability of a material that has been pressed or stretched by a force to return to its original shape when the force is removed. The theory of elasticity deals with the relations between the forces acting on a body and the resulting changes in configuration. It is, therefore, relevant in many aspects of science and technology, including the

design of structures, the theory of vibration and sound and the study of the forces between atoms in crystal lattices. 2. In mathematics, the quantity e , defined at a point x for a function f as the absolute value of the derivative of $\log f(x)$ with respect to $\log x$. The function is elastic, when $e > 1$ and inelastic, when $e < 1$. In economics, it is known as elasticity of demand (Ed) or the price elasticity of demand and measures how responsive demand is to changes in price of a given good. If a good is subject to demand $d(p)$ that is dependent upon the price p and so yields revenue $pd(p)$, then if $d(p)$ is elastic, the revenue increases as the price p increases; if it is inelastic, the revenue decreases as p increases.

ELASTIN: A fibrous protein that is the major constituent of the yellow elastic fibres of connective tissue. It enables many tissues in the body and the skin to return to their original position after stretching, contracting, pinching or poking. It is also an important load-bearing tissue in the bodies of mammals and used in places where mechanical energy is required to be stored.

ELASTOMER: Any material, usually a polymer with elastic or viscoelasticity properties, which stretches easily and then quickly returns to its original length when released. Elastomers are amorphous polymers existing above their glass transition temperature. In view of this, significant segmental motion is possible. They have generally low Young's modulus and high yield strain compared with other materials and each of the monomers which link up to form the polymer is usually made of carbon, hydrogen, oxygen and/or silicon. Examples are natural rubber and synthetic rubbers such as, neoprene, butyl rubber or nitrile rubber that are produced by polymerisation of butadiene, isoprene and other unsaturated hydrocarbons.

ELATER: Any of the numerous elongated helically thickened cells intermixed with the spores in the capsule of most liverwort sporophytes. As the mature capsule dries out, tension builds up in the thickening of the elater. This tension is released when the capsule dehisces, dispersing the spores over a distance.

ELBOW: The joint or bend of the arm between the forearm and the upper arm. The ulna forms the part of the joint that allows the hinge motion of the elbow. The elbow joint is very stable because of the wrench shape of the trochlear notch of the ulna, which fits around the pulley-shaped trochlea of the humerus. The joint is also strengthened by strong medial and lateral collateral ligaments.

ELECTRA COMPLEX: A term coined by Austrian physician and psychologist, Sigmund Freud (1856-1939) that defines an antagonistic behaviour of a girl to her mother, whom she sees as a rival for her father's affection.

ELECTRET: A permanently electrified substance or body that has quasi-opposite charges at its extremities or a quasi-permanently polarised dielectric material. Its electric field is similar to the magnetic field of a permanent magnet; it has a positive charge at one end and a negative charge at the other. They resemble permanent magnets in many ways and they can be made by cooling certain waxes in a strong electric field. They generate internal and external electric fields and they are the electrostatic equivalent of permanent magnets.

ELECTRIC ARC (ARC DISCHARGE): A continuous electric discharge of high current between two electrodes, giving out a brilliant light and heat or an electrical breakdown of a gas which produces an ongoing plasma discharge, resulting from a current flowing through normally nonconductive media such as air. It was first described by Vasily V. Petrov, a Russian scientist, in 1802.

ELECTRIC ARC FURNACE: A furnace used in melting metals to make alloys, especially in steel manufacturing in which the heat source is an electric arc. They differ from induction furnaces in that the charged material is directly exposed to an electric arc and the current in the furnace terminals passes through the charged material. Those used in research laboratories and by dentists may have a capacity of only a few grams while others range in size from small units of approximately one ton capacity, used in foundries for producing cast iron products

up to about 400 ton units used for secondary steelmaking. Their temperatures can be as high as 1800 °C (2073 K).

ELECTRIC BELL: A bell that makes use of electromagnetism. It consists of a coil of wire wound on an iron core (an electromagnet) which, when a direct current flows through it attracts an iron armature and produces a repetitive buzzing or clanging sound. Currently replaced mostly by electronic buzzers or beepers, they were once widely used in fire and burglar alarms, doorbells and alarms in industrial plants.

ELECTRIC CELL: A combination of two or more electrochemical cells used to convert stored chemical energy into electrical energy. An example is the dry cell battery. There are two types: primary batteries or disposable batteries, which are designed to be used once and discarded when they are exhausted and secondary batteries or rechargeable batteries, which are designed to be recharged and used many times. There are also miniature cells which are used to power devices such as hearing aids and wristwatches. Larger batteries provide standby power for telephone exchanges, computer data centres, etc.

ELECTRIC CHARGE: The fundamental property of matter that allows one object to exert an electric force on another object when near each other without physically touching. The amount of electric charge is defined by the word equation, electric charge (Q) = current (I) $\times$ time (t in seconds). The SI unit of electric charge is the coulomb (C). It occurs in nature in two forms and conventionally described as positive and negative in order to distinguish between them in terms of the kinds of interactions they can have. The proton has a positive charge and the electron has a negative charge is a property of the electron. Two positively or negatively charged substances experience a mutual repulsive force, while positively charged objects and negatively charged objects experience an attractive force.

ELECTRIC CIRCUIT: A configuration of electronics or electrical components through which electricity is made to flow. Common circuit elements include the resistor, inductor, capacitor, battery and diode. A simple circuit might consist of a single strand of metal wire linking the positive and negative terminals. A more realistic circuit possesses multiple branch points, so that current can take many different paths between the two terminals.

ELECTRIC CONSTANT: The absolute permittivity of empty space, with the symbol ϵ_0 . It is equal to 1 in centimetre-gram-second electrostatic units and to $\frac{1}{c_0^2 \mu_0} = \frac{1}{35950207149.4727056\pi}$ farads per

metre, where c_0 is the speed of light in free space and μ_0 is the vacuum permeability). Numerically, it is equal to 8.854×10^{-12} farad per metre.

ELECTRIC CURRENT (CURRENT): The flow of electrically charged particles through a conducting circuit due to the presence of potential difference. Depending on the context, the term means a flow of electric charge, which is a phenomenon or the rate of flow of electric charge, which is a quantity. Current carries electrical energy from a power supply, such as the battery of electrical cells to the components of the circuit. This flowing electric charge is typically carried by moving electrons, in a conductor such as a wire. In an electrolyte, it is carried by ions, while in plasma it is carried by both. The SI unit for measuring the rate of flow of electric charge is the ampere, with the symbol I, which is the charge flowing through some surface at the rate of one coulomb per second. Electric current is measured using an ammeter.

ELECTRIC CURRENT DENSITY: A measure of the density of an electric current, defined as a vector whose magnitude is the electric current per cross-sectional area. In SI units, the current density is measured in amperes per square metre. Its symbol is J . By magnitude it is the electric current divided by the cross sectional area: $J = I/A$, where I is the current flowing through the conductor given in Amperes and A is the cross sectional area in m^2 .

ELECTRIC DIPOLE: An object whose centres of positive and negative charge do not coincide. For example, a hydrogen chloride (HCl) molecule is an electric dipole because bonding electrons are on average closer to the chlorine atom than the hydrogen, producing a partial positive charge on the H end and a partial negative charge on the Cl end.

ELECTRIC DIPOLE MOMENT: A measure of the separation of positive and negative electrical charges in a system of charges, that is, a measure of the charge system's overall polarity. The SI unit is Coulomb-metre (C m).

E

ELECTRIC DISPLACEMENT (ELECTRIC FLUX DENSITY): The charge per unit area that would be displaced across a layer of a conductor placed across an electric field. Its symbol is D and it stands for displacement. The electric displacement field D is, therefore, defined as $D \equiv \epsilon_0 E + P$, where ϵ_0 is the permittivity in free space, E is the actual average value of the electric field and P is the (macroscopic) density of the permanent and induced electric dipole moments in the material, known as the polarisation density. It is a vector field whose divergence depends on the density of bound charge in the material. It is responsible for the effects of free charges within materials. In free space, the electric displacement field is equivalent to flux density, a concept that lends understanding to Gauss's law.

ELECTRIC DOG: An electric wire at the side of the fence at the entrance to a milking parlour, which encourages cows to go into the parlour.

ELECTRIC FIELD: A region within which a particle possessing electric charge experiences a force owing to the distribution of other charges; or a region around an electric charge in which an electric force is exerted on another charge. Its strength, E at any point is defined as the electric force, F exerted per unit positive electric charge, Q at that point: $E = \frac{F}{Q}$. It is a type of electromagnetic field, having both

magnitude and direction and can be represented by lines of force or field lines, that start on positive charges and terminate on negative charges. The electric field is stronger, where the field lines are closer than where they are farther apart. The value of the electric field has dimensions of force per unit charge and is measured in units of newtons per coulomb.

ELECTRIC FIELD STRENGTH (ELECTRIC FIELD INTENSITY; ELECTRIC INTENSITY): A vector quantity, which is a measure of the strength of an electric field at a given point in space. It is equal to the force the field would induce on a unit electric charge at that point. Its symbol is E with a standard unit of volt per metre (V/m or V m^{-1}) or newton/coulomb: Force (newton)/Charge (coulomb): Its strength, E at any point is defined as the electric force F exerted per unit positive electric charge Q at that point: $E = \frac{F}{Q}$. The field strength

E at a point is defined as the force per unit charge acting on a small positive charge placed at the point. The field strength at a particular distance from an object is directly proportional to the electric charge (in coulombs) on that object and inversely proportional to the distance from a charged object.

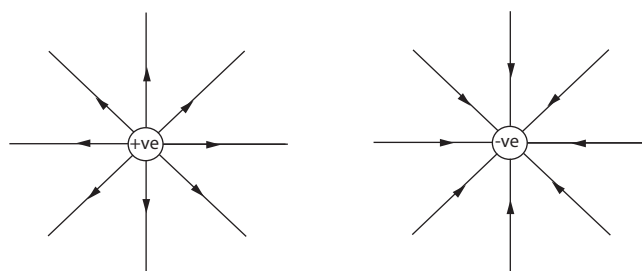
ELECTRIC FLUX: The strength of electric field acting on a particular area, defined as the area size multiplied by the component of the field acting at right angles to the area, expressed as $\phi = EA$, where ϕ (phi) is the electric flux, E is the strength of the electric field and A is the area size.

ELECTRIC INSULATOR: A material or device that reduces or prevents the flow of electric current across it, due to the absence of free electrons. In most materials even the outermost electrons are so tightly bound that there is essentially zero electron flow through them with ordinary voltages. An insulator can be a solid, liquid or gaseous and it provides insulation from electric shock. It is, therefore, used to insulate electric conductors from one another and to confine electric currents to specified pathways, for example,

as in the insulation of wires, electric switchgear and electronic components. The electrical function of an insulator is characterised by its resistivity, dielectric strength or breakdown voltage and dielectric constant or relative permittivity. It also provides a mechanical and heat-dissipation function. Some materials, which are particularly good insulators and characterised by their high resistivities are glass with a resistivity of 10^{12} ohm metre ($\Omega\cdot\text{m}$) and mica with a resistivity of 9×10^{13} ohm metre ($\Omega\cdot\text{m}$). Copper, a very good conductor has a resistivity of 1.7×10^{-8} ohm metre ($\Omega\cdot\text{m}$).

ELECTRIC LIGHTING: Illumination provided by electric currents through devices such as the light bulb and the fluorescent tube. These lights are normally powered by the electric grid, but some run on local generators and emergency generators serve as backups. Battery-powered lights, usually called "flashlights" or "torches", are used for portability and as backups when the main lights fail.

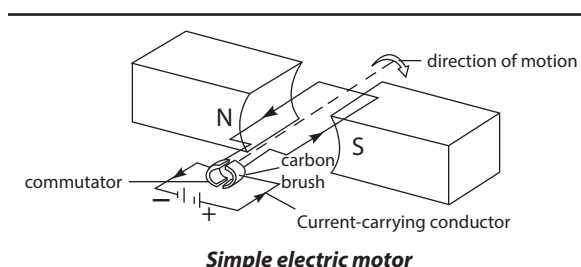
ELECTRIC LINES OF FORCE (ELECTRIC FLUX LINE): Imaginary lines of force by means of which electric field is visualised. They are drawn so that each segment of the line is parallel to the direction of the electric field or of the electric displacement at that point. The density of the set of lines is proportional to the electric field or electrical displacement.



Electric lines of force around positively and negatively charged particles

ELECTRIC MOBILITY: The ability of charged particles such as electrons or protons to move through a medium in response to an electric field that is pulling them. The separation of ions according to their mobility in gas phase is called **ion mobility spectrometry**, in liquid phase it is called **electrophoresis**. It is expressed as the speed of ions divided by the electric field strength in the phase concerned.

ELECTRIC MOTOR: A machine for converting electrical energy into mechanical energy, by exploiting the force on a current-carrying conductor in a magnetic field. It works on the principle that a current passing through coils of wire within a magnetic field will experience a torque that causes the coil to rotate. Fleming's left-hand rule is used to determine the direction of motion or rotation of the coil in relationship with the current and the magnetic field. In order for the coil to rotate continuously, the direction of the current must be reversed every half turn by using a conducting ring split in two, called a split ring or commutator. Two carbon brushes supply the current to the slip rings. The reverse process, which produces electrical energy from mechanical energy, is accomplished by an alternator, a generator or dynamo. Electric motors may be classified by the source of electric power, by their internal construction, by their application or by the type of motion they give. Motors are also used in industrial fans, blowers and pumps, machine tools, household appliances, power tools, disk drives and many other diverse ways. They may be powered by direct current such as from a battery powered portable device or motor vehicle or by alternating current from a central electrical distribution grid. The largest electric motors are used for propulsion of large ships and for such purposes as pipeline compressors, with ratings in the millions of watts. Medium-sized motors provide convenient mechanical power for industrial uses while the smallest motors may be found in electric wrist watches.



ELECTRIC ORGAN: 1. A specialised tissue organ, often derived from muscles, which are present on the body of certain fish such as the electric ray and electric eel, which gives an electric shock when touched and is used either to stun prey or predators or to maintain a weak electric field in the surrounding water for navigation and communication. Other uses include finding enemies, obstacles and food in murky water and, in some species, for defence against attack. 2. A musical instrument, which is a type of organ, whose sound is produced or amplified by means of electricity.

ELECTRIC POLARISATION (POLARISATION DENSITY; POLARISED DENSITY): The vector field that expresses the density of permanent or induced electric dipole moments in a dielectric material or slight relative shift of positive and negative electric charge in opposite directions within an insulator or dielectric, induced by an external electric field. It occurs when an electric field distorts the negative cloud of electrons around positive atomic nuclei in a direction opposite the field. The polarisation vector P is defined as the dipole moment per unit volume V of a polarised material, $P = \frac{p}{V}$. The SI unit is coulombs per square metre.

ELECTRIC POTENTIAL: 1. Also known as **electrostatic potential**, it is the work per unit of charge required to move a charge from infinity to a point in an electric field. Its unit is joules per coulomb or volts.

Electric potential energy or **electrostatic potential energy** is the energy that a charged particle has because of its own electric charge and its position in an electric field. Its symbol is U_E . **Electric potential difference** or **electrostatic potential difference** is the difference in electric potential (V) between two points in an electric field when work is done upon a charge to change its potential energy. Its symbol is V or U . 2. **Electric scalar potential** is used in electrodynamics when time-varying electromagnetic fields are present, which cannot be simply interpreted as the ratio of potential energy to charge. 3. In biology, it is the difference in the amount of electric charge between a region of positive charge and a region of negative charge. The establishment of electric potentials across the plasma membrane and across organelle membranes makes possible a number of phenomena, including the chemiosmotic synthesis of adenosine triphosphate (ATP), the conduction of nerve impulses and muscle contraction.

ELECTRIC POWER: The rate at which electrical energy is supplied; or the rate at which electrical energy is dissipated in a system. For a direct current circuit, it is measured by the product of the current passing through a system and the potential difference across it: $P = IV$. In alternating-current circuits, the power is given by $VI\cos\phi$, where V and I are the RMS values and ϕ is the phase angle. $\cos\phi$ is called the power factor of the circuit. The SI unit of power is the watt and the term wattage is used to mean electric power in watts. Electricity can be produced mechanically by generation or chemically or by direct conversion from light in photovoltaic cells, also it can be stored chemically in batteries. When electric current flows in a circuit, it can transfer energy to do mechanical or thermodynamic work,

with devices converting electrical energy into many useful forms, such as heat (electric heaters), light (light bulbs), motion (electric motors), sound (loudspeaker) or chemical changes. With resistive loads, Joule's law can be combined with Ohm's law ($I = V/R$) to yield another equation for dissipated power: Given that R is the electrical

resistance, then $P = I^2 R = \frac{V^2}{R}$.

ELECTRIC QUADRUPOLE: A charge distribution that produces an electric field equivalent to that produced by two electric dipoles whose dipole moments have the same magnitude but point in opposite directions and which are separated from each other by a small distance.

ELECTRIC SECTOR: In mass spectrometry, it is an arrangement of two conducting sheets forming a capacitor and giving a radial electrostatic field, which is used to deflect and focus ion beams of different energies. The capacitor may, for example, be cylindrical, spherical or toroidal.

ELECTRIC SPACE PROPULSION: A form of advanced rocket propulsion that uses electrical energy for heating and/or directly ejecting propellant. Electric propulsion (EP) provides much lower thrust levels than conventional chemical propulsion (CP) does, but much higher specific impulse. This means that an EP device must thrust for a longer period to produce a desired change in trajectory or velocity; however, the higher specific impulse enables a spacecraft using EP to carry out a mission with relatively little propellant and, in the case of a deep-space probe, to build up a high final velocity.

ELECTRIC SPARK: The transient passage of an electric current through a gas between two points of high opposite potential, with the emission of light and sound. Lightning consists of a spark between a cloud and earth or between two oppositely charged parts of the same cloud.

ELECTRIC SHOCK (SHOCK): The physical state caused by electric current passing through the body, characterised by pain and muscular spasm. It can affect the respiratory system and heart rhythm and if electric contact is not broken as quickly as possible, it can be fatal.

ELECTRIC SUSCEPTIBILITY (DIELECTRIC SUSCEPTIBILITY): A dimensionless parameter, which is a measure of how easily a material polarises in response to an electric field, which is equal in mks units to the ratio of the polarisation to the product of the electric field strength and the vacuum permittivity. It determines the electric permittivity of the material and thus influences many other phenomena in that medium, from the capacitance of capacitors to the speed of light. Its symbol is χ_e .

ELECTRICAL ARC: A self-sustaining electrical discharge between at least two electrodes, characterised by a comparatively small cathode fall voltage, a low burning voltage and a relatively high current density. The burning voltage of an arc is the voltage across the electrode gap during an arc discharge.

ELECTRICAL CIRCUIT: A path or closed loop through which charges can continuously move taking electrons from a voltage or current source to a load such as a bulb or motor. It comprises of an energy source such as a battery, conductors to carry electrons, a switch installed in the conductor to control the flow of electrons, and a load. When the switch is closed the electrons begin to flow and current is supplied to the load. When the switch is opened, the circuit is in an open circuit condition and no electrons flow and current flow to the load is interrupted. The point where current enters an electrical circuit is called the source, and the point at which it leaves an electrical circuit is called the return path.

ELECTRICAL CONDUCTANCE: A measure of the capacity of an electrical element to conduct current. It is the inverse quantity of electrical resistance with the symbol G and expressed as $G = \frac{1}{R}$, where R is the resistance. Its unit is Ω^{-1} (inverse ohms) or Siemens.

ELECTRICAL CONDUCTIVITY (SPECIFIC CONDUCTANCE): The reciprocal quantity of electrical resistivity, which measures a material's ability to conduct an electric current. It is commonly represented by the Greek letter σ (sigma), but κ (kappa) (especially in electrical engineering) or γ (gamma) are also occasionally used. Its SI unit is siemens per metre ($\text{S}\cdot\text{m}^{-1}$). It is expressed as $\sigma = \frac{1}{\rho}$, where ρ is the resistivity. For a body of length L , resistance R and cross sectional area A , the electrical conductivity is given by $\sigma = \frac{L}{RA}$.

ELECTRICAL CONDUCTOR: A material or substance in which electrical charge carriers, usually electrons, move easily from atom to atom with the application of voltage. The outer electrons of the atoms in a conductor are loosely bound and free to move through the material. Metals are classified as conductors because their outer electrons are not tightly bound. In copper, for example, the valence electrons are essentially free and strongly repel each other. Any external influence which moves one of them will cause a repulsion of other electrons which propagates, through the conductor. Resistance increases with increasing temperature in conductors. Pure elemental silver is the best electrical conductor. Copper, steel, gold, aluminium and brass are also good conductors of electricity. Some liquids are good electrical conductors. Mercury is an excellent example. A saturated salt-water solution acts as a fair conductor. Gases are normally poor conductors because the atoms are too far apart to allow a free exchange of electrons. However, if a sample of gas contains a significant number of ions, it can act as a fair conductor. Good conductors are characterised by low resistivities, while good insulators are characterised by high resistivities. For example, copper, a very good conductor has a resistivity of 1.7×10^{-8} ohm metre ($\Omega\cdot\text{m}$) and glass, a good insulator has a resistivity of 10^{12} ohm metre ($\Omega\cdot\text{m}$) and mica has a resistivity of 9×10^{13} ohm metre ($\Omega\cdot\text{m}$).

ELECTRICAL DOUBLE LAYER (EDL; DOUBLE LAYER; DL): A model of the interface between an electrode and the solution close to it or a structure that appears on the surface of an object when it is placed into a liquid or the two parallel layers of charge surrounding such object. The object could be a solid particle, a gas bubble, a liquid droplet or a porous body. In this model a sheet of one type of the opposite charge surrounds the first sheet in the solution. The first layer, the surface charge, which may be positive or negative, is made up of ions adsorbed directly onto the object due to several chemical interactions. The second layer which is made up of ions attracted to the surface charge through the coulomb force electrically screens the first layer. This second layer, referred to as diffuse layer, is often associated with the object because it is made of free ions which move in the fluid under the influence of electric attraction and thermal motion, rather than being firmly anchored.

ELECTRICAL ENERGY: The flow of electric charges as current along a conductor or a circuit to create energy that is converted into other forms of energy. It is a form of power created by the movement of electrons. Electrical energy is a secondary source of energy, which is obtained through the conversion of other forms of energy known as the primary sources of energy. The primary sources from which electrical energy is obtained can be either non-renewable forms of energy or renewable forms of energy. Examples are hydro or water, coal, nuclear energy, natural gas, oil, wind, from chemical reactions as in cells or batteries, etc. Power, measured in watts and time measured in seconds are factors of electrical energy, expressed as: Energy = Power $\times$ Time. The unit of energy is the joule. In terms of the charge of electron, e and applied voltage, V , electrical energy, E can be expressed as $E = eV$. Therefore, electrical power that has been converted into work in a certain time or power that has been used over time is called electrical energy. Electrical energy is expressed in kilowatt-hour (kWhr): Power (kw) $\times$ time (hour).

ELECTRICAL HYGROMETER: A hygrometer whose sensitive element has electrical properties which vary with the humidity of the gas which traverses the hygrometer.

ELECTRICAL RELAY: An electro-magnetic switch typically operated by a low voltage that controls a higher voltage circuit and turns it on or off. Where it is necessary to control a circuit by a low-power signal or where several circuits must be controlled by one signal, relays are often used and the type of relay that handles the high power required to directly drive an electric motor is known as a **magnetic contactor**. Relays are also extensively used in telephone exchanges and early computers to perform logical operations. Instead of using a semiconductor device to perform switching, solid-state relays are used to control power circuits with no moving parts. Relays with calibrated operating characteristics and sometimes multiple operating coils are also used to protect electrical circuits from overload or faults. Presently, these functions are performed by digital instruments which are known as protection relays.

ELECTRICAL-RESISTANCE STRAIN GAGE: A vibration-measuring device consisting of a grid of fine wire cemented to the vibrating object to measure fluctuating strains.

ELECTRICALLY ERASABLE PROGRAMMABLE READ-ONLY MEMORY (EEPROM): A non-volatile rewritable memory chip that maintains its contents without electrical power. Contents can be erased and reprogrammed. EEPROMs are typically used on circuit boards to store small amounts of instructions and data.

ELECTRICITY: Any effect resulting from the existence of stationary or moving electric charges. It is a physical phenomena caused by the behaviour of electrons and protons by the attraction of particles with opposite charges and the repulsion of particles with the same charge. These include lightning, static electricity, electromagnetic field and electromagnetic induction. If charged particles collect and stay in one place, the effects are called **static electricity** but if they flow steadily, the effects are that of an **electric current**. Electricity was discovered by Benjamin Franklin in 1752. The electric generator was invented by Michael Faraday in 1831; and Thomas Edison invented the electric light bulb in 1879.

ELECTRIFIED INTERPHASE: An interphase between phases containing free charged components which are usually accumulated or depleted in the surface region thus giving rise to net charges on the phase.

ELECTRO ORGANIC REACTION: Synthesis of organic compounds usually difficult to carry out under normal conditions in electrolytic cells.

ELECTRO-OLFACTOGRAM (EOG): A reading of the electrical activity of smell receptors in the nose or other olfactory organs as detected by an electrode placed on the olfactory mucosa as it is exposed to stimuli with an odour.

ELECTRO-OPTICAL EFFECT: An effect in which the presence of an electric field alters an optical system in an observable way. Examples include the Kerr effect and Pockels effect.

ELECTRO-OSMOSIS (ELECTROOSMOSIS; ELECTROENDOSMOSIS): First documented by F.F. Reuss in 1809, it is the drift of a polar liquid through a membrane or other porous structure, generally along charged surfaces of any shape and also through non-macroporous materials which have ionic sites and allow for water uptake (chemical porosity) under the influence of electrical energy passed through it. It has growing applications in microfluidics. For example, it is used as an alternative or supplemental explanation for the movement of polar liquids via the phloem that differs from the cohesion-tension theory supplied in the mass flow hypothesis and others, such as cytoplasmic streaming.

ELECTRO-OSMOTIC HOLD-UP TIME: The time required for a liquid in a capillary to move due to electro-osmosis through the effective length of the capillary, L_{eff} . Its symbol is t_{eo} . This time is usually measured as the migration time of a neutral compound, called an electro-osmotic flow marker which is assumed to have an

electro-osmotic mobility that is negligible compared to that of the analyte.

ELECTRO-OSMOTIC PRESSURE: The pressure difference across the membrane, plug, etc., needed just to stop electro-osmotic volume flow. The pressure difference is positive if the higher pressure is on the high potential side.

ELECTRO-OSMOTIC VOLUME FLOW: The volume flow per unit time per unit field strength through the whole plug.

ELECTROACTIVE SUBSTANCE: A species in solution that can take part in an electrode reaction or that can be adsorbed on the electrode.

ELECTROCAPILLARITY: The dependence of the interfacial tension on the electrical state of the interphase.

ELECTROCARDIOGRAM (ECG; EKG): The graphic recording of the electrical changes in the heart muscle, as detected by electrodes placed on the chest or a transthoracic interpretation of the electrical activity of the heart over time as captured and externally recorded by skin electrodes. It is used in the diagnosis of heart diseases. It operates by detecting and amplifying the tiny electrical changes on the skin that are caused when the heart muscle depolarises during each heart beat. A healthy heart will have an orderly progression of a wave of depolarisation that is triggered by the cells in the sinoatrial node during each heartbeat, spreads out through the atrium, passes through intrinsic conduction pathways and then spreads all over the ventricles. This is detected as tiny rises and falls in the voltage between two electrodes placed at either side of the heart, which is displayed as a wavy line on a screen or paper. This display or chart shows the overall rhythm of the heart and weaknesses in different parts of the heart muscle. Expert analysis of the chart can provide a diagnosis of a fault in the functioning of the heart.

ELECTROCHEMICAL: Of or relating to chemical reactions brought about by electricity.

ELECTROCHEMICAL CELL: A combination of two electrodes arranged so that an overall oxidation-reduction reaction produces an electromotive force; or a device that uses spontaneous redox reaction to produce electricity or uses electrical energy to cause non-spontaneous redox reaction. There are two types, galvanic or voltaic cell and electrolytic cell. An example of a galvanic cell is lead-acid battery, which consists of six cells with a cell voltage of 2 volts is a standard 1.5 volt battery.

ELECTROCHEMICAL DETECTOR: An electrochemical cell which responds to certain substances in the carrier gas eluting from the column. The electrochemical process may be an oxidation, reduction or a change in conductivity.

ELECTROCHEMICAL EQUIVALENT: The mass of a given element liberated from a solution of its ions in electrolysis by one coulomb of charge or the mass of a substance produced or consumed by electrolysis with 100% current efficiency during the flow of a quantity of electricity equal to 1 farad (96,487 coulombs). This is given by the gram-atomic or gram-molecular mass of this substance divided by the number of electrons involved in the electrode reaction. For example, the electrochemical equivalent of zinc for which two electrons are required in order to deposit one atom is $65.37/2\text{g (Zn/2)}$. Electrochemical equivalents are essential in the calculation of the current efficiency of an electrode process. Its symbol is z .

ELECTROCHEMICAL GRADIENT: A type of potential energy that accounts for both the concentration difference of the ion across a membrane and its tendency to move relative to the membrane potential. For example, the electrical gradient of K^+ , a positive ion, also tends to drive it into the cell, but the concentration gradient of K^+ tends to drive K^+ out of the cell. In biological processes, the direction an ion moves by diffusion or active transport across a membrane is determined by the electrochemical gradient.

ELECTROCHEMICAL METHOD OF DETECTION: A method in which either current or potential is measured during an electrochemical

reaction. The gas or liquid containing the trace impurity to be analysed is sent through an electrochemical cell containing a liquid or solid electrolyte and in which an electrochemical reaction specific to the impurity takes place.

ELECTROCHEMICAL POTENTIAL: A thermodynamic function characterising the state of some component consisting of charged particles (electrons or ions) in a phase of a given solution.

ELECTROCHEMICAL SERIES (ACTIVITY SERIES; ELECTROCHEMICAL SERIES): A series of chemical elements arranged in descending order of their chemical reactivity or electrode potentials, with the most active at the top and the less reactive or more noble metals at the bottom. The series shows the order in which metals replace one another from their salts. For example, electropositive metals will replace hydrogen in acids; zinc will displace hydrogen from hydrogen chloride. Basically, such an activity series need not be limited to the metals but may be carried on through the electronegative or nonmetallic elements as well. (refer to page 25 for table).

ELECTROCHEMISTRY: The branch of chemistry that studies the chemical changes observed upon the passage of electric current through solutions or molten substances and production of electrical energy by a spontaneous redox reaction. Basically, it is concerned with the relationship between electricity and chemical change.

ELECTROCHROMATOGRAPHY (ELECTROPHOREOGRAPHY): A method of resolving and separating mostly large biomolecules such as proteins by the use of an electric potential to produce an electric differential. It is a combination of size exclusion chromatography (gel filtration chromatography) and gel electrophoresis. These separation mechanisms operate essentially in superposition along the length of a gel filtration column to which an axial electric field gradient has been added. The molecules are separated by size due to the gel filtration mechanism and by electrophoretic mobility due to the gel electrophoresis mechanism. Additionally there are secondary chromatographic solute retention mechanisms.

ELECTROCONVULSIVE THERAPY, ECT (ELECTROSHOCK THERAPY): A treatment mainly for severe depression, given under anaesthesia and with a muscle relaxant in which small electric currents are passed through the brain, intentionally triggering a brief seizure. If high doses of electricity are administered without anaesthesia, it can cause distress, loss of concentration and memory, fractured bones and other serious side effects.

ELECTROCUTION: The often fatal effect of passing a high-energy electric current through the body. Electricity passing through the body fluid, which acts as a resistor, causes burns at sites of connection and along the electrical pathway. Convulsions and rhythm disturbance in the heart are usual; the latter are the cause of immediate death. Artificial respiration with cardiac massage must be started immediately if resuscitation is to be successful.

ELECTROCYCLIC REACTION: A type of cyclic rearrangement in which a sigma bond is formed between two terminal carbon atoms of a conjugated molecule, resulting in a decrease of one in the number of pi bonds present.

ELECTRODE: A substance that conducts electricity into or out of an electrolyte in electrolysis or any terminal by which an electric current passes in or out of a conducting substance. It is an electrical conductor used to make contact with a nonmetallic part of a circuit such as a semiconductor, an electrolyte or a vacuum. An example is the anode and cathode in a battery.

ELECTRODE POTENTIAL (REDUCTION POTENTIAL): The potential difference produced between the electrode of a given chemical element and the solution of a half cell or in which it is immersed. It is measured in volts; however, it is not possible to measure it directly since any measurement involves completing the circuit with another dry cell.

ELECTRODE PROCESS: The totality of changes occurring at or near a single electrode during the passage of current.

ELECTRODE REACTION: the net oxidation or reduction process that takes place at an electrode. This reaction may take place in a single electron-transfer step or as a succession of two or more steps. The substances that receive and lose electrons are called the electroactive species. For an electrode process at a given potential, the current is controlled by the kinetics of a number of steps which include the transport of reactants to and from the interface and the interfacial reaction itself. The latter, which is called the electrode reaction, must always include at least one elementary step in which charge is transferred from one phase to the other, but may also involve purely chemical steps within the interfacial region.

ELECTRODEPOSITION: The deposition of dissolved or suspended material by an electric field on an electrode.

ELECTRODEPOSITION ANALYSIS: An electro analytical technique in which an element is quantitatively deposited on an electrode by electrolysis as in electroforming and electroplating.

ELECTRODIALYSIS (ED): A method of obtaining pure water from water containing a salt as in desalination. ED processes differ, compared to distillation techniques and other membrane based processes such as reverse osmosis, because dissolved species are moved away from the feed stream rather than the reverse. The process is utilised to move salt ions from one solution through exchange membranes to another solution in a setup called an electrodialysis cell, under the influence of an applied electric potential difference. The cell is made of a feed or diluate compartment and a concentrate or brine compartment, formed by anion and cation exchange membranes arranged between two electrodes. In most processes, multiple electrodialysis cells are put into a setup known as electrodialysis stack, with alternating anion and cation exchange membranes forming the multiple electrodialysis cells. Since the quantity of dissolved species in the feed stream is much lower than that of the fluid, ED has the advantage of giving a much higher feed recovery in many applications.

ELECTRODYNAMICS: 1. The study of electric charges in motion, the forces created by electric and magnetic fields and the relationship between them especially with respect to the theory and design of generators and motors or the branch of physics dealing with electric currents and associated magnetic forces. Since a moving charge produces a magnetic field, it is concerned with effects such as magnetism, electromagnetic radiation and electromagnetic induction, including such practical applications as the electric generator and the electric motor. When the velocities of the charged particles under consideration become comparable with the speed of light, corrections involving the theory of relativity must be made, leading to a theory known as relativistic electrodynamics, which is used in studies involving particle accelerators and electron tubes that are subject to high voltages and carry heavy currents. 2. Energy that moves as a transverse wave at a velocity of about $3 \times 10^8 \text{ m/s}$.

ELECTRODYNAMOMETER: An instrument for measuring electric current by indicating the levels of magnetic repulsion or attraction between a fixed and movable coil, one of which carries the current to be measured.

ELECTROENCEPHALOGRAPH (EEG; ENCEPHALOGRAPH): In medicine, it is the graphic measuring and recording of the electrical activity in the brain by placing between 16-20 electrodes on the scalp which are connected to an electroencephalograph and electronic instrument. It is helpful in the diagnosis of some brain disorders, such as epilepsy, tumours or brain damage as well as in establishing clinical death.

ELECTROFORMING: A method of forming intricate metal articles or parts by the electrodeposition of the metal on the removable conductive mould. The part is made by plating a metal skin onto a base form called a **mandrel**, which is removed after plating. This method differs from electroplating in that the plating is much thicker and can exist as a self-supporting structure when the mandrel is removed.

ELECTROFUGE: A leaving group that does not carry away the bonding electron pair. For example, in the nitration of benzene by NO_2^+ , H^+ is the electrofuge.

ELECTROGENIC: A term describing an ion transport protein or other cell mechanism that generates an electrochemical potential across a cell membrane by creating a net redistribution of charge ions.

Electrogenic pump is an active ion transport protein generating voltage across the cell membrane during facilitated diffusion. An example is the sodium potassium pump, which allows three sodium ions (Na^+) to be moved out of the cell and two potassium ions (K^+) to be moved into the cell in each cycle, creating a net negative charge inside the cell. It is the main method by which nerve impulses travel along the neurons and movement of ions across cell wall of fungi, plants and bacteria.

ELECTROGRAPHITE: Synthetic graphite made by electrical heating of graphitisable carbon.

ELECTROKINETIC POTENTIAL (ZETA POTENTIAL): The potential difference between the dispersion medium and the stationary layer of fluid attached to the dispersed particle or the electric potential associated with an electrical double layer around a colloid at the radius shear relative to the value of the bulk of the solution far from the colloid, with the radius of shear being the radius of the entity made of the colloid and the rigid layer of ions at the surface of the colloid. Alternatively, it may be defined as the electric potential in the interfacial double layer (DL) at the location of the slipping plane versus a point in the bulk fluid away from the interface. It is usually denoted using the Greek letter zeta, hence ζ -potential.

ELECTROLUMINESCENCE: An optical and electrical phenomenon in which a material emits light in response to the passage of an electric current or to a strong electric field. This is distinct from black body light emission resulting from heat (incandescence), from a chemical reaction (chemiluminescence), sound (sonoluminescence) or other mechanical action (mechanoluminescence).

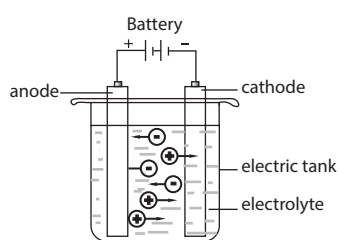
ELECTROLYSIS: The chemical breakdown of a substance by a direct electric current (dc) or the production of chemical changes by passing a direct electric current (dc) through a solution or molten salt (the electrolyte), resulting in the movement of ions to the electrodes, with positive ions (cations) to the negative electrode (cathode) and the negative ions (anions) to the positive electrode (anode). Important examples of electrolysis are the decomposition of water into hydrogen and oxygen and bauxite into aluminium and other chemicals. The process is commercially important as a stage in the separation of elements from naturally occurring sources such as ores, using an electrolytic cell.

ELECTROLYTE: A solution or molten substance capable of conducting an electric current due to the presence of positive and negative ions. It is composed of negatively charged species known as anions and positively charged species called cations. For example, sodium chloride (NaCl) is composed of sodium cations (Na^+) and chlorine anions (Cl^-), with the ratio of the two ions being always such that the substance is electrically neutral. If two wires connected to a light bulb and to a power source are placed in a beaker of water, it will not glow. However, if the sodium chloride is dissolved in it and the two wires are connected to the light bulb and to a power source the light bulb will glow because the solution can now conduct electricity. The amount of electric current that can be carried by an electrolyte solution is proportional to the number of ions dissolved in it, implying that the bulb will glow more brightly if the amount of sodium chloride in the solution is increased. Electrolytes can be strong or weak, based on the extent to which the substance ionises in a solution. Some of the factors that, therefore affect the conductivity of electrolytes are as follows: (i) The higher the concentration of ions in a solution, the higher its conductivity; (ii) the type of substance that dissolves in water; (iii) temperature, as the warmer the solution, the higher the solubility of the material being dissolved and therefore the higher the conductivity.

ELECTROLYTIC CAPACITOR (ELECTROLYTIC CONDENSER): A type of capacitor that uses an electrolyte, an ionic conducting liquid, as one of its plates, to achieve a larger capacitance per unit volume than other types. One plate is coated through electrolysis with an oxide to serve as the dielectric, while the other plate is replaced by

an electrolyte. They are often referred to in electronics usage simply as electrolytics. They are used in relatively high-current and low-frequency electrical circuits, particularly in power supply filters, where they store charge needed to moderate output voltage and current fluctuations in rectifier output. They are also widely used as coupling capacitors in circuits where AC should be conducted but DC should not. There are two types of electrolytics; aluminium and tantalum. Electrolytic capacitors are capable of providing the highest capacitance values of any type of capacitor. However, they have drawbacks which limit their use. The voltage applied to them must be polarised; one specified terminal must always have positive potential with respect to the other. Therefore, they cannot be used with AC signals without a DC bias. They also have very low breakdown voltage, higher leakage current and inductance, poorer tolerances and temperature range and shorter lifetimes compared to other types of capacitors.

ELECTROLYTIC CELL: A cell in which electrolysis occurs or a cell in which current is passed through the electrolyte from an external source. It decomposes chemical compounds by means of electrical energy, in a process called electrolysis, with the result that the chemical energy is increased.



Electrolytic cell

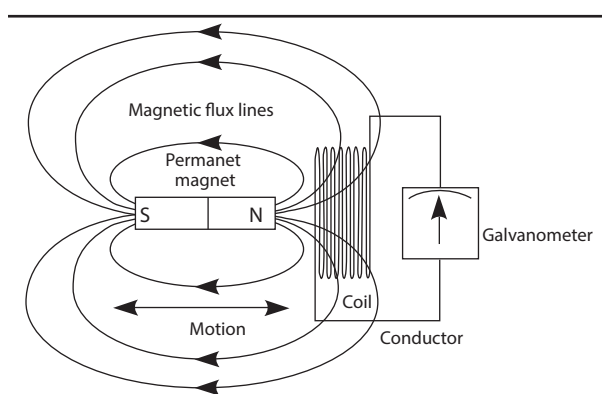
ELECTROLYTIC CONDUCTION:

Transportation of electrical current by ions through a solution or pure liquid or the conduction of electric charges (ions).

ELECTROMAGNETIC ENERGY: A form of energy in the form of transverse wave that travels with the speed of light and does not require a medium or matter to carry it (it travels through space). In nature, the main sources of electromagnetic energy are the Sun, the Earth and the ionosphere. Examples of electromagnetic energy are radio waves, microwaves, infrared radiation, visible light, ultraviolet light, x-rays and gamma radiation.

ELECTROMAGNETIC INDUCTION (MAGNETIC INDUCTION):

The production of an electromotive force across an electrical conductor by its motion across a magnetic field or by a change in magnetic flux in a magnetic field. Practically, it is used in transformers, electric motors and generators.



Electromagnetic induction

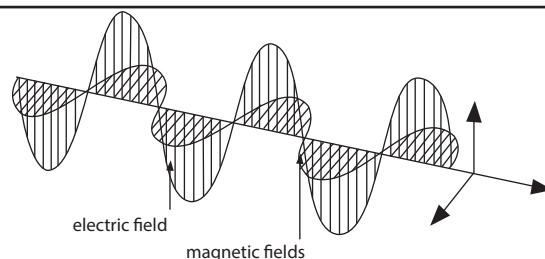
ELECTROMAGNETIC INTELLIGENCE (ELINT): The interception and analysis of electromagnetic signals of other countries including radar, radio, telephony and microwave transmissions. It can be carried out effectively using satellites. ELINT is a sub-category of SIGINT (signals intelligence).

ELECTROMAGNETIC POTENTIALS: Name for a scalar field, ϕ and a vector field, A , in terms of which the electric field and magnetic field,

E and B respectively, may be expressed completely. This results in a degree of simplification in the use of Maxwell's equations, particularly in their relativistic formulation.

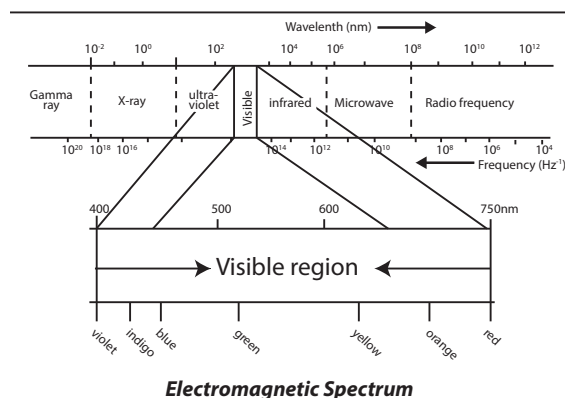
ELECTROMAGNETIC PUMP: A pump used for moving liquid metals such as liquid-sodium coolant in a fast reactor. The conductive liquid is made to move through a pipe by sending a large current transversely through the liquid. This current reacts with a magnetic field that is at right angles to the pipe and to current flow, to move the current-carrying liquid conductor.

ELECTROMAGNETIC RADIATION (E-M RADIATION; EMR): A form of energy that is transmitted through space at a very high speed or self-propagating waves in a vacuum or in matter, made of electric and magnetic field components, which oscillate in phase perpendicular to each other and to the direction of energy propagation. It has both wave and corpuscular nature. It can be characterised by such parameters as velocity, wavelength and amplitude as well as particles or packets of energy known as photons. It is classified according to the frequency of its wave and in order of increasing frequency and decreasing wavelength. The various types include radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, x-rays and gamma rays. The quantum of the electromagnetic interaction and the basic unit of light and all other forms of electromagnetic radiation is the photon and it is also the force carrier for the electromagnetic force.



Electromagnetic radiation showing the electric/magnetic fields

ELECTROMAGNETIC SPECTRUM: The range of wavelengths over which electromagnetic radiation extends or the characteristic distribution of electromagnetic radiation emitted or absorbed by a particular object. Only a small portion of this spectrum (390 to 700 nm) is detectable by the human eye. The spectrum extends from low frequencies used for modern radio to gamma radiation at the short-wavelength end and covers wavelengths from thousands of kilometres down to a fraction of the size of an atom. Although, in principle the spectrum is infinite and continuous, the long wavelength limit is the size of the universe itself, while it is believed that the short wavelength limit is in the region of the Planck's length. The main segments of electromagnetic spectrum are gamma, x-ray, ultraviolet, visible, infrared, microwave and radiowave radiations.



Electromagnetic Spectrum

ELECTROMAGNETIC UNIT: Now replaced by SI units, it is any unit in the centimetre-gram-second (cgs) system of units for measuring

electricity and magnetism, that gives a value of 1 to the magnetic constant. They are based on a system of equations in which the permeability of free space is taken as unity and by means of which the abampere is defined as the fundamental unit of current. Examples are the abampere and the abvolt.

ELECTROMAGNETIC WAVES: Oscillating electric and magnetic fields travelling together through space at a speed of nearly 300,000 km/s (3×10^8 m/s). They may be confined in tubes, such as wave guides or guided by transmission lines. The term hertzian or electric wave is often applied to electromagnetic waves in the radar and radio range. These waves were predicted by J. C. Maxwell in 1864 and verified experimentally by H. Hertz in 1887.

ELECTROMERIC EFFECT: A molecular polarisability effect occurring by an intramolecular electron displacement characterised by the substitution of one electron pair for another within the same atomic octet of electrons.

ELECTROMETALLURGY: The range of metallurgical processes in which electricity has a key role. Examples include electroplating and the use of arc furnaces.

ELECTROMETER: An electrical instrument for determining a voltage difference (electric charge or electrical potential difference) without drawing an appreciable current from the source. They vary from historical hand-made mechanical instruments through high-precision electronic devices to modern electrometers based on vacuum tube or solid state technology which can be used to make voltage and charge measurements with very low leakage currents, down to 1 femtoampere.

ELECTROMIGRATION: The transport of particles under the influence of an electric field.

ELECTROMOTIVE FORCE (EMF): The highest (maximum) potential difference, voltage or energy that can be generated by a particular source of electric current in driving a unit charge around an electrical circuit. The unit is the volt (V).

ELECTROMOTIVE SERIES (ELECTROCHEMICAL SERIES): A series of chemical elements arranged in order of their electrode potentials. The hydrogen electrode ($H^+ + e \rightarrow \frac{1}{2}H_2$) is taken as having zero electrode potential. Elements that have a greater tendency than hydrogen to lose electrons to their solution are taken as **electropositive**; those that gain electrons from their solution are below hydrogen in the series and are called **electronegative**.

ELECTROMYOGRAPHY (EMG): In medicine, it is a diagnostic procedure of recording the electrical activity of muscle tissue, using electrodes that are attached to the skin or inserted into the muscle to transmit or detect the signals. It can be displayed visually or as audible signal. It can be used to diagnose conditions such as nerve and muscle disorders. The record produced by this procedure is called **electromyogram**.

ELECTRON: A stable, negatively charged elementary particle of rest mass 9.1×10^{-31} kg (about $\frac{1}{1836}$ of the mass of proton or neutron) and carries a charge of 1.6×10^{-19} coulombs. It moves at a great speed around the nucleus. It is the lightest known particle which possesses an electric charge. It was discovered in 1895 by J. J. Thomson in the form of cathode rays and it was the first elementary particle to be identified.

ELECTRON ACCEPTOR: 1. A chemical entity that accepts electrons transferred to it from another compound. It is, therefore, an oxidising agent, that is, by virtue of its accepting electrons, it is itself reduced in the process. Examples of electron acceptors include oxygen, nitrate, iron(III), manganese(IV), sulfate and carbon dioxide. 2. A molecule that forms part of the electron transport system that transfers electrons ejected by chlorophyll during photosynthesis. Part of the energy carried by the electrons is transferred to adenosine triphosphate (ATP), part is transferred to NADPH and part is lost in the transfer system.

ELECTRON AFFINITY: The energy change occurring when an atom or a molecule gains an electron to form a negative charge or the energy evolved or liberated when one mole of an element in the form of gaseous atom forms negative ions. In other words, it is an enthalpy change when one electron is added on to the outermost shell of an isolated gaseous atom to form a negative ion. In terms of a chemical equation, the electron affinity of a species M is equal to the exothermicity of the reaction $e + M \rightarrow M^-$, where the negative ion M^- is left in its lowest electronic, vibrational and rotational state.

ELECTRON ATTACHMENT: The transfer of an electron to a molecular entity, resulting in a molecular entity of (algebraically) increased negative charge. It is not an attachment.

ELECTRON BEAM: Electrons released from a hot cathode by thermionic emission, all moving through vacuum with the same acceleration from cathode to anode. The speed at which the electron arrive at the anode can be determined by equating the kinetic energy gained to electrical potential energy lost: $\frac{1}{2}mv^2 = eV$; $v = \sqrt{\frac{2eV}{m_e}}$, where

m is the mass of electrons, e the electronic charge, v the speed of electrons and V is the potential difference across the terminals.

ELECTRON BIPRISM: Arrangement of fields that split a beam of electrons or other charged particles in an analogous way to an optical biprism (a glass prism that produces a double image of a single object).

ELECTRON BOMBARDMENT THRUSTERS (ELECTROSTATIC ION THRUSTERS): One of the most promising forms of electric space propulsion and one of only two forms of ion propulsion currently employed aboard spacecraft, the other being the Hall effect thruster.

ELECTRON CAPTURE (INVERSE BETA DECAY): 1. A radioactive transformation process for isotopes that have too many protons in the nucleus and insufficient energy to emit a positron. In this case, one of the orbital electrons, usually from the K or L shell is captured by a proton in nucleus, thus forming a neutron and a neutrino (neutral elementary particle). Since the proton is changed to a neutron the atomic number decreases by 1, the nuclide is transformed into a new element. But the atomic mass number remains unchanged. Examples of nuclides that undergo electron capture are ^{41}Ca , ^{51}Cr , ^{56}Ni , etc. 2. The formation of a negative ion by an atom or molecule when it acquires an extra free electron or a radioactive transformation in which a nucleus acquires an electron from an inner orbit of the atom thereby transforming initially into a nucleus with the same mass number but an atomic number one less than that of the original nucleus. It is accompanied by emission of an x-ray photon or Auger electron as the vacancy in the inner orbit is filled by an outer electron. If the energy difference between the parent atom and the daughter atom is less than 1.022 MeV, positron emission is forbidden and electron capture is the sole decay mode. For example, Rubidium-83 will decay to Krypton-83 only by electron capture (the energy difference is about 0.9 MeV).

ELECTRON CAPTURE DETECTOR: A small radioactive source containing ^3H or ^{63}Ni that ionises the molecules of the carrier gas (nitrogen or argon-methane) and a potential difference creates a small current. This current is reduced when an electronegative substance such as a halocarbon is introduced. The reduction in current is a measure of the concentration of the electronegative substance. The detection limit (threshold) varies greatly according to the substances to be analysed and can reach a mixing ratio of 10^{-12} . The linear dynamic range may be 10^4 but the maximum measuring value generally lies below 1 ppmv.

ELECTRON CARRIER: A protein or coenzyme that can reversibly gain and lose electrons and that serves the function of carrying electrons from one site to another or a molecule that accepts electrons from electron donors and donates them to electron acceptors, creating an energy-producing electron transport chain such as occurs in respiration and photosynthesis. Examples are iron-sulfur proteins, blue copper proteins and cytochromes.

ELECTRON CHARGE: The fundamental negative electric charge which appears on the electron or univalent ions. It is equal to

$1.6021765 \times 10^{-19}$ coulomb or $4.80320451 \times 10^{-10}$ electrostatic unit (esu or statcoulomb). Its symbol is e . In addition to the electron, all freely existing charged subatomic particles thus far discovered have an electric charge equal to this value or some whole-number multiple of it.

ELECTRON CLOUD: A region of negative charge around the nucleus of an atom where electrons are likely to be found.

ELECTRON CONFIGURATION: The electron configuration of an atom gives the distribution of electrons in energy levels or shells around the nucleus of an atom. Numbers (1, 2, 3, etc.) are used to identify the energy levels. Electrons that are in the first energy level (energy level 1) are closest to the nucleus and will have the lowest energy. Electrons further away from the nucleus will have a higher energy level. Within each energy level, the electrons move in spaces or subshells called **orbitals**, which are identified by the letters s, p, d and f .

ELECTRON CORRELATION: The adjustment of electron motion to the instantaneous (as opposed to time-averaged) positions of all the electrons in a molecular entity.

ELECTRON COUNTING RULES: Rules for counting the numbers of electrons that give rise to stable structures in atoms, molecules and solids. Electron counting is a formalism used for classifying compounds and for explaining or predicting electronic structure and bonding.

ELECTRON CYCLOTRON MASER (ECM): A type of maser in which the operation is based on the stimulated emission of radiation caused by the motion of high-energy electrons moving, such as in a cyclotron. It constitutes a cornerstone of relativistic electronics, a discipline that has emerged from our understanding and utilisation of relativistic effects for the generation of coherent radiation from free electrons. There are many practical applications of the device including plasma heating in fusion research and telecommunications. This mechanism of maser emission may account for short intense pulses in solar microwave radiation and for certain ionospheric radio emissions.

ELECTRON DEFICIENT COMPOUNDS: Compounds in which there are fewer electrons forming the chemical bonds than required in normal electron-pair bonds or compounds with at least one atom, except H, that shares less than eight electrons. Such compounds use multicentre bonds. Examples include boron trifluoride (BF_3), boron trichloride (BCl_3), borane (BF_3) and aluminium chloride (AlCl_3).

ELECTRON DEGENERACY PRESSURE: Electron degeneracy pressure occurs when electrons are compressed into a very small volume. Since their positions are well-known, they gain a large momentum in accordance with Heisenberg's uncertainty principle. This momentum creates a pressure which is temperature-independent and can result in helium flash in stars.

ELECTRON DEGENERATE MATTER: A form of degenerate matter in which the weight of overlying material tries to force all of the electrons surrounding the atomic nucleus into the lowest energy quantum state. The electrons resist, because of the Paul exclusion principle and so exert a pressure, known as degenerate electron pressure or electron degeneracy pressure, that halts further collapse. This pressure is sufficient up to the Chandrasekar limit of about 1.4 solar masses. However, beyond this critical mass, gravity overwhelms the degeneracy pressure and results in further collapse.

ELECTRON DENSITY: A number that is used for the relative probability of finding an electron at a particular point in space. In quantum mechanics, an electron can be viewed as a stationary wave or, cloud of negative charge. Therefore, the electron density is a number that tells us how much charge is located at each point in the cloud. In other words, electron density is a representation of the probability of finding an electron in a specific location around an atom or molecule. Electron dense structures deflect most of the electron beam and appear dark, whereas electron lucent features do not deflect many electrons and appear bright. Such differences are often accentuated by the use of electron stains that render parts of the specimen electron dense that would otherwise appear electron lucent. In general, the electron is more likely to be found in regions with high electron density.

ELECTRON DIFFRACTION: The diffraction of a beam of electrons by atoms or molecules, which refers to the wave structure of electrons. Due to the dependence of the diffraction pattern on the distances between the atoms, it is an important tool for the study of the structure of crystals and of free molecules, in much the same way as the use of x-rays for these purposes. It also presents the most obvious confirmation of quantum mechanics.

ELECTRON DONOR: A substance that donates electrons in an oxidation-reduction reaction. It is a reducing agent in that, by virtue of its donating electrons, it is itself oxidised in the process. In most cases, the transfer of electronic charge to an electron acceptor may be only fractional, implying that an electron is not totally transferred, but results in an electron resonance between the donor and acceptor. This leads to the formation of charge transfer complexes in which the components mostly retain their chemical identities. Typical reducing agents undergo permanent chemical alteration through covalent or ionic reaction which results in the complete and irreversible transfer of one or more electrons.

ELECTRON EMISSION: The liberation of electrons from the surface of a substance. It can occur as a result of thermionic emission, light (photoelectric effect, high electric field (field emission) or bombardment by ions or other electrons (secondary emission). It may also occur from radioactive. In all cases, the electrons must acquire energy from the outside source in excess of the work function of the substance. Most electron tubes and cathode-ray tubes depend on thermionic emission, whereas photoelectric cells rely on photoelectric emission. Field emission is important in the field-emission microscope and secondary emission is the principle used by electron multipliers and storage tubes.

ELECTRON ENERGY: The potential difference through which electrons are accelerated before they are used to bring about electron ionisation.

ELECTRON ENERGY LOSS SPECTROSCOPY (EELS): A technique for studying atoms, molecules or solids in which a substance is bombarded with monochromatic electrons and the energies of scattered electrons are measured to determine the distribution of energy loss. It consists of three parts: (i) Zero-loss peak at 0 eV, which mainly contains electrons that still have the original beam energy E_0 (they have only interacted elastically or not at all with the specimen). (ii) Low-loss region ($< 100\text{eV}$), where the electrons that have induced plasmon oscillations occur. Since the plasmon generation is the most frequent inelastic interaction of electron with the sample, the intensity in this region is relatively high. Intensity and number of plasmon peaks increases with specimen thickness. (iii) High-Loss region ($> 100\text{eV}$), in which for the ionisation of atoms, a specific minimum energy, the critical ionisation energy E_c or ionisation threshold, must be transferred from the incident electron to the expelled inner-shell electron, which leads to ionisation edges in the spectrum at energy losses that are characteristic for an element. Thus, EELS is complementary to X-ray spectroscopy and it can be utilised for qualitative and quantitative element analysis as well, in particular, the detection of light elements. In the high-loss region, the amount of inelastically scattered electrons drastically decreases with increasing energy loss, thus small peaks are superimposed on a strongly decreasing background.

ELECTRON FLOW: The transfer of electron along a series of carrier molecules in the electron transport chain or electrical current produced by the movement of free electrons towards a positive terminal.

ELECTRON GAS: 1. A model of the electrons in a metal or a plasma in which they are regarded as forming a gas that interacts with a uniformly distributed background of positive charge to ensure that the system is electrically neutral. 2. A concentration of electrons whose behaviour is, in first approximation, not governed by forces.

ELECTRON GUN (ELECTRON EMITTER): A device used in cathode ray tubes, electron microscope, etc, to produce a steady, narrow beam of electrons that have precise kinetic energy or an electrical component in many electronic devices consisting of a series of electrodes including

a cathode for producing an electron beam. They may be classified by the type of electric field generation (DC or RF), emission mechanism (thermionic, photocathode, cold emission or plasma source), focusing (pure electrostatic or with magnetic fields) or by the number of electrodes. They play important roles in television tubes and monitors which use cathode ray tube technology, as well as in other instruments such as electron microscopes and particle accelerators.

ELECTRON IN A BOX MODEL: A model of the atom useful for explaining the origin of atomic energy levels. The model assumes that, if an electron is confined to move in one dimension by a box, the de Broglie waves associated with the electron will be standing

waves of wavelength $\frac{2L}{n}$, where L is the length of the box and n is a positive integer. Further, the kinetic energy of the electron in the box is $\frac{n^2 h^2}{8m_e L^2}$.

ELECTRON KINETIC ENERGY: The difference between the energy of the exciting photon and the ionisation energy. Mathematically, kinetic energy (K.E.) = $hf - eV$.

ELECTRON MAGNETIC RESONANCE, EMR (ELECTRON PARAMAGNETIC RESONANCE; EPR): It is the magnetic behaviour of an electron immersed in an external magnetic field. In so doing it uses the concept of a "bound" as opposed to a "free" electron.

ELECTRON MASS: The mass of the electron or the electron's mass, written as m_e , is $9.109\,382\,15(45) \times 10^{-31}$ kg.

ELECTRON MICROPROBE (EMP; ELECTRON PROBE MICROANALYSER; EPMA): An analytical tool used to non-destructively determine the chemical composition of small volumes of solid materials. It is an electron microscope fitted with an x-ray spectrum analyser, which examines the high-energy x-rays emitted by the sample when it is bombarded with electrons. The identity of different atoms or molecules can be determined from their x-ray emissions, so the electron probe analyser not only provides a magnified image of the sample, but also information about the sample's chemical composition.

ELECTRON MICROSCOPE: Any of a class of microscopes that use electrons rather than visible light to produce magnified images, especially of objects having dimensions smaller than the wavelengths of visible light, with linear magnification approaching or exceeding a million (10^6). It was first constructed by Max Knoll and Ernst Ruska around 1930. The instrument consists typically of an evacuated column of energetic lenses with a 20–1,000 kV electron gun at the top of a fluorescent screen or photographic plate at the bottom. The various lenses allow the operator to see details almost at the atomic level (0.3 nm) at up to a million times magnification to obtain diffraction patterns from very small areas.

ELECTRON OPTICS: The study of the use of electron lenses in electron microscope lenses, the electron microscope cathode ray tubes and other similar devices. The focusing of beams of positive or negative ions also relies on these methods. It also deals with the focusing and deflection of electrons using magnetic and/or electrostatic fields.

ELECTRON PAIR PRODUCTION: The formation or materialisation of two electrons, one negative and the other positive (positron), from a pulse of electromagnetic energy travelling through matter, usually in the vicinity of an atomic nucleus. Pair production is a direct conversion of radiant energy to matter. It is one of the principal ways in which high-energy gamma rays are absorbed in matter. For pair production to occur, the electromagnetic energy, in a discrete quantity called a photon, must be at least equivalent to the mass of two electrons.

ELECTRON PARAMAGNETIC RESONANCE (EPR; ELECTRON SPIN RESONANCE, ESR): 1. A spectroscopic method of locating electrons within the molecules of a paramagnetic substance in order to provide information regarding its bond and structure. 2. Magnetic resonance arising from the magnetic moment of unpaired electrons in a paramagnetic substance or in a paramagnetic centre in a diamagnetic substance.

ELECTRON PROBE MICROANALYSIS (EPM; ELECTRON MICROPROBE, EMP): A method of analysing a very small quantity (about 10^{-13} gram) of a substance. It involves directing a very finely focused beam of electrons on to the sample to produce the characteristic x-ray spectrum of the elements present. Quantitatively, it can be used for elements with atomic numbers in excess of 11. It can also be used to non-destructively determine the chemical composition of small volumes of solid materials and it works like a scanning electron microscope. The sample is bombarded with an electron beam and signals that come from the sample are collected. This enables the elements present within sample volumes of 10-30 cubic micrometres or less to be determined. For example, the concentrations of elements from boron to plutonium can be measured at levels as low as 100 parts per million (ppm).

ELECTRON PROBE X-RAY MICROANALYSIS (EPXMA): Any analytical technique which uses the excitation and evaluation of the characteristic X-ray spectrum of a solid specimen by a focused electron beam (typically of a diameter of less than $1\mu\text{m}$). Qualitative and quantitative X-ray analysis is the main feature of EPXMA. For obtaining supplementary information the secondary, back scattered and absorbed electrons are also frequently observed. Qualitative and quantitative X-ray analysis is, however, the main feature of EPXMA.

ELECTRON QUANTUM NUMBERS: Each electron has a set of four numbers that specify it completely; no two electrons in the same atom can have the same four. They are n (principal quantum number), l (azimuthal quantum number), m_l (magnetic quantum number) and m_s (spin quantum number).

ELECTRON RADIUS: The classical value r of $2.8179403 \times 10^{-13}$ centimetre for the radius of an electron; obtained by equating mc^2 for the electron to e^2/r , where e and m are the charge and mass of the electron respectively; any classical model for an electron will have approximately this radius.

ELECTRON REST MASS: Atomic fundamental physical constant used as atomic unit of mass. Its symbol is m_e , which is $= 9.1093897(54) \times 10^{-31}$ kg.

ELECTRON SCATTERING: Deflection of the path of electrons as they pass through a solid (typically a metal, semiconductor or insulator). Deflections or collisions, are caused by electrostatic forces operating between the negatively charged electrons and atoms within the solid. These forces reduce the speed of the electrons, thereby limiting the performance of electronic devices based on transistors and integrated circuits. The deflection of a beam of electrons by a target is also called electron scattering and has been used to probe the size and charge distribution of atomic nuclei. In the early 1970s, electron scattering helped to confirm that protons and neutrons are made of still more elementary subatomic particles known as quarks.

ELECTRON SPECTROSCOPY: Any of a number of techniques in which information is obtained by analysing the energy spectrum of electrons. Photoelectron is a typical example. Electron spectroscopy is an analytical technique to study the electronic structure and its dynamics in atoms and molecules. In general, an excitation source such as x-rays, electrons or synchrotron radiation will eject an electron from an inner-shell orbital of an atom. Detecting photoelectrons that are ejected by x-rays is called x-ray **photoelectron spectroscopy** (XPS) or electron spectroscopy for chemical analysis (ESCA). Detecting electrons that are ejected from higher orbitals to conserve energy during electron transitions is called **Auger electron spectroscopy** (AES). Experimental applications include high-resolution measurements on the intensity and angular distributions of emitted electrons as well as on the total and partial ion yields. Ejected electrons can escape only from a depth of approximately 3 nanometres or less, making electron spectroscopy most useful to study surfaces of solid materials. Depth profiling is accomplished by combining an electron spectroscopy with a sputtering source that removes surface layers.

ELECTRON STOPPING POWER: The average energy loss per unit distance travelled along an electron path, $\frac{dE}{dx}$. When a beam of

electrons strikes a target or specimen, there are three ways in which the electrons may lose energy: low energy collisions, X-ray production and formation of a spectral continuum.

ELECTRON TEMPERATURE (KINETIC TEMPERATURE): The temperature determined by the mean kinetic energy of free electrons in plasma. Inside stars, electron temperature is likely to match the temperature as measured by other methods, but in the rarefied environment of a nebula or the outer atmospheres of stars, it may be quite different.

ELECTRON TRANSFER: The transfer of an electron from one molecular entity to another or between two localised sites in the same molecular entity.

ELECTRON TRANSFER PHOTOSENSITISATION/PHOTOSENSITIZATION: Photochemical process in which a reaction of a non-absorbing substrate is induced by electron transfer (not energy transfer) with an excited light-absorbing sensitiser. The overall process must be such that the sensitizer is recycled. Depending on the action of the excited sensitiser as electron donor or acceptor the sensitisation is called reductive or oxidative.

ELECTRON TRANSFER REACTION (ELECTRON-TRANSFER REACTION): A chemical reaction, usually non-stoichiometric in which electron(s) are transferred from either one species to the other or from one part of a molecule to another part leading to a disproportion or rearrangement. It involves a transfer, addition or removal of electrons and complexes of transition metals are often involved. In such complexes one general mechanism for electron transfer is the inner-sphere mechanism, in which two complexes form an intermediate, with ligand bridges enabling electrons to be transferred from one complex to another complex. The other main mechanism is the outer-sphere mechanism, in which two complexes retain all their ligands, with electrons passing from one complex to the other. The rates of electron-transfer reactions vary considerably and they can be explained in terms of the way in which molecules of the solvent solvating the reactants rearrange so as to solvate the products in the case of the outer-sphere mechanism. In the case of the inner-sphere (ligand-bridged) reactions the rate of the reaction depends on the intermediate and the way in which the electron is transferred.

ELECTRON TRANSITION: The passage of an electron from one orbital (allowed energy state) in an atom or molecule to another orbital. An electron transition is accompanied by the emission or absorption of energy corresponding to the difference in energy between the two orbitals.

ELECTRON TRANSPORT CHAIN (RESPIRATORY CHAIN): A sequence of biochemical reduction-oxidation reactions by which free energy contained within hydrogen derived from the Krebs cycle is released so that it can be used to synthesise adenosine triphosphate (ATP) during the final stage of aerobic respiration. Each reaction involves a specific electron-carrier molecule, which has a particular affinity for hydrogen or an electron derived from hydrogen. The carriers are organised in a sequence of increasing affinity and the final link in the electron transport chain is oxygen that combines with the hydrogen and electrons to form water. Examples of electron transport chains occur in mitochondria.

ELECTRON TUBE (VALVE): An evacuated glass or metal tube which may contain gas at low pressure, through which electrons flow between two or more electrodes. A heated filament, the cathode, emits electrons which are attracted to the positively-charged anode. By varying the voltage on intermediate (grid) electrodes, the electron flow can be regulated and the tube made to work as an amplifier.

ELECTRON VOLT (ELECTRONVOLT): A unit of energy used for convenience in atomic systems, which is equal to the work done on an electron or energy gained by the electron accelerated through a potential difference of one volt. Its symbol is eV and it is equal to 1.602×10^{-19} joule, used as a measure of particle energies although it is not an SI unit. It may also be defined as the unit of mass of elementary

particles, measured as a function of energy. In this case, it is usually expressed in terms of mega electron volts (MeV).

ELECTRON-COUNTING RULES: Rules establishing correspondence between the topology of a molecular structure and the number of electrons which may be placed into its bonding molecular orbitals. In the origin of various electron counting rules lies a general assumption that the completeness of the valence electron shell of a molecular entity belonging to a certain structural type serves as the major criterion of structural stability.

ELECTRON-DEFICIENT BOND: A single bond between adjacent atoms that is formed by less than two electrons, as in diborane (B_2H_6) or dialuminium hexachloride (Al_2Cl_6).

ELECTRON-NUCLEAR DOUBLE RESONANCE (ENDOR): A magnetic double resonance technique involving excitation of both electron spins and nuclear spins. It permits greatly enhanced resolution, in which a material is simultaneously irradiated at one of its electron paramagnetic resonance frequencies and by a second oscillatory field, whose frequency is swept over the range of nuclear frequencies.

ELECTRON-SPIN RESONANCE (ESR; ELECTRON PARAMAGNETIC RESONANCE, EPR): A spectroscopic method of locating electrons within the molecules of a paramagnetic substance in order to provide information regarding its bonds and structure. It involves studying chemical species that have one or more unpaired electrons, such as organic and inorganic free radicals or inorganic complexes possessing a transition metal ion. Its basic physical concepts are similar to those of nuclear magnetic resonance (NMR), however, it is electron spins that are excited instead of atomic nuclei spins. Since most stable molecules have all their electrons paired, this technique is of great specificity, particularly since ordinary chemical solvents and matrices do not give rise to EPR spectra. It was first observed by the Soviet physicist, Yevgeny Zavoisky in 1944 and developed independently at the same time by Brebis Bleaney.

ELECTRONEGATIVITY: A measure of the tendency of an atom to attract a bonding pair of electrons to itself to form negative ions. US chemist Linus Pauling (1901-1994) devised an electronegativity scale, called the Pauling scale to indicate the relative power of attraction of elements for electrons. Fluorine, the most non-metallic element, has a value of 4.0 on this scale; oxygen, the next most non-metallic, has a value of 3.5; caesium and francium which are the least electronegative has a value of 0.7. In a covalent bond between two atoms of different electronegativities, the bonding electrons will be located close to the more electronegative atom, creating a dipole. For example, chlorine atom has a higher electronegativity than the hydrogen atom, so the bonding electrons will be closer to the Cl than to the H in the HCl molecule.

ELECTROENCEPHALOGRAPH (EEG; ENCEPHALOGRAPH): In medicine, it is the graphic measuring and recording of the electrical activity in the brain by placing between 16-20 electrodes on the scalp which are connected to an electroencephalograph and electronic instrument. It is helpful in the diagnosis of some brain disorders, such as epilepsy, tumours or brain damage as well as in establishing clinical death.

ELECTRONEGATIVITY: The tendency of atoms to attract electrons. The electronegativity of the elements starts from about 0.7 (Francium (Fr) and goes up to 4 (Fluorine (F)).

ELECTRONEUTRALITY PRINCIPLE: The principle expresses the fact that all pure substances carry a net charge of zero.

ELECTRONIC BALANCE: A balance (weighing scale) that generates a current proportional to the displacement of the pan. It is used to find accurate measurements and is commonly used in laboratories for weighing precise measurement of chemicals for use in various experiments.

ELECTRONIC BUSINESS (E-BUSINESS; E-COMMERCE; ELECTRONIC COMMERCE): Businesses or commercial transactions conducted over the Internet. Examples are web sites such as Amazon.com, Buy.com and eBay.

ELECTRONIC CHEMICAL POTENTIAL: The quantity that measures the escaping tendency of electrons from a species in its ground state. It is the negative of the absolute electronegativity.

ELECTRONIC DATA INTERCHANGE (EDI): A set of standards for exchanging business data that helps specify how electronic data such as orders, confirmations, tracking shipments and electronic invoice should be handled or formatted between organisations. It is commonly used for e-commerce purposes. The standard is ANSI X12, which was developed by the Data Interchange Standards Association. An EDI message contains a string of data elements called a **data segment**, each of which represents a singular fact, such as a price, product model number, etc, separated by delimiter. Its purpose is to help eliminate paperwork and at the same time speed up the process of exchanging data.

ELECTRONIC DATA INTERCHANGE FOR ADMINISTRATION, COMMERCE AND TRANSPORT (EDIFACT): The United Nations rules that comprise a set of internationally agreed standards, directories and guidelines for the electronic interchange of structured data and in particular that related to trade in goods and services between independent, computerized information systems.

ELECTRONIC DATA PROCESSING (EDP): In computing, it is the abbreviation for electronic data processing, which is the use of computers in recording, classifying, manipulating and summarising data.

ELECTRONIC EFFECT (STEREOELECTRONIC EFFECT): The manner by which the reactivity of an organic compound is influenced by displacement or distribution of its electrons. It influences the structure, reactivity or properties of molecule but is neither a traditional bond nor a steric effect.

ELECTRONIC ENERGY MIGRATION (ELECTRONIC ENERGY HOPPING): The movement of electronic excitation energy from one molecular entity to another of the same species or from one part of a molecular entity to another of the same kind, for example, excitation migration between the chromophores of an aromatic polymer. The migration can happen via radiative or radiationless processes.

ELECTRONIC FILTER: An arrangement of electronic components used in a circuit to transmit signals within a given frequency range, rejecting others. It typically consists of an inductor in series with a capacitor shunted across the load to short out high-frequency current. A low-pass filter transmits frequencies below a specified cut-off and blocks out higher frequencies. In a high-pass filter the inductor and capacitor are interchanged and only high frequencies are transmitted. Electronic filters are used in radios, television receivers, telephone systems, etc.

ELECTRONIC FLASH: A discharge tube that produces a high-intensity flash of light used for photography in dim conditions. It produces a flash of artificial light (typically 1/1000 to 1/200 of a second) at a colour temperature of about 5226.85 °C (5500 K) to help illuminate a scene. A major purpose of a flash is to illuminate a dark scene and capturing quickly moving objects or changing the quality of light.

ELECTRONIC GEOMETRY: The geometric distribution of orbitals containing the shared and unshared electron pairs surrounding the central atom of a molecule or polyatomic ion.

ELECTRONIC LENS: A device used to focus an electron beam. It is analogous to an optical lens.

ELECTRONIC MAIL (E-MAIL): Correspondence sent via a computer network. It includes messages, documents, texts, graphics, sounds, animated images, etc., from one user of a computer system to another. E-mail is part of the standard TCP/IP set of protocols. Sending messages is typically done by SMTP (Simple Mail Transfer Protocol) and receiving messages is handled by POP3 (Post Office Protocol 3) or

IMAP (Internet Message Access Protocol). IMAP is the newer protocol, allowing users to view and sort messages on the mail server, without downloading them to the hard drive.

ELECTRONIC PERSONAL DOSIMETER (EPD): In radiation protection, it is a type of active dosimeter. It is designed to provide real time information on dose and dose rate.

ELECTRONIC PUBLISHING (E-PUBLISHING; DIGITAL PUBLISHING): Distribution of information such as of e-books and digital magazines using computer-based media such as diskettes, CD-ROMs, DVDs, portable document files (PDF) or over the internet.

ELECTRONIC SERIAL NUMBER (ESN): A 32 bit long unique identification number embedded on a microchip in mobile phones to check the validity of a call to prevent fraud. When a call is made, the ESN is automatically transmitted to a base station.

ELECTRONIC SPECTRA OF MOLECULES: The spectra associated with transitions between the electronic states of molecules. These transitions correspond to the visible or ultraviolet regions of the electromagnetic spectrum. They are used to obtain information about energy levels in molecules, interatomic distances, dissociation energies of molecules and force constants of chemical bonds.

ELECTRONIC STABILITY: Unavailability of another electronic structure (different electronic state) of lower energy with the same number of electrons.

ELECTRONIC STATE: An arrangement allowed by the laws of quantum mechanics of electrons within an atom, molecule (or system of molecules).

ELECTRONIC TRANSITION: The transfer of an electron from one energy level to another. They take place when valence electrons in a molecule are excited from one energy level to a higher energy level. The energy change associated with this transition gives information on the structure of a molecule. It also determines many molecular properties, including colour. The relationship between the energy involved in electronic transition and the frequency of radiation is given by Planck's law. The electronic transitions of molecules in solution may highly depend on the type of solvent, with additional bathochromic or hypsochromic shifts. The electronic transitions in organic compounds can be determined by UV/VIS spectroscopy provided that transitions in the UV or visible range of the electromagnetic spectrum exist for this compound. Electronic transitions also have 'bands' associated with them. These include the R-band (which is radical-like), the K-band (which is conjugated), the B-band from benzoic and the E-band from ethylenic. Spectral lines are associated with atomic electronic transitions and polyatomic gases have their own absorption band system.

ELECTRONICALLY EXCITED STATE: A state of an atom or molecular entity which has greater electronic energy than the ground state of the same entity.

ELECTRONICS: The branch of science and technology that uses the controlled motion of electrons through different media and vacuum. It involves the study and design of control, communication and computing devices that rely on the movement of electrons in circuits containing semiconductors, thermionic valves, resistors, capacitors and inductors. The ability to control electron flow is usually applied to information handling or device control, thus electronics is distinct from electrical science and technology, which deals with the generation, distribution, control and application of electrical power.

ELECTROORGANIC REACTION: An organic reaction produced in an electrolytic cell. These reactions are used to synthesise compounds that are difficult to produce by conventional techniques. An example of such reaction is Kolbe's method of synthesising alkanes.

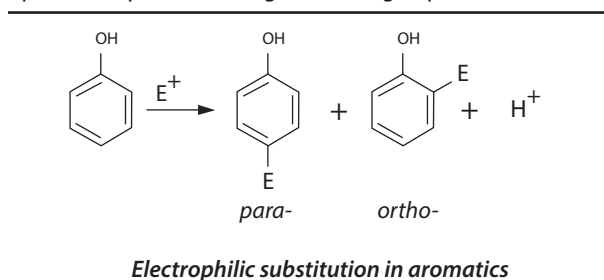
ELECTROPHILE: An electron-deficient reactive species, ion or molecule which is capable of accepting a lone pair electron or

attacking molecules of high electron density. Examples are H^+ , H_2O^+ , NO_2^+ , carbocation and molecules having partially positively charged centre. An electrophilic reagent is a reagent that forms a *bond* to its reaction partner (the nucleophile) by accepting both bonding electrons from that reaction partner. An electrophilic substitution reaction is a heterolytic reaction in which the reagent supplying the entering group acts as an electrophile. Electrophilic reagents are Lewis acids. Electrophilic catalysis is a catalysis by Lewis acids. The term 'electrophilic' is also used to designate the apparent polar character of certain radicals as inferred from their higher relative reactivities with reaction sites of higher electron density.

ELECTROPHILIC ADDITION: A chemical reaction that involves the addition of a molecule to an unsaturated organic compound across a double or triple bond. In this reaction, a π bond is broken and two new σ bonds are formed in a chemical compound. The sustaining force for this reaction is the formation of an electrophile X^+ that forms a covalent bond with an electron-rich unsaturated $\text{C}=\text{C}$ bond. The positive charge on X is transferred to the carbon-carbon bond, with the formation of a carbocation.

ELECTROPHILIC REAGENT: A reagent that attacks molecules of high electron density. Electrophiles frequently seen in organic syntheses are cations such as hydrogen ion (H^+) and nitrosonium ion (NO^+), polarised neutral molecules such as hydrochloric acid, alkyl halides, acyl halides and carbonyl compounds, polarisable neutral molecules such as Cl_2 and Br_2 , oxidising agents such as organic peracids, chemical species that do not satisfy the octet rule such as carbenes and radicals and some Lewis acids such as BH_3 and DIBAL.

ELECTROPHILIC SUBSTITUTION: A reaction that involves the substitution of an atom or group of atoms in an organic compound. The attacking substituent is an electrophile and it displaces a group in a compound, typically but not always hydrogen. Electrophilic aromatic and electrophilic aliphatic substitutions are the main reaction types, with the former being characteristic of aromatic compounds as well as an important way of introducing functional groups onto benzene rings.



ELECTROPHORESIS (CATAPHORESIS): A technique for the analysis and separation of colloids based on the diffusion of charged particles through a fluid under the influence of an electric field. It is used extensively in studying mixtures of proteins, nucleic acids, carbohydrates, enzymes, etc. In clinical medicine, for example, it is used for determining the protein content of body fluids. This electrokinetic phenomenon was first observed by Reuss in 1807. He noticed that the application of a constant electric field caused clay particles dispersed in water to migrate. This was caused by the presence of a charged interface between the particle surface and the surrounding fluid. Strictly speaking, however, electrophoretic procedures are not genuine chromatographic methods, since no partition of the analyte occurs between a mobile phase and a stationary phase. The method is useful in the analysis of proteins, polynucleotides and other biopolymers, since these are charged.

ELECTROPHORETIC EFFECT: 1. The effect in which the mobility of ions in a solution moving under the influence of an applied electric field is affected by the flow of opposite charge in the opposite direction. This drags the solvent molecules along because of the attractive

forces between ions and solvent molecules. It slows the motion of ions within a solution. 2. The inclination of an applied potential to move the ionic atmosphere itself. Due to this, the central ion at the centre of the ionic atmosphere is influenced to move towards the pole opposite its ionic atmosphere. This tends to retard its motion.

ELECTROPHORETIC VELOCITY: The velocity of a charged analyte under the influence of an electric field relative to the background electrolyte. It is also called rate of migration in electrophoresis.

ELECTROPHORUS: A capacitive generator used to produce electrostatic charge by means of electrostatic induction or a device that produces electric charges as a result of the friction between a metal plate and a disk. A dielectric plate such as plastic is used together with a metal plate with an insulating handle. The dielectric plate is charged through a triboelectric effect by rubbing it with fur or cloth. The metal plate is placed onto the dielectric plate. The electrostatic field of the charged dielectric causes the charges in the metal plate to separate. Next, the side facing up is temporarily grounded (which can be done by touching it with a finger), draining off the negative charge. Finally, the metal plate, now carrying only one sign of charge (positive in this example), is lifted.

ELECTROPHOTOGRAPHY: Processes of photo imaging which are based on photo-induced changes of electric fields (photoconductive or photo-electrostatic effects).

ELECTROPLATING: A technique of plating one metal with another by electrodeposition. The metallic objects to be plated are made the cathode of an electrolytic cell and a rod or bar of the plating metal is made the anode. Electroplating is used for covering metals with a decorative, more expensive or corrosion-resistant layer of another metal. For example, silver plating is used on tableware, electrical contacts and engine bearings, while gold plating is mostly used on jewellery and watch cases. Zinc coatings prevent the corrosion of steel articles and nickel and chromium plates are used on automobiles and household appliances.

ELECTROPLAX: One of the cells or structural units found in the electric organs of certain fish. It is made of thin, flattened plates of modified muscle that appear as two large, wafer-like, roughly circular or rectangular surfaces.

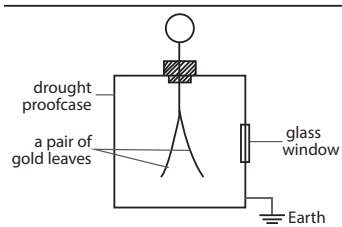
ELECTROPORATION (ELECTROPERMEABILIZATION): A significant increase in the electrical conductivity and permeability of the cell plasma membrane caused by an externally applied electrical field. It is a method of introducing foreign DNA into the protoplast or pollen with a strong burst of electricity and it is used in creating genetically engineered plants, introducing substances into a cell, such as loading it with a molecular probe or drug that can change the cell's function, etc. It depends on the local trans-membrane voltage at each point on the cell membrane.

ELECTROPOSITIVITY: A measure of the ability of elements to lose or donate electrons to form positive ions. The greater the metallic character, the more electropositive the element. Therefore, the alkali metals are most electropositive of all. This is because they have a single electron in their outer shell and, as this is relatively far from the nucleus of the atom, it is easily lost; in other words, these metals have low ionisation energies. While electronegativity increases along periods in the periodic table and decreases down groups; electropositivity decreases along periods and increases down groups. Examples of electropositive elements are alkali metals, such as sodium (K) and titanium (Li). Since they generate measurable voltages in a seawater electrolyte relative to a shark, electropositive metals are also used as shark repellents.

ELECTRORECEPTOR: An organ which is specialised for detecting electric currents. Such organs are fairly common in cartilaginous fishes such as sharks, rays and chimaeras, which use them for detecting prey or positional attackers.

ELECTORHEOLOGICAL FLUID: A class of liquid which stiffens into a semi-solid when subjected to an electric field. An example is Lithium polymethacrylate. They are most commonly colloidal suspensions and their stiffening under an electric field is reversible. Under the electric field, they form fibrous structures which are parallel to the applied field and can increase in viscosity by a factor of up to 105. The stiffening is sometimes called the **Winslow effect** after its first investigator, Willis Winslow in 1949. Electrorheological fluids can be characterised by their Mason number. The effect has been proposed as a method of constructing shock absorbers on magnetically levitated trains.

ELECTROSCOPE: An instrument invented by the British physicist, William Gilbert around 1600 and used for detecting a charged object and the polarity or sign of the charge. Examples include pith-ball electroscope and the gold-leaf electroscope. Another type is used in the quartz fibre radiation dosimeter. Since the electric potential or voltage of an object equals its charge divided by its capacitance, so electroscopes can be regarded as crude voltmeters. Electroscopes generally give only a rough, qualitative indication of the magnitude of the charge. An instrument that measures charge quantitatively is called an electrometer.



Gold-Leaf Electroscope

ELECTROSHOCK THERAPY (ELECTROCONVULSIVE THERAPY, ECT): A treatment mainly for severe depression, given under anaesthesia and with a muscle relaxant in which small electric currents are passed through the brain, intentionally triggering a brief seizure. If high doses of electricity are administered without anaesthesia, it can cause distress, loss of concentration and memory, fractured bones and other serious side effects.

ELECTROSPRAY IONISATION (ESI): A technique for producing ions for mass spectrometry, used especially for obtaining ions from large molecules, which would be likely to produce fragment ions in electron-impact ionisation sources. It was developed by John Bennett Fenn in 2002. Mass spectrometry using ESI is known as electrospray ionisation mass spectrometry (ESI-MS) or electrospray mass spectrometry (ES-MS).

ELECTROSTATIC ATTRACTION: The attraction between atoms of opposite charge that holds the atoms together in ionic bonds. If the bodies are oppositely charged, they are attracted toward one another and if they are similarly charged, the force between them is repulsive. According to Coulomb's law, the electrostatic force between two charged bodies is proportional to the product of the amount of charge on the bodies divided by the square of the distance between them. When combined with principles of quantum physics, Coulomb's law helps describe the forces that bind electrons to an atomic nucleus, that bind atoms together into molecules and that hold together solids and liquids. Coulomb's law is exactly applicable only when the charged bodies are much smaller than the distance separating them and hence can be regarded approximately as point charges.

ELECTROSTATIC FIELD: The electric field that surrounds a stationary charged body or the force field produced by static electrical charges.

ELECTROSTATIC FILTER: Filters for which an electrostatic charge is applied to the filter element. A fibrous filter material is often pleated in between V-shaped supports consisting of electrostatically charged, metal rods which are insulated from the supporting frame.

ELECTROSTATIC FORCE (COULOMB FORCE; COULOMB INTERACTION): Attraction or repulsion of particles or objects because of their electric charge. One of the basic physical forces, the electric force is named after a French physicist, Charles-Augustin de Coulomb, who in 1785 published the results of an experimental investigation into the correct quantitative description of this force. Two

like electric charges, both positive or both negative, repel each other along a straight line between their centres. Two unlike charges, one positive, one negative, attract each other along a straight line joining their centres. The electric force is operative between charges down to distances of at least 10^{-16} metre or approximately one-tenth of the diameter of atomic nuclei. Because of their positive charge, protons within nuclei repel each other, but nuclei hold together because of another basic physical force, the strong interaction or nuclear force, which is stronger than the electric force.

ELECTROSTATIC GENERATOR (ELECTROSTATIC MACHINE):

A device used to build up electric charge to an extreme potential, usually for experimental purposes. It operates by using manual or other power to transform mechanical work into electric energy and it uses moving plates, drums or belts to carry electric charge to a high potential electrode. The charge is generated by either of two techniques: electrostatic induction or triboelectric effect (friction).

ELECTROSTATIC PRECIPITATION: A method of removing solid and liquid particles and other pollutants from suspension in a gas and from air. The process involves the suspension passing through an electric discharge (corona discharge) area where ionisation of the gas occurs. The ions produced collide with the suspended particles and confer on them an electric charge. The charged particles drift toward an electrode of opposite sign and are deposited on the electrode where their electric charge is neutralised. The phenomenon could also be termed electrodeposition from the gas phase.

ELECTROSTATIC PRECIPITATOR (COTTRELL PRECIPITATOR;

ELECTROSTATIC AIR CLEANER): A highly efficient filtration device used in electrostatic precipitation to remove fine particulate matter such as dust and smoke from industrial waste gases by attracting them to charged grids or wires. The particles are charged before passing through the field and migrate to an oppositely charged electrode. They are used in removing particles from power plant plumes and in other industrial applications.

ELECTROSTATIC PROPULSION: A form of electric propulsion in which the thrust is produced by accelerating charged particles in an electrostatic field. It includes three types of device: electron bombardment thrusters, contact ion thrusters and field emission/colloid thrusters. The first two involve the production and acceleration of separate ions and are therefore forms of ion propulsion. The third type involves the production and acceleration of charged liquid droplets. Only electron bombardment thrusters have been used operationally aboard spacecraft.

ELECTROSTATIC UNITS (E.S.U.): A system of electrical units in the centimetre-gram-second system of electric and magnetic units in which the unit of charge is that charge which exerts a force of 1 dyne on another unit charge when separated from it by a distance of 1 centimetre in vacuum. The e.s.u. of electric charge is the statcoulomb.

ELECTROSTATICS: The branch of physics concerned with the study of static electricity or electric charges that do not move because they are on insulators like polythene or acetate. The combined electric field in a given region depends on the location, magnitude and polarity of the charges in that region. Applications of electrostatics range from the analysis of phenomena such as thunderstorms to the study of the behaviour of electron tubes.

ELECTROSTRICTION: A change in the dimension of a body as a result of reorientation of its molecules or the elastic deformation of a dielectric, when it is placed in an electric field. It applies to those components of strain which are independent of reversal of the field direction. It is a property of all dielectrics and is different from piezoelectric effect, which is a field-induced strain that changes sign upon field reversal and which occurs only in piezoelectric materials. The effect is employed for commercial purposes in electromechanical transducers for sonic and ultrasonic applications.

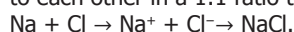
ELECTROTHERMAL ATOMIZER: A device which is heated to the temperature required for analyte atomisation by the passage of electrical current through its body. In open atomizers the sample vapour, together with analyte atoms, leaving the atomizer surface may move freely into the surrounding space. In confined atomizers

the sample vapour is restricted by the atomizer wall except for the openings where the optical beam enters and leaves the atomizer and the sample introduction hole.

ELECTROTHERMAL PROPULSION: A form of electric propulsion in which electrical energy is used to heat a suitable propellant causing it to expand through a supersonic nozzle and generate thrust. Two basic types of electrothermal thrusters are in use today: the resistjet and the arcjet. In both, material characteristics limit the effective exhaust velocity to values similar to those of chemical rockets. A third, experimental type is the microwave plasma thruster which potentially could achieve somewhat higher exhaust velocities.

ELECTROTYPE: A printing plate made by the electrode-position of copper in a mould often made from paper or plastic that has been pressed onto metal type.

ELECTROVALENT BOND (IONIC BOND): A type of chemical bond formed by the complete transfer of one or more electrons from one atom to another, so that oppositely charged ions are produced or a chemical bond in which the constituent species are held together by electrostatic force of attraction between a cation (positively charged ion) and an anion (negatively charged ion). For example, when sodium (Na) and chlorine (Cl) are combined, the sodium atoms each lose an electron, forming cations (Na^+) and the chlorine atoms each gain an electron to form anions (Cl^-). These ions are then attracted to each other in a 1:1 ratio to form sodium chloride (NaCl):



ELECTROVALENT CRYSTAL (IONIC CRYSTAL): A crystal in which the ions are linked by a bond resulting from electrostatic attraction or a group of crystals, such as sodium chloride (NaCl) and calcium chloride (CaCl_2), in which the lattice-site occupants are charged ions held together primarily by their electrostatic interaction. They are characterised by strong absorption of infrared radiation, good ionic conductivity at high temperatures and the existence of planes along which the crystals cleave easily. Solids which are ionic crystals are formed by compounds of strongly electropositive and strongly electronegative elements. Examples are the alkali halides, other monovalent metal halides and the alkaline-earth halides, oxides and sulfides. Crystals in which some of the ions are complex, such as metal carbonates, metal nitrates and ammonium salts, may also be classed as ionic crystals. However, the ideal ionic crystal is approached most closely by the alkali halides.

ELECTROVISCOUS EFFECTS: For dispersions of charged particles, these are those components of the viscosity connected with the charge of the particles.

ELECTROWEAK FORCE: The unification of the electromagnetic and weak interactions described by the Weinberg-Salam theory. It was proposed by physicists Sheldon Glashow, Abdus Salam and Steven Weinberg which unifies the electromagnetic and the weak force under conditions of extreme temperature prevalent much earlier in the history of the universe. The symmetry corresponding to this unification has since been "broken," with the ensuing expansion and drop in temperature of the universe yielding two of the four fundamental forces known today. When the electroweak unification energy (around 100 GeV) is achieved locally, for example in particle accelerators, the distinction between the electromagnetic force and the weak force disappears.

ELECTROWEAK THEORY (QUANTUM FLAVOUR DYNAMICS): A gauge theory that gives a unified description of the electromagnetic and weak forces. In this theory the gauge group is non-Abelian and the gauge symmetry is a broken symmetry. Though the forces appear to be different, they are actually different facets of a more fundamental force. It was formulated in the 1960s by Sheldon Glashow, Steven Weinberg and Abdus Salam. It was validated in the 1980s with the discovery of the W and Z particles, which it had predicted.

ELECTRUM: 1. An alloy of gold and silver containing 55 – 88% of gold. 2. A German silver alloy containing 52% copper, 26% nickel and 22% zinc. Its colour varies from white-gold to brassy, depending on

the percentages of the major constituents and copper. It was used to make the first known coins in the Western world.

ELEMENT: 1. In geometry, a point, line, plane or other parts of a figure. 2. Also known as **member**, one of the objects or numbers that together form a set or class; if there is a structure imposed on the set, it may be a group, ring, field, etc. The p -elements of a group are those of order p^α , where p is a prime and α is any positive integer. 3. Any of the terms in the array constituting a determinant or matrix. 4. Any one of the rectangles summed in an integral and represented by the expression following the integral sign; for example, in $\int_a^b f(x)dx$, where $f(x)dx$ is an element of area. 5. A pure substance

made up of atoms of the same kind and with the same atomic number that cannot be split chemically into simpler substances or a uniform substance which cannot be further decomposed by ordinary means. The total number of electrons in an element is equal to the total number of protons, but the number of neutrons may be different. Elements are classified in families or groups in the periodic table and they are often classified also as metals and non-metals. A metallic element is one whose atoms form positive ions in a solution and a non-metallic element is one whose atoms form negative ions in a solution. Some of the elements, including oxygen, nitrogen, the noble gases (helium, neon, argon, krypton, xenon and radon), sulfur, copper, silver and gold are found in nature in the free (uncombined) state. Most of the elements in nature are combined with other elements in the form of compounds. The most abundant element on the Earth is oxygen and the most abundant element in the universe is hydrogen. 6. In biology, a place or condition best suited or most favourable for an individual, for example, water is the element of fishes. 7. In astronomy, any of the numerical quantities, such as the major axis or eccentricity, used in describing the orbit of a planet, satellite, etc. 8. In electricity, the resistance wire in an electrical appliance such as an electric iron or oven.

ELEMENTARILY EQUIVALENT: Of models, such that every sentence in one model is true in the other. Two structures M and N of the same signature, σ are elementarily equivalent if every first-order sentence (formula without free variables) over σ is true in M if and only if it is true in N. This is a weaker relation than isomorphism; that is isomorphic models are elementarily equivalent, but not necessarily vice versa. For example, the non-standard real numbers and the ordinary real numbers are elementarily equivalent.

ELEMENTARY: 1. In logic, of a theory, involving axioms using only finitary first-order logic, without reference to set theory or using any axioms which have consistency strength equal to set theory. For example, saying that a theory is elementary is a weaker condition than saying it is algebraic. 2. Pertaining to or consisting of elementary functions or operations. 3. In number theory, not using techniques involving complex numbers such as complex analysis. Elementary proofs are not necessarily simple.

ELEMENTARY COLUMN OPERATION: An elementary matrix operation on the columns of a matrix.

ELEMENTARY DIVISOR: Any of the distinct linear factors of the characteristic polynomial of a matrix.

ELEMENTARY ENTITY: Any countable object or event, but usually a molecule, an ion or a specified group of atoms.

ELEMENTARY FUNCTION: A member of the class of functions built up from a finite combination of constant functions, field operations such as addition, multiplication, division and root extractions; and algebraic, exponential and logarithmic functions and their inverses under repeated compositions. For example, $\log[\tan^{-1}[\sqrt{(\exp(x^2) + 1)}]]$ is constructed by successively applying each of the following operations to the result of the preceding: squaring, exponentiating, adding 1, extracting the square root, finding the inverse of the sine taking the logarithm. The simplest elementary functions include logarithm, exponential function (including the hyperbolic functions), power function and trigonometric functions.

ELEMENTARY MATRIX (E-MATRIX): A matrix obtained by applying an elementary matrix operation to the appropriate identity matrix. Any non-singular matrix is a product of such matrices and each elementary row or column matrix represents respectively an elementary row or column matrix operation.

ELEMENTARY MATRIX OPERATION (E-OPERATION): Any one of the three operations of multiplying each element in a row or column by a non-zero number; multiplying a row or column by a non-zero number and adding the result to another row or column; or transposing two rows or columns; used in finding the inverse of a matrix, solving linear systems and also used for other calculations. Rows on which these operations are performed are called **elementary row operations**; and when they are performed on columns, they are called **elementary column operations**.

ELEMENTARY OPERATION: Any one of the mathematical operations of addition, subtraction, multiplication, division and extraction of integral roots.

ELEMENTARY PARTICLES (FUNDAMENTAL PARTICLES): Subatomic particles that are not made up of smaller particles and so can be considered one of the fundamental constituents of all the matter in the universe. In the Standard Model, the quarks, leptons and gauge bosons are examples of elementary particles. However, elementary particles are grouped into families. For example, electron (e^-) and muon (μ) belong to the Lepton family; Bosons family may be either elementary, like photons and gluons or composite, like mesons; positive pion (p^+) and negative pion (p^-) belong the Mesons family and (proton (p^+) and neutron (n) belong to the Baryons family. The unit of the mass of an elementary particle is MeV (million electron volts), which 1.8×10^{-27} g.

ELEMENTARY PROCESS: A chemical process, which takes place in a single step without going through any intermediate. The order with respect to any of the species is the stoichiometric coefficient of a balanced chemical equation for these types of reactions.

ELEMENTARY REACTIONS (ELEMENTARY STEP; ELEMENTARY REACTIONS): A reaction which involves reactants being transformed into products in a single step without reaction intermediates, but with a single transition state.

ELEMENTARY REDUCTION: The operation that forms a word from another by replacing any term of the form xx^{-1} or $x^{-1}x$ by the empty word; or the word so formed.

ELEMENTARY STEPS: A series of simple reactions that represent the progress of the overall reaction at the molecular level.

ELEMENTS OF AN ORBIT (ORBITAL ELEMENTS): Six parameters that can be used to define the path of a celestial body. There are many different ways to mathematically describe the same orbit, but certain schemes each consisting of this set of six parameters are commonly used in astronomy and orbital mechanics. In celestial mechanics these elements are generally considered in classical two-body systems, where a Kepler orbit derived from Newton's laws of motion and Newton's law of universal gravitation are used.

ELEPHANT: Any of three ungulate species belonging to the order Proboscidea and family Elephantidae, characterised by their large size, long trunk, tusks, massive legs, large ears and huge head. There are two species of the African elephant, *Loxodonta africana* (savanna elephant) and *Loxodonta cyclotis* (forest elephant) and one species of Indian Elephant (*Elephas maximus*). All species are greyish to brown, with sparse, coarse body hair. The trunk is used for breathing, drinking and reaching for food. They eat grasses, leaves and fruit. The African savanna or bush elephant (*Loxodonta africana*), of sub-Saharan Africa, is the largest living land animal, weighing up to 7,500 kg and standing 3-4 m tall at the shoulder. All species are considered endangered.

ELEPHANT GRASS (MISCANTHUS): Tall coarse grass or a similar plant. There are about 15 species of this perennial grasses, native to subtropical and tropical regions of Africa and southern Asia, with one species (*M. sinensis*) extending north into temperate eastern Asia.

ELEPHANTIASIS: A disease in which there is massive swelling and hypertrophy of the skin and subcutaneous tissue of the legs or scrotum, due to the obstructed flow of lymph. This may be a congenital disease, due to trauma, cancer or infection with filariasis, tuberculosis and some venereal diseases. Recurrent secondary bacterial infections are common and chronic skin ulcers may form.

ELEVATION: 1. The vertical distance of a point from the horizontal. 2. The height to which something is elevated above a point of reference such as the ground. 3. A scale drawing of the side, front or rear of a structure. 4. The upward or vertical movement of a part of the body, for example, raising the arm vertically upwards.

ELEVATION OF ANGLE: An angle through which the eyes are elevated to view an object.

ELEVATION OF BOILING POINT (MOLAL ELEVATION BOILING POINT): The increase in the boiling point of a liquid above that of the pure solvent, when an impurity is dissolved in it or the ratio of the elevation of the boiling point of a solvent caused by dissolving a solute, to the molality of the solution, taken at extremely low concentrations. For example, when salt is added to boiling water, the water ceases to boil because its boiling point has been elevated.

ELEVATOR: 1. Also known as **lift**, it is a platform, compartment or cage raised or lowered in a vertical shaft to transport persons or goods in a building. 2. A mechanical hoist for raising something such as grain or coal, often consisting of a chain of scoops linked together on a conveyor belt. 3. In biology, it is any muscle that raises a part of the body. 4. In agriculture, it is a large granary equipped with an elevator that usually facilitates for cleaning and grading the grain. 5. One of the three basic control surfaces of an airplane, used to control pitch. Elevators are usually pivoted on the stabiliser.

ELFIN FOREST (KRUMMHOLZ): A forest of short and stunted trees found in the zone just above the timber line on tropical mountains. Continual exposure to fierce freezing winds causes vegetation to become stunted and deformed. Some trees showing krumholtz formation include balsam fir, red spruce, subalpine larch and lodgepole pine.

ELICITOR: A substance or other stimulus that triggers the hypersensitive response in a plant. Most elicitors are polysaccharides, small proteins or lipids associated with the fungal or bacterial cell wall.

ELIMINANT: The relationship between coefficients that results from eliminating the variable from a set of simultaneous equations. For example, in the equations $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_3x + b_3y + c_3 = 0$; the eliminant is given by the matrix of coefficients:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

ELIMINATION: 1. The process of removing one of the unknowns in an algebraic equation, for example, by the substitution of variables or cancellation. Alternatively, removing a common element from the numerator and denominator of a fraction or from both sides of an equation, thereby simplifying it and yielding an equivalent expression. 2. The process through which small molecules such as solvents of crystallisation or two atoms or molecules attached to adjacent atoms such as H and OH are lost from a molecule. An **elimination reaction** is a type of organic reaction in which two substituents are removed from a molecule in either a one or two-step mechanism or a reaction in which one molecule decomposes into two, with one much smaller than the other. Either the unsaturation of the molecule increases as in most organic elimination reactions or the valence of an atom

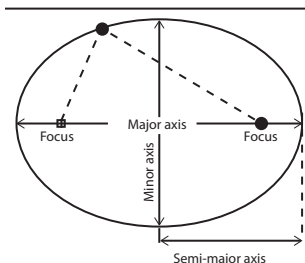
in the molecule decreases by two, a process known as **reductive elimination**. A major category of elimination reactions are those involving alkyl halides or alkanes in general.

ELIMINATION RULE: In logic, any syntactic rule specifying the conditions under which a formula or statement containing a specified operator may permit the valid derivation of others that do not contain it. For example, **conjunction elimination** is the rule permitting the derivation of either P alone or Q alone from P & Q; **universal elimination** permits the derivation of Fa from $(\forall x)Fx$.

ELINVAR: An alloy of chromium, iron and nickel, containing manganese and tungsten in varying amounts and having a low thermal expansion and almost invariable modulus of elasticity. It is used for mechanical watches and chronometer balances and springs for gauges and other instruments.

ELLINGHAM DIAGRAM: Plots of change in standard free energy with respect to temperature for various reactions such as the formation of oxides and sulfides of various elements. They are basically straight lines except at very high and very low temperatures. This diagram is used to forecast the equilibrium temperature between a metal, its oxide and oxygen and by extension, reactions of a metal with sulfur, nitrogen and other non-metals. They are also used to show the conditions under which a metal oxide can be reduced to a metal.

ELLIPSE: 1. A regular oval shaped curve joining all points around two fixed points such that the sum of the distances from those points is always constant. Assuming that the distance between two points R, R' of a plane is denoted by $2c$, $c > 0$, let $2a$ be a constant, with $a > c$. Then the ellipse with foci R and R' and major axis $2a$ is the locus of points P of the plane such that

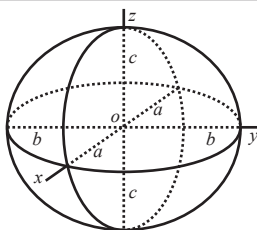


An ellipse with its various parts

that $PR + PR' = 2a$, where PF denotes the distance of P and R. One property of an ellipse is that the reflection off its boundary of a line from one focus

will pass through the other. The standard equation of an ellipse is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. 2. A conic formed by the intersection of a plane with a right circular cone so that the plane is inclined to the axis of the cone at an angle of the cone.

ELLIPSOID (SPHEROID): A solid body formed when an ellipse is rotated about an axis. It is a closed type of quadric surface that is a higher dimensional analogue of an ellipse. The equation of a standard axis-aligned ellipsoid body in an xyz-Cartesian coordinate system is $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, where, a and b are the equatorial radii along the x and y axes and c is the polar radius along the z -axis, all of which are fixed positive real determining the shape of the ellipsoid. **Elliptic** refers to possessing the shape of an ellipse; broad at the middle but narrowing at the two equal ends.



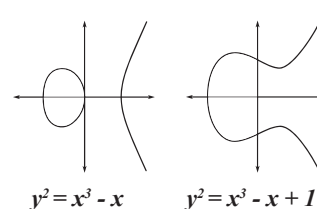
An ellipsoid on a three dimensional plane

ELLIPSOIDAL JOINT: A type of synovial joint such as the wrist, in which there is an elliptical convex articular surface that fits into an elliptical concave articular surface. The movements of flexion, extension, abduction and adduction can take place, but rotation is impossible.

ELLIPSOID METHOD: One of a class of polynomial-time algorithms for linear programming that finds an optimal solution in a finite number of steps, by generating a sequence of ellipsoids whose volume uniformly decreases at every step, thus enclosing a minimizer of a convex function.

ELLIPSOMETRY: A device that uses polarised light to measure the thickness of very thin films. Linearly polarised light is made to reflect or transmit off a surface and the polarisation of the resulting elliptically polarised light is then examined.

ELLIPTIC CURVE: The set of solutions to a type of cubic equation whose solutions lie on a torus (a donut-shaped surface). The particular type of cubic equation whose solutions lead to elliptic curves takes the form:



ELLIPTIC EQUATION (ELLIPTIC PARTIAL DIFFERENTIAL EQUATION):

An elliptic curve on a Cartesian plane

A second order partial differential equation, which includes Laplace's equation, having the form

$au_{xx} + bu_{xy} + cu_{yy} + du_x + eu_y + fu = h$, with the discriminant $b^2 - 4ac$ negative.

ELLIPTIC FUNCTION: In complex analysis, it is a function in which one of the forms expresses the relation of the arc of an ellipse to the straight lines to which it is connected. While a singly periodic function like sine has a number, $a = 2\pi$, so that $\sin(x + a) = \sin(x)$, a doubly periodic function f has the property that there are two numbers a, b , not rational multiples of each other, so that $f(x + a) = f(x + b) = f(x)$.

ELLIPTIC GEOMETRY (RIEMANNIAN GEOMETRY): One of the two most important types of non-Euclidean geometry, the other is hyperbolic geometry. In elliptic geometry, no line is parallel to any other line through any given point; lines intersect in 1 point and the angles of a triangle always add up to more than 180 degrees. In spherical geometry, all lines intersect in 2 points.

ELLIPTIC INTEGRAL: A definite integral not usually evaluable in closed form by antidifferentiation, having the form $\int_c^u R(x, y) dx$, where

R is a rational function of x and y , y^2 is a quartic polynomial in x and c is a fixed constant. It is a generalisation of the inverse trigonometric functions and provides solutions to more problems. Elliptic functions are obtained by inverting elliptic integrals to obtain generalisations of the trigonometric functions. The integral is called a **complete elliptic integral** if the range of integration is maximal; if not, it is an **incomplete elliptic integral**. The integral is named elliptic because one of the integrals expresses the length of an arc of an ellipse.

ELLIPTIC PARABOLOID: A paraboloid that can be put into a position such that its sections parallel to one coordinate plane are ellipses, while its sections parallel to the other two coordinate planes are parabolas. In other words, the vertical cross sections are all parabolas, while the horizontal cross sections are ellipses. In a suitable coordinate system with three axes x, y and z , it can be represented by the

equation $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$, where a and b are constants that dictate the level of curvature in the x - z and y - z planes respectively. With $a = b$ an elliptic paraboloid is a paraboloid of revolution: a surface obtained by revolving a parabola around its axis. This is an elliptic paraboloid which opens upward for $c > 0$ and downward for $c < 0$. It is the shape of the parabolic reflectors used in mirrors, antenna dishes, etc.; and is also the shape of the surface of a rotating liquid, a principle used in liquid mirror telescopes and in making solid telescope mirrors.

ELLIPTICAL GALAXY: A galaxy having an approximately ellipsoidal shape and a smooth, nearly featureless brightness profile. They range

in shape from nearly spherical to highly flat and in size from hundreds of millions to over one trillion stars. They can be the result of two galaxies colliding. In the nucleus of some galaxies, active sources of relativistic particles emit radio waves and x-rays as well as visible light with luminosities ranging up to 100 times that of all the stars in the galaxy. Most elliptical galaxies are composed of older, low-mass stars, with a sparse interstellar medium and minimal star formation activity. They are surrounded by large numbers of globular clusters.

ELLIPTICAL POLARISATION: A phenomenon observed in electromagnetic radiation in which the electric field vector magnitude varies and rotates in a circle around the direction of wave propagation. An elliptically polarised wave may be resolved into two linearly polarised waves in phase quadrature, with their polarisation planes at right angles to each other. Since the electric field can rotate clockwise or counterclockwise as it propagates, elliptically polarised waves exhibit chirality. An elliptically polarised wave may be resolved into two linearly polarised waves in phase quadrature, with their polarisation planes at right angles to each other.

ELLIPTICITY (OBLATENESS): A measure of the amount by which an ellipsoidal shape or path, such as a planet or a galaxy, deviates from a circle or perfect sphere, obtained as the ratio of the major to minor axes. For a planet and galaxy, it is found by subtracting the polar diameter from the equatorial diameter, then dividing by the equatorial diameter. For a planet and galaxy, ellipticity is typically an indication of how fast a body is rotating. The ellipticity of Earth is about 1/299.

ELODEA (ANACHARIS): An aquatic plant belonging to the genus *Elodea* and often called waterweeds that grows submerged in ditches and ponds. It is commonly used in oxygenating aquariums. The introduction of some species of *Elodea* into certain geographical locations has created a significant problem and it is now considered a noxious weed in these areas.

ELONGATION: 1. The phase in which amino acids are linked together by sequentially formed peptide bonds to form a polypeptide chain. 2. The angular distance between two points in the sky as measured from a third point. For example, the elongation of a planet is normally measured as the angular distance from the Sun to the planet as measured from the Earth. It is measured east (eastern quadrature) or west (western quadrature) from the Sun. A planet is said to be in conjunction when it lies on a line drawn from the Earth to the Sun, in which case its elongation is 0°. A planet with an elongation of 90° is said to be in **quadrature** and when its elongation is 180°, it is said to be in **opposition**.

ELUATE: The solution of solvent and dissolved matter resulting from elution or the effluent or solvent from a chromatographic column removing the sample components from the stationary phase.

ELUENT (ELUANT): A substance used as a solvent in separating materials in elution or a solvent used for elution or a solution obtained from elution as for example, the liquid in chromatography.

ELUTION: To extract one material from another, usually by means of a solvent or the process of removing an adsorbed material (adsorbate) from an adsorbent by washing it in a liquid (eluent). The solution consisting of the adsorbate dissolved in the eluent is the eluate.

ELUTION CHROMATOGRAPHY: A procedure for chromatographic separation in which the mobile phase is passed through the chromatographic bed after the application of the sample.

ELUTRIATION (AIR CLASSIFICATION): The technique of separating lighter particles from heavier ones based on their size, shape and density using a vertically-directed stream of liquid or gas. In other words, it is the process of suspending finely divided particles in an upward flowing stream of air or water to wash and separate them into sized fractions or particles. Since their terminal velocities are lower than the velocity of the rising fluid, the smaller/lighter particles rise to the top or overflow. It is generally used for particles with sizes greater than 1 µm.

ELUVIAL HORIZON: A soil horizon in which minerals, humus, or plant nutrients have been lost through the process of eluviation. In geology, **eluviation**, also called **leaching** is the process of removal of materials from geological or soil horizons. In soil science, eluviation is the transport of soil material from upper layers of soil to lower levels by downward precipitation of water across soil horizons, and accumulation of this material in lower levels is called **illuviation**. Eluviation occurs when precipitation exceeds evaporation.

ELUVIUM: An accumulation of dust and soil particles caused by the weathering and disintegration of rocks in place, or deposited by the wind.

ELYTRA: 1. Also called **shard**, it is the thickened horny forewings of certain insect orders, such as beetles (Coleoptera) or leathery cockroach (Dictyoptera) and true bugs (Hemiptera), which cover and protect the membranous hind wings when the insect is at rest. To fly, the insect opens the elytra and then extends the hind wings and flies while still holding the elytra open. However, in some ground beetles such as those belonging to the family Carabidae the elytra are fused together, rendering the insect flightless. 2. The hard scales of some polychaete worms, such as the Polynoidae (scale worms). These outgrowths of the body wall are distinguished from the chaeta, which grows from follicles and thus possess roots. The plural of elytra is **elytron**.

EMANATION (RADIOACTIVE GAS): Any gas formed by a radioactive decay or any of several radioactive gases that are isotopes of radon.

EMANATION THERMAL ANALYSIS: A thermo analytical technique in which the release of radioactive emanation from a substance (and/or its reaction products) is measured as a function of temperature whilst the substance is subjected to a controlled temperature programme.

EMARGINATE: Of a leaf, petal or sepal, indented at the tip or having a v-shaped indentation at the tip or apex, as shown in the diagram.



Emarginate leaf

EMASCULATION: 1. The castration or removal of testicles, as done with some bulls to turn them into steers to produce tender meat. 2.

The removal of the anthers or male flowers in order to prevent self-pollination or the undesirable pollination of neighbouring plants. An example is the removal of tassels from corn (*Zea mays*) for hybridisation.

EMBASE (EXCERPTA MEDICA DATABASE): A bibliographic database containing citations, abstracts, indexing terms and codes covering over 5,000 biological, pharmacological and medical journals from more than 70 countries, especially Europe. It is produced by Elsevier and it contains over 11 million records, with some half a million added each year and has a particular emphasis on drug-related fields. It is available online through a number of database vendors.

EMBDEN-MEYERHOF-PARNAS PATHWAY (GLYCOLYSIS): The metabolic pathway by which glucose is anaerobically degraded to pyruvic acid. In the glycolysis of one molecule of glucose, two molecules of adenosine triphosphate (ATP) are used in the phosphorylation reactions at the beginning of the pathway and four molecules of ATP are formed later giving a net yield of two ATP molecules. Glycolysis is an example of an anaerobic fermentation. Under aerobic conditions, however, it is important principally not as an energy-supplying pathway but as a preparation of glucose for entry into the tricarboxylic acid (TCA) cycle.

EMBEDDED COMPUTER (EMBEDDED SYSTEMS): A device which has its own computing power dedicated to specific functions, usually consisting of a microprocessor and firmware. The computer becomes an integral part of the device as opposed to devices which are controlled by an independent, stand-alone computer. It implies software that integrates operating system and application functions.

EMBEDDED FIGURE: A figure entirely enclosed within another figure. It is a test used to measure cognitive functioning and style in

an individual by having them find geometric shapes within a larger complex image. This test can also be used to look at analytical style, social behaviours, problem solving styles and defense mechanisms.

EMBEDDED OS (EMBEDDED OPERATING SYSTEM): A specialised operating system used for embedded systems or for hardwares that have very little computing power, with the programs stored in the ROM of the device rather than the hard disk. Examples of embedded operating systems are Embedded Linux, BlackBerry OS, Palm OS and Windows Mobile.

EMBEDDING: 1. Putting one mathematical object inside another, such as a subgroup within a group or one topological space inside another, while preserving all topological properties. 2. A stage in the preparation of a sample for examination by microscopy that involves impregnation of the sample with wax of plastic following dehydration.

EMBOLISM: The presence of substances other than liquid blood in the circulatory system, causing obstruction in arteries or interfering with the pumping of the heart. The commonest embolism is from atheromatous plaques or thrombosis on a blood vessel or the heart walls. Fat globules may form emboli from bone marrow after major bone fractures and amniotic fluid may cause embolism during childbirth. Stroke or transient cerebral episodes, pulmonary embolism, coronary thrombosis and obstruction of limb or organ blood supply with consequent cell death are common results, some of them fatal.

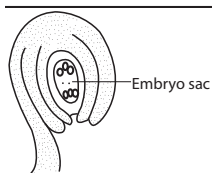
EMBOSSING: Mechanical production of a raised pattern on a surface. Suitable materials are plastic, paper, leather, fabric, etc. A male die is machined such that the pattern is raised and a female die so that the mirror image of the pattern is engraved into it. When the two dies are forced together the pattern is embossed on the material between.

EMBRYO: 1. A multicellular diploid eukaryote in its earliest stage of development, from the time of first cell division until birth, hatching or germination or an organism in its early stage of development from a fertilised egg or seed or a young animal developed from a zygote as a result of repeated cell division or a young flowering plant developed from a fertilised ovule. In humans, it is called an embryo until about eight weeks after fertilisation and from then it is called a **foetus**. The development of the embryo is known as **embryogenesis**. 2. In chemistry, embryo is an aggregate of a small number of atoms, molecules or ions. An embryo of such a size that the Gibbs energy at constant pressure and temperature is a maximum is called a **critical embryo**. An embryo larger than a critical embryo is called a **homogeneous nucleus**.

EMBRYO CULTURE: The growth of isolated plant embryos on suitable media in vitro. The technique is useful in plant breeding as it enables certain hybrids to be raised that would abort if left on the plant, either because of endosperm breakdown or endosperm incompatibility.

EMBRYO RESEARCH: The study of human embryos at an early stage, in order to improve in vitro fertilisation (IVF) methods, to understand the earliest stages of human development and to detect hereditary disease and genetic defects and to develop cures for them. Examples of diseases that can be tested include cystic fibrosis, Duchenne muscular dystrophy, Lesch-Nyhan syndrome, Tay-Sachs disease and haemophilia A.

EMBRYO SAC (MEGAGAMETOPHYTE): A large oval cell in the ovules of flowering plants in which the female gametes are located and which gives rise to the embryo and the endosperm nuclei.



EMBRYO TRANSFER: A process that removes fertilised eggs from a female and places them in another female who carries them until birth. It is a step in the process of in vitro fertilisation (IVF) in which one or several embryos are placed into the uterus of the female with the intent to establish a pregnancy. The embryos that are transferred may be fresh from fertilised egg cells of the same menstrual cycle or frozen from preceding cycle, cryo preserved and thawed just prior to the transfer.

EMBRYOBIONTA (EMBRYOPHYTA): In certain classifications, one of two sub-kingdoms of plants, the other being the Thallophyta. It contains the bryophytes and vascular plants. These plants are grouped together because they all have an embryonic stage that is dependent on the parent plant for a greater or lesser period.

EMBRYOGENY (EMBRYOGENESIS, EMBRYONY): The development of an embryo, normally from a fertilised egg cell. The first divisions of the zygote are at right angles to the long axis of the archegonium or embryo sac. In angiosperms and in some gymnosperms and pteridophytes, these initial divisions give rise to a chain of cells, the suspensor. Embryogeny often lags behind the development of other parts of the seed so the embryo may grow at the expense of the previously formed endosperm. Certain cells of the embryo may be polyploid if DNA synthesis outstrips nuclear division. The term embryogenesis is also applied to the development of embryos from diploid somatic cells in suspension cultures.

EMBRYOLOGY: The study of the changes undergone by an organism from its conception as a fertilised ovum or egg to its emergence into the world at hatching or birth.

EMBRYONIC DISK: The early embryo before the formation of somites. It is a flat disk of tissue bounded dorsally by the amniotic cavity and ventrally by the yolk sac. The formation of the primitive streak and archenteron in the embryonic disk determines the orientation of the embryo, which then becomes progressively elongated.

EMBRYONIC STEM CELL: A stem cell derived from embryos that develop from eggs that have been fertilised in vitro and then donated for research purposes with the consent of the donors. They are not derived from eggs fertilised in a woman's body. The embryos from which human embryonic stem cells are derived are typically four or five days old and are a hollow microscopic ball of cells called the blastocyst.

EMBRYOPHYTE (LAND PLANTS): A plant that develops from an embryo and, therefore, is a complex multicellular eukaryote with specialised reproductive organs. It is a true plant and the term underlines the distinction between plants and algae, which lacks embryo. They live primarily in terrestrial habitats, in contrast with the related green algae that are primarily aquatic. Examples include trees, flowers, ferns, mosses and various other green land plants. Most get their energy through photosynthesis and they synthesise their food from carbon dioxide.

EMERALD: A clear, green gemstone variety of the mineral beryl, with the chemical formula $\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$ and Mohs scale hardness 7.5-8. Emeralds are notorious for their flaws. Flawless stones are very uncommon and have great value. Its toughness or resistance to breakage is classified as generally poor.

EMERGENCE: The arising of new, unexpected structures, patterns or processes in a self-organising system. These emergents can be understood as existing on a higher level than the lower level components from which the emergents emerged. The term was first used by the 19th-century philosopher G. H. Lewes and came into greater currency in the scientific and philosophical movement known as emergent evolutionism in the 1920s and 1930s.

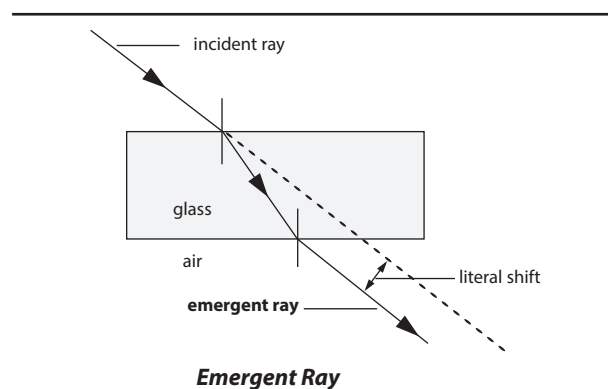
EMERGENCY/SAFETY SHOWER: Safety equipment used to splash water over the entire body at one time if a toxic or corrosive chemical is spilled on the skin or clothing. An eye wash is used to irrigate and flush the eyes and does not cover the entire body.

EMERGENT: 1. A forest tree that stands taller than surrounding trees, withstanding the hot temperatures and strong winds in some areas. The emergent layer contains a small number of very large trees called emergents, which grow above the general canopy, reaching heights of 45-55 metres, although a few species can grow to 70-80 metres tall. 2. A plant that has its roots under water but its upper part above the surface. Collectively, such plants form emergent vegetation. The various species include the reed (*Phragmites*), *Cyperus papyrus*, *Typha*

species, flowering rush and wild rice species. Some species, such as purple loosestrife, may grow in water as emergent plants but they are capable of flourishing in fenland or marshy land. Often, the main aerial feature is the flower and the related reproductive process and pollination by wind or by flying insects is a common characteristic. Their leaves can photosynthesise more efficiently above the shade of cloudy water and competition from submerged plants.

EMERGENT RAY: A light ray that emerges from a propagation medium. The angle made by the emergent ray with the normal to the surface is called angle of emergence.

E



EMERY: A natural mixture of corundum (aluminium oxide, Al_2O_3) consisting of iron oxides such as magnetite (Fe_3O_4) or hematite (Fe_2O_3), with specific gravity of 3.7 to 4.3, depending on the relative amounts of the constituent minerals. The aggregate has a grey-to-black colour and is extremely tough and difficult to break. It is used as an abrasive and polishing material by lapidaries and in the manufacture of lenses, prisms and other optical equipment. Emery wheels, emery paper and emery cloth are also used by machinists in the grinding and polishing of steel.

EMETIC: Any agent such as a strong solution of salt, mustard water, powdered ipecac and ipecac syrup that causes vomiting. They are used when a poison such as an overdose of tablets has been recently ingested. They should not be used, when the poison taken causes damage to the oesophagus or lungs.

EMIRP: The backwards spelling of "prime", it is a prime number that becomes a different prime number, when its digits are reversed. The first twenty emirps are 13, 17, 31, 37, 71, 73, 79, 97, 107, 113, 149, 157, 167, 179, 199, 311, 337, 347, 359 and 389. A palindromic prime gives the same prime, when reversed.

EMISSION: 1. The total rate at which a solid, liquid or gaseous pollutant is emitted into the atmosphere from a given source; usually expressed as mass per unit time. Primary emissions are those substances such as nitrogen monoxide (NO) and sulfur dioxide (SO_2), which are emitted directly to the atmosphere. Secondary emissions are formed from the primary emissions through thermal or photochemical reactions, for example, ozone, aldehydes, ketones, sulfuric acid and nitric acid. The point or area from which the discharge takes place is called the source. The area in which the emission or its transformed products, for example, in the case of aerosols and acidic deposition, may be deposited is called the receptor area or sink. Emission may be applied to noise, heat, etc., as well as pollutants. 2. Radiative deactivation of an excited state; transfer of energy from a molecular entity to an electromagnetic field.

EMISSION COMPUTED TOMOGRAPHY: Tomography using emissions from radionuclides and a computer algorithm to reconstruct the image.

EMISSION CONTROL EQUIPMENT: Air pollution control equipment, which either converts the pollutant chemically to a non-polluting substance or collects the pollutant by some means including gravity settling chambers, inertial separators, cyclonic separators, filters, electrical precipitators, scrubbers, (spray towers, jet scrubbers, Venturi scrubbers, inertial scrubbers, mechanical scrubbers and

packed scrubbers). Certain gases and odoriferous compounds are controlled by combustion, absorption (spray chambers, bubble cap or sieve plate contactors and packed towers) and adsorption units (packed beds or fluidized beds).

EMISSION ELECTRON: The liberation of electrons from a surface in a process like evaporation or beta particle resulting from radioactive decay.

EMISSION FACTOR: The relationship between the amount of pollution produced and the amount of raw material processed. For example, an emission factor for a blast furnace making iron would be the number of pounds of particulates per ton of raw materials.

EMISSION INTENSITY: The average emission rate of a given pollutant from a given source relative to the intensity of a specific activity, for example, grams of carbon dioxide released per megajoule of energy produced or the ratio of greenhouse gas emissions produced to GDP. Emission intensities are used to derive estimates of air pollutant or greenhouse gas emissions based on the amount of fuel combusted, the number of animals in animal husbandry, on industrial production levels, distances travelled or similar activity data. Emission intensities may also be used to compare the environmental impact of different fuels or activities.

EMISSION LINE: A bright line in the spectrum of a luminous object caused by the emission of light at a particular wavelength. Emission lines may appear on their own, as in the spectrum of a nebula energised by radiation from a nearby hot star or they may be superimposed on an absorption spectrum, as happens when a star is surrounded by hot gas.

EMISSION SPECTRA: 1. The relative intensity of each frequency of radiation emitted by an element's atoms or a compound's molecules when they are returned to a ground state or a range of electromagnetic energies arranged in order of increasing or decreasing wavelength or frequency. Since each element's emission spectrum is unique, spectroscopy can be used to identify them in matter of unknown composition. Similarly, the emission spectra of molecules can be used in chemical analysis of substances. Emission spectra may form a continuous spectra (contain an unbroken sequence of frequencies over a relatively wide range) or line spectra (exhibit discontinuous lines as produced by excited atoms and ions as they fall back to a lower energy level). 2. A distribution of entities or properties arrayed in order of increasing or decreasing magnitude.

EMISSION SPECTROSCOPY: The study of electromagnetic radiation spectra given off by atoms or molecules that undergo a transition to a lower energy level. Such a process is known as fluorescence or, under certain conditions, phosphorescence. It deals with visible light and shorter wavelengths since fluorescence is less likely to happen with long wavelengths. Examples include fluorescence spectroscopy, flame emission spectroscopy, x-ray fluorescence spectroscopy and stella spectroscopy.

EMISSIONS PERMIT: The non-transferable or tradable allocation of entitlements by an administrative authority (intergovernmental organisation, central or local government agency) to a regional (country, sub-national) or a sectoral (an individual firm) entity to emit a specified amount of a substance.

EMISSIONS QUOTA: The portion or share of total allowable emissions assigned to a country or group of countries within a framework of maximum total emissions and mandatory allocations of resources.

EMISSIONS REDUCTION UNIT (ERU): A unit issued under a Joint Implementation project in terms of the Kyoto Protocol. An ERU represents a reduction of greenhouse gases under the Joint Implementation mechanism, where it represents one metric ton (tonne) of CO_2 equivalent reduced.

EMISSIONS SCENARIO: A plausible representation of the future development of emissions of substances that are potentially radiatively active such as greenhouse gases and aerosols, based on a coherent and internally consistent set of assumptions about driving forces

(such as demographic and socio-economic development, technological change) and their key relationships.

EMISSIONS TRADING: A government, market-based approach to controlling pollution by providing economic incentives for those reducing greenhouse gas emissions below what is required, to use or trade the excess reductions to offset emissions at another source inside or outside the country.

EMISSIVITY: The ability of a surface to emit radiation, measured as the ratio of the energy radiated by a surface to that radiated by a blackbody at the same temperature or the ratio of the radiation intensity of a nonblack body to the radiation intensity of a blackbody. Its symbol is ϵ . Its value is always less than or just equal to one. It varies with temperature and throughout the spectrum. Published values can readily be obtained for most substances.

EMITTANCE (EMISSIVE POWER; RADIATING POWER): The power per unit area radiated by a surface. The SI unit is watts per square metre (W/m^2). It is often used in astronomy.

EMITTER FOLLOWER (BJT VOLTAGE FOLLOWER; COMMON-COLLECTOR AMPLIFIER): An amplifying circuit using a bipolar junction transistor with a common-collector connection. It is one of three basic single-stage bipolar junction transistor (BJT) amplifier topologies, typically used as a voltage buffer. The base terminal of the transistor in this circuit serves as the input, the emitter and the output and the collector is common to both. It may be tied, for example, to ground reference or a power supply rail, hence its name. A similar field-effect transistor circuit is the common-drain amplifier.

EMITTER-COUPLED LOGIC (ECL): A set of integrated logic circuits or a digital circuit composed of bipolar transistors in which the emitter ends are wired together, the input part consists of an emitter-coupled transistor and the output is through an emitter follower. ECL gates switch faster than transistor-transistor logic (TTL) gates, but use more power.

EMITTERS: 1. A transistor region from which charge carriers that are minority carriers in the base are injected into the base or layers of semiconductor materials from which charge carriers such as electrons originate and control the flow of current through the collector. Its symbol is E and it is similar to the cathode of an electron tube. 2. A time pulse generator found in some equipment such as card punch.

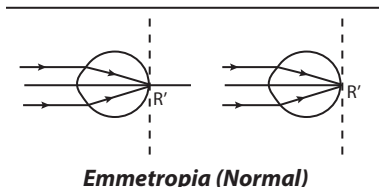
EMMENAGOGUES: Agents or substances containing properties which promote menstrual flow. They normalise and tone the female reproductive system. They include herbs such as mugwort, yarrow, black cohosh and blue cohosh. Such herbs are good for vagina douching.

EMMETROPIA: It's a condition of the normal eye, which focuses parallel rays on the retina when at rest. Its point is at infinity.

EMMOLIENTS: Agents or substances containing properties which are used to soothe and protect dry, irritated skin or mucous membrane. They are applied topically in the form of balm, gel, etc. They are used to treat serious skin disorders such as psoriasis and eczema, as well as applied to dry, irritated skins. Emollients include herbs such as plantain, aloe vera, acacia, neem, shea nut butter, sunflower and chickweed.

EMOTICON: A contraction of emotion and icon, which in computing is apictorial representation of a facial expression used in e-mail to indicate an emotion or attitude using punctuation marks and letters. Examples are: :-) for a smiley face, and :(for a sad face.

EMPHYSEMA: A degenerative lung disease characterised by shortness of breath. Often associated with smoking, the tissues necessary to support the physical shape and function of the lungs are destroyed irreversibly. It is included in a group of diseases called chronic obstructive pulmonary disease (COPD) because the destruction of lung tissue around smaller airways, called alveoli, makes these airways unable to hold their functional shape upon exhalation.



Emmetropia (Normal)

EMPIRICAL: A term denoting a result that is obtained by experiment, observation or experience, rather than from theory. A key concept of science and the scientific method is that all evidence must be empirical or empirically based or dependent on evidence or consequences that are observable by the senses. For example, empirical data is data that is produced by experiment or observation. The term empirical also refers to the process of elemental analysis, which is a method applied in analytical chemistry for determining the relative amounts of each element in a chemical compound.

EMPIRICAL EVIDENCE: Observations, measurement and other types of data acquired that scientists record and analyse to support and evaluate scientific explanations. Scientists need empirical evidence for a hypothesis to gain acceptance in the scientific community.

EMPIRICAL FORMULA: The chemical formula showing the relative or simplest proportion of atoms of elements in a compound instead of their structural arrangement or molecular weights. For example, the formula HO is the empirical formula for H_2O_2 , which means there is one H for every O atom. The empirical formula for ethanoic acid is CH_2O and it represents the atoms in their simplest whole number ratio. It makes no reference to isomerism, structure or absolute number of atoms and it is used as a standard for most ionic compounds, including CaCl_2 and for macromolecules, such as SiO_2 . Empirical formula of a compound is the formula which expresses only the relative number of atoms of each element in the compound. Empirical formulas are obtained from experimental data and are useful to determine the molecular mass (or weight) of compounds whose molecular masses have not been known; and also to determine the molecular masses of compounds whose molecular masses are variable, even though they have a definite percentage composition (i.e., for different compounds formed from the same elements). On the other hand, the molecular formula gives types and numbers of atoms present, for example, the molecular formula for ethanoic acid is $\text{C}_2\text{H}_4\text{O}_2$. The structural formula indicates the way the atoms are arranged and it is commonly written by dividing the formula into groups. For example, ethanoic acid can be written as CH_3COOH . The structure can also be drawn to show how the elements are bonded or linked to one another.

EMPIRICAL PROBABILITY (POSTERIOR PROBABILITY): A probability based on previous experience or experiment. It is the ratio of the number of times an event has occurred to the total number of trials performed.

EMPTY: In logic, of a name or description, vacuous, having no reference.

EMPTY SET (NULL SET): A set that has no elements or members. The symbols $\{ \}$ or ϕ are often used to denote the empty set. It is a unique set and its size is zero. Many possible properties of sets are trivially true for the null set.

EMPTY WORD: In group theory, any entity that is not a non-empty word; multiplication is extended to the empty word, denoted **1**, by providing that $1.u = u$, for any word u .

EMULATOR: A term used in computing to denote an item of software or firmware that allows one device to imitate the functioning of another system, so that the second system behaves like the first system. This allows a user to select from a wider range of application programs.

Emulation in computing, refers to a model such as software that accepts the same inputs and produces the same outputs as a given system.

EMULSIFICATION: The breakdown of fat globules in the duodenum into tiny droplet, which provides a larger surface area on which the enzyme pancreatic lipase acts to digest the fats into fatty acids and glycerol. This process is facilitated by the action of bile salts. Fat, which is not emulsified usually passes through the intestines and is eliminated in faeces.

EMULSIFIER: A surfactant, when present in small amounts facilitates the formation of an emulsion or enhances its colloidal stability by decreasing either or both of the rates of aggregation and coalescence. In other words, it is a substance which prevents an emulsion from

separating into layers or any ingredient used to bind together normally **noncombinitive** substances such as oil and water. They are widely used in foods, cosmetics and medicines. For example, egg yolks contain a natural emulsifier, lecithin, used to thicken and bind sauces such as hollandaise, as well as to bind ingredients in baking. Also xanthan gum, which is a commercial emulsifier, is used in numerous foods like salad dressings and dairy products. Some commercial emulsifiers also inhibit baked goods from going stale.

EMULSION: A fluid colloidal system in which liquid droplets and/or liquid crystals are dispersed in a liquid. The droplets often exceed the usual limits for colloids in size. In other words, it is a stable dispersion of a liquid in another liquid or a mixture of two or more liquids in which one is dispersed in the other as microscopic/ultramicroscopic droplets or a colloid made from two liquids that do not mix. An example is a suspension of oil in water. Many products we use in everyday life are oil-in-water (o/w) or water-in-oil (w/o) emulsions. Examples are milk (oil in water) and butter (water in oil) emulsion. Emulsions are stabilised by agents called emulsifiers and more-stable emulsions can be destroyed by inactivating the emulsifier, by freezing or by heating, while less-stable emulsions eventually separate spontaneously into two liquid layers. Polymerisation reactions are often performed in emulsions.

ENABLE: In computing, it is to activate a device or system, or make a program or device operational. In running a program, or using a device some features may be disabled by the original settings. Enabling these features allow the computer to make use of them.

ENAMEL: 1. A material made up of calcium phosphate that forms a covering over the crown of a tooth. 2. A glassy substance which is melted and put onto objects made of metal, glass or clay, which hardens to form a decoration or protection. 3. A point which is used especially on wood to produce a very shiny surface.

ENAMEL CLEAVAGE: The splitting of enamel in a plane parallel to the direction of the enamel rods.

ENAMEL ORGAN (DENTAL ORGAN): A mass of soft pulpy tissue that develops on the surface of a dental papilla. It lies above a condensation the cells of the ectomesenchyme called the dental papilla and consists of a mesh of fibre, fluid albumin and ameloblasts, which deposit a layer of protective enamel on the outer surface of the dentine.

ENAMEL ROD (ENAMEL FIBRE; ENAMEL PRISM): One of the calcified, microscopic rods radiating from the surface of the dentin, forming the substance of the enamel of a tooth. Its scientific name is *Prismata adamantina*.

ENANTIOMER: Each of a pair of optical isomers or stereoisomers that are mirror images of each other and cannot be superimposed on the mirror image of the other. They have identical chemical and physical properties, when present in a symmetric environment, except for their ability to rotate plane-polarised light (+/-) by equal amounts but in opposite directions. However, it is relevant to note that the polarised light can be considered an asymmetric media. A mixture of equal parts of an optically active isomer and its enantiomer is termed **racemic** and has a net rotation of plane-polarised light of zero. **Enantiomeric groups** are chiral groups that are mirror images of one another. **Enantiomeric units** are two configurational units that correspond to the same constitutional unit are considered to be enantiomeric if they are non-superposable mirror images. **Enantiomerism** is having chiral molecules that are mirror images of each other that are not superimposable.

ENANTIOMERICALLY ENRICHED (ENANTIOENRICHED): A sample of a chiral substance whose enantiomeric ratio is greater than but less than 50:50 but less than 100:0.

ENANTIOMORPH: One of a pair of chiral objects or models that are non-superposable mirror images of each other. The term, enantiomorph is also applied to mirror-image related groups within a molecular entity. The term also applies in Euclidean geometry and enantiomorph refers to a pair of asymmetric figures such that each is the non-superposable mirror images of the other, but a reflection

of either is congruent to the other. The pair are said to be of opposite handedness, for example, a pair of gloves. A pair of enantiomorph figures in n dimensions are congruent in $n + 1$ dimensions.

ENANTIOTOPIC: Denoting a ligand attached to a prochiral centre, in which the replacement of this ligand with ligand d, which is different from the ligands a, b, c attached to the prochiral centre, gives rise to a pair of enantiomers C_{abcd} . For example, the two hydrogen atoms attached to the second carbon in butane are enantiotopic. If one of the hydrogen atoms is replaced with a bromine atom, it will produce (R)-2-bromobutane. Replacement of the other hydrogen atom with a bromine atom will produce the enantiomer (S)-2-bromobutane.

ENANTIOTROPIC MESOPHASE: A mesophase that is thermodynamically stable over a definite temperature or pressure range.

ENATION: 1. An outgrowth, usually from a leaf, as shown in the diagram. Enations may occur in response to virus infection, for example, cotton leaf curl virus. 2. A small lateral outgrowth of the stem that is produced by certain primitive vascular plants and may be an early stage in microphyll evolution.



ENCAPSULATED POSTSCRIPT (EPS): An image description format that is compatible with PostScript printers and is often used for transferring files between various graphics applications. It is supported by most graphic design and page layout programs.

ENCAPSULATION: 1. In telecommunication, it is the process of translating data from one protocol into another, so that the first data is selectively hidden temporarily in the second protocol during transmission across a network. For example, a TCP/IP-formatted data packet can be encapsulated within an ATM frame. 2. In programming, it is the inclusion of information within a module or other packaged section of code that enables the program or programmer to use the code while keeping it intact, so that other programs defined within its class can also access it. It forms one of the three pillars of object-oriented programming (OOP). 3. In chemistry, **molecular encapsulation** is the confinement of an individual molecule within a supramolecule or larger molecule. 4. In biology, **cell encapsulation** is a technology used to overcome the existing problem of graft rejection in tissue engineering applications. It involves placing the transplanted cells within a biocompatible membrane in attempt to isolate the cells from the host immune attack and enhance or prolong their function in vivo. 5. **Micro-encapsulation** is the coating of microscopic particles with another material.

ENCEPHALIN (ENKEPHALIN): Any of a class of endorphins consisting of five amino-acids or a pentapeptide found principally in the brain and secondarily in the gut and adrenal glands. They bind onto opiate receptors and imitate the analgesic effects of morphine.

ENCEPHALITIS: Inflammation of the brain, usually caused by a direct viral infection or a hypersensitivity reaction to a virus or foreign protein. Brain inflammation caused by a bacterial infection is sometimes called **cerebritis**. When both the brain and spinal cord are involved, the disorder is called **encephalomyelitis**. An inflammation of the brain's covering or meninges, is called **meningitis**.

ENCEPHALISATION/ENCEPHALIZATION QUOTIENT, EQ (BRAIN-TO-BODY WEIGHT RATIO): A measure of the ratio of the relative weight of the brain of a particular species compared with the expected value for members of the group to which it belongs or that of a typical animal size. $EQ = w(\text{brain})/Ew(\text{brain})$. The formula for the expected weight of the brain varies, but is usually $Ew(\text{brain}) = 0.12w(\text{body})^{2/3}$, though for some classes of animals the power is $3/4$ rather than $2/3$. The value is used to estimate the comparative intelligence of animals, particularly in anthropology. It is believed that the larger an organism is, the more brain weight is required for basic survival tasks such as breathing, thermoregulation, senses, motor skill, etc. The larger the brain is relative to the body, the more brain weight might be

available for more complex cognitive tasks. This method, as opposed to the method of simply measuring brain weight alone, puts humans closer at the top of the list. Also, reflecting the evolution of the recent cerebral cortex, different animals have different degrees of brain folding, which increase the surface of the cortex, which is positively correlated to intelligence in humans.

ENCEPHALOGRAM (ELECTROENCEPHALOGRAM, EEG): In medicine, it is the graphic measuring and recording of the electrical activity in the brain by placing between 16-20 electrodes on the scalp which are connected to an electroencephalograph and electronic instrument. It is helpful in the diagnosis of some brain disorders, such as epilepsy, tumours or brain damage as well as in establishing clinical death.

ENCEPHALOMYELITIS: A serious disease that is characterised by the inflammation of the brain and the spinal cord. It is caused by agents such as bacteria, viruses, fungi and malignant cells.

ENCODING: In computer science, it refers to the various processes of converting data from one form to another. It includes image encoding, audio and video encoding and character encoding.

ENCOUNTER: A set of collisions between reactant molecules in solids, liquids or dense gases, occurring in rapid succession as a result of the cage effect.

ENCOUNTER COMPLEX: A complex of molecular entities produced at an encounter-controlled rate and which occurs as an intermediate in a reaction mechanism. When the complex is formed from two molecular entities it is called an **encounter pair**. A distinction between encounter pairs and (larger) encounter complexes may be relevant in some cases, such as for mechanisms involving pre-association.

ENCRYPTION: The coding of information to prevent unauthorised eavesdropping along the transmission line so that it can only be decoded and read by someone who has the correct decoding key. Encryption is used in secure web sites as well as other mediums of data transfer.

END ORGAN: The structure at the end of a peripheral nerve or the specialised end of a sensory or motor nerve.

END PLATE: The area of the plasma membrane of a muscle cell that lies immediately beneath a motor nerve ending at a neuromuscular. Release of a neurotransmitter at the end plate induces contraction of the muscle fibre.

END POINT: 1. The point at which a chemical reaction is complete or the point in a titration at which no more titrant should be added. For example, the end point of a titration is when all or one of the reactants has been used up and the indicator changes colour. The endpoint can be identified by an indicator through a colour change, for example, by the appearance of a precipitate or with an instrument such as a pH meter or a conductivity meter. 2. In mathematics, **endpoint** is either of two points that mark the ends of a line segment or a point that marks the end of a ray.

END PRODUCT INHIBITION (FEEDBACK INHIBITION): The inhibition of the first enzyme in a pathway by the end product of that pathway. In sequential enzyme systems, a control mechanism in which accumulation of final product from a metabolic reaction causes the inhibition of product formation.

END USER: In information technology, it is the individual, who uses a computer software program or hardware device rather than someone who writes the programs. It is an abstraction of the group of persons who will ultimately operate the piece of software. Most computer users are considered end-users.

END USER LICENSE AGREEMENT (EULA): A legal agreement between a software application author or publisher and the user of that application stating the terms of usage. It is included with a product or displayed on the screen at the time of installation and requires that the user click "Accept" or "I Agree," in order to install the software successfully.

END-OF-FILE (EOF): A symbolic constant placed by a computer after a file's last byte of data where no more data can be read from a

data source or when an instruction or command resets the file pointer to the end. EOF marks are helpful in data transmission and storage. Files are stored in blocks and the end marker helps the computer know it has allocated enough space to store the file.

END-PLATE POTENTIAL (EPP; END PLATE SPIKES): The electric potential produced by depolarisation of the end plate at a neuromuscular junction in response to stimulation by the motor neuron. In vertebrate skeletal muscle, a single EPP, typically about 10–20mV in amplitude, triggers an action potential in the muscle fibres which propagates through the fibre causing it to contract.

ENDANGERED SPECIES: A categorisation of threatened species by the International Union for Conservation of Nature (IUCN) and defined as a species that is likely to become extinct. Conservation biologists use the IUCN Red List, to categorise endangered and critically endangered species. There are currently 3079 animals and 2655 plants endangered worldwide. An example of an endangered animal species is the Javan rhinoceros. The main factors that cause species to become endangered are habitat destruction, invasive species, pollution and over-exploitation.

ENDARCH: Describing xylem maturation in which the older cells (protoxylem) are nearer the centre of the axis than the younger cells. Development is thus centrifugal.

ENDECADIC: Of or relating to eleven.

ENDEMIC: 1. Plants or animals restricted to an area or few localities in its distribution such as a specific island, habitat type, nation or other defined zone. For example, all species of lemur are endemic to the island of Madagascar, while the orange-breasted sunbird is exclusively found in the Fynbos vegetation zone of southwestern South Africa. 2. Disease or pest always present in an area. Physical, climatic and biological factors can contribute to endemism. There are two subcategories of endemism: **paleoendemism**, which are species that were formerly widespread but are now restricted to a smaller area; and **neoendemism**, which are species that have recently arisen such as species that have diverged and become reproductively isolated or those that have formed following hybridisation and are now regarded as separate species.

ENDERGONIC REACTION (ENDOTHERMIC REACTION; UNFAVOURABLE REACTION; NONSPONTANEOUS REACTION):

A chemical or biochemical reaction requiring an external source of free energy, which is usually heat or a reaction with a positive free energy change, in which case, energy is absorbed in the form of heat. Examples are photosynthesis, melting ice, cracking of alkanes, evaporation of water, thermal decomposition reactions, reaction of barium hydroxide octahydrate crystals with dry ammonium chloride, splitting a gas molecule apart and dissolving ammonium chloride in water. In endothermic reactions reaction, there is a loss in the total amount of energy; it takes more energy to start the reaction than what comes out of it, so that the total energy is a negative net result. Under constant temperature and pressure, the change in the standard Gibbs free energy would be positive for the reaction at standard pressure (1 bar) and standard concentrations (1 molar) of all the reagents.

ENDOCARDIUM: The thin serous membrane, made of endothelial cells, that lines the inner wall of the heart. It also covers the arteries, veins and valves. The cusps of the valves are formed from folds of the endocardium.

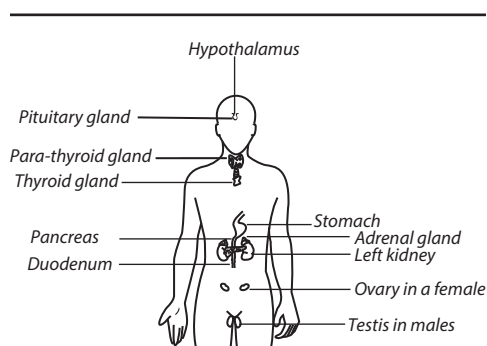
ENDOCARP: The innermost layer of the fruit wall or the innermost of the three layers of the pericarp of a fruit.

ENDOCHONDRAL OSSIFICATION: One of the two essential processes during foetal development of the mammalian skeletal, by which human bones form from cartilage; the other process, in which cartilage is not involved is ossification. It is also an important process during the primary formation of long bones, the growth of the length of long bones and the natural healing of bone fractures.

ENDOCRINE GLAND (DUCTLESS GLAND): A gland that produces a hormone and releases it straight into the blood circulation of an animal to regulate body processes or discharge secretions. In humans, it is usually about the size of a pea and consists of two

main parts that include the following: (i) The front part, called the anterior pituitary that synthesises and secretes hormones such as prolactin that stimulates breast milk production after childbirth and also affects sex hormone levels from ovaries in women and from the testes (testicles) in men, as well as fertility; growth hormone (GH) that stimulates growth; adrenocorticotropin (ACTH) that stimulates the production of cortisol by the adrenal glands; cortisol, that helps maintain blood pressure and blood glucose (sugar) levels; thyroid-stimulating hormone (TSH) that stimulates the thyroid gland to produce thyroid hormones that regulate the body's metabolism, energy balance, growth, and nervous system activity; luteinizing hormone (LH) that stimulates testosterone production in men and egg release (ovulation) in women; and follicle-stimulating hormone (FSH). (ii) The back part is called the posterior that is responsible for producing antidiuretic hormone (ADH), also known as vasopressin that regulates water balance in the body and oxytocin that causes milk to flow from the breasts in breastfeeding women.

ENDOCRINE SYSTEM: The system that, along with the nervous system, functions in the regulation of body activities. The endocrine system acts through chemical messengers called hormones that influence growth, development and metabolic activities. It includes the thyroid and parathyroids, anterior and posterior pituitary, pancreas, adrenals, pineal and gonads. The action of the endocrine system is measured in minutes, hours or weeks and is more generalised than the action of the nervous system. The nervous system acts through electrical impulses and neurotransmitters to cause muscle contraction and glandular secretion. The effect is of short duration, measured in seconds and localised. The pancreas is located under the stomach.



The endocrine glands

ENDOCRINOLOGY: The study of the structure and functions of the endocrine glands and of the hormones they produce. The glands are ductless and the hormones or chemical messengers are secreted directly into the blood to be transported to the target tissue or organ. The hormones are highly specific and their action may be selective or generalised.

ENDOCYTOSIS: The process through which a material enters a cell without passing through the plasma membrane or the process by which cells absorb molecules such as proteins from outside the cell by engulfing them with their cell membrane. The membrane folds around material outside the cell, resulting in the formation of a saclike vesicle into which the material is incorporated. It is used by all cells of the body because most substances important to them are large polar molecules that cannot pass through the hydrophobic plasma or cell membrane. On the other hand, exocytosis is the release to a cell surface of substances such as waste or secretions through vesicles.

ENDODERM (ENTODERM): A germ layer lying remote from the surface of the embryo that gives rise to internal tissues such as the gut or the internal layer of cells of the gastrula, which develops into the alimentary canal, digestive glands of the adult and part of the respiratory system.

ENDODERMIS: The innermost, single layer of plant cells that is located between the cortex and the vascular (xylem and phloem) tissues. It developed in roots and subaerial stems. Its functions include

absorption of water, selection of solutes and ions and production of oils, antibiotic phenols and acetylenic acids. It has extra sets of chromosomes as compared with cortical and other cells in the plant.

ENDOENZYME (INTRACELLULAR ENZYME): An enzyme that functions within the cell in which it was produced. Majority of enzymes fall within this category.

ENDOERGIC: A term used to describe a chemical or nuclear process that absorbs energy.

ENDOGAMY: The fusion of reproductive cells or gametes from closely related parents or inbreeding or fertilisation resulting from pollination among flowers of the same plant.

ENDOGENOUS: A substance, stimulus organ, etc., that originates from within an organism, tissue or cell. Endogenous processes include circadian rhythms. Endogenous retroviruses are caused by ancient infections of germ cells in humans, mammals and other vertebrates. Their proviruses remain in the genome and are passed on to the next generation. In certain biological systems, endogeneity refers to the recipient of DNA, usually in prokaryotes.

ENDOGENOUS RHYTHMS: Sequential physical or biochemical processes that occur in a plant or plant part in response to internal stimuli. An example is the apparently autonomic component of the nyctinastic movements of certain plants, the leaflets of which continue to follow a periodic folding and unfolding sequence even if kept in continuous darkness.

ENDOLITHS: Organisms that live inside rock or the tiny pores between interlocking mineral grains. Thousands of different species are known, including representatives of bacteria, archaea and fungi. Endoliths have been found inhabiting the Earth's crust at depths of up to nearly 3 km. Many are autotrophs which manufacture their own organic compounds from inorganic chemicals in the rock. Other varieties of endolith feed on the organics produced by the autotrophs, giving rise to microscopic underground communities known as SLiMEs (Subsurface Lithotrophic Microscopic Environments). The extreme depth at which endoliths can exist has yet to be established.

ENDOLYMPH (SCARPA'S FLUID): The fluid that fills the membranous labyrinth of the vertebrate's inner ear. The endolymph has high potassium content, which is secreted from the stria vascularis. In the cochlear duct, fluid waves in the endolymph stimulate the receptor cells, which in turn translate their movement into nerve impulses that the brain perceives as sound. In the semicircular canals, angular acceleration of the endolymph stimulates the vestibular receptors of the endolymph. The semicircular canals of both inner ears act in concert to coordinate balance.

ENDOMEMBRANE SYSTEM: The set of membranes that form a single functional and developmental unit, either being connected together directly or exchanging material through the vesicle transport. There is the hypothetical concept that the membranes of many apparently discrete cell structures are part of an interrelated system. It is composed of the different membranes that are suspended in the cytoplasm within a eukaryotic cell and divide the cell into functional and structural compartments or organelles. In eukaryotes, the organelles of the system are made up of the nuclear envelope, the endoplasmic reticulum, the golgi apparatus, lysosomes, vacuoles, vesicles and the cell membrane, but does not include the membranes of mitochondria or chloroplasts.

ENDOMETRIOSIS: In medicine, it is a condition in women in which some tissues of the endometrium (the lining of the womb) are found outside the uterus or womb. Signs and symptoms include painful periods, pelvic pain and cramping, pain during or after intercourse, etc. It may lead to infertility.

ENDOMETRIUM: The mucous or inner membrane that lines the uterus of mammals. It comprises an upper mucus-secreting layer, which is shed during menstruation and a basal layer, which proliferates to form the upper layer. It has an average thickness of 6.7 mm and it is central and detectable using ultrasound scanners (echogenic). Its functions include providing a lining for the uterus and preventing adhesions between the opposed walls of the myometrium, thereby

maintaining the patency of the uterine cavity. During the menstrual or oestrous cycle, the endometrium grows to a thick, blood vessel-rich, glandular tissue layer which gives an optimal environment for the implantation of a blastocyst upon its arrival in the uterus. The glands and blood vessels in the endometrium increase in size and number during pregnancy and vascular spaces fuse and become interconnected, forming the placenta, which supplies oxygen and nutrition to the embryo and foetus.

ENDOMITOSIS: The replication of chromosomes in the absence of cell or nuclear division, resulting in numerous copies within each cell (polyploidy). Many rounds of endomitosis give rise to the giant polytene chromosomes of Dipteran salivary glands, notably in the salivary glands of *Drosophila*.

ENDOMORPHISM: Often abbreviated **End**, it is a homomorphism that maps a mathematical structure into itself. For example, an endomorphism of a vector space V is a linear map $f: V \rightarrow V$ and an endomorphism of a group G is a group homomorphism $f: G \rightarrow G$.

ENDOMYCETALES: An order of the Hemiascomycetes containing about 200 species in some 50 genera. It includes the family Saccharomycetaceae, the yeasts, which contains the commercially important species *Saccharomyces cerevisiae* (brewer's yeast), used in bread and beer making and *S. ellipsoideus*, used in wine making. The yeasts reproduce asexually by budding or by dividing into two equal cells as in *Schizosaccharomyces* (fission yeasts).

ENDONUCLEASE: One of a large group of enzymes that catalyse the internal cleavage of nucleic acids or an enzyme that breaks a phosphodiester linkage at some point within a polynucleotide chain. This is in contrast to exonucleases, which cleave phosphodiester bonds at the end of a polynucleotide chain. These enzymes are often used in genetic engineering to make recombinant DNA for introduction into bacterial, plant or animal cells. Restriction endonucleases (ENases) are products of bacteria and can be used to map a piece of DNA.

ENDOPARASITE: A parasite such as tapeworm that lives inside the body of its host. Endoparasites derive their nutrition from feeding on the host's blood. They can exist in one of two forms: (i) **intercellular parasites**, such as protozoa, bacteria or viruses, which inhabit spaces in the host's body; or (ii) **intracellular parasites**, which inhabit cells in the host's body. Examples of endoparasites are roundworm and tapeworm as shown in the diagrams. Note that proglottids are segments containing sexual reproductive systems.

ENDOPEPTIDASE: A protein digesting enzyme or any of a large group of enzymes that catalyse the hydrolysis of peptide bonds of a polypeptide chain at specific sites between amino acids or at an internal peptide or protein linkage.

ENDOPHENOTYPE: A component part of a broad and complex phenotype which are microscopic, internal and are, therefore, not apparent to the casual observer. It has been applied in medicine and psychiatry in efforts to unravel the genetic and physiological basis of multifactorial diseases such as schizophrenia and heart disease. For example, cortisol secretion is an endophenotype of anxiety.

ENDOPHYTE: An organism such as bacterium or fungus, which may be parasitic or symbiotic, but lives on or within a plant for at least part of its life cycle without causing apparent disease.

ENDOPLASM: The inner liquid part of a cell's cytoplasm. It contains most of the cell's structures and is distinguishable in certain cells, such as motile cells.

ENDOPLASMIC RETICULUM (ER): An organelle in a cell consisting of a system of membranes that forms an interconnected network of tubules, vesicles and cisternae within the cytoplasm of plant and animal cells or a membranous system of tubes, channels and flattened

sacs that form compartments within eukaryotic cells. Generally, it stores and transports proteins and lipids within cells and also carries various enzymes needed for the synthesis of fats and proteins. However, there are various types with various functions. For example, the rough endoplasmic reticulum which has ribosomes attached to it synthesises proteins while the smooth endoplasmic reticulum which does not have ribosomes attached synthesises lipids and steroids, metabolises carbohydrates and steroids and regulates calcium concentration, drug detoxification and attachment of receptors on cell membrane proteins. Sarcoplasmic reticula regulate calcium levels.

ENDOPTERYGOTE (ENDOPTERYGOTA; HOLOMETABOLA): Any winged insect having holometabolous development; the eggs hatch into larvae that are dissimilar to the adults and lack wings. They are of the subclass Pterygota, which go through distinctive larval, pupal and adult stages and undergo a radical metamorphosis, complete metamorphism or holometabolism, with the larval and adult stages differing considerably in their structure and behaviour. They are among the most diverse insect superorders, with approximately 850,000 living species divided between eleven orders, containing insects such as butterflies, fleas, bees, ants and beetles. The Endopterygota (internally formed wings) are distinguished from the Hemipterodea or Exopterygota (externally formed wings) by the way in which their wings develop, with the latter not going through a pupal stage.

ENDOREDUPPLICATION: Repeated duplication of the entire complement of genes within a single nucleus without mitosis. It occurs in many actively metabolising plant tissues, where it is characterised by the enlargement of the cell nuclei. In animals, it is most frequent in Appendicularia, where endoreduplicated epithelial cells are used to produce feeding structures and in arthropods, where the highest frequency of endopolyploidy occurs in the Malpighian tubules and salivary glands of various dipters.

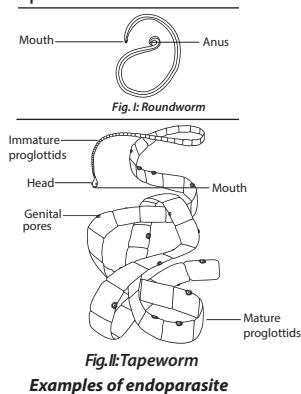
ENDORPHIN: Endogenous opioid peptide neurotransmitters produced by the pituitary gland and the hypothalamus in vertebrates during exercise, excitement, pain, consumption of spicy food and orgasm. They resemble the opiates in their abilities to produce analgesia and a feeling of well-being. The term implies a pharmacological activity, similar to the activity of the corticosteroid category of biochemicals, contrary to a specific chemical formulation. When a nerve impulse reaches the spinal cord, endorphins are released which prevent nerve cells from releasing more pain signals; and immediately after injury, endorphins allow animals to feel a sense of power and control over themselves that allow them to persist with activity for an extended time.

ENDOSCOPIC EMBRYOGENY: Embryogeny in which the embryo develops from the inner of the two cells that results from the first division of the zygote. The outer cell often forms the suspensor. Endoscopic embryogeny is seen in the Lycopsida, most of the Filicinae and all the Gymnospermae and Angiospermae.

ENDOSCOPY: The examination of the internal organs or tissues with an instrument called endoscope, which is a tube with a collection of optical fibres using lasers to look inside the human body. Its advantage is that it permits diagnosis without the need for exploratory surgery. It can also refer to using a borescope in technical situations, where direct line-of-sight observation is not feasible.

ENDOSKELETON (INTERNAL SKELETON): The skeleton lying within an animal's body or the internal supporting structure of vertebrates made up of bone, cartilage and mineralised tissue. It provides shape, support (as in the case of sponges) and protection. It also serves as an attachment site for muscle and a mechanism for transmitting muscular forces. In collaboration with muscles, they produce movement. Endoskeletons of various complexity are found in three phyla and one subclass of animals: Chordata, Echinodermata, Porifera and Coleoidea. True endoskeletons are derived from mesodermal tissues and are found in echinoderms and chordates.

ENDOSMOSIS: Osmosis in which fluid is absorbed into a cell from a surrounding fluid or the inward flow of a fluid through a permeable membrane toward a fluid of greater concentration. Its opposite is **exosmosis**.



ENDOSOME: A membrane bound compartment or vesicle formed within a eukaryotic cell, during certain types of endocytosis, in which the material to be ingested first binds to receptor sites on the cell surface. It is a compartment of the endocytic membrane transport pathway from the plasma membrane to the lysosome and, therefore, represents a major sorting compartment of the endomembrane system in cells. It is approximately 500 nm in diameter when fully mature and provides an environment for materials to be sorted before they reach the degradative lysosome. For example, Low-Density Lipoprotein (LDL) is taken into the cell by binding the LDL receptor at the cell surface.

ENDOSPERM: The nutritive tissue such as starch, oil and protein in the seeds of most flowering plants that surrounds the developing embryo in the seed. This makes it a major source of nutrition in human diet as well. For example, wheat endosperm is ground into flour for bread while barley endosperm is the main source for beer production. Corn and coconut are other examples of plants whose endosperms form the bulk of the edible portion. Some plants like the orchids, however, lack endosperm in their seeds.

ENDOSPORE: A dormant, tough and non-reproductive structure produced by certain bacteria from the firmicute phylum or the resisting body of certain bacteria, formed within the cell in response to adverse conditions. It consists of the bacterium's DNA and part of its cytoplasm, enclosed in a tough resistant spore coat and when conditions become favourable, the spore germinates, producing a cell that emerges from the spore investment. It is important when the bacterium is experiencing an environment that is deleterious to its usual vegetative state. It is often found in soil and water, where it may survive for long periods of time. It can survive without nutrients and it is resistant to ultraviolet radiation, desiccation, high temperature and chemical disinfectants. Common anti-bacterial agents that work by destroying vegetative cell walls do not work on endospores. Some classes of bacteria can turn into exospores, known as microbial cysts, instead of endospores. Exospores and endospores are two kinds of dormant stages seen in some classes of microorganisms. Not all types of bacteria can change to the endospore form.

ENDOSPORIC GAMETOPHYTE: A gametophyte that develops within a spore. For example, the female gametophyte of *Selaginella* is contained within the megaspore and at maturity only a portion, bearing the archegonia, is exposed through the spore wall. Such gametophytes are better able to withstand dry conditions and this pattern of development may be seen as a step in the evolution of the seed habit.

ENDOSTYLE (HYPOPHARYNGEAL GROOVE): A longitudinal ciliated groove in the ventral wall of the pharynx of lancelets, some tunicates, urochordates, cephalochordates and larval lampreys that contain mucus-secreting glands for gathering food particles. It aids in transporting food to the oesophagus. The endostyle in larval lampreys (ammocetes) metamorphoses into the thyroid gland in adults and is regarded as being homologous to the thyroid gland in vertebrates. Since it is found in the three branches of chordates, it is believed to have originated from the common ancestor of these taxa, along with a shift to internal feeding for extracting suspended food from the water.

ENDOSYMBIOTIC THEORY: A theory stating that the mitochondria of eukaryotes and the chloroplasts of green plants and flagellates originated as free-living prokaryotes that invaded primitive eukaryotic cells and became established as permanent symbionts in the cytoplasm or eukaryotic organisms evolved from symbiotic association between prokaryotic. Free-living aerobic bacteria and chloroxy-bacteria became incorporated inside larger nucleated prokaryotic cells, where they acted as forerunners of the mitochondria and chloroplasts seen in modern eukaryotes. According to this theory, blue green algae and bacteria were not digested; they became symbiotic instead. For example, There are two prokaryotic cells, one cell "engulfs" the other cell and a double membrane can be found inside. The theory was propounded by the American biologist, Lynn Margulis (1938 –2011), now widely accepted, particularly the finding that mitochondria and

chloroplasts have DNA similar to that of eubacteria and that they contain prokaryotic-type ribosomes.

ENDOTHECIUM: 1. The subepidermal layer of the anther wall in angiosperms. The cells of the endothecium often develop thickenings on the anticlinal and inner tangential walls as the anther matures, which are believed to aid in anther dehiscence. 2. The inner layer of cells in the young sporophyte of bryophytes, giving rise to the columella and (except in *Sphagnum*) sporogenous tissue.

ENDOTHELIAL CELLS: A squamous (flattened) cell, large numbers of which make up the lining of blood vessels and lymphatic vessels. Endothelial cells line the entire circulatory system, from the heart to the smallest capillary. A layer of epithelial cells is called an endothelium. An individual endothelial cell has a central nucleus, is very flat (only about 1-2 μm thick) and measures some 10-20 μm in diameter. The intercellular junctions between endothelial cells overlap, thus helping to make a tight seal, for example in the wall of a blood vessel. Endothelial cells play a key role in many phenomena to do with blood and blood vessels, including: (i) formation of new blood vessels (angiogenesis); (ii) vasoconstriction and vasodilation and hence the control of blood pressure; (iii) blood clotting (thrombosis and fibrinolysis); (iv) inflammation and swelling (oedema). Endothelial cells also control the movement of substances and the passage of white blood cells (leukocytes), into and out of the bloodstream.

ENDOTHELIN (ET): Any of several related peptides, one of which (endothelin) is released by the endothelium of blood vessels and is a potent vasopressor, causing constriction of blood vessels and raising blood pressure or a 21-amino acid vasoconstricting peptides produced primarily in the endothelium and having a key role in homeostasis. They are normally kept in balance by other mechanisms, but when they are over-expressed, they contribute to high blood pressure or hypertension and heart disease.

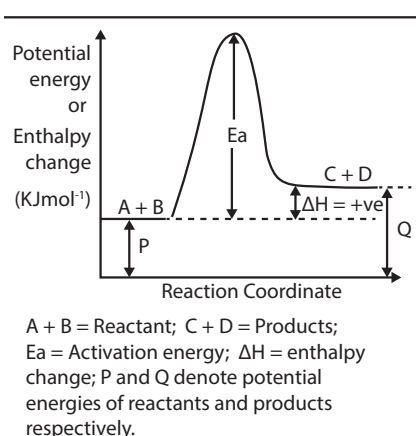
ENDOTHELIUM: A layer of smooth, flattened cells called endothelial cells lining the heart, blood vessels and lymph vessels. It is derived from embryonic mesoderm. The corneal endothelium is the innermost layer of the cornea.

ENDOTHERM (HOMEOTHERM): An animal such as a bird or mammal that uses metabolic energy to maintain a constant body temperature. These animals are commonly referred to as "warm-blooded." These organisms capable of using metabolic energy to maintain and regulate a constant internal body are known as **homiothermic organisms**.

ENDOTHERMIC: 1. Maintaining a constant body temperature despite changes in the temperature of the environment. It is a term used of or relating to an organism that generates heat to maintain its body temperature, typically above the temperature of its surroundings. Animals which exhibit such characteristics are said to be warm blooded or homeothermic. For example, birds and mammals are endothermic. 2. Characterised by or causing the absorption of heat (endoergic), for example, in a chemical reaction.

ENDOTHERMIC REACTION (ENDERGONIC REACTION; UNFAVOURABLE REACTION; NONSPONTANEOUS REACTION):

A chemical or biochemical reaction requiring an external source of free energy, which is usually heat or a reaction with a positive free energy change, in which case, energy is absorbed in the form of heat. Examples are photosynthesis, melting ice, cracking of alkanes, evaporation of water, thermal decomposition reactions, reaction of barium hydroxide octahydrate crystals with dry ammonium chloride, splitting a gas molecule apart and dissolving ammonium chloride in water. In endothermic reactions reaction, there is a loss in the total amount of energy; it takes more energy to start the reaction than what comes out of it, so that the total energy is a negative net result. Under constant temperature and pressure, the change in the standard Gibbs free energy would be positive for the reaction at standard pressure (1 bar) and standard concentrations (1 molar) of all the reagents.



**Reaction profile diagram
for endothermic reaction**

ENDOTOXIN: In biology, heat-stable complex of protein and lipopolysaccharide that is produced following the death of certain gram-negative bacteria, such as *Brucella*, *Neisseria* and *Vibrio* species and released upon destruction of the bacterial cell. In large quantities endotoxin can cause haemorrhagic and severe diarrhoea. It also causes fever in small quantities.

ENDOTROPHIC: Nourished or receiving nourishment from within; for example, fungi or their hyphae receive nourishment from plant roots in a mycorrhizal association.

ENDOTROPHIC MYCORRHIZA (ENDOPHYTIC MYCORRHIZA):

A form of mycorrhizal association in which the fungus lives between and within the cells of the cortex and growth on the outside of the root is limited. Such associations are found on many herbaceous species, especially orchids and heathers and on certain woody plants, such as rhododendrons. The fungi involved are usually species of *Rhizoctonia* on orchids or *Phoma* on heathers. The hyphae form tightly coiled masses, termed pelotons, within the cells of the outer cortex. Plants that commonly develop such associations cannot grow normally without the appropriate fungus.

ENDOW: To define a relation or function (on a given structure) in order to consider them together as a single mathematical entity. For example, the metric space (S, δ) consists of the set S endowed with the metric δ .

ENDPOINT: 1. A maximal or minimal point of a segment on a line or interval; for example, the endpoints of the interval $[0, 1]$ are the points 0 and 1. 2. Either of the ends of a curvilinear arc; that is, the image of either of the endpoints of the unit interval under a given mapping. 3. In chemistry, **end point** is the point in a titration usually noting the completion of a reaction and marked by, for example, a colour change in an indicator or by the appearance of a precipitate. 4. The point at which an antibody or antiserum solution is at the most dilute and can still detectably react with the antigen. 5. In information technology, in asynchronous transfer mode (ATM) a switch or other device at the end of the ATM network that serves as the source (transmitter) and sink (receiver) of data in ATM cell format.

ENEMA: A procedure in which liquid or gas is injected into the lower bowel through the rectum, to expel its contents, to introduce drugs, for X-ray imaging or for other therapeutic or diagnostic purposes.

ENERGETIC GAMMA RAY EXPERIMENT TELESCOPE (EGRET):

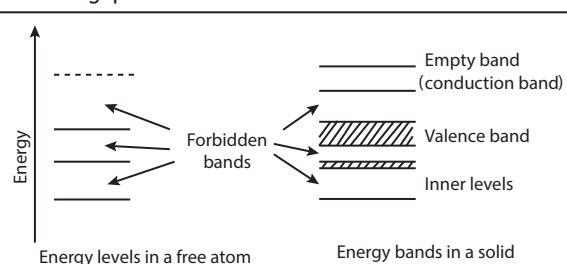
One of the four main instruments installed on board National Aeronautics and Space Administration's (NASA) Compton Gamma-ray Observatory to detect high energy gamma ray from 30 MeV to 30 GeV. The Compton Gamma-ray Observatory operated from April 1991 to June 2000.

ENERGY: 1. The ability or capacity to do work. It is a quantity that can be assigned to every particle, object or system of objects as a result of its state. The SI unit of energy is joules, but in some fields other units such as kilowatt-hours and kilocalories are also used. The different types of energy include kinetic, potential, thermal/heat, gravitational, sound, elastic, light, electrical/electromagnetic, chemical and nuclear energies. It is a scalar physical quantity. In living organisms, work involves movement, feeding, reproduction, growth, respiration and sensitivity. The German physicist, Hermann von Helmholtz observed that all forms of energy are equivalent: energy in one form can disappear but the same amount of energy will appear in another form, subject to the conservation law. When it is in a form other than thermal energy, it may be transformed with perfect efficiency to any other type of energy. Although the total energy of a system does not change with time, its value may depend on the frame of reference. For example, a seated passenger in a moving plane has zero kinetic energy relative to the plane, but non-zero kinetic energy and higher total energy relative to the Earth. 2. **Energy of a radiation** is the energy of the individual particles or photons of which a radiation consists.

ENERGY BALANCE (GLOBAL ENERGY BALANCE): The balance between the incoming radiation from the Sun and the sum of the outgoing reflected solar radiation and the infrared radiation emitted by the climate system. The global energy balance regulates the state of the Earth's climate, and modifications to it as a result of natural and man-made climate forcing (radiative forcing), cause the global climate to change. Averaged over the globe and over longer time periods, the energy budget of the climate system must be in balance.

ENERGY BALANCE EQUATION: The result that the rate of change of the sum of the kinetic energy and internal energy of a sub-body is equal to the sum of the power and the heating of that sub-body.

ENERGY BAND (ENERGY GAP): The energy difference measured in electron volts (eV) between the bottom of the conduction band and the top of the valence band of the electrons in a solid material. A material is considered to be a semiconductor if the value of the energy band is below 2eV; above this value, it is an insulator. Therefore, semiconductors have a very small band gap, whilst insulators have a wider band gap.



Energy band

ENERGY CHANGE: The heat energy released from reactants (exothermic) or taken in by reactants (endothermic), when chemical reactions take place. It may also be defined as a measure of the total combined energies within a system, derived from heats of transformation, disorder and other forms of internal energy such as electrostatic charges. Energy changes are measured in joules or kilojoules. The change in free energy between an initial and a final state is useful in evaluating certain thermodynamic processes and can be used to judge whether transformations will occur spontaneously.

ENERGY CONSERVATION: The methods of reducing energy or going without a service to save energy. Examples of methods used include turning off electrical appliances such as bulbs and refrigerators; through insulation of houses to prevent energy in the form of heat or air-conditioning from escaping outside; and increasing energy efficiency by replacing incandescent bulbs or electrical appliances such as refrigerators, washing machines and office devices with newer more energy-efficient ones, which provide the same or better service

but consuming less energy. Energy conservation and energy efficiency are similar as both of them reduce energy use and preserve resources for the future and reduce environmental pollution such as greenhouse gas emissions. The difference is that some of the methods used in conservation demand going without some services such as turning off light bulbs, heaters or air conditioners that may reduce comfort or safety. Energy efficiency, however, maintains the same level of output but uses less energy to do it through using more advanced energy efficient products.

ENERGY CONVERSION: Change of energy from one form to another. For example, nuclear energy can be converted into heat energy or solar energy into electrical energy.

ENERGY CROPS: Crops which are grown for energy rather than for food or fibre and used to make biofuels, such as bioethanol. Such crops include oilseed rape, soybeans, sunflower, wheat, barley, maize, rye, sugar beet, sugarcane and eucalyptus.

ENERGY DENSITY: The amount of energy stored in a given system or region of space per unit volume and is most commonly denoted u . It, therefore, has units of energy per length cubed. General relativity provides a way of calculating the energy density associated with a given curvature of spacetime.

ENERGY EFFICIENCY: Means of using less energy that does not cut back on the level of service produced by using advances in science and technology to provide services and products that use less energy. Examples include replacing older incandescent light bulbs or electrical appliances such as refrigerators, washing machines and office devices with newer more energy-efficient ones, which provide the same or better service but consuming less energy.

ENERGY FLOW (CALORIFIC FLOW): The flow of energy that occurs along a food chain. A general energy flow is as follows: solar energy is fixed by the photoautotrophs (primary producers) like green plants, used by primary consumers. They absorb most of the stored energy in the plant through digestion and convert it into the form they require (adenosine triphosphate) through respiration. Some of the energy herbivores received is changed to bodily heat as a result of respiration, which is radiated away and lost from the system. The expulsion of egesta, which contains undigested energy compounds also results in some energy losses. Secondary consumers then use the primary consumers resulting in energy that had been used by the primary consumers for growth and storage being absorbed by them through digestion. They also transform this energy into a more suitable form adenosine triphosphate (ATP) during respiration. Since energy, which the primary consumers had used for respiration cannot be utilised by the secondary consumers, some energy is again lost from the system. As tertiary consumers consume the secondary consumers, most of the energy is passed along, while some is again lost in the ways described. The last link in the food chain is decomposers (saprotrophic bacteria and fungi) which break down the organic matter of the tertiary consumers and release nutrients into the soil. Decomposers play a pivotal role in the nitrogen and carbon cycles. The energy is passed on from trophic level to trophic level and each time most (about 90%) of the energy is lost in the form of heat into the environment, an effect of respiration, egesta, etc., implying that the top consumer of a food chain receives the least energy, as a lot of it is lost between trophic levels. This loss of energy at each level limits typical food chains to only 4-6 links.

ENERGY FLUX: The rate of transfer of energy through a surface. The quantity is defined in two different ways, depending on the context.

ENERGY FLUX DENSITY: For mono-directional radiation, the energy traversing in a time interval over a small area perpendicular to the direction of the energy flow, divided by that time interval and by that area.

ENERGY GRADIENT: First derivatives of the total energy with respect to nuclear coordinates, *i.e.*, negative values of forces on the nuclei. Evaluation of energy gradient plays a central role in searching potential energy surfaces for stationary points. A widespread technique is based on the minimisation of the gradient norm (the square of energy gradient).

ENERGY INTENSITY: The ratio of energy consumption to economic or physical output. It is defined as the ratio between Gross Inland Energy Consumption (GIEC) and Gross Domestic Product (GDP), calculated for a calendar year. GIEC is calculated as the sum of the gross inland consumption of the five sources of energy: solid fuels, oil, gas, nuclear and renewable sources.

ENERGY LANDSCAPE: A multi-dimensional surface in which the energy of a system, typically Gibbs free energy is plotted against the physical states or parameters characterising that system.

ENERGY LEVEL: A definite or fixed energy that a system such as a molecule, an atom, an electron or a nucleus can have. For example, an atom has a fixed energy corresponding to the orbital in which its electrons move around the nucleus. The energy spectrum of a system with energy levels is said to be quantised and energy levels are said to be degenerate, if the same energy level is shared by more than one quantum mechanical state, in which case they are called degenerate energy levels. An **energy level diagram** is a diagram that shows the changes in energy during a chemical reaction.

ENERGY OF ACTIVATION (ACTIVATION ENERGY): The minimum amount of energy such as heat, electromagnetic radiation, electrical or other energy required to activate atoms or molecules to a condition in which it is equally likely that they will undergo a chemical reaction. In other words, it is the minimum amount of energy required to initiate a chemical reaction. It varies from reaction to reaction and it is determined by experiments that measure it as the constant of proportionality in the equation describing the dependence of reaction rate on temperature. Catalysts lower the activation energy by changing the transition state. It must be noted that the activation energy is unrelated to the heat of reaction.

ENERGY OF ATOMISATION: The energy change which takes place when 1 mole of the free gaseous atoms is formed from the element in its standard state under standard conditions.

ENERGY OF REACTION (HEAT OF REACTION; ENTHALPY OF REACTION): The energy released or absorbed during a chemical reaction to keep all substances involved at the same temperature. A positive value or heat being added implies that the reaction is endothermic, while a negative value or heat being given off implies that the reaction is exothermic. Accurate heat of reaction values are required for better design of equipment used in chemical processes. These are often calculated from compiled tables of thermodynamics data, heats of formation and heats of combustion of many known materials.

ENERGY PRINCIPLE: The consequence of the laws of motion that in a system of discrete particles with all forces conservative or doing no work, the sum of the kinetic energy and potential energy is constant.

ENERGY PYRAMID: A graphical model that compares the amount of energy available at each position, or level, in the feeding order. It is used to show the amount of energy available to living things in an ecosystem. The different levels represent different groups of organisms that might compose a food chain. From the bottom to the top, they are as follows: (i) primary producers which are organisms such as plants that make their own food from sunlight or bring energy from non-living sources into the community; (ii) primary consumers which are mostly herbivores that eat the producers; (iii) secondary consumers which are carnivores that eat the primary consumers; and (iii) tertiary consumers such as lion, man, hawk, tiger, etc., that eat the secondary consumers. Secondary consumers consist of carnivores, who prey on other animals and omnivores, which feed on both plants and animals.

ENERGY RESOLUTION: A measurement, at given energy, of the smallest difference between the energies of two particles or photons capable of being distinguished by a radiation spectrometer.

ENERGY SOURCE: Any source such as fossil fuels (*e.g.*, coal, oil and natural gas), uranium, water power, wind and the Sun from which energy can be obtained to provide heat, light and power. Plants use the Sun's energy and convert it into food and oxygen.

ENERGY THRESHOLD: The limiting kinetic energy of an incident particle or energy of an incident photon below which a specified process cannot take place.

ENERGY TRANSFER: Movement of energy from one system to another or from one substance to another, for example, through the transfer of heat, work or mass transfer. In a car engine, the chemical energy of the fuel is converted to kinetic (motion), heat and sound energy. In chemistry, examples include energy transfer through molecular collisions or through fluorescence resonance energy transfer.

ENERGY TRANSFORMATION: The change from one form of energy, such as the energy in fossil fuels, to another, such as electricity. Examples are thermoelectric that transforms heat to electric energy; geothermal power transforms heat to electric energy; and heat engines, such as the internal combustion engine transforms heat to mechanical energy.

ENGINE: 1. A machine or system that changes the chemical energy in fuel into mechanical energy, often resulting in motion or a device for converting stored energy into mechanical or useful work. Examples are heat engine and steam engine. The main features of an engine are its capacity to deliver appreciable mechanical power, as contrasted to a mechanism such as a clock, whose significant output is motion. An engine usually burns or otherwise consumes a fuel. A hydraulic turbine, for example, is not an engine, although it competes with the engine as a prime source of mechanical power. 2. In computing, it is a software or hardware system that performs a core or essential function for other programs. A database search engine, for example, accepts user input and returns a list of the documents where matches between the user input and the database records are found. The term is also used for systems like Google, Bing and Yahoo!, which enable users to search for documents on the World Wide Web. **Game engine** is the core software used as the basis for building a game in video games that powers the entire game. It includes input devices such as keyboard, mouse, gamepad and touch, joystick and steering wheel; and others such as sound effects, graphics, scripting, animation and artificial intelligence allowing users to manipulate the various moves repeatedly until the game is won or lost.

ENGINE OIL: A type of viscous liquid or liquefiable substance not miscible with water and enhanced with additives that is used to lubricate the moving parts or sliding surfaces of internal combustion engines to reduce wear; it also cleans moving parts from the sludge, inhibits corrosion, improves sealing, and cools the engine by carrying heat away from moving parts.

ENGINEERING: The field in which the concepts of pure science are applied to real life problems such as the application of science to the design, construction and maintenance of works, machinery, roads, railways, bridges, harbour installations, engines and ships that realise a desired objective or invention. The main divisions of engineering include aerospace, chemical, civil, electrical, electronic, gas, marine, materials, mechanical, mining, production, radio and structural.

ENGINEERING DRAWING: Technical drawing that forms the plans for the design and construction of engineering components and structures to show graphically the ideas and information necessary for the construction or analysis of machines, structures and systems. It usually includes such features as various types of lines, dimensions, lettered notes, sectional views and symbols. They may be in the form of freehand sketches or carefully planned and checked mechanical drawings and usually a sketch precedes the mechanical drawing. The basis for much engineering drawing is orthographic representation (projection). Objects are depicted by front, top, side, auxiliary or oblique views or combinations of these. The complexity of an object determines the number of views shown and at times, pictorial views are depicted.

ENGRAVING: Various craft and technological processes for producing blocks or plates from which to print illustrations, banknotes, etc.; also, an individual print made by one of these processes. **Line engraving** refers to preparing a plate by scratching its smooth surface with a

highly-tempered steel tool called a **burin** or **graver**. If the desired design is left standing high as is common with woodcuts and linocuts, this is known as a **relief process**. If the ink is transferred to the paper from lines incised into the plate, the surface of the ink plate having been wiped clean, this is known as **intaglio**.

ENHANCED DYNAMIC RANDOM ACCESS MEMORY (EDRAM):

A high-performance access memory that includes a small amount of static RAM (SRAM) inside a larger amount of DRAM. EDRAM is sometimes used as L1 and L2 memory and, together with Enhanced Synchronous Dynamic DRAM, known as cached DRAM.

ENHANCED (ANTHROPOGENIC) GREENHOUSE EFFECT:

Human activities that increase the greenhouse effect. This is caused mainly by the burning of fossil fuels that release extra carbon dioxide into the atmosphere, thereby amplifying the greenhouse effect.

ENHANCEMENT REACTION: A reaction which enables a favourable measurement to be made in which the constituent which is finally measured has not been magnified in any way.

ENHANCED RHIZOSPHERE DEGRADATION: Enhanced activity of micro-organisms involved with biodegradation of contaminants near plant roots which is brought about by compounds exuded by the plant roots.

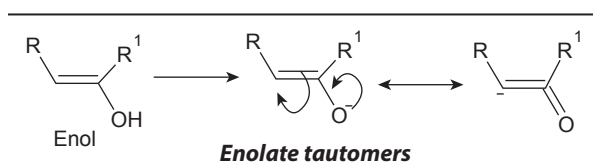
ENHANCER SEQUENCE: A DNA sequence that can bind protein factors (the trans-acting factors, much like a set of transcription factors) and stimulates transcription at an appreciable distance from the site where it is located. It acts in orientation either upstream or downstream from the promoter. For example, in eukaryotic cells, the structure of the chromatin complex of DNA is folded in a way that functionally resembles the supercoiled state characteristic of prokaryotic DNA, thus in spite of the fact that the DNA is far from the genes in nucleotides, it is geometrically close to the promoter and gene. This allows it to interact with the general transcription factors and RNA polymerase II. Thus, while enhancers are usually cis-acting, they do not need to be particularly close to the genes they act on and need not be located on the same chromosome.

ENKEPHALIN (ENCEPHALIN): Any of a class of endogenous opioids consisting of five amino acids, found principally in the central nervous system or a pentapeptide and involved in regulating nociception in the body. They bind to opioid receptors, chiefly in the brain and their release controls levels of pain and other sensations, for example, by blocking the release of substance P, a neurotransmitter involved in nociception. They are termed endogenous ligands (endorphins) since they are internally derived and bind to the body's opioid receptors. The two types are the one containing leucine and that containing methionine. Both belong to the proenkephalin gene.

ENLARGEMENT TRANSFORMATION: A transformation that multiplies all sides of a figure by a simple number or a scale factor greater than one without changing its shape.

ENNEA- (NONA-): A prefix denoting nine. For example, an **ennead** is a sequence of nine elements; an **enneagon** is a polygon with nine sides; and an **enneahedron** is a polyhedron with nine faces.

ENOLATE ION: A negative ion obtained from an enol, by removal of a hydrogen atom. They can have two forms: one with a single C-C bond and the negative charge on the beta carbon atom and the other with a double C-C bond and the negative charge on the oxygen atom.



ENOLS (ALKENOLS): Compounds containing the group -CH=C(OH)- in their molecules. They are alkenes with a hydroxyl group affixed to one of the carbon atoms composing the bond. The C=C double bond with adjacent alcohol gives enols and enediols their chemical

characteristics, by which they present keto-enol tautomerism. In keto-enol tautomerism, enols interconvert with ketones or aldehydes. Alkenes with a hydroxyl group on both sides of the double bond are known as **enediols**, while deprotonated anions of enols are called **enolates**. A compound that has an enediol structure with an adjacent carbonyl-group is called **reductone**.

ENORMOUS THEOREM: The largest theorem in mathematics; it concerns the classification of finite simple groups and encapsulates the work of hundreds of mathematicians over many years.

ENOSSES: Monosaccharides having a carbon-carbon double bond anywhere in the backbone chain. Unsaturated hexoses, for example, are called hexenoses. An example is the hex-1-enopyranose derived from D-glucopyranose.

ENRICHED URANIUM: Uranium in which the content U-235 is increased above its natural value of around 0.7% by weight. The enrichment process will also yield depleted uranium. Enriched uranium is fissile and can undergo nuclear fission under certain conditions. It is therefore used in nuclear power production and nuclear weapons.

ENRICHMENT: 1. The process of increasing the abundance of a specified isotope in a mixture of isotopes. In a nuclear reactor or weapon, it is normally applied to increase the proportion of U-235 or the addition of Pu-239 to natural uranium for use by the reactor. 2. In analytical chemistry, the process of increasing the concentration of trace components of a solution by preconcentration. Metal ions in natural waters can be enriched using cation exchange chromatography.

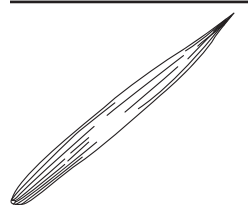
ENRICHMENT CULTURE: A technique in which environmental, including nutritional, conditions are controlled to favour the development of a specific organism or group of organisms over others. This is generally done by introducing nutrients or environmental conditions that only allow the growth of the specific organism of interest.

ENRICO FERMI (1901-1954): Italian-born American physicist who got his Ph.D in physics at Normale Superiore de Pisa in Italy in 1922. He also studied in Germany under Max Born at the University of Göttingen and under Paul Ehrenfest at the State University of Leiden in Netherlands. In 1924, he became a lecturer in mathematical physics at the University of Florence and later became the chair in physics at the University of Florence. He discovered the statistical laws (Fermi-Dirac statistics) independently with English physicist P. A. M. Dirac in 1926. In 1927, he was appointed professor of theoretical physics at the University of Rome. He showed that nuclear transformation occurs in almost every element which is bombarded with a neutron. He was awarded the 1938 Nobel Prize for Physics for his work in radioactivity. He was appointed professor of physics at Columbia University in 1939. In 1942, he led a team of scientists in Chicago to build the first nuclear reactor (atomic pile), which was the first controlled nuclear chain reaction. He worked on the Manhattan Project for the development of the atomic bomb during World War II. Fermium, element number 100 was named after him.

ENSEMBLE: 1. A set of particles used to describe a single system in statistical mechanics. An ensemble of system can be represented as a set of points in phase space with each system of the ensemble represented by a point. 2. An idealisation consisting of a large number of mental copies of a system, considered all at once, each of which represents a possible state that the real system might be in.

ENSIFORM: In botany, it describes plant parts such as leaves that are shaped like a sword blade with sharp edges and tapering to a slender point, as shown in the diagram. An example of a plant with a leaf with an ensiform structure is Iris.

ENTAIL: In logic, to have as a necessary consequence. **Entailment** is a statement to the effect that one statement is



Ensiform leaf

a necessary consequence of another. In other words, it is the relationship between sentences in which one sentence is true if all the others are also true. It is usually denoted by the fishhook symbol $\supset$. An example statement is: "Although Vivian is a non-male human" implies that she is a woman; it does not entail that conclusion, since it is merely an accidental fact rather than a necessary truth that all humans are either male or female. Such a proposition is usually symbolised by $P \supset Q$, where P and Q are the component statements.

ENTATIC STATE: A state of an atom or group which occurs due to its binding in a protein and has its geometric or electronic condition adapted for function.

ENTER KEY (RETURN KEY): The keyboard key that is pressed to move the cursor to the beginning of the next line or to start a new paragraph. It usually signals the computer to act on the information that has just been entered on the keyboard, for example, to submit requests online or search an engine without using the cursor to point to the search button.

ENTERIC BACTERIA: A large group of facultative aerobic gram-negative rod-shaped bacteria that is present in the intestinal tract of humans and other animals. They may be physiologic or pathologic.

ENTERING GROUP: An atom or a group that forms a bond to what is considered to be the main part of the substrate during a reaction. An example is the attacking nucleophile in a bimolecular nucleophilic substitution reaction.

ENTERITIS: Inflammation of the intestine, most commonly of the small intestine. It is caused by the ingestion of substances contaminated with pathogenic microorganisms and its symptoms include abdominal pain, cramping, diarrhoea, dehydration and fever.

ENTERO-: Referring to the intestine. For example, enterovirus is a group of RNA viruses which typically occur in the gastrointestinal tract of mammals, sometimes spreading to the central nervous system or other parts of the body. Some of them cause diseases such as polio and hepatitis A. Anything situated in or relating to the intestine is **enteric**.

ENTEROGASTRIC REFLEX: A nervous reflex in which stretching of the wall of the duodenum results in inhibition of gastric motility and reduced rate of emptying the stomach. It is a feedback mechanism to regulate the rate at which partially digested food leaves the stomach and enters the small intestines.

ENTEROKINASE (ENTEROPEPTIDASE): An enzyme in the ileum or small intestine secreted by the upper intestinal mucosa that activates trypsinogen to trypsin for the digestion of proteins.

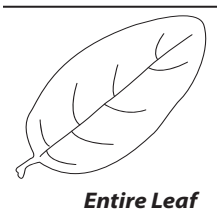
ENTEROVIRUS: A virus which lives in the intestine of mammals. It is a genus of (+)ssRNA viruses associated with several human and mammalian diseases. About 66 human enterovirus serotypes have been distinguished on the basis of antibody neutralisation tests. Those isolated more recently are named with a system of consecutive numbers: EV68, EV69, EV70, EV71, etc.

ENTERPRISE BUDGET: A type of farm financial report that owners or managers may use to project costs and returns to produce a product (enterprise). Enterprise budget helps to estimate the costs and income associated with the production of a specific agricultural product. It also helps to allocate land, labour, and capital, which are limited, to the most appropriate use.

ENTHALPY: A thermodynamic property of a system denoted by H and defined by $H=U+pV$, where U is the internal energy of the system, p its pressure and V the volume. It is used to calculate the heat transferred during a quasi static process taking place in a closed thermodynamic system under constant pressure (isobaric process). In an endothermic reaction, the enthalpy change is positive, while it is negative in an exothermic reaction. It cannot be measured directly but when chemical substances react, the difference in the heat energy between the reactants and products can be measured. This difference

is the enthalpy change and is given by the symbol ΔH . Specific enthalpy is denoted as h .

ENTIRE: 1. In botany, not having any indented margins, not toothed or divided, for example, an entire leaf, as shown in the diagram. 2. In zoology, not castrated, such as a male animal that has not been castrated. 3. In mathematics, of a complex function, analytic at all points of the finite complex plane. For example, $f(z) = e^z$ is an entire function.



Entire Leaf

ENTIRE SURD (PURE SURD): A surd in which all terms are irrational numbers. In other words, it is a surd having no rational factor except unity. Examples are $\sqrt{3}$, $3\sqrt{9}$, $\sqrt{7}$, $\sqrt{10}$ and $\sqrt[3]{50}$.

ENTOMOLOGY: A branch of zoology concerned with the scientific study of insects, including their taxonomy, morphology, physiology and ecology. Applied aspects of entomology, such as the harmful and beneficial impact of insects on humans, are also studied.

ENTOMOPHAGOUS: Feeding on insects or insectivorous.

ENTOMOPHILY: Pollination of a flower in which the pollen is carried on and distributed by insects, such as bees, Lepidoptera such as butterflies and moths; flies and beetles. Entomophilous species frequently evolve mechanisms to become more appealing to insects, for example, flowers are usually bright coloured, scented or have appealing shapes and patterns to attract insects and often secrete nectar. Their pollen grains are generally larger than the fine pollens of anemophilous or wind-pollinated plants; and often of more nutritional value to insects, which may use them for food and subsequently spread them to other flowers. **Entomophilous plants** are flowering plants that are pollinated by insects.

ENTOPROCTA: A phylum of small, sessile, aquatic and often colonial invertebrates, typically having looped gut with both mouth and anus situated inside a circlet of tentacles, a pseudocoelomate or cup-shaped body cavity, with no hardened skeleton and being attached to a substrate by means of a stalk. It is divided into two orders namely Loxosomatida or Solitaria, with about 100 species; and Pedicellinida (or Coloniales), with about 30 species.

ENTRAINMENT: 1. The act of forming a mist or fog or droplets of a liquid carried off by the vapours of a boiling liquid or from a liquid through which bubbles of gas or vapour are passing rapidly. 2. A term used in photostimulated radical nucleophilic substitution to designate the induction of the reaction of a less reactive (or non-reactive) nucleophile with a substrate by the addition of catalytic amounts of another nucleophile, more reactive at initiation. Often, the non-reactive nucleophile at initiation is, however, quite reactive at propagation. 3. A term used in photobiology in connection with the setting of the circadian clock by biological photosensors such as phytochromes and cryptochromes in plants and cryptochromes in mammals and insects.

ENTRANCE CHANNEL: The region of a potential-energy surface or hypersurface that corresponds to molecular configurations that are closer in geometry to those of the reactants than to those of the products.

ENTRANCE PUPIL: The optical aperture of a telescope, eyepiece, binoculars or other optical system through which light initially enters.

ENTROPY: 1. A measure of the unavailability of a system's energy to do work or a measure of the microscopic disorder within a system. An increase in entropy is accompanied by a decrease in energy available in a closed system. It is an important part of the second law of thermodynamics, which states that the thermodynamic entropy is a measure of the amount of energy which does no work during energy conversions. As a matter of convention, entropy is measured in units of energy per temperature, joules per Kelvin (SI unit). If thermodynamic systems are described using thermal energy instead of temperature, then entropy is just a number by which the thermal energy in the system is multiplied and the resulting energy is an energy

for which no information is available, which would be required to convert the energy in technical systems from one form, for example, electrical into mechanical. In technical applications, machines are basically energy conversion devices thus they can only be driven by convertible energy. This also applies to biological organisms. 2. Also known as **Shannon entropy**, it is a random variable X with values in a finite set X , defined as $H(X) \equiv -\sum_x P(x) \log_2[P(x)]$, where $P(X)$

is the probability that X is in the state x and $P \log_2 P$ is defined as 0 if $P = 0$. The joint entropy of variables $(X_1, \dots, X_n)$ is defined by

$$H(X_1, \dots, X_n) \equiv -\sum_{x_1, \dots, x_n} P(x_1, \dots, x_n) \log_2[P(x_1, \dots, x_n)],$$

3. The macroscopic variable representing the degree of disorder within a system and corresponding to a statistical description of the system by the relation that the entropy is equal to the expected value of $k \log P$, where k is a constant that relates the mean kinetic energy and absolute temperature of a system and P is the coefficient of probability of the system.

ENTROPY OF A CHEMICAL SYSTEM: A measure of the degree of disorderliness or randomness of a chemical system.

ENTROPY OF INFORMATION: A measure of the average uncertainty in a single random variable. It is the number of bits on the average required to describe the random variable. Mathematically, the entropy of a random variable with a probability mass function $p(x)$

$$\text{is defined by } H(X) = -\sum_{x \in \Omega} p(x) \log_2 p(x).$$

ENTRY: Any of the elements constituting a matrix, determinant, vector or array, usually regarded in terms of its position, so that a_{ij} is the entry at the intersection of the i^{th} row and the j^{th} column. In other words, an entry in a matrix is identified by writing a subscript of the respective entry's row followed by that of the column. The leading entry of a matrix in all the entire rows of matrix is considered as the first non-zero entry in that row.

ENUCLEATION: In medicine, it is the surgical removal of a complete organ or tumour. An example is the removal of the eye from its socket using local or general anaesthesia.

ENUMERABLE (DENumerable): Capable of being put into one-to-one correspondence with the positive integers.

ENUMERABLE SET (COUNTABLY INFINITE SET; DENumerable SET): A set is enumerable if its elements can be put in one-to-one correspondence with the set of natural numbers or contains a finite number of elements or if there exists a partial recursive function such that this set is its range of values. In other words, a set A of integers is said to be recursively enumerable if it constitutes the range of a recursive function or if there exists a recursive function that can eventually generate any element in A .

ENVELOPE: 1. In mathematics, it is a curve that touches or is tangent to all the members of a family of lines or curves at some point. For example, a family of three equal circles all touching each other and forming a triangular pattern has two envelopes. 2. In biology, it is an enclosing structure or cover, such as a membrane or the outer coat of a virus. 3. The bag containing the gas in a balloon or airship. 4. The set of limitations within which a technological system, especially an aircraft, can perform safely and effectively. 5. In aeronautics, it is the coma of a comet. 6. The sealed glass or metal housing of a valve, electric light, etc. 7. The outer shape of a modulated wave, formed by the peaks of successive cycles of the carrier wave.

ENVELOPE CONFORMATION: The conformation (of symmetry group) of a five-membered ring in which four atoms are coplanar and one atom (the flap) projects out of the plane.

ENVIRONMENT: The physical, chemical and biological condition of the area in which an organism lives and which influence the distribution and success of organisms or the sum of all external factors, both biotic or living and abiotic or nonliving, to which an organism is exposed. Biotic factors include influences by members of the same and other

species on the development and survival of the individual. Primary abiotic factors are light, temperature, water, atmospheric gases and ionising radiation, influencing the form and function of the individual.

ENVIRONMENT VARIABLE: A set of dynamic named values that can define some aspect of a user's or a program's environment that can vary. The value of an environment variable is a string of characters. It helps programs know what directory to install files in, where to store temporary files, where to find user profile settings, etc. It is usually part of the login procedure and establishes some component of the user's working environment, such as the default printer, browser or text editor to be used. There are two types of environment variables: (i) **System environment variables**, which define the behaviour of the global operating system environment and are preset in the operating system and available to all Windows processes. Only users with administrative privileges can change system variables. These variables are most commonly used in login scripts. (ii) **Local environment variables** define the behaviour of the environment of a process and only available when the user for whom they were created is logged on to the computer. Examples of environment variables include: PATH, a list of directory paths. For example, when the user types a command without providing the full path, this list is checked to see whether it contains a path that leads to the command; HOME (Unix-like) and USERPROFILE (Microsoft Windows), which indicate where a user's home directory is located in the file system; HOME/{.AppName} (Unix-like) and APPDATA\{DeveloperName\AppName} (Microsoft Windows) used for storing application settings; MAIL (Unix-like) used to indicate where a user's mail is to be found; etc.

ENVIRONMENT-HEREDITY CONTROVERSY (NATURE-NURTURE CONTROVERSY): A long-standing dispute among philosophers and psychologists over whether nature or nurture is more important in the development of living organisms, especially human beings. Nurture is defined as the relative importance of environment including upbringing, experience and learning; and nature is defined as the physical and personality traits determined by the genes of an organism.

ENVIRONMENTAL AUDIT (GREEN AUDIT): The inspection of an organisation to assess its environmental compliance and management system in accordance with environmental laws and regulations in order to prevent or minimize harm to the environment.

ENVIRONMENTAL BENCHMARKING (BENCHMARKING): A tool for analysing environmentally related practices and indicators that leads to superior environmental performance, while also enhancing economic performance. It includes all areas of activities and not restricted solely to those activities that have an obvious environmental impact. Therefore, it may include an assessment of environmental management systems (EMS), management performance, environmental accounting, resource and waste management, product environmental quality, environmental education and training, customer relations and emergency response.

ENVIRONMENTAL BURDEN: Any activity affecting the environment or any consequence of such activity which, exclusively or simultaneously, has caused or continues to cause environmental pollution, environmental risk or the use of a natural asset.

ENVIRONMENTAL DECONTAMINATION: In radiation protection, it refers to the systematic clean-up of radioactive contamination within the workplace or wider environment.

ENVIRONMENTAL ENGINE: A reaction engine that obtains the reactants or propellants it needs from its environment. Examples include the conventional jet engine used by aircraft and the speculative interstellar ramjet.

ENVIRONMENTAL EXPOSURE: In radiation protection, it refers to ionising radiation exposure within the workplace or wider environment.

ENVIRONMENTAL GRAIN: A term used in ecology for the effect of spatial variation or patchiness, relative to the size and behaviour of an organism.

ENVIRONMENTAL IMPACT ASSESSMENT (EIA): A process or a set of activities designed to evaluate the potential environmental consequences of the implementation of projects (e.g., the construction of a power station, a dam), programmes and policies. This ensures that the potential problems are discussed by government, non-governmental organisations and the general public before a decision is made on whether or not the project can proceed.

ENVIRONMENTAL INCIDENT: A specific occurrence or set of circumstances of limited geographical scope, as a consequence of pollution, contamination or degradation in the quality of the environment. It does not cover general environmental risks unconnected with a specific event, such as climate change or atmospheric pollution.

ENVIRONMENTAL ISSUES: The harmful effects relating to the effects of human activity on the biosphere, their causes and the search for possible solutions. Examples of some environmental issues currently include climate change, pollution, environmental degradation, natural resource depletion, overpopulation, deforestation, ocean acidification, ozone layer depletion, acid rain and water pollution. Some of international organisations that are concerned with environmental issues include the United Nations Environment Programme and International Union for Conservation of Nature, which through conventions, protocols and agreements provide a basis for co-operation among countries at bilateral, regional and global levels for the management of environmental risks, control of pollution and conservation of natural resources. International non-governmental organisations such as Greenpeace and Friends of the Earth and World Wide Fund for Nature address environmental issues through advocacy, education and activism. At the national level, governments regulate activities of enterprises and people in the light of environmental objectives and enforce environmental law.

ENVIRONMENTAL LAPSE RATE: The rate at which atmospheric temperature changes with altitude. Usually, because the atmosphere is heated from below, temperatures usually decrease at an average rate of 0.6°C per 100 metres. The rate varies from place to place and according to the time of day.

ENVIRONMENTAL MONITORING (AMBIENT MONITORING): Continuous or repeated measurement of agents in the (working) environment to evaluate environmental exposure and possible damage by comparison with appropriate reference values based on knowledge of the probable relationship between ambient exposure and resultant health effects.

ENVIRONMENTAL POLLUTION: Contamination of the Earth's environment with materials mainly as a result of human activity that interfere with human health, the quality of life or the normal functions of the ecosystem. It causes instability, disorder, harm or discomfort to the ecosystem and it takes the form of chemical substances or energy, such as noise, heat or light. Pollutants are often classed as point source or non-point source pollution. They can be naturally occurring or due to foreign substances or energies. When naturally occurring and exceeds natural levels they are considered contaminants. According to the Blacksmith Institute and Green Cross Switzerland's published report for the year 2013: "The Top Ten Toxic Threats: Cleanup, Progress and Ongoing Challenges", the world's ten worst polluted places were located in Agbogbloshie, Ghana; Chernobyl, Ukraine; Citarum River, Indonesia; Dzershinsk, Russia; Hazaribagh, Bangladesh; Kabwe, Zambia; Kalimantan, Indonesia; Matanza Riachuelo, Argentina; Niger River Delta, Nigeria; and Norilsk, Russia. Note that the list is not ranked.

ENVIRONMENTAL RESISTANCE: The sum total of the factors that prevent populations from continually growing or the effect of physical and biological factors in preventing a species from reproducing at its maximum rate. They, therefore, tend to keep populations at constant levels.

ENVIRONMENTAL SELECTION: Natural selection within a population resulting from the influence exerted by the environment. It leads to a change in the composition of genes within a population.

ENVIRONMENTAL STATISTICS: Procedures for determining how quality of life is affected by the environment, in particular by such factors as air and water pollution, solid wastes, hazardous substances, foods and drugs.

ENVIRONMENTAL STRESS CRACKING: Cracking caused by the combined actions of mechanical stress and chemical agents or radiation or both.

ENVIRONMENTALISM: A theory emphasising the primary influence of an environment on the development of groups or individuals. It is a broad philosophy and social movement regarding concerns for environmental conservation and improvement of the state of the environment and such concerns are often represented by the colour green.

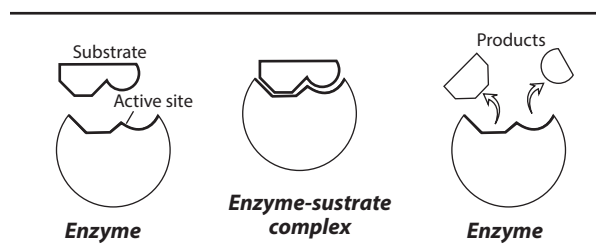
ENVIRONMENTALLY SOUND TECHNOLOGIES (ESTs): Technologies that protect the environment and capable of reducing environmental damage relative to other technologies. Such technologies are less polluting, use processes and materials that generate fewer potentially damaging substances, use all resources in a more sustainable manner, recycle more wastes and products, and handle residual wastes in a more acceptable manner.

ENYL COMPLEX: A type of complex in which there is a link between the metal atom or ion and the pi electrons of a double bond.

ENZOOTIC ABORTION: A virus infection of sheep causing abortion about two weeks before lambing.

ENZOOTIC PNEUMONIA: A disease of pigs, caused by the coccobacilli *Pasteurella multocida* and *P. haemolytica*. The symptoms include cough and stunted growth. The term also applies to diseases caused by any *Pasteurella* species.

ENZYME: A chemical substance made from proteins and contained in living cells that speed up the rate of chemical and biochemical reactions. Most metabolic reactions cannot occur without enzyme catalysis and like all catalysts, they work by lowering the activation energy (E_a^+) for a reaction, dramatically increasing the rate of the reaction. The lack of one enzyme in the body can cause problems. For example, people who lack lactase cannot digest milk and are called lactose intolerant. The molecules at the beginning of the process of enzymatic reactions are called **substrates** (reactants), while the different molecules into which the enzyme converts them are known as **products**. Enzymes are very specific to the substrates and reactions that they can catalyse. The active site has a fixed structure, the lock, which exactly matches the structure of a specific substrate, the key. Once the enzyme identifies the right substrate, they fit into each other exactly. This results in the formation of the enzyme-substrate complex and further reactions continue at the active site until the substrate is transformed to the products. Once the reaction is complete and the products are released, the active site remains the same and is ready to react with new substrates. Almost all processes in a biological cell need enzymes to occur at significant rates but since enzymes are selective for their substrates and speed up only a few reactions from among many possibilities, the set of enzymes made in a cell determines which metabolic pathways occur in that cell. Enzymes are known to catalyse around 4,000 biochemical reactions and their reaction rates are millions of times faster than those of similar un-catalysed reactions. As with all catalysts, they are not consumed by the reactions they catalyse, nor do they alter the equilibrium of these reactions, however, they differ from most other catalysts by being much more specific.



ENZYME INDUCTION: 1. A process in which an (inducible) enzyme is synthesised in response to a specific molecule. The inducer molecule (often a substrate that needs the catalytic activity of the inducible enzyme for its metabolism) combines with a repressor and thereby prevents the blocking of an operator by the repressor. 2. A process in which an enzyme is synthesised in response to the presence of a specific substance or to other agents such as heat or a metal.

ENZYME KINETICS: The study of enzyme-catalysed reactions, including the effects of varying the conditions of the reaction. Rates of reactions are usually measured by using the purified enzyme *in vitro* with the substrate and then observing the formation or disappearance of the substrate. Such studies can reveal the catalytic mechanism of an enzyme, its role in metabolism, how its activity is controlled and how a drug or a poison might inhibit the enzyme.

ENZYME REPRESSION: The mode by which the synthesis of an enzyme is prevented by repressor molecules. In many cases, the end product of a synthesis chain such as an amino acid acts as a feed-back corepressor by combining with an intracellular aporepressor protein, so that this complex is able to block the function of an operator. As a result, the whole operon is prevented from being transcribed into mRNA and the expression of all enzymes necessary for the synthesis of the end product amino acid is abolished.

ENZYME SUBSTRATE ELECTRODE: A sensor in which an ion-selective electrode is covered with a coating containing an enzyme which causes the reaction of an organic or inorganic substance (substrate) to produce a species to which the electrode responds. Alternatively, the sensor could be covered with a layer of substrate which reacts with the enzyme, co-factor or inhibitor to be assayed.

ENZYME TECHNOLOGY: The use of isolated purified enzymes as catalysts in industrial processes. The enzymes used are normally extracellular enzymes with no requirements for complex cofactors. Examples are proteases, amylases, cellulases and lipases extracted from bacterial or yeast cultures. Cell-free enzymes have certain distinct advantages over the use of microorganisms, a major one being that only the desired enzymatic reaction will occur and substrate is not wasted in the formation of unwanted by-products or microbial biomass.

ENZYME-LINKED IMMUNOSORBENT ASSAY, ELISA (ENZYME IMMUNOASSAY, EIA): A sensitive biochemical technique used mainly in immunology for accurately determining the amount of protein or other antigen in a given sample by means of an enzyme catalysed colour change. It is used as a diagnostic tool in medicine and plant pathology, as well as a quality control check in various industries. Antibody specific to the test protein is adsorbed onto a solid substrate, such as a PVC sheet and a measured amount of the sample is added; all molecules of the test protein in the sample are bound by the antibody. A second antibody specific for a second site on the test protein is added; this is conjugated with an enzyme, which catalyses colour change in the fourth reagent, added to the sheet. The colour change can be measured photometrically and compared against a standard curve to give the concentration of protein in the sample.

ENZYME-SUBSTRATE COMPLEX: The intermediate formed when substrate molecules interact with the active site of an enzyme. The substrate molecule undergoes a chemical reaction and is converted into a new product, following the formation of an enzyme-substrate complex.

ENZYMIC DECOMPOSITION: Decomposition of organic materials such as starch, sugars and proteins, in which the enzyme converts a high molecular mass compound into lower molecular mass species. The process can be regarded as an example of enzymic degradation.

EOCENE: The second geological epoch of the tertiary period; the second oldest of the five major worldwide divisions or epochs of the Tertiary Period (Cenozoic Era), the interval of time extending from the end of the Paleocene period to the beginning of the Oligocene Period. The Eocene period was the time of rising temperatures and an abundance of mammals, including the first horses, bats and whales.

E

EAON (AEON): 1. A large division of geological time consisting of several eras. 2. A term for a unit of geologic time, which is equal a thousand million (10^9) years.

EOPHYLL: The first few leaves with green, expanded lamina developed by seedlings; or transitional type leaves developed before formation of adult leaves.

EOSIN: One of the series of acidic dyes used in optical microscopy, that colours cytoplasm pink and cellulose red. It is a fluorescent red dye resulting from the action of bromine on fluorescein and it is used often as a counter stain with haematoxylin for colouring tissue smears and sections of animal and plant tissues. It can be used to stain cytoplasm, collagen and muscle fibres for examination under the microscope. Structures that stain readily with it are termed **eosinophilic**.

EOSINOPHIL (EOSINOPHIL GRANULOCYTE; ACIDOPHIL): A type of white blood cell with a granular cytoplasm that is one of the immune system components responsible for combating multicellular parasites and certain infections in vertebrates. They are granulocytes that develop during haematopoiesis in the bone marrow before migrating into the blood. Their functions involve the regulation of allergic responses and asthma in conjunction with mast cells, as well as producing an enzyme capable of destroying parasites.

EOTVOS: Unit for measuring small changes in the gravity of the Earth with horizontal distance. It is equal to 10^{-9} galileo per horizontal centimetre. Its symbol is E.

EPAPPOSE: Bearing a pappus.

EPHEDRINE: An alkaloid with the chemical formula $C_6H_5CH(OH)CH(CH_3)NHCH_3$, found in plants of the genus Ephedra. It was formerly used in the treatment of asthma, but now used as a stimulant, appetite suppressant and illegally in the production of methamphetamine. Other uses include relieve of nasal congestion originating from allergic conditions, such as hay fever or from bacterial or viral infection of the upper respiratory tract, raising blood pressure. It is slow to take effect and of mild potency and long duration.

EPHEMERAL: 1. A plant such as groundsel and an insect such as mayfly that live only for a short period of time. 2. Wetland, spring, stream, river, pond, lake or other water body that only exists for a short period following precipitation or snowmelt. Examples include the Ugab River in Southern Africa and a number of small watercourses that drain Talak in northern Niger. They are different from seasonal or intermittent water bodies, which exist for longer periods, but not all year round.

EPHEMERIS: A calculated and tabulated future position of the Sun, Moon and planets with other useful information for navigators and astronomers. For example, a lunar ephemeris may give the right ascension and declination of the Moon for every hour of a particular year.

EPHEMERIS TIME, ET (TERRESTRIAL DYNAMICAL TIME, TDT): A time system that has a constant uniform rate as opposed to other systems that depend on the Earth's rate of rotation which has inherent irregularities. It is used to calculate the future positions of the Sun and the planets and it is calculated from the positions of the Sun and Moon relative to the Earth, assuming that Newton's laws are adhered to exactly. By convention, the standard seasonal year is

taken to be A.D. 1900 and to contain 31,556,925.9747 seconds of ephemeris time. Another time, known as the **Barycentric Dynamical Time (TDB)**, which is based on the orbital motion of the Sun, Moon and planets is considered identical to the Ephemeris Time, because they differ by only milliseconds.

EPHEMEROPTERA: An order of exopterygote insects that comprises the mayflies, in which the adult stage lasts for only a few hours. The nymphs live for up to a number of years and are mainly herbivorous but some possess mandibles for feeding on animal prey. They are aquatic and live in clean, fresh waters during their immature or nymphal lives. They constitute a year-round basic food supply for carnivores in their environment, particularly for trout and other fishes, during their nymphal stage.

EPIBASAL CELL: The outer of the two cells formed by the first division of the zygote. It gives rise to the embryo in plants showing exoscopic embryogeny and to the foot in those showing endoscopic embryogeny.

EPIBLAST: A small flap of tissue forming the outermost layer of an embryo.

EPICALYX: A ring of bracts below a flower that resembles a calyx. Examples are seen in okra, hibiscus and carnation.

EPICARP (EXOCARP): The outer layer of the fruit wall (pericarp). An example is the skin of some fruits such as oranges, mangoes and coconuts.

EPICENTRE: The point on the surface of the Earth, which is directly above the focus or hypocentre of an earthquake or the point above or below a nuclear explosion. It is the point where the earthquake or underground explosion originates and in most cases, it is the area of greatest damage. However, in larger events, the length of the fault rupture is much longer and damage can be spread across the rupture zone.

EPICOTYL: The apical end of the axis of an embryo above the cotyledon(s), which gives rise to the stem and associated organs.

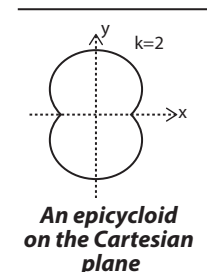
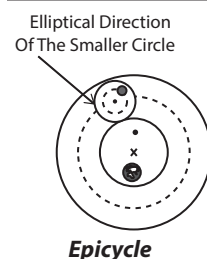
EPICYCLE: A small circle whose centre rolls around the circumference of a larger fixed circle. Proposed by Apollonius at the end of the 3rd century BC and formalised by Ptolemy, it was a geometric model used to explain the variations in speed and direction of the apparent motion of the Moon, Sun and planets. The curve which is traced out by a point on the epicycle is called **epicycloids**.

EPICYCLIC GEAR: A gear system that consists of one or more gear wheels travelling around the inside or outside of another gear whose axis is fixed. It is found in bicycle hub gears and in automatic gear boxes.

EPICYCLOID: A curve described by a point on the circumference of a circle that rolls round the outside of the circumference of another circle or a plane curve produced by tracing the path of a chosen point of a circle known as an epicycle, which rolls without slipping around a fixed circle. An epicycloid is a special type of roulette and it becomes a cardioid if its diameter is equal to the diameter of the inner circle.

EPIDEMIC: A large number of cases of the same infectious disease, occurring suddenly during a single period of time in a community or region, or occurrence of cases of disease in excess of what is usually expected for a given period of time in a community or region. They include infectious diseases such as measles, polio or streptococcal sore throat.

EPIDEMIOLOGY: The study of disease that affects a large number of organisms or a study of disease distribution in populations, which often



takes a historical perspective. **Descriptive epidemiology** examines a population to determine what segments such as age, sex, ethnic group and occupation are affected by a disorder, follows changes or variations in its incidence or mortality over time and in different locations and helps identify syndromes or suggest associations with risk factors. **Analytic epidemiology** conducts studies to test the conclusions of descriptive surveys or laboratory observations.

EPIDERMAL CELL: A cell of the epidermis such as dendritic cells that synthesise tyrosinase.

EPIDERMAL GROWTH FACTOR: A small protein or polypeptide produced in animals that stimulates and sustains the replication of epidermal cells or acts as a cytokine and stimulates the division of cells in the skin and connective tissue by binding to receptors on the cell surface. It originates from the ectoderm or endoderm and its human equivalent is the urogastrone.

EPIDERMIS: 1. The relatively thin, tough, outer layer of the skin. Most of the cells in the epidermis are keratinocytes, which originate from cells in the deepest layer of the epidermis called the basal layer. New keratinocytes slowly migrate up toward the surface of the epidermis. Once the keratinocytes reach the skin surface, they are gradually shed and are replaced by younger cells pushed up from below. The outermost portion of the epidermis, known as the **stratum corneum**, is relatively waterproof and, when undamaged, prevents most bacteria, viruses and other foreign substances from entering the body. The epidermis (along with other layers of the skin) also protects the internal organs, muscles, nerves and blood vessels against trauma. In certain areas of the body such as the palms of the hands and the soles of the feet that require greater protection, the outer keratin layer of the epidermis, called **stratum corneum** is much thicker. 2. In botany, epidermis is the outermost cellular layer which covers the whole plant structure (roots, stem, leaves, flowers and fruit). 3. An open-source GTK application software for changing the appearance of Linux desktops.

EPIDIASCOPE (OPAQUE PROJECTOR; EPIDIOSCOPE; EPISCOPY): An optical instrument used for projecting an enlarged image of either a translucent object such as a slide or a transparency or an opaque object such as a diagram or printed page on to a screen. It displays opaque materials by shining a bright lamp onto the object from above. A system of mirrors, prisms and/or imaging lenses is used to focus the image of the material onto a viewing screen. Since they must project the reflected light, they need brighter bulbs and larger lenses than overhead projectors. The materials could be damaged by the heat generated by the light source, if care is not taken.

EPIDIDYMIS: A long coiled tube attached to each testis in vertebrates in which spermatozoa are stored and passed on to the vas deferens. In reptiles, birds and mammals, it is attached at one end of the testis and opens into the sperm duct. A similar, but probably non-homologous, structure is found in cartilaginous fishes.

EPIGAMIC: Of or relating to a trait or characters serving to attract a mate. An example is the bright plumage of some male birds or big antlers of some animals.

EPIGASTRIUM: The upper and central region of the abdomen located between the sternum and umbilicus. **Epigastric pain** is a type of pain that is felt within that region, which may be caused by inflammation of the stomach's lining, inflammation of the pancreas and inflammation of the peritoneal membrane.

EPIGEAL (EPIGEAN; EPIGEIC; EPIGEOUS): An organism is said to be epigeal, epigeic or epigeous if it crawls (epigeal), creeps like a vine (epigeal) or grows (epigeous) on the soil surface or more generally in animals, which neither burrows nor swims nor flies. It describes an organism's activity above the soil surface, such as a seed germination in which the seed leaves (cotyledons) emerge or appear above the ground level and function as true leaves. Examples are found in sunflower, castor, mustard, neem and tomato. The other type of germination is the **hypogeal** in which the cotyledons remain

inside the seed shell underground without photosynthesis. Examples of such germinations are found in dicotyledon plants such as pea, gram, broad bean, scarlet runner bean, mango and jack-fruit; and in monocotyledons such as maize, rice, wheat and coconut.

EPIGENESIS: The theory that an organism develops from a fertilised egg by a gradual series of interdependent physiological and physical changes, brought about by the genes that result in an increase in the complexity of its separate parts. It refutes the earlier preformation theory, which stated that either the male or female gamete contained a miniature version of the adult and that only an increase in size followed thereafter.

EPIGENETIC: Describing heritable changes that are not the result of changes in DNA sequence. An example is paramutation.

EPIGENOUS (EPIGAEUS): Growing on the upper surface of an organ. An example is a fungus growing on the surface of a leaf.

EPIGEOUS: Emerging near the surface of the ground.

EPIGLOTTIS: A flap of elastic cartilage tissue covered with a mucus membrane and attached to the root of the tongue. It projects obliquely upwards behind the tongue and the hyoid bone and pointing dorsally. It closes over the wind pipe during swallowing to prevent food from entering and causing choking.

EPIGRAPH: In mathematics, the epigraph of a real-valued function is the set of points (x, y) that lie on or above its graph: such that $y \geq f(x)$. A function is convex if and only if its epigraph is a convex set. A function is lower semicontinuous if and only if its epigraph is closed.

EPIGYNY: A floral arrangement in which the ovary is completely enclosed by the receptacle so that the stamens and perianth stand above it from the top of the receptacle.

EPIHYPERIGYNY: The condition in which the sepals, petals and stamens are attached to the floral tube or hypanthium surrounding the ovary; a combination of perigyny and partly inferior ovary.

EPIHYPOGYNY: The condition in which the sepals, petals and stamens are attached about half-way from the base of the ovary to the partly adnate hypanthium tube; a half-inferior insertion of parts.

EPILEPSY: A neurological disorder characterised by unusual behaviour, sensory or cognitive malfunction and sometimes loss of consciousness and sudden recurrent episodes of convulsive seizures. It may be caused by abnormal electrical discharges in the cerebral hemispheres of the brain brought about by head injuries, brain tumours, lead poisoning, genetic and infectious illnesses or in some cases no cause is found. Epilepsy can be controlled with a number of anticonvulsant drugs.

EPIIMNION: The uppermost circulating layer of warm water in a lake with different temperatures at different levels in which mixing occurs as a result of wind action and convection currents.

EPIMATIUM: A specialised type of ovuliferous scale that bears and completely encloses a single inverted ovule. It is found in the monkey-puzzle tree (*Araucaria araucana*) and in certain of the Podocarpaceae.

EPIEDIUM (BARRENWORT; HORN GOAT WEED): A genus of flowering herbaceous perennial plants in the family Berberidaceae, found mostly in Asia, with one or two species including *Epimedium alpinum* being found in the Mediterranean regions of Europe. The Asian species include *E. pubescens*, *E. wushanense*, *E. myrianthum*, *E. sagittatum*, *E. koreanum* and *E. brevicornu* and *E. grandiflorum*. The stem of *Epimedium* develops from an underground rhizome and may be solitary or in multiples growing closer together. The leaves are compound with spiny margins. The inflorescence is an open raceme or panicle, the number of flowers varying by species. The plant contains icariin, which is a PDE5 inhibitor like sildenafil, the active ingredient of Viagra. It regulates blood flow in the penis and increases the production of nitric oxide and used in the treatment for erectile dysfunction.

EPIMERASE: An enzyme that catalyses the transfer of a hydroxyl group from one position to another within a molecule. Epimerases

are a form of isomerase and are important biologically in the interconversion of sugars.

EPIMERS: Two stereoisomers with more than one chiral centre that differ in configuration at one of their chiral centres. In other words, it is a kind of isomer in which the difference between the two compounds is the relative position of the hydrogen group and hydroxyl group on the last asymmetric carbon atom of the chain. An example is found in the sugars D-glucose and D-mannose. **Epimerism** is having stereoisomers with more than one chiral centre that differ in configuration at one of their chiral centres.

E

EPIMORPHISM (EPIC MORPHISM; EPI): Of an arrow, $h: a \rightarrow b$, having the property that its compositions with distinct arrows are necessarily distinct; that is, for any arrows g_1 and g_2 , $b \rightarrow c$, if $g_1 \circ h = g \circ h$, then $g_1 = g_2$. Epimorphisms are analogues of surjective functions, but they are not exactly the same. The dual of an epimorphism is a monomorphism.

EPINASTY: A nastic movement in which the resultant bending of the plant part is downwards, due to increased growth on the upper side of an organ. This occurs in the opening of many flowers, when the bracts and perianth parts curl downwards to expose the sexual organs. Epinasty can also occur in leaves, where the petiole bends so that the leaf points to the ground rather than upwards.

EPINEPHRINE (ADRENALINE): A hormone produced by the adrenal medulla and secreted under stress. It serves as a neurotransmitter, increasing the rate and force of heart contractions, blood output and pressure and also stimulates the breakdown of glycogen to glucose in the liver, thus raising the blood glucose levels. Together with norepinephrine both hormones increase the level of circulating free fatty acids. These actions prepare the body for action in times of stress or danger as well as times requiring increased alertness or exertion. It, therefore, contributes to the "fight or flight" response. Epinephrine is used in medical situations, including cardiac arrest, asthma and acute allergic reaction.

EPIPERIGYNY: The condition in which the sepals, petals, stamens are attached to the floral or hypanthium cup above the ovary with the lower part of the hypanthium completely adnate to the ovary.

EPIPETALOUS: Describing stamens that arise from the petals, as occur in many flowers with tubular corollas.

EPIPETIOLAR: Having branches arising from buds on the petiole.

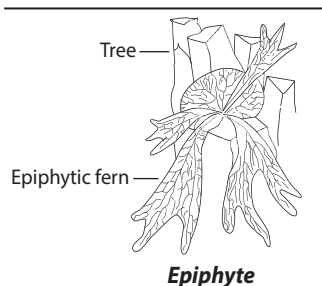
EPIPETRIC: Developing on a rock as found in many ferns.

EPIPHYSIS: The rounded end of a long (growing) bone, such as the limb bone in mammals that fuses with the shaft of the bone at the point where it was previously separated by cartilage to allow bone growth. Once the epiphysis fuses, no further growth of long bones is possible.

EPIPHYTE (AIR PLANT): A plant which is one of the subdivisions of the Raunkiaer system, that lives and grows on another plant, such as a tree, for physical support, but is neither parasitic on it nor rooted in the ground or soil. It sometimes grows upon objects such as buildings or telegraph wires, derives its moisture and nutrients from the air and rain and sometimes from debris accumulating around it. It is found in the tropics (many ferns, cacti orchids and bromeliads) and in the temperate zones (many mosses, liverworts, lichens and algae). There are many aquatic species of algae, including seaweeds that are also epiphytes on other aquatic plants (seaweeds or aquatic angiosperms).

EPIPHYTIC: A widespread outbreak of disease among plants.

EPISEPALOUS: Any part attached to the sepals.



EPSILON: 1. In mathematics, the Greek letter, ϵ , which is the symbol conventionally used to denote a strictly positive infinitesimal quantity, whose limit is usually taken as $\epsilon > 0$. 2. A small but strictly positive quantity, especially the maximum difference between any pair of members of a set, for example, ϵ -sequence and ϵ -neighbourhood. 3. The symbols $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ for the signature of a permutation. It is a generalisation of Kronecker's delta.

EPSILON NET (ϵ -NET): A finite or infinite set of points in a metric space such that each point of the space is within distance ϵ of some point in the set.

EPSILON-DELTA NOTATION: The standard notation used in the definitions of limits, continuity and related concepts. The principal notion is that a function tends to a limit at a given point if its value lies within some small ϵ (epsilon) of the limit whenever the independent variable is within a small δ (delta) of the given argument. Formally, a real function $f(x)$ has a limit L at a point p , at which it is defined, if it satisfies the condition that for every $\epsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - L| < \epsilon$ for all x such that $|x - p| < \delta$.

EPISTEMIC LOGIC: The branch of modal logic that is concerned with the concepts of knowledge, belief and ignorance.

EPISOME: A genetic element consisting of DNA that can exist and replicate either independently of its host cell chromosomes or as an integrated part of the chromosomes. They may be attached to the bacterial cell membrane or become part of the chromosome. They are capable of giving selective advantage to the bacteria in which they occur. Cells with episomes act like males during conjugation. Experiments involving gene transfers from cells in which episomes have been incorporated in the chromosomes have been used to determine the locations of genes on the chromosome.

EPISTASIS: The masking of the effects of one gene by the action of another or several others, known as modifier gene(s). The gene whose phenotype is expressed is said to be epistatic, while the phenotype altered or suppressed is said to be hypostatic.

EPISTAXIS (NOSEBLEED): Bleeding or haemorrhage from the nose, usually due to rupture of small blood vessels in the nose that may be brought about by injury to the nose or face, infection, tumours, irritation from foreign bodies, hypertension, alcohol abuse, inherited bleeding problems or excessive nose blowing or picking during a cold.

EPITAXY (EPITAXIAL GROWTH): The growth from gaseous or liquid precursors of a layer of one substance on a single crystal of another or the technique of depositing a monocrystalline film on a monocrystalline substrate, such that the crystal structure in the layer is the same as that in the substrate. Since the substrate acts as a seed crystal, the deposited film takes on a lattice structure and orientation similar to those of the substrate. It is used for making semiconductor devices.

EPITHELIAL TISSUE: A layer of cells that serves as a protective covering over some surface, such as the lining of the wall of a cavity in the body or the outside of an organ.

EPITHELIUM: Living tissue in vertebrates consisting of closely packed layers of cells in a sheet with little intercellular material and covering internal and external surfaces. Examples are the skin and the lining of the breathing, digestive and urinogenital organs. Its functions are protective, absorptive, secretory, transcellular transport and detection of sensation. Due to this, they often give extensive apical-basolateral polarity and specialisation.

EPITHERMAL NEUTRON: A neutron with energy in excess of that associated with a thermal neutron (thermal agitation) but less than that of a fast neutron, with energy range of 0.01 to 100 eV or a neutron that has energy of between 10^{-2} and 10^2 electron volts (eV).

EPITOPE (ANTIGENIC DETERMINANT): The structure of the surface of an antigen that is recognised by the immune system, specifically by antibodies, B cells or T cells and can bind to a specific

antibody. Its shape is generally complementary to that of the antibody's antigen-binding site. Examples are a group of sugar units of the polysaccharide component of a bacterial cell wall and a cluster of amino acids on the exposed surface of a protein molecule. The section of an antibody that recognises the epitope is known as a **paratope**.

EPITROCHOID: A curve traced out by a point that is a distance c from the centre of a circle of radius b , where $c < b$, that is rolling around the outside of another circle of radius a . It is described by the parametric equations:

$$x = (a + b) \cos(t) - c \cos[(a/b + 1)t],$$

$$y = (a + b) \sin(t) - c \sin[(a/b + 1)t].$$

Closely related to the epitrochoid are the epicycloid, hypocycloid and the hypotrochoid.

EPIZOO: An organism such as lice, ticks, many acari, the lerneans or fish lice that lives parasitically on the external surface of an animal.

EPOCH: 1. A subdivision of a geological period in the geological time scale. Examples are Palaeocene, Eocene, Oligocene and Miocene. A given epoch is also divided into periods of early, middle or late. For example, Eocene is often divided into Early (56 million to 47.8 million years ago), Middle (47.8 million to 38 million years ago) and Late (38 million to 33.9 million years ago). 2. In astronomy, it is an arbitrarily fixed moment in time, usually the beginning of a century or half century, used as a reference point in some time-varying astronomical quantity, such as a point of reference for specification of celestial coordinates. 3. In physics, it is the displacement from zero at zero time of an oscillating or vibrating body.

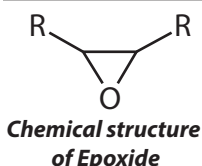
EPOIETIN: A hormone that regulates the production of red blood cells in the body. It is used medically to treat anaemia and also used illegally by some athletes to enhance oxygen transfer and improve endurance.

EPOXIDE: An organic compound whose molecules include a three-member ring that includes oxygen. This ring approximately looks like an equilateral triangle, which makes it highly strained and more reactive than other ethers. Simple epoxides are named from the parent compound ethylene oxide or oxirane, such as in chloromethyloxirane, however, as a functional group, they feature the epoxy prefix such as in the compound 1,2-epoxycycloheptane which is also known as cycloheptene epoxide or cycloheptene oxide.

EPOXY COMPOUNDS: Compounds in which an oxygen atom is directly attached to two adjacent or non-adjacent carbon atoms of a carbon chain or ring system; thus cyclic ethers. The term epoxides represents a subclass of epoxy compounds containing a saturated three-membered cyclic ether; thus oxirane derivatives. Examples are 1,2-epoxypropane or 2-methyloxirane (an epoxide); 9,10-epoxy-9,10-dihydroanthracene (an epoxy compound).

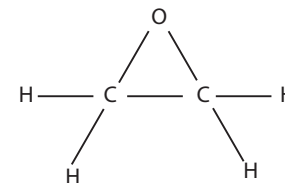
EPOXY (EPOXY RESIN): Any of a class of synthetic resins, containing ether linkages and epoxy groups, formed by the copolymerisation of an epoxide, for example, 1-chloro-2,3-epoxypropane and a polyphenol such as 2,2-bis(4-hydroxyphenyl)propane. When the resin with epoxide rings at the ends of its molecules is mixed with certain accelerators or a curing agent such as an amine or an anhydride, they react and after curing, result in a complex network with ether groups linking the monomers; producing stable, hard, clear, corrosive resistant, thermosetting polymers that are used as adhesives and coatings and for embedding materials for microscopy, especially electron microscopy. Epoxy is also used in the construction of airplanes and cars.

EPOXY+: A prefix used in chemical nomenclature, which indicates the presence in a molecule of an oxygen atom directly attached by single covalent bonds to two adjacent or nonadjacent carbon atoms of a carbon chain or ring system. Epoxy compounds are cyclic ethers.



EPOXYETHANE (ETHYLENE OXIDE; OXIRANE; EPOXYETHANE; DIMETHYLENE OXIDE; OXACYCLOPROPANE): A

colourless, flammable gas with a faintly sweet odour and the molecular formula C_2H_4O , density 0.882 g/ml, melting point $-111.3^\circ C$ (161.85 K) and boiling point $10.7^\circ C$ (283.85 K). It is the simplest epoxide and isomeric with acetaldehyde. It is produced industrially by the direct oxidation of ethylene or ethene in the presence of silver catalyst. It is used



Structure of Epoxyethane

for producing many chemicals and intermediates, such as ethylene glycol, ethanolamines, simple and complex glycols, polyglycol ethers and other compounds. It is also a common gas-phase disinfectant which is widely used in hospitals to sterilise heat-sensitive tools and equipment. It is used as a main component of thermobaric weapons. However, it is a strong poison to humans, showing such effects as carcinogenic, mutagenic, irritating and narcotic.

EPROM (ERASABLE PROGRAMMABLE READ-ONLY MEMORY):

Acronym for read-only memory device in a computer in which stored data can be erased by the use of high-intensity ultraviolet light but which can be reprogrammed repeatedly using suitable voltage pulses. They have been largely replaced by EEPROMs and flash memory, both of which can be erased in place on the circuit board.

EPSILON: The Greek letter for e, written as ϵ and normally used to denote a small quantity.

EPSOMITE (MAGNESIUM SULFATE; EPSOM SALTS): A mineral form of magnesium sulfate heptahydrate, with the chemical formula $MgSO_4 \cdot 7H_2O$, Mohs scale hardness 2.0-2.5 and relative density of 1.68. It has a salty bitter taste and is soluble in water. It occurs in clear, needle like orthorhombic crystals and it is massive or fibrous, with a lustre which varies from vitreous to milky. It is used as a relaxant and laxative and added to baths to soothe the skin. It is also used as a purgative and in leather tanning, mordant dyeing and as filler in cotton goods and paper.

EPSTEIN-BARR VIRUS, EBV (HUMAN HERPESVIRUS 4, HHV-4):

A virus that causes glandular fever. It is also associated with certain cancers such as Hodgkin's lymphoma, Burkitt's lymphoma, gastric cancer and nasopharyngeal carcinoma. EBV was discovered in 1964 by English microbiologist Michael Epstein and his assistant Y.M. Barr. EBV infection is characterised by fatigue and general malaise. Infection with EBV is common, carried by 90% of people and is normally temporary and minor; however, in some individuals EBV it can trigger some chronic diseases, including immune and lymphoproliferative syndromes.

EQUAL CHANCE: Two events having the same probability.

EQUAL CLEAVAGE: A holoblastic cleavage producing blastomeres of approximately equal size. An example is the holoblastic cleavage in amphioxus egg where there is little amount of yolk.

EQUAL FRACTION: The pairs (a_1, b_1) , (a_2, b_2) , (a_3, b_3) , ..., (a_n, b_n) are said to be proportional if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n}$, then each of

the fractions is equal to $\frac{l_1 a_1 + l_2 a_2 + \dots + l_n a_n}{l_1 b_1 + l_2 b_2 + \dots + l_n b_n}$, where $l_1, l_2, \dots, l_n$ are

constants such that $l_1 b_1 + l_2 b_2 + l_3 b_3 + \dots + l_n b_n \neq 0$.

EQUAL GROUP PROBLEMS: Problems that require students to count or organise quantities using equal groups to find the number of groups, group size, or product.

EQUAL GROUPS: Sets with the same number of elements.

EQUAL-GROUPS NOTATION: A way to denote a number of equal-size groups. The size of each group is shown inside square brackets

and the number of groups is written in front of the brackets. For example, 3 [6s] means 3 groups with 6 in each group. In general, n [bs] means n groups with b in each group.

EQUALITY: Two or more expressions which mean the same thing. Thus $3 + 4 = 7$ is an equality. For constant non-zero a, b, c , the equality $a = b$ implies $b = a$; $a + c = b + c$; $ac = bc$ and if $b = c$ then $a = c$.

EQUALITY OF MATRICES: Two matrices are equal if each matrix has the same number of rows and columns. The corresponding elements

in each matrix are equal. For example, if $x = \begin{pmatrix} a & 14 \\ 7 & t \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 4 & 6 \end{pmatrix}$ then $a = 3$ and $t = 6$.

EQUALITY OF SETS: Consider the sets X and Y , if X and Y are equal, then, they comprise of the same elements. This means $X = Y$ or instead $X \subseteq Y$ or $Y \subseteq X$.

EQUALLY LIKELY OUTCOMES: A probability theory that tells the equal likelihood of two or more events happening. For example, when a coin is tossed, it has two possible outcomes called head and tail. But if two or more coins are tossed, it is assumed that all the heads and tails have the same chance of occurring.

EQUALS SIGN (EQUAL SIGN): 1. The symbol '=' used between two expressions to indicate the identity of their references or values. 2. Symbol used for definitions, usually written either with 'df' subscript to the right, with a small triangle superscript or with the word 'definition' or 'defn' in the same line. The equals sign written with a colon to the left is used both for definitions and in computing for the assignment of values to variables, where $x := x + y$ means that the new value of the variable labelled x is to be the sum of the previous values of x and y .

EQUATION: 1. An expression usually involving constants and/or variables that represents the equality of two expressions or that two expressions on either side of an equal sign (=) have the same value. There are two forms of equation: (i) **identical equation**, known as an **identity** that is true for any values of the variables, for example, $x^2 - 1 = (x + 1)(x - 1)$ or (ii) **conditional equation** that is only true for certain values of the variables, called the roots of the equation. For example, $x^2 - 1 = 3$ with roots $x = \pm 2$. 2. The representation of a chemical reaction by symbols and numbers. The equation can be a word equation or a formula or symbol. 3. To **equate** is to form an equation by joining two separate expressions with an equal sign (=).

EQUATION OF A CIRCLE: A circle is the set of points equidistant from a point $C(h, k)$ called the centre. The fixed distance r from the centre to any point on the circle is called the radius. The standard equation of a circle with centre $C(h, k)$ and radius r is given by $(x - h)^2 + (y - k)^2 = r^2$. For example, to find the equation of a circle whose centre is at $(2, -4)$ and radius 5, the solution is:

$(h, k) = (2, -4)$ and $r = 5$; substituting h, k and r in the standard equation $(x - 2)^2 + (y - (-4))^2 = 5^2$, therefore $(x - 2)^2 + (y + 4)^2 = 25$.

The general equation of a circle is given by $x^2 + y^2 + 2gx + 2fy + c = 0$, where g, f and c are constants.

EQUATION OF A PARABOLA: The equation of a parabola is given by $(y - k)^2 = 4p(x - h)$, the directrix is parallel to the y -axis, p is the length of the focus and (h, k) is the vertex. When the directrix is parallel to the x -axis the equation of the parabola is given by $(x - h)^2 = 4p(y - k)$.

EQUATION OF STATE: An equation that relates the pressure, volume and thermodynamic temperature of an amount of substance or defines the physical state of a homogeneous substance such as gas, liquid or solid, by relating volume to pressure and absolute temperature for a given mass of the material. The simplest is the ideal gas law.

EQUATION OF TIME: The length of time that must be added to the mean solar time as shown on a clock to give the apparent solar time, as shown by a sundial. The amount varies during the year, being at a minimum of -14.2 minutes in February and a maximum of +10.4 minutes in October. Apparent solar time has annual variation as a

result of the Sun's inclination in the ecliptic and the eccentricity of the Earth's elliptical orbit.

EQUATIONS OF MOTION: Equations that define the velocity of a body, a collection of particles or a fluid in terms of its motion as a function of time. The three equations of motion are: (i) $v = u + at$, (ii) $s = ut + 1/2(at^2)$, (iii) $v^2 = u^2 + 2as$, where v is the final velocity, u is the initial velocity, a is the acceleration, s is the distance travelled by a body and t is the time taken.

EQUATOR: The intersection of a sphere's surface with the plane perpendicular to the sphere's axis of rotation and containing the sphere's centre of mass. For the Earth's Equator, it is an imaginary line drawn round the Earth halfway between the North and South Poles at Latitude 0° or the great circle whose plane is perpendicular to the Earth's axis (the imaginary line joining the poles) that divides the Earth into the northern and southern hemispheres. The equators of other planets and astronomical bodies are defined analogously, however, the capitalised term Equator refers to the Earth's equator. Anything related to or near the equator is described as **equatorial**.

EQUATORIAL BULGE: The extension of the equator of a spinning body and the subsequent flattening of the poles, caused by rotation. It is most conveniently thought of as being due to an outward centrifugal force (although this is really a fictitious force). Equatorial bulges are very noticeable in the case of the planets Jupiter and Saturn, which are large, spin rapidly and are composed of fluids that are easily deformed. The Earth also has an equatorial bulge, one effect of which is to contribute to a variation in the acceleration due to gravity with latitude. The bulge also causes the planes of satellite orbits inclined to the equator (but not polar) to rotate slowly around the Earth's axis.

EQUATORIAL CLEAVAGE: A cleavage in which the plane of cytoplasmic division is at right angles to the axis of the ovum or passing through the embryo's midsection. For example, in mammals, the first and second cleavage are vertical while the third cleavage is equatorial or at the midsection of the embryo.

EQUATORIAL COORDINATES: In astronomy, it is a system for measuring the position of astronomical objects on the celestial sphere in terms of its declination and right ascension with reference to the plane of the Earth's Equator. Declination and right ascension correspond directly to geographic latitude and longitude as projected outward onto the celestial sphere. Declination is measured in degrees north or south of the celestial equator, the same as geographic latitude, but right ascension is measured in hours, minutes and seconds eastward along the celestial equator from the point of the vernal equinox.

EQUATORIAL ORBIT: An orbit in the same plane as the Earth's equator or the equator of some other celestial object.

EQUI: A prefix denoting equality.

EQUIANGULAR: 1. A geometric figure having all of its angles being equal. 2. Of a pair of figures, each having angles that are equal to the corresponding angles of the other.

EQUIANGULAR POLYGON: A polygon with all its interior angles equal. If the lengths of the sides are also equal such as in an equilateral triangle, then it is a regular polygon. Rectangles and squares are the only equiangular four-sided figures. According to the equiangular polygon theorem, for an equiangular n -gon each angle is $180^\circ - 360^\circ/n$.



Equiangular polygon

EQUIANGULAR SPIRAL (GROWTH SPIRAL; LOGISTIC SPIRAL; LOGARITHMIC SPIRAL): A type of spiral that is seen commonly in the natural world. Examples are found in the shells of some molluscs, such as that of the nautilus and the fossil ammonites and also in spider webs. The logarithmic spiral can be constructed from equally spaced rays by starting at a point along one ray and drawing the perpendicular to a neighbouring ray. The logarithmic spiral is an example of a transcendental plane curve.

EQUICHORDAL POINT: A point inside a closed convex curve in a plane, if all chords through that point have the same length.

EQUICONTINUOUS: 1. In mathematics, of a family, F of real functions between metric spaces at a point c , such that for every $\varepsilon > 0$ there exists a $\delta > 0$, such that $|f(x) - f(c)| < \varepsilon$ for all x such that $|x - c| < \delta$; which means that δ depends only on ε , and does not vary from function to function. 2. Of a family, F of real functions from a topological space, X to a metric space, at a point, p of X , such that for every $\varepsilon > 0$, there exists an open neighbourhood U of p , such that for all functions f in F , $d[f(x)], f(p) < \varepsilon$, whenever $x \in U$.

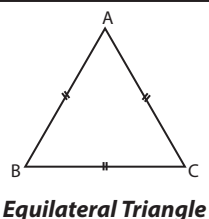
EQUIDISTANT: Situated at the same distance from two or more points or places.

EQUIFINALITY (ISOTELESIS): The principle that in open systems a given end state can be reached by many potential means or similar results may be achieved with different initial conditions and in many different ways. The term is attributed to Ludwig von Bertalanffy (1901-1972), who is the founder of the General Systems Theory. In open systems such as biological and social systems, the principle may be stated as the ability of certain structures or behaviour patterns to be formed by more than one route in a developing embryo or young animals.

EQUILATERAL: Referring to a geometrical figure such as squares and rhombuses which has all sides of equal length. The term is often applied to a triangle with all three sides and angles equal.

EQUILATERAL POLYGON: A polygon with all its sides and angles equal.

EQUILATERAL TRIANGLE: A triangle with three equal sides. In Euclidean geometry, equilateral triangles are also equiangular, all three internal angles are also congruent to each other and are each 60° . They are regular polygons and can therefore also be referred to as **regular triangles**.



EQUILIBRANT: A force which is able to balance out another force and produces equilibrium. It could be a single force which cancels the vector sum of a given system of forces acting on a rigid body and whose torque cancels the sum of the torques of the system. For example, for three forces, F_1 , F_2 and F_3 to be in equilibrium, there must be another force that can balance the three forces. In this case,

$$F_1 + F_2 + F_3 + F = 0, \text{ therefore the equilibrant force,}$$

$$F = -(F_1 + F_2 + F_3).$$

EQUILIBRATION: The operation by which a system of two or more phases is brought to a condition, where further changes with time do not occur. This term is not the same as pre-equilibrium and should not be used in that sense.

EQUILIBRIUM: 1. An object is said to be in equilibrium if its momentum is constant or if a vertical line drawn from its centre of gravity stays within its base or a state of balance where an object remains in the same plane, state of balance or rest. This is true only if there is no resultant force and torque. 2. A reversible chemical reaction is said to be in equilibrium (dynamic equilibrium) if the rate of formation of the products is equal to the rate of formation of the reactants. 3. In ecology, it is the state of an ecosystem in which no further net change of species population occurs.

EQUILIBRIUM AND TRANSIENT CLIMATE EXPERIMENT: An equilibrium climate experiment is an experiment in which a climate model is allowed to fully adjust to a change in radiative forcing. Such experiments provide information on the difference between the initial and final states of the model, but not on the time-dependent response. If the forcing is allowed to evolve gradually according to a prescribed emission scenario, the time-dependent response of a climate model may be analysed. Such an experiment is called a **transient climate experiment**.

EQUILIBRIUM CONSTANT: The product of the molar concentrations or pressure in terms of gases of the products of a chemical reaction

divided by the product of the concentrations of the reactants, with each concentration term raised to its stoichiometric coefficient, at a given temperature. It is denoted by the symbol, K .

EQUILIBRIUM DIALYSATE: The colloid-free solution obtained at equilibrium in dialysis. Its composition approaches that of the dispersion medium (more precisely, the limit to which the composition of the dispersion medium tends at large distances from the particles). In the dialysis equilibrium an osmotic pressure difference exists between sol and equilibrium dialysate.

EQUILIBRIUM FILM: A liquid film of thickness or thicknesses, at which it is stable or metastable with respect to small thickness changes. Unless the area of the film is small, its composition may not be the same over its area and the (metastable) equilibrium thickness may be characteristic of the local condition only.

EQUILIBRIUM GEOMETRY: Molecular geometry that corresponds to the true minimum on the respective potential energy surface. While information relating to the equilibrium geometry is provided by calculations within the adiabatic approximation (minimisation of the total energy with respect to any independent geometrical parameter), various experiments yield some effective geometries for the molecule which are averaged over molecular vibrations.

EQUILIBRIUM POINT: A constant solution to a differential equation. Formally, a point y^ε such that, for a given ordinary differential equation $y' = f(y)$, $f(y^\varepsilon) = 0$.

EQUILIBRIUM QUOTIENT: The term on the right-hand side of the mathematical form of the law of mass action.

EQUILIBRIUM REACTION: When the reactants in a chemical reaction are initially in a Boltzmann distribution the reaction is referred to as an equilibrium reaction.

EQUILIBRIUM SEDIMENTATION: A method by which the distribution of the concentration of the solute or dispersed component in a dilute solution or dispersion along the centrifuge cell is measured at sedimentation equilibrium and the results are interpreted in terms of molar masses or their distribution or both.

EQUILIBRIUM SENSE (VESTIBULAR SENSE): The mammalian ability to remain upright in relation to the gravity and to detect changes in position and momentum. The principal organs of equilibrium are contained in the inner ear – the utricle, which transmits orientation information and the semicircular canals, which are concerned with acceleration and deceleration. These systems help humans locate their bodies in space.

EQUILIBRIUM SPECIES: Species characterised by low reproduction rates, long development times, large body size and a long adult life with repeated reproductions. These species show characteristics which are consonant with a stable niche. Persistence of individuals enables the species to survive. Dispersal is less important and perseverance is more significant than recovery from adverse conditions. For example, large desert plants exhibit some of these properties. They grow rapidly in the rains and, unlike opportunist species, put most of their energy into growth and conservation of resources. Reproduction is a rare event; some species may set seed only once in several years as with cacti. see r-selection.

EQUILIBRIUM VAPOUR PRESSURE: The vapour pressure measured under dynamic equilibrium of condensation and evaporation.

EQUIMEASURABLE: Of a pair of functions, f and g , such that both functions are real-valued and measurable and $\mu(\{f(x) : f(x) > y\}) = \mu(\{g(x) : g(x) > y\})$ for all real y . Functions that agree almost everywhere are equimeasurable.

EQUIMOLAR MIXTURE: A mixture of substances in equal molecular proportions.

EQUINOCTIAL: 1. Bearing flowers that open and close regularly at specific times. 2. Pertaining to an equinox or the equinoxes or to the equality of day and night. 3. A violent storm of wind and rain occurring at or near the time of the equinox.

EQUINOX: Either of the two times in a year around 21 March and 22 September, when the sun crosses the equator and day and night are of equal length or either of the two points on the celestial sphere at which the ecliptic intersect the celestial equator or either of the two instants at which the centre of the Sun appears to cross the celestial equator. The two types are vernal and autumnal equinoxes.

EQUINUMEROUS: Of two classes, having the same number of members. Formally, two sets A and B are equinumerous if there exists a one-to-one correspondence or a bijection between them, which means that if there exists a function from A to B such that for every element y of B there is exactly one element x of A with $f(x) = y$. It is written mathematically as $A \approx B$ or $A \sim B$ or $|A| = |B|$.

EQUIPARTITION OF ENERGY (EQUIPARTITION THEOREM; LAW OF EQUIPARTITION; EQUIPARTITION): The theory states that the energy of gas molecules in a large sample under thermal equilibrium is equally divided among their available degrees of freedom, the average energy for each degree of freedom being $\frac{1}{2}kT$, where k is the Boltzmann constant and T is the thermodynamic temperature. It is a general formula that relates the temperature of a system with its average energies.

EQUIPMENT: A set of tools, devices, kit, etc., assembled for a specific purpose. For example, chemistry laboratory equipment includes beaker, dropper, gas jar, flasks, measuring cylinder and test tube.

EQUIPOLLENT (EQUIPOTENT): 1. Of statements or propositions, deducible from one another. 2. Of two classes, having the same cardinality. For example, two sets A and B are equipollent if and only if there is a one-to-one correspondence or a bijection from A onto B .

EQUIPOTENTIAL CURVE: 1. Also known as **contour line**, it is a curve in two dimensions on which the value of a function $f(x, y)$ is a constant. 2. Also known as **contour**, it is a line drawn on a map that joins points with equal elevation above or below a level, usually mean sea level.

EQUIPOTENTIAL LINES: Lines that join points of equal potential in a gravitational or electric field. They are always at rightangles to the field lines. No work is done in moving a charged particle or mass along an equipotential lines.

EQUIPOTENTIAL SURFACE: 1. A surface on which the electric potential is the same at every point. 2. A surface characterised by the potential being constant everywhere on it for the attractive forces concerned. 3. A surface which is always normal to the lines of force of a field and on which the potential is everywhere the same.

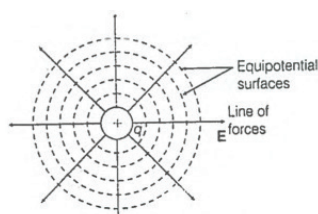


Fig. 1: Equipotential surface due to a point charge, and a uniform field.

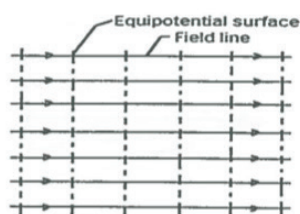


Fig. 2: Equipotential surface due to a point charge, and a uniform field.

EQUIPROBABLE: Of a number of events, having the same logical or mathematical probability. For example, in a fair game of dice with a single die, each face is an equiprobable outcome. The principle of indifference is that in the absence of any reason to the contrary, all elementary events should be considered equiprobable.

EQUISETOID: Any plant of the phylum or division Equisetophyta (or Sphenopsida), including the horsetails and many extinct species of the Carboniferous Period.

EQUITANT: A leaf having the base folded around the stem so that it overlaps to form a fan-like arrangement in two ranks, as seen in the leaves of iris.

EQUIVALENCE: 1. The relation that holds between two statements when each implies the other. 2. Also called **biconditional**, having a compound statement that one sentence is true if and only if the other sentence is true. Its symbol is $\leftrightarrow$ or $\Leftrightarrow$. Symbolically, this is written as $p \leftrightarrow q$ or $p \Leftrightarrow q$. For example, "A triangle has two equal sides if and only if it has two equal angles" is a biconditional statement. 3. The relation that holds between two statements when the sentence so formed is true.

EQUIVALENCE CLASS: A grouping of elements that are related to one another by an equivalence relation such that every element of the set is a member of one and only one cell of the partition.

EQUIVALENT NORMS: Two norms for which the associated distance functions are equivalent. In other words, two norms $\| \cdot \|_1$ and $\| \cdot \|_2$ are equivalent if there are positive constants M and N such that $M\| \cdot \|_1 \leq \| \cdot \|_2 \leq N\| \cdot \|_1$. All norms on a given Euclidean space are equivalent and so give rise to the same topology.

EQUIVALENCE PRINCIPLE OF GRAVITATION: It states that the gravitational force caused by an external object is the same for all objects of the same mass, regardless of their compositions.

EQUIVALENCE PRINCIPLE OF SPECIAL RELATIVITY: A principle central to general relativity stating that a gravitational field is locally indistinguishable from the effects of inertial forces. For example, according to the equivalence principle, it is impossible for someone in a box who experiences a force pushing him to the bottom of the box to know, from the force alone, whether that force is the result of a gravitational field (the box is standing on the surface of a planet) or an acceleration (the box is being pushed by a rocket).

EQUIVALENCE REACTION: A binary relation that has the following properties: reflexive, symmetric and transitive.

EQUIVALENCE RELATION: A relation $\sim$ on a set X if and only if it is reflexive, symmetric and transitive. In other words, a relation that holds between two elements if and only if they are members of the same cell within a set that has been partitioned into cells such that every element of the set is a member of one and only one cell of the partition.

EQUIVALENT: 1. In chemistry, a unit of amount of substance, defined as the entity of a chemical species that in a specified reaction would either react with or supply one mole of hydrogen ions (H^+) in an acid-base reaction or one entity of electrons in a redox reaction. It is the arithmetic product of the molar mass of the chemical species and its equivalence factor. 2. Equal in amount, volume, importance, meaning, etc, for example, 420g is equivalent to 0.420kg. 3. In mathematics, the term is used for two sets capable of being put into a one-to-one relationship or having the same cardinality. Alternatively, relating to corresponding elements under an equivalence relation. Its symbol is given by $\equiv$. 4. In geometry, it defines two geometric figures, having some particular property in common, such as similar triangles or parallelograms of the same height on the same base. 5. In logic, the term is used for two statements or propositions in which each implies the other, being either both true or both false. 6. In terms of two equations or inequalities, having the same solution set. 7. Of two fractions, reducible to the same proper fraction and so representing the same rational number. For example, $5/10$ and $10/20$ are equivalent since both can be reduced to the rational number $1/2$. 8. Of two matrices A and B , such that there exist non-singular matrices C and D with $A = CBD$. 9. Of two metrics on a given set, generating the same

topology, in that the same topology results from using the open balls for either metric as a base. 10. Of two ideals in an integral domain, related in such a way that there are elements a and b for which $(a)I = (b)J$, where I and J are the given ideals and (a) and (b) are the principal ideals generated by a and b , respectively. 11. Also known as **equipollent**, it defines two systems, having the same vector sum and the same torque about the same point.

EQUIVALENT CHAIN: A hypothetical freely jointed chain with the same mean-square end-to-end distance and contour length as an actual polymer chain in a theta state.

EQUIVALENT CARBON DIOXIDE/CO₂: The concentration of carbon dioxide that would cause the same amount of radiative forcing as a given mixture of carbon dioxide and other greenhouse gases.

EQUIVALENT DIAMETER: According to the IUPAC definition, the equivalent diameter of a non-spherical particle is equal to a diameter of a spherical particle that exhibits identical properties (e.g., aerodynamic, hydrodynamic, optical, electrical) to that of the investigated non-spherical particle. For particles in non-turbulent motion, the equivalent diameter is identical to the diameter encountered in the Stokes' law.

EQUIVALENT DOSE: A measure of the local radiation damage caused to biological tissue through exposure to intense ionising radiation, such as alpha particles. Local damage to DNA is harder for the body to repair than the more diffuse damage from gamma rays and beta particles. This relative biological effectiveness gives rise to the use of equivalent dose, which is the absorbed dose multiplied by an appropriate radiation weighting factor (WR). The value of WR is 20 for alpha particles and 1 for both beta particles and gamma rays. The resulting unit of equivalent dose is the sievert (Sv). Submultiples are the millisievert and microsievert. Prior to 1990, the weighting factor used was referred to as the quality factor (QF).

EQUIVALENT EQUATIONS: Equations with the same solution. For example, $2 + x = 4$ and $6 + x = 8$ are equivalent equations with the common solution 2.

EQUIVALENT EXPRESSIONS: Numerical or algebraic expressions that give the same result after substituting the variables by any value or using different mathematical phrases, for example, $3a + 3b = 3(a + b)$, $3 + 4 + 7 = 7 + 7$.

EQUIVALENT FRACTIONS: Two fractions are equivalent if the ratio of the numerator and the denominator is the same in both cases, for example, $2/4$ and $4/8$ are both equal to 1.

EQUIVALENT NAMES: Different ways of naming the same number. For example, $2 + 6$, $4 + 4$, $12 - 4$, $18 - 10$, $100 - 92$, $5 + 1 + 2$, eight, VIII, and $///\backslash ///$ are all equivalent names for 8.

EQUIVALENT NUMBERS: Numbers such that the sums of their aliquot parts (proper divisors) are the same. For example, 159, 559 and 703 are equivalent numbers because their aliquot parts all sum to 57.

EQUIVALENT POINT: In a titration, the point at which an amount of the standard solution which is chemically equivalent to the analyte concentration has been added. For example, the equivalent point in the titration of sodium chloride with silver nitrate is when exactly one mole of silver ion has been added to one mole of chloride ions. The equivalent point is a theoretical concept and can be estimated only by observation of a physical change associated with it in solution.

EQUIVALENT PROPORTIONS: A law referring to a situation, where two elements form chemical compounds with a third element, a compound of the first two contains them in the relative proportions they have in compounds with the third one.

EQUIVALENT RATIOS: Two ratios that express the same relationship between numbers. It is in effect, equivalent fractions, for example, $1/2$ and $4/8$ are equivalent ratios. Equivalent ratios can be created by multiplying or dividing both the numerator and denominator of a given ratio by the same number.

EQUIVALENT THICKNESS: A measure of column density written in terms of the height of a column under standard temperature and pressure conditions that would contain the same number of molecules.

EQUIVALENT WEIGHT: The number of parts by mass of an element that can combine with or displace one part by mass of hydrogen. For all elements, the atomic weight is equal to the equivalent weight times the valence of the element. An element can have more than one valence and therefore, more than one equivalent weight. Relations are now mostly expressed in terms of balanced stoichiometric chemical equations and relative numbers of moles reacting, instead of equivalent weight and gram-equivalent weight.

EQUIVALENT WIDTH: A measure of the strength of a spectral line. The equivalent width is the width of a rectangle centred on a spectral line that, on a plot of intensity against wavelength, has the same area as the line. On a plot of intensity against wavelength, a spectral line appears as a curve with a shape defined by the line profile.

ERA: A unit of geologic time usually lasting hundreds of millions of years. Eras are divided into smaller units of geologic time called periods.

ERADICANT FUNGICIDE: A fungicide applied after a disease infection has occurred.

ERAMOUS STEM: Having no branched stems; erect or upright.

ERASABLE PROGRAMMABLE READ-ONLY MEMORY (EPROM): A read-only memory device in a computer in which stored data can be erased by the use of high-intensity ultraviolet light but which can be reprogrammed repeatedly using suitable voltage pulses. type of memory chip that retains its data when its power supply is switched off. EPROMs have been largely replaced by EEPROMs and flash memory, both of which can be erased in place on the circuit board.

ERATOSTHENES OF CYRENE (ABOUT 275-195 BC): A Greek astronomer and mathematician, who first calculated the size of the Earth by measuring the angle of the Sun at two different places, a known distance apart. He also measured the tilt of the Earth axis. He is credited with the method called sieve of Eratosthenes used to obtain prime numbers by sifting out the composite numbers from the set of natural numbers so that only prime numbers remain.

ERBIUM: A soft, lustrous, greyish metallic element in the lanthanide series, with the symbol Er, atomic number 68, relative atomic mass 167.3, melting point 1497 °C (1 770 K) and boiling point 2900 °C (3173 K). It is used as an alloy or a pigment. Its key uses involve its pink-coloured Er^{3+} ions, which have optical fluorescent properties and particularly useful in certain laser applications. For example, erbium-doped glasses or crystals can be used as optical amplification media, where erbium (III) ions are optically pumped at around 980 nm or 1480 nm and then radiate light at 1530 nm in stimulated emission.

ERECT: 1. In mathematics, to construct or draw a line or figure on a given base. 2. Describing or referring to an optically produced image that is upright. 3. Swollen and stiff as a result of being filled with blood, for example, the erection of the penis when sexually aroused.

ERECTING PRISM (INVERTING PRISM): A system of glass prism used in an optical instrument to convert an inverted image, formed by most types of astronomical telescopes into an erect image as in prismatic binoculars.

ERG: The centimetre–gram–second (cgs) unit of work or energy, which is equal to the work done when the point of application of a force of one dyne is displaced through a distance of one centimetre in the direction of the force. $1 \text{ erg} = 10^{-7} \text{ joule}$. The International System of Units (SI) is the currently used system of units, and the joule has replaced the erg.

ERGASTIC MATTER: Non-protoplasmic substances that are produced as by-products of protoplasmic activity. It includes crystals, starch grains, tannins and oil droplets. Many ergastic substances are regarded as waste matter, some are storage products, while the functions of other materials are unknown.

ERGODIC: Pertaining to the condition that, in an interval of sufficient duration, a system will return to states that are closely similar to previous ones. This is the basis of statistical methods used in modern dynamics and atomic theory.

ERGODIC HYPOTHESIS: A hypothesis in statistical mechanics concerning phase space which states that the orbit of the representative point in phase space eventually goes through all points on the surface, the time spent by a particle in some region of the phase space of microstates with the same energy is proportional to the volume of this region over long periods of time. Systems such as spin glasses in which ergodicity is thought not to hold are described as having broken ergodicity.

ERGODIC SET: A minimal set of states of a Markov chain such that the probability of exiting from the set is zero, a minimal stochastically closed set of states. For a Markov chain to be ergodic, it is required that its states and the non-zero transition probabilities meet two conditions: **irreducibility**, which ensures that there is a sequence of transitions of non-zero probability from any state to any other; and **aperiodicity** that ensures that the states are not partitioned into sets such that all state transitions occur cyclically from one set to another.

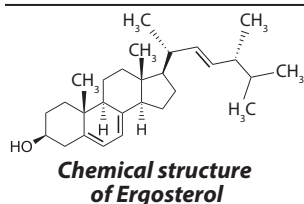
ERGODICITY: A property of a system that obeys ergodic hypothesis or a system that tends in probability to a limiting form that is independent of the initial conditions. It is an attribute of stochastic systems.

ERGONOMICS: The study of the engineering aspect of the relationship between workers and their working environment. It is the science of designing the job, equipment and workplace to fit the worker and it is necessary to prevent repetitive strain injuries, which can develop over time and can lead to long-term disability. It is employed to fulfill the two goals of health and productivity.

ERGONOMICAL: Of or relating to the design of equipment or devices to fit the human body's control, position, movement and environment.

ERGOSPHERE: An ellipsoidal region around a rotating Kerr black hole in which the hole's rotation drags the spacetime continuum round with it, so that frames of reference are not stationary with respect to the rest of the universe. This phenomenon is known as the **Lense-Thirring effect** or **frame dragging**. The ergosphere is bounded on the outside by the static or stationary surface and on the inside by the outer event horizon. If an object passes from normal space into the ergosphere it can still be ejected from the black hole by gaining energy from the hole's rotation. However, if an object crosses the event horizon, it will be sucked into the black hole and never escape.

ERGOSTEROL ([22E]-ERGOSTA-5,7,22-TRIEN-3 β -ol; (22E)-(24R)-24-METHYLCHOLESTA-5,7,22-TRIEN-3 β -OL): A sterol occurring in fungi, bacteria, algae and higher plants, with the chemical formula $C_{28}H_{44}O$, molar mass 396.65 g/mol, melting point 160°C (433 K) and boiling point 250°C (523 K). It is converted into vitamin D₂ by the action of ultraviolet light and then converted into ergocalciferol, which is a form of vitamin D. It is a component of fungal cell membranes, serving the same function that cholesterol serves in animal cells. It is, therefore,

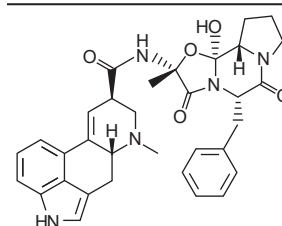


a useful target for antifungal drugs. Since it is also present in the cell membranes of some protists, such as trypanosomes, some antifungals are used against the African sleeping sickness (trypanosomiasis).

ERGOT: 1. A disease of cereals, especially rye, caused by the parasitic fungus *Claviceps purpurea* that forms compact black masses of branching filaments that replace many of the grains of the host plant. An ear of rye infected with ergot exudes sweet, yellowish mucus. Taking an overdose of ergot-derived medications or eating flour milled from ergot-infected rye can cause ergotism in humans and livestock. Symptoms may include convulsions, miscarriages in females and dry gangrene and may result in death. 2. The dried sclerotia of an ergot fungus containing physiologically active substances. It is used in the treatment of migraines, in initiation of labour in pregnancy and in controlling postpartum haemorrhage. Lysergic acid, from which the powerful hallucinogen LSD is synthesised, comes from ergot.

ERGOTAMINE: One of the alkaloids in ergot that causes ergotism. The others are ergometrine, ergometrinine and ergotaminine.

ERGOTISM: Poisoning by eating ergot contaminated cereals or bread.



Chemical structure of Ergotamine

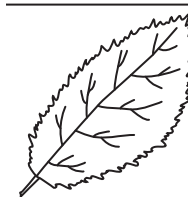
ERICOID MYCORRHIZAE: The type of mutualistic symbiosis formed between the plants in the Ericaceae family and several species of fungi in which they create and exchange the required nutrients that each needs to survive. This makes the plants and the various fungi species adapt to the acidic and nutrient poor soils such as bogs and boreal forests that they inhabit. Examples of plants in the Ericaceae family are bilberry, cranberry, blueberry, and azaleas.

ERLANG: A dimensionless unit in telecommunications traffic that describes carried load in telephony.

ERLANG PROGRAMMING LANGUAGE: A general purpose language used in building scalable soft real time systems which require high availability with built in support for concurrency, for example, telecoms, banking and e-commerce.

ERLANGEN PROGRAMME (ERLANGER PROGRAMME): The view put forward by a German mathematician Felix Klein that each geometry can be characterised by a group of transformations. An influential research programme and manifesto was published in 1872 by Felix Klein; called Erlangen Programme because Klein was then at Erlangen, a city in south central Germany.

EROSE: With the margin irregularly toothed, as if gnawed, as occurs in some leaves, as shown in the diagram.



An erose leaf

EROSION: The undesirable displacement of top soil or the wearing away of the Earth's surface, caused by the breakdown and transportation of particles of rocks and soil, by the action of weather, wind, sea, ice, rivers, rain water and organisms such as plants, animals and man. There are different types of water erosion as follows: (i) **accelerated erosion** is an erosion caused primarily by weather and climatic factors together with human or animal activities. It occurs much more rapidly than other types of erosion. (ii) **Geological erosion** sometimes known as **natural erosion** is the normal or natural erosion caused by geological processes acting over long geologic periods and resulting in the wearing away of mountains, the dissection of plains, and the building up of floodplains and coastal plains. (iii) **Gully erosion** is the erosion process whereby water accumulates in narrow channels and, over short periods, removes the soil from this narrow area to various depths, from about 0.3 m to as much as 30 m. (iv) **Natural erosion** is the wearing away of the earth's surface by water, ice, or other natural agents under natural

environmental conditions such as climate and vegetation, undisturbed by man. This can also be described as geological erosion. (v) **Normal erosion** is the gradual erosion of land used by man. It does not greatly exceed natural erosion. (vi) **Rill erosion** is an erosion process in which many small channels a few centimetres deep are formed; it occurs mainly on recently cultivated soils. (vii) **Sheet erosion** is the removal of a fairly uniform layer of soil from the land surface by runoff water. (viii) **Splash erosion** is the spattering of small soil particles caused by the impact of raindrops on very wet soils. The loosened and spattered particles may, or may not, be subsequently removed by surface runoff.

EROSION CLASSES: A grouping of erosion conditions based on the degree of erosion or on the characteristic patterns. The classes apply to accelerated erosion, but not to normal, natural, or geological erosion. Four erosion classes are recognised for water erosion and three for wind erosion.

EROSION PAVEMENT: A layer of coarse fragments such as sand, gravel, cobbles, or rock fragments that remains on the surface of the ground after the removal of fine particles by erosion protecting the underlying soil from being removed.

EROSIONAL FEATURES: Land surfaces shaped by the action of natural processes like the removal and relocation of weathered materials. For example, ice, water, waves, and wind break down and carry away rock. Examples of such features are caves and bays.

ERRATIC: 1. Inconsistent or changeable in behaviour; unpredictable. 2. Deviating from the usual or proper course; eccentric. 3. In geology, it is a piece of displaced rock that has been transported by a glacier or some other natural force to a site of different geological composition. All erratic blocks in northern Germany, for example originated in Scandinavia.

ERROR: 1. A property of a measured value to differ from the true value, often expressed as an absolute or relative range, such as ± 5 mm or $\pm 5\%$. The common types of errors include random, systematic and erratic errors. An example in applied mathematics is the difference between a value and an estimate of that value. In statistics, a common example is the difference between the mean age of a given group of people and that of a sample drawn from the group. In numerical analysis, an example of round-off error is the difference between the true value of π and commonly substituted expressions like $22/7$ and shorter versions like 3.14159. Truncation error results from using only the first few terms of an infinite series. Relative error is the ratio of the size of an error to the size of the quantity measured and percentage error is relative error expressed as a percent. 2. A fault or mistake or bug, either in a software or on the part of the user that causes a program to stop running or produce unexpected results.

ERROR DETECTION (DEBUGGING): The techniques that enable a program to detect incorrect data or to find and remove errors in a system, especially a computer program or a piece of electronic hardware, thus making it behave as expected.

ERROR MESSAGE: A message produced by a computer to inform the user that an error in the software or hardware has occurred.

ERROR-CORRECTING CODE: In information technology, a coding system that can recognise and immediately correct data that is being read or transmitted.

ERTHROPHLEUM: A genus of plants in the *Caesalpinaceae* family containing diterpenoid alkaloids, found in Africa, Asia and Australia. The stipules are small and caducous. The leaves are 2-pinnate and the leaflets alternate. The inflorescence of small flowers is in a pedunculate spike-like raceme, usually aggregated into a panicle. The pod is oblong or oblong-elliptic, flattened, woody and dehiscent. One of the species, *Erthrophleum suaveolens* commonly known as forest ordeal tree and ordeal tree is a medium to large tree, which grows from 9 to 25 metres high. It is found in Madagascar, Mozambique, Zimbabwe, some other tropical countries of Africa, Asia and Australia. It is used traditionally

to treat heart problems, headaches and migraines, etc. The dried leaves are mixed with stored grains and pulses to repel or kill storage insects. The fallen leaves promote nitrogen fixation in agroforestry and the large amount of leaf litter are advantageous to intercrops. The tree produces quite hard, heavy and durable wood, which is used for joinery, flooring, railway sleepers, harbour and dock work, turnery, construction and bridges, boat building, etc. *Erythrophleum africanum* with the common name the African blackwood, produces a gum similar to gum arabic. It has been used successfully to treat dysentery, diarrhoea and passive haemorrhages. *Erythrophleum ivorense* is a large tree, usually irregular and often buttressed. Seed and bark are poisonous if ingested. The wood is used for flooring, heavy construction, railway cross-ties, harbour and dockwork.

ERUPTION: 1. The ejection of molten rock, steam, etc., as from a volcano or geyser. During a volcanic eruption, for example, lava, tephra (ash, lapilli, solid chunks of rock) and various gases are expelled from a volcanic vent or fissure. 2. The sudden appearance of spots, etc. on the skin.

ERYSIPELAS (ST. ANTHONY'S FIRE): A skin infection, usually affecting the face, caused by certain types of streptococcus. Erysipelas is common in infancy and middle age. Erythema and swelling spread with a clear margin and cause blistering. It is a short illness with fever; if it affects the trunk it may however cause prostration and can prove fatal.

ERYSIPHALES: An order of the Plectomycetes containing about 150 species (20 genera) of obligate parasites that cause powdery mildew diseases of plants. The diseases are so named because the superficial hyphae produce masses of powdery conidiospores.

ERYTHEMA: A redness of the skin due to increased capillary blood flow. Erythema occurs in inflammation including dermatitis and numerous rashes.

ERYTHROBLAST: Any of the cells in the myeloid tissue of red bone marrow that develops into red blood cells. It is the immediate precursor of a normal erythrocyte.

ERYTHROCYTE SEDIMENTATION RATE, ESR (SED RATE; SEDIMENTATION RATE; WESTERGRENN SEDIMENTATION RATE): In medicine, it is a non-specific blood test that is a measure of the rate at which red blood cells (erythrocytes) settle out of suspension in blood plasma. ESR is useful in detecting inflammation associated with conditions such as serious infections, cancers and autoimmune diseases.

ERYTHROCYTES (RED BLOOD CELLS): Biconcave blood cells which originate in the bone marrow and contain the red pigment haemoglobin which combines reversibly with oxygen to form oxyhaemoglobin and are responsible for oxygen transport. Carbon dioxide can also combine with haemoglobin. They are the most numerous types of blood cells, lack nucleus and shaped like flat discs at maturity. The red blood cells have a limited life span of about 120 days. Because of this the bone marrow manufacture new ones at the rate of about one and half millions per second (1.5 millions/second) to replace the destroyed ones.

ERYTHROMYCIN: An antibiotic that is produced by a red-pigmented soil bacterium, *Streptomyces erythreus*. Its chemical formula is $C_{37}H_{67}NO_{11}$. It is used in the treatment of many infectious diseases such as respiratory tract infections, skin infections, chlamydia infections and syphilis caused by a wide range of Gram-positive and some Gram-negative bacteria. Its effects are similar to that of penicillin and is useful in treating patients who are allergic to penicillin.

ERYTHROPOIESIS: The formation of red blood cells (erythrocytes), which occurs in the red bone marrow. It is stimulated by decreased oxygen delivery to the kidneys, which then secrete the hormone erythropoietin. This activates increased erythropoiesis in the haemopoietic tissues.

ERYTHROPOIETIN (ERYTHROPOETIN; ERITHROPOYETIN; HAEMATOPOIETIN; HAEMOPOIETIN): A hormone that is released from the kidney in response to low concentration of oxygen in the tissue. It is produced by the peritubular capillary endothelial cells in the kidney and liver. It speeds up the process of erythropoiesis and is the means by which the rate of red cell production is controlled. It also plays an important role in the brain's response to neuronal injury, as well as involved in the healing process of wounds.

ERYTHROSE: A four-carbon aldose sugar. In the phosphorylated form, erythrose 4-phosphate, is an intermediate in the Calvin cycle. In the shikimic acid pathway, the first reaction step is the joining of erythrose 4-phosphate with phosphoenolpyruvate to form a seven-carbon keto sugar acid.

ESAKI DIODE (TUNNEL DIODE): A type of semiconductor diode which is capable of very fast operation, well into the microwave frequency region, by using quantum mechanical effects. It was invented in August 1957 by Leo Esaki, Japanese physicist, when he developed a diode with electrical junction insulations of 10 billionths of a metre thick through which electrons could tunnel through. The diodes have a heavily doped p-n junction only some 10 nm (100 Å) wide. The heavy doping results in a broken bandgap, where conduction band electron states on the n-side are more or less aligned with valence band hole states on the p-side. He was jointly awarded the 1973 Nobel Prize in Physics with Brian Josephson, for discovering the electron tunnelling effect used in these diodes. Since the tunnel diode exhibits a negative incremental resistance with a rapid response, it is capable of serving as an active element for amplification, oscillation and switching in electronic circuits at high frequencies.

ESCALATOR: Electrically-powered moving stairs, driven by chain and sprocket and held in the correct plane by two tracks. It is usually inclined at 30° and limited to a total rise of up to 18 metres. As the steps approach the landing, they pass through a protective comb-like device. Escalators are used to carry pedestrians between floors in busy areas such as large stores, office buildings and underground railway stations.

ESCAPE KEY (ESCAPE): In computing, it is a key, usually identified as **Esc** or **esc** and found on most computer keyboards and used for any of various functions, such as to abort or cancel the current process or running program or to close a pop-up window.

ESCAPE SEQUENCE: In computing, it is a string of special characters that is started with the Escape key (Esc) and used to change the state of computers and their attached peripheral devices to perform a function or command. Examples of commonly used and available escape sequences used in programming languages and some text editors include \' for Single quote, \" for Double quote, \a for Audible bell, \b for Backspace and \n for Newline. For example, an escape sequence of Hello\nThis is an example of\nescape sequences, would be displayed by a computer as **"Hello"**. This is an example of escape sequences.

ESCAPE TRAJECTORY: The path a body must follow to escape a central force field, such as Earth's gravity.

ESCAPE VELOCITY (ESCAPE SPEED; PARABOLIC VELOCITY): The minimum speed needed by a space vehicle, rocket, etc., to escape from the gravitational field of the Earth, Moon or other celestial bodies or the minimum velocity with which an object must be projected for it to escape from the gravitational pull of a planetary body. It is the speed at which the kinetic energy plus the gravitational potential energy of an object is zero. This velocity can be calculated by equating the kinetic energy of the moving body to the work necessary to overcome the gravity at the surface of the parent. The escape velocity from the Earth is 11.2 km/s and that from the Moon is 2.370km/s. Earth retains an atmosphere because the escape velocity is considerably higher than the mean velocity of the gas molecules in its atmosphere.

ESCAPE-VELOCITY ORBIT: The orbit achieved around the Sun by a spacecraft escaping from Earth's gravity.

ESCAPEMENT: 1. A device in a clock or watch that controls the transmission of power from the spring or falling weight to the hands. It is usually based on a balance wheel or pendulum. It is found mainly in timepieces but may be used wherever oscillating motion is required. 2. A ratchet device such as the spacing mechanism of a typewriter that permits motion in one direction only in equal steps. 3. In instruments such as a piano, it is the mechanism that allows the hammer to clear the string after striking, so that the string can vibrate. 4. In civil engineering, it is an overflow channel.

ESCARPMENT (CUESTA): A large ridge created by the erosion of dipping sedimentary rocks. It has one steep side and one gently sloping side.

Escherichia coli (E. coli): A species of gram-negative aerobic flagellated bacteria in vertebrate intestine also occurring in freshwater bodies and soil; and existing as numerous strains, some of which are responsible for diarrhoea. It is the bacteria most widely used in microbiological and genetic research. Some types can cause food poisoning, but proper cooking of meat and washing of produce can prevent infection from contaminated food sources.

ESCRIBED CIRCLE (EXCIRCLE): A circle lying outside a triangle, but tangent to one side of the triangle and to the extensions of the other sides.

ESCULENT: Something edible, especially a plant.

ESERINE (PHYSOSTIGMINE): A crystalline alkaloid, with the molecular formula $C_{15}H_{21}N_3O_2$, derived from the calabar bean plant (*Physostigma venenosum*), that inhibits cholinesterase by covalently binding with it. It is used to treat glaucoma. It is also used in medicine as a miotic and cholinergic agent and to enhance memory in patients with Alzheimer's disease.

ESKER: A feature found in formerly glaciated areas consisting of a long steep walled narrow ridge often sinuous and sometimes branching. They are thought to be formed by the deposits of streams running through tunnels underneath melting stagnant ice.

ESPALIER: A method by which a fruit tree is trained, making its branches to grow flat against a wall or other support. Examples of plants suitable as espaliers are evergreens such as pear trees, loquat, fire thorn, sweet bay magnolia and upright yew.

ESSENTIAL: 1. In mathematics, of a singularity of a function of a complex variable, noting that the Laurent series at the point has an infinite number of terms with negative powers. 2. Of a discontinuity, noting that the function is discontinuous and has no limit at the point. 3. In logic, of a property, holding of its subject in every possible world, so that nothing could be that subject if it lacked that property. For example, if having the atomic number 79 is an essential property of gold, nothing can be gold unless it has that atomic number. 4. Of a substance such as amino acid that is required for normal functioning but cannot be synthesized by the body and therefore must be included in the diet.

ESSENTIAL AMINO ACIDS: Any of the naturally occurring amino acids that are indispensable for optimum animal growth that an organism is unable to synthesise in sufficient quantities and must therefore be obtained from the diet to maintain growth. They include arginine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine. All of these amino acids are available in proteins produced in the seeds of plants, but because plant sources are often weak in lysine and tryptophan, the diet must be supplemented with animal protein from meat, eggs and milk, which contain all the essential acids. An example of a disease caused by amino acid deficiency is Kwashiorkor, which occurs in children living in developing countries.

ESSENTIAL DOMAIN: The subset of a given universe of discourse on which a partial function, relation or predicate is defined. This term

is useful to distinguish the sense of domain in which it is correlative to range from that in which it is correlative to co-domain.

ESSENTIAL ELEMENT (DIETARY MINERAL): Any of the metals and salts that animals must take in with their food to remain healthy. For example, calcium is required to make bones and teeth strong and some amount of cobalt, zinc, copper and iron are needed to make some types of proteins and enzymes.

ESSENTIAL FATTY ACID (EFAs): A fatty acid which is needed in the human diet because the body does not produce any or enough of it. There are four main essential fatty acids, most of which are found in vegetable oils. They include alpha-linolenic acid, an omega-3 fatty acid and linoleic acid, an omega-6 fatty acid. Gamma-linolenic acid (omega-6), lauric acid (saturated fatty acid) and palmitoleic acid (monosaturated fatty acid) are also considered as conditionally essential. Deficiency can cause weight loss, dermatosis, irregular oestrus, etc.

ESSENTIAL OIL (ETHERAL OIL): Natural oil with a distinctive scent secreted by the glands of certain aromatic plants. Most essential oils are terpenoids, while some are derivatives of benzene. Many xerophytic plants produce essential oils as a means of reducing their transpiration rates. In hot conditions the evaporating oils contribute to the density of the boundary layer at the leaf surface. Examples are citrus oils, flower oils and oils of cloves. They are extracted from plants by steam and distillation, etc. Some essential oils repel insects, while others, such as those of *Myoporum deserti*, are toxic to grazing animals. A great number of chemical compounds have been identified from essential oils. They are used commercially for flavouring food and in perfumes; some have medicinal properties.

ESSENTIAL SINGULARITY: An isolated singularity of a complex function that is neither a removable singularity nor a pole. A function f , has an isolated singularity at infinity if $f(1/z)$ has one at the origin.

ESSENTIALLY BOUNDED: Of a measurable function, having the property that there exists a constant C such that the set $\{x: |f(x)| > C\}$ has zero measure. The infimum of such constants is called the **essential supremum** of $|f|$. For example, if $f(x) = 1/x$ for rational x and $f(x) = \sin x$ for all other real x , then $f(x)$ is essentially bounded.

ESTER: An organic compound such as ethyl acetate ($C_4H_8O_2$) or methyl methacrylate ($C_5H_8O_2$) formed by the reaction between an alcohol and a carboxylic acid or its derivative, with the elimination of small molecules such as water. It is usually catalysed by mineral acids. The reaction is known as **esterification**. Various kinds of esters are used in many everyday products. For example, amyl acetate, ethyl acetate and cyclohexanol acetate are used as solvents in lacquers; some types are used in the manufacture of drugs for conditions such as high blood pressure, leprosy, asthma and epilepsy. Other uses include perfumes, flavourings and essence. **Esterification** is a reversible reaction, thus many esterification reactions are equilibrium reactions and, therefore, need to be driven to completion according to Le Chatelier's principle that a change affecting a chemical equilibrium is offset by compensatory changes in other components of the equilibrium, thus producing little overall effect. It is among the simplest and most often performed organic transformations.

ESTIMATE: 1. Approximation or a guess about an answer to a numerical problem, usually with some thought or calculation involved. Rounding off a number to the nearest place value is a kind of estimation. 2. In statistics, to assign a value or a range of values derived from a sample to a population parameter. 3. Any statistic that is intended as an approximation to the correct value of a parameter of a distribution; in particular, a sample statistic.

ESTIMATE OF MEAN: Analysing grouped frequency distributions when the actual mean is not known and the midpoint of each class is used to estimate the mean. For example, the estimate of mean $(\sum fx) / (\sum f) = (\text{Sum of products of frequency} \times \text{midpoint for each class}) \div \text{total frequency}$.

ESTIMATION: In statistics, it is the procedure that is used to determine the value of a particular parameter associated with a

population. To estimate the parameter, a sample is drawn from the population and the value of the estimator for the unknown parameter is calculated. Estimation is divided into two large categories: point estimation and interval estimation.

ESTIMATOR: Any statistical function of a sample, which is used to estimate an unknown parameter of the population. Any value obtained from this estimator is an estimate of the parameter of the population.

ESTIPELLATE: Having no stipels.

ESTIPULATE: Having no stipules.

ESTIVATION/AESTIVATION: 1. In botany, also known as prefoliation, it is the arrangement of the parts of a flower bud especially of the sepals and petals before it has opened. It can be an important taxonomic diagnostic. 2. The opposite of hibernation, it is a state of animal dormancy in response to high temperatures and conditions, characterised by inactivity and a lowered metabolic rate.

ESTUARY: Ecological system including bays, streams and tidal areas influenced by brackish water or a river mouth widening into the sea, where fresh water mixes with salt water and tidal effects are felt. They are partly enclosed coastal body of water in which river water is mixed with seawater, defined by salinity rather than geography. They are rich in various life forms and provide breeding grounds for many species. Some of the oldest continuous civilisations such as the Nile delta have flourished in estuarine environments and cities such as London (River Thames), New York (Hudson River) and Montreal (St. Lawrence River) developed on estuaries.

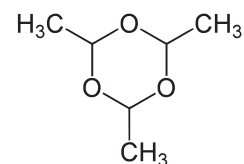
ETAERIO: A cluster of fruits formed from the infused carpel of a single flower. Examples are achenes (dandelions and sunflowers), drupes (apricots, plums, cherries and almonds) or follicles (larkspur and milkweed).

ETCHING: An engraving technique in which acid is used to put lines into a metal plate which is then printed, usually intaglio. The plate, usually copper or zinc, is first coated with a resin through which the design is drawn with a needle. Only the exposed metal is etched away. Different line thicknesses can be obtained by selective stopping out and repeated exposure to the acid.

ETHANAL (ACETALDEHYDE; ALDEHYDE; ACETIC ALDEHYDE; ETHYL ALDEHYDE): A colourless, inflammable liquid with the molecular formula C_2H_4O , density 0.788 g cm^{-3} , melting point -123.5°C (150 K) and a boiling point of 20.8°C (293.95 K). It is a member of the group of organic compounds called aldehydes and it is formed by the oxidation of ethanol or ethane. It is produced by plants as part of their normal metabolism and it occurs naturally in ripe fruits, coffee and bread. It is also produced by oxidation of ethanol. It is an important precursor to pyridine derivatives, pentaerythritol and crotonaldehyde. It combines with urea to give a useful resin.

ETHANAL TRIMER (PARALDEHYDE; 2,4,6-TRIMETHYL-1,3,5-TRIOXANE): A colourless flammable liquid with a pungent odour and unpleasant taste having the chemical formula $C_6H_{12}O_3$, molar mass $132.16 \text{ g mol}^{-1}$, melting point 12°C (285 K) and boiling point 124°C (397 K). It is produced by treating acetaldehyde with a small amount of sulfuric acid. It is slightly soluble in water and highly soluble in alcohol. It is used as a sedative and hypnotic drug (the safest hypnotic drug); in the manufacture of synthetic resins; as a preservative; and in preparing leather.

ETHANAMIDE (ACETAMIDE; ACETIC ACID AMIDE): A colourless solid, with the chemical formula CH_3CONH_2 , density 1.16 g/cm^3 , melting point $79-81^\circ \text{C}$ (352-354 K) and boiling point 222°C (495 K). It crystallises in the form of long white crystals. It has the characteristic smell of a mouse. It is made by the dehydration of ammonium ethanoate or by the action of ammonia on ethanol



Chemical Structure of Paraldehyde

chloride, ethanoic anhydride or ethyl ethanoate. It is used as a solvent and wetting agent and in lacquers and explosives.

ETHANE: A colourless, odourless gas at standard temperature and pressure with the chemical formula C_2H_6 , density $1.3562 \text{ mg cm}^{-3}$ (at 0°C , 273.15 K) and 0.5446 g cm^{-3} (at -89.15°C , 184 K), melting point -183°C (90.4 K) and boiling point -89°C (184.6 K). It is the second member of the alkane series of hydrocarbons (paraffins). It is isolated industrially from natural gas and as a by-product of petroleum refining. It is used as a plasticiser and as an industrial solvent, however, *N,N*-dimethylacetamide (DMA), which is a related compound is more widely used. It is also used as petrochemical feedstock for ethylene production. It is the only two-carbon alkane that is an aliphatic hydrocarbon.

ETHANE-1,2-DIOL (ETHYLENE GLYCOL; GLYCOL; ETHYLENE ALCOHOL; HYPODICARBOUS ACID; MONOETHYLENE GLYCOL): A colourless, odourless, syrupy, sweet-tasting and viscous hygroscopic liquid, with the molecular formula $C_2H_6O_2$, molar mass, 62.07 g mol^{-1} , density 1.1132 g/cm^3 , melting point -12.9°C (260 K) and boiling point 97.3°C (470 K). It is made by hydrolysis of epoxyethane from ethene and used as antifreeze and raw material for making polyesters. It is toxic and ingestion can result in death.

ETHANEDIOIC ACID (OXALIC ACID): A colourless poisonous dicarboxylic acid, with the molecular formula $H_2C_2O_4$, density 1.90 gcm^{-3} (anhydrous), 1.653 gcm^{-3} (dihydrate) and melting point $189-191^\circ\text{C}$ ($462-464 \text{ K}$). It typically occurs as a dihydrate with the chemical formula $C_2O_4H_2 \cdot 2H_2O$. It is obtained from plants or made synthetically. It is about 3,000 times stronger than acetic acid. Its conjugate base, known as oxalate ($C_2O_4^{2-}$), is a reducing agent as well as a chelating agent for metal cations. It is used for dyeing, bleaching and cleaning.

ETHANENITRILE (ACETONITRILE; METHYL CYANIDE; CYANOMETHANE; ETHYL NITRILE; METHANECARBONITRILE): A colourless, poisonous liquid, with the chemical formula CH_3CN , density 0.787 g/ml liquid, melting point -45°C (228 K) and boiling point 82°C (355 K). It is made by dehydrating ethanamide or from ammonia and ethyne. It is produced mainly as a by-product of acrylonitrile manufacture. It has a dipole moment of 3.84 D and it is a good polar solvent used for dissolving a wide range of ionic and nonpolar compounds when water cannot be used. In the laboratory, it is used as a medium-polarity solvent that is miscible with water and has a convenient liquid range. It is mainly used as a polar aprotic solvent in the purification of butadiene.

ETHANOATE (ACETATE): A negative ion ($C_2H_3O_2^-$), salt, ester or acylal derived from ethanoic (acetic) acid or any salt containing this ion. It is relevant in the biochemical synthesis of fats from carbohydrates in plants and animals and industrially. Metal acetate is used in printing, vinyl acetate in plastic production, cellulose acetate in photographic films and textiles and volatile organic ester as a solvent.

ETHANOIC ACID (ACETIC ACID; METHANE CARBOXYLIC ACID): A liquid carboxylic acid with the chemical formula CH_3CO_2H , density 1.049 g cm^{-3} , melting point $16-17^\circ\text{C}$ ($289-290 \text{ K}$) and boiling point $118-119^\circ\text{C}$ ($391-392 \text{ K}$), whose salts and esters are acetates. In its pure or glacial state, it is a clear, syrupy, corrosive liquid that mixes readily with water. It occurs naturally as a metabolic intermediate in body fluids and plant juices. Its dilute solution, from fermentation and oxidation of natural products is vinegar. Industrial production is either synthetic, from acetylene; or biological, from ethanol. Industrial chemicals made from it are used in printing and as plastics, photographic films, textiles and solvents.

ETHANOIC ANHYDRIDE (ACETIC ANHYDRIDE): A colourless liquid with a sharp, pungent odour. Its molecular formula is $C_4H_6O_3$, boiling point 139.8°C (412.9 K) and a melting point of -73.1°C (200.1 K). It is produced by carbonylation of methyl acetate. It is mainly used in converting cellulose to cellulose acetate, a coating material used in the production of plastics, photographic film and fabrics. It is also used in the production of aspirin (acetylsalicylic acid) and as a wood preservative. It can be safely used to substitute ethanoyl (acetyl) group (CH_3CO-) into organic compounds.

ETHANOL (ALCOHOL; GRAIN ALCOHOL; FERMENTATION ALCOHOL; ETHYL ALCOHOL; ETHYL HYDROXIDE; METHYLCARBINOL): A colourless, limpid, volatile, flammable, toxic liquid with a pleasant odour and a pungent taste and the molecular formula C_2H_5OH , relative density 0.7851 at 20°C , melting point at -112.3°C (160.85 K) and boiling point at 78.4°C (351.55 K). It is soluble in water and most organic liquids. It is produced by chemical synthesis and by fermentation or biosynthetic processes. It is the alcohol found in alcoholic drinks such as beer, wine and spirits. Apart from being a key input into alcoholic beverages, it is used as a solvent, extractant, antifreeze and intermediate in the synthesis of innumerable organic chemicals.

ETHANOLAMINE (2-AMINOETHANOL, MONOETHANOLAMINE; β -AMINOETHANOL; β -HYDROXYETHYLAMINE; β -AMINOETHYL ALCOHOL; GLYCINOL; OLAMINE): A toxic, flammable, corrosive, colourless, viscous liquid with ammonia-like odour and the chemical formula $NH_2(CH_2)_2OH$, density 1.012 g/cm^3 , melting point 10.3°C (283 K) and boiling point 170°C (443 K). It is an organic chemical compound that is both a primary amine, due to an amino group in its molecule and a primary alcohol, due to a hydroxyl group. The term also refers to any of three low melting hygroscopic bases which absorb water readily to form viscous liquids. It is the second-most-abundant head group for phospholipids, substances found in biological membranes. It is used in the purification of petroleum, as a solvent in dry cleaning and as an ingredient in paints and pharmaceuticals.

ETHANOYL CHLORIDE (ACETYL CHLORIDE; ACYL CHLORIDE): A colourless pungent liquid acid chloride with the chemical formula CH_3COCl , density 1.104 g/ml (liquid), melting point -112°C (161 K) and boiling point 52°C (325 K). It is derived from acetic acid. It is made by reacting ethanoic acid with a halogenating agent such as phosphorus(III) chloride, phosphorus(V) chloride or sulfur dichloride oxide. It is used to introduce ethanoyl groups into organic compounds containing $-OH$, $-NH_2$ and $-SH$ groups. It does not exist in nature, because contact with water would hydrolyse it into acetic acid and hydrogen chloride.

ETHANOYLATING AGENT (ACETYLATED AGENT): A chemical reagent used to introduce an ethanoyl group ($-COCH_3$) instead of hydrogen in an organic compound. Examples include ethanoic anhydride and ethanoyl chloride (acetyl chloride, CH_3COCl).

ETHENE (ETHYLENE): A colourless, flammable, gaseous hydrocarbon with the molecular formula C_2H_4 which is derived from natural gas and petroleum. It is the first member of the alkene series of hydrocarbons. It is made by cracking hydrocarbons from petroleum and is a major raw material for making other organic chemicals such as ethanal, ethanol and ethane-1,2-diol. It can be polymerised to polyethene. It occurs naturally in plants, in which it acts as a growth substance. Its best-known effect is the stimulation of fruit ripening. Fruits such as bananas, apple and avocados naturally produce ethylene during the later stages of their ripening and ethylene gas is used to induce ripening in fruits, which are picked green. It also suppresses flowering, except in members of the pineapple family, where flowering may be synchronised by releasing ethylene into the growing plant. It is used as a source of many organic compounds, in welding and cutting metals, to colour citrus fruits and as an anaesthetic.

ETHER: Any of a series of organic chemical compounds having an oxygen atom linked groups, for example, an oxygen atom bonded to two carbon atoms. It is produced by the reaction of concentrated sulfuric acid and ethanol. Another example is dimethyl ether (CH_3-O-CH_3). One of the most important ethers is diethyl ether ($C_2H_5OC_2H_5$). This is used as an anaesthetic and as an external cleansing agent before surgical operations. It is also used as a solvent and in the extraction of oils, fats, waxes, resin, alkaloids and for substances which do not dissolve in water.

ETHEREAL OIL PRODUCING GLANDS: Aromatic compound producing glands found on various parts of a plant, without definitely known functions.

ETHERNET: The global standard (IEEE 802.3) method of connecting computers to a local area network (LAN) using coaxial or twisted pair cables with protocols to control the passing of information and to avoid simultaneous transmission by two or more systems. The Ethernet port can be used to connect a computer to another computer, a local network or an external DSL or cable modem. Ethernet can handle about 100 GB/s and can be used with almost any kind of computer.

ETHICAL HACKING: Hacking performed to help a company or individual identify potential threats on the computer or network. Ethical hackers use the same methods as unethical hackers, but report problems instead of taking advantage of them to improve the system security, in an effort to minimise, if not eliminate, any potential unethical hacker attacks.

ETHMOID BONE: An irregularly shaped bone of the skull which lies between the orbital plates of the frontal bone and anterior to the sphenoid bone. The ethmoid articulates (joins) with the sphenoid, frontal, maxillary, lacrimal and palatine bones, the inferior nasal concha and the vomer. It also enters into the formation of the anterior cranial fossa, the orbits and the nasal cavity.

ETHOGRAM: In animal behaviour, it is a catalogue or table consisting of a detailed record of an animal's normal behaviour over a long period of time.

ETHOLOGY: The comparative study of the behaviour of living organisms or animals in their natural setting. It was first outlined in the 1920s by Konrad Lorenz and Nikolaas Tinbergen. Ethologists study natural processes that range across all animal groups, including courtship, mating and self-defence. Ethologists regard the naturalistic approach as very important because it reveals the environmental and social circumstances in which the behaviour originally evolved and prepares the way for more realistically designed laboratory experiments.

ETHOXYETHANE (DIETHYL ETHER; ETHER; ETHYL OXIDE; 3-OXAPENTANE): A colourless flammable volatile liquid with a low boiling point and a characteristic odour. Its chemical formula is $C_2H_5OC_2H_5$, density 0.7134 g/cm³ (liquid), melting point -116.3 °C (156.9 K) and boiling point 34.6 °C (307.8 K). It is sparingly soluble in water (6.9 g/100 mL). It is the most common member of a class of chemical compounds known generically as ethers. It is an anaesthetic and useful organic solvent.

ETHYL 3-OXOBUTANOATE (ETHYL ACETOACETONATE; ACETOACETIC ACID ETHYL ESTER; ETHYL ACETYLACETATE 3-OXOBUTANOIC ACID ETHYL ESTER): A colourless liquid ester with a pleasant odour and the molecular formula $C_6H_{10}O_3$, density 1.021 g/cm³ (liquid), melting point, -45 °C (228 K) and boiling point 180.8 °C (454 K). It can be prepared by reacting ethyl ethanoate ($CH_3COOC_2H_5$) with sodium or sodium ethoxide. The compound shows keto-enol tautomerism and contains about 7% of the enol form, $CH_3C(OH):CHCOOC_2H_5$, under normal conditions. It is mainly used as a chemical intermediate in the production of compounds, such as amino acids, analgesics, antibiotics, antimalarial agents, antipyrene and aminopyrine and vitamin B₁. It is used as a flavouring for food; it is also used in the manufacture of dyes, inks, lacquers, perfumes, plastics and yellow paint pigments.

ETHYL ALCOHOL (ETHANOL; ALCOHOL; GRAIN ALCOHOL; FERMENTATION ALCOHOL; ETHYL HYDROXIDE; METHYLCARBINOL): A colourless, limpid, volatile, flammable, toxic liquid with a pleasant odour and a pungent taste and the molecular formula C_2H_5OH , relative density 0.7851 at 20 °C, melting point at -112.3 °C (160.85) and boiling point at 78.4 °C (351.55

K). It is soluble in water and most organic liquids. It is produced by chemical synthesis and by fermentation or biosynthetic processes. It is the alcohol found in alcoholic drinks such as beer, wine and spirits. Apart from being a key input into alcoholic beverages, it is used as a solvent, extractant, antifreeze and intermediate in the synthesis of innumerable organic chemicals.

ETHYL BROMIDE (BROMOETHANE): A colourless flammable liquid with the chemical formula C_2H_5Br , density 1.47 g/ml and boiling point 39 °C (312.15 K). It is prepared from ethane and hydrogen bromide and used as a refrigerant.

ETHYL ETHANOATE (ETHYL ACETATE; ETHYL ESTER; ACETIC ESTER; ESTER OF ETHANOL): A colourless liquid with a characteristic sweet smell, with the molecular formula $C_4H_8O_2$, density 0.897 g/cm³, melting point -83.6 °C (190 K) and boiling point 77.1 °C (350 K). It is produced by reacting ethanol and ethanoic acid and used as a solvent in glues and paints.

ETHYL GROUP: The organic group, with the chemical formula CH_3CH_2 , derived from ethane (C_2H_6), often abbreviated -Et. Ethyl is the IUPAC organic chemistry nomenclature for an alkane or alkyl molecule, using the prefix "eth-" to indicate the presence of two carbon atoms in the molecule.

ETHYLAMINE (ETHAMINE; MONOETHYLAMINE; AMINOETHANE; 1-AMINOETHANE): A colourless, flammable and volatile liquid, with the chemical formula $C_2H_5NH_2$, density 0.689 g/cm³, melting point -81 °C (192 K) and boiling point 16.6 °C (290 K). It is a primary amine made by reacting chloroethane with ammonia. It is used in petroleum refining, in detergents and in organic synthesis.

ETHYLBENZENE (ETHYLBENZOL; EB; PHENYLETHANE): A colourless flammable liquid, with the chemical formula $C_6H_5CH_2CH_3$, density 0.8665 g/ml, melting point -95 °C (178 K) and boiling point 136 °C (409 K). It is produced commercially by combining benzene and ethylene in an acid-catalysed chemical reaction:

$C_6H_6 + C_2H_4 \rightarrow C_6H_5CH_2CH_3$. It is an aromatic hydrocarbon, whose greatest use (over 99 percent) is as an intermediate in the production of styrene monomer. It is also used as a solvent for coatings and in the production of synthetic rubber and cellulose acetate.

ETHYLENE (ETHENE): A colourless, flammable, gaseous hydrocarbon, with the molecular formula C_2H_4 . It is derived from natural gas and petroleum. It is the first member of the alkene series of hydrocarbons. It is made by cracking hydrocarbons from petroleum and is a major raw material for making other organic chemicals such as ethanal, ethanol, ethane-1,2-diol. It can be polymerised to polyethene. It occurs naturally in plants, in which it acts as a growth substance. Its best-known effect is the stimulation of fruit ripening. Fruits such as bananas, apple and avocados naturally produce ethylene during the later stages of their ripening and ethylene gas is used to induce ripening in fruits picked green. It also suppresses flowering, except in members of the pineapple family, where flowering may be synchronised by releasing ethylene into the growing plant. It is used as a source of many organic compounds, in welding and cutting metals, to colour citrus fruits and as an anaesthetic.

ETHYLENE CHLOROXYDRIN (2-CHLOROETHANOL; ETHYLENE CHLORHYDRIN): A chlorinated derivative of ethylene glycol, with the chemical formula CH_2ClCH_2OH , used to break bud dormancy.

ETHYLENEDIAMINE TETRA-ACETIC ACID (EDTA): A compound that reversibly binds iron, magnesium and other positive ions (it acts as a chelating agent). If used in culture media complexed with iron it slowly releases iron into the medium as required. It may also act as a noncompetitive inhibitor of certain enzymes that require metal ions as cofactors.

ETHYNE (ACETYLENE): A colourless gas or chemical compound that is an unsaturated hydrocarbon as well as the simplest alkyne, with the molecular formula C_2H_2 , density 1.097 kg m⁻³, melting point -80.8 °C

(192.4 K - triple point and boiling point -75.0 °C (98 K). It is made by adding water to solid calcium carbide. It burns in pure oxygen with a flame temperature of over 3000 °C (3273 K), hot enough to weld and cut steel. It is unstable in the pure form, thus it is usually handled as a solution. It is unsaturated because its two carbon atoms are bonded together in a triple bond. The carbon-carbon triple bond puts all of its atoms in the same straight line, with CCH bond angles of 180° and since it is a linear symmetrical molecule, it possesses the $D_{\infty h}$ point group. It is widely used as a fuel and a chemical building block.

ETIOPLAST: A microscopic membranous sac with no pigment that occurs inside the cells of plants that are deprived of light. If they are exposed to light they change into chloroplasts.

EUATMOTIC REACTION: An isothermal, reversible reaction between two (or more) solid phases during the heating of a system as a result of which a single vapour phase is produced.

EUBACTERIA (BACTERIA): The true bacteria, which have unicellular cells with rigid peptidoglycan cell wall. They are prokaryotic and some possess flagella for movement. They show a wide variety of biochemical diversity as a group and their numerous members are found in virtually every habitat on Earth. Besides the Archaea and the Eukarya, they are one of the three main domains of life. Even though they are responsible for many human diseases, they also help maintain health, as well as making an important aspect of all ecosystems on Earth.

EUBACTERIALES: An order containing easily recognised gram-positive spherical or rod-shaped forms. Some are motile with numerous flagella. In some classifications it is considered an order of Schizomycetes.

EUCALYPTUS: A genus of trees and some shrubs of the Myrtaceae family, which includes over 800 species and forms the preponderant vegetation and most valuable timber trees of western Australian forests. The wood is especially strong, dense and heavy because of the high proportion of tracheids and fibres. Eucalyptus species produce gum and oils, such as lemon scented gum and Eucalyptus citriodora (lemon-scented gum, similar to but smaller than the spotted gum and having lemon-scented leaves). The oil obtained by distillation of the leaves of some species is used medically.

EUCAMPTODROMOUS: It defines leaves having a single primary vein, the secondary veins curved upward and gradually diminishing distally within the margin and interconnected by a series of cross-veins without forming conspicuous marginal loops.

EUCARPIC: Denoting a fungus in which the thallus is differentiated into vegetative and reproductive region.

EUCHROMATIN: An expanded region of chromatin in the interphase nucleus, which stains lightly with basic dyes. Metabolically active nuclei show large amounts of euchromatin. Autoradiographic investigations support the view that messenger RNA is transcribed in the euchromatic regions, but the amount of euchromatin is related to the number of genes being transcribed rather than the amount of mRNA produced.

EUCLID: Greek mathematician, who lived around third-century B.C. He wrote the book, Elements, in which he set the deductive principles of logic to geometry, thereby deriving statements from clearly defined axioms. Euclid's book, Element, influenced mathematicians and scientists for over two thousand years.

EUCLID NUMBERS: The even perfect numbers of the form $E_n = p_n \# + 1$, where $p_n \#$ is the product of the first n primes. An Euclid number can not be a square. This is implied by Euclid numbers always being congruent to 3 mod 4. The first few Euclid numbers are 3, 7, 31, 211, 2311, 30031, 510511.

EUCLID'S AXIOM: Any of five axioms that are generally recognised as the basis for Euclidean geometry, as listed: (i) a straight line can be drawn between any two points; (ii) any terminated straight line can be projected indefinitely; (iii) a circle with any radius can be drawn around any point; (iv) all right angles are equal; and (v) only one line can be drawn through a point parallel to another line.

EUCLID'S POSTULATES: The five postulates, which together with 23 definitions and five "common notions" form the basis of Euclid's great work on geometry. The postulates are: (i) a straight line may be drawn from any one point to any other point; (ii) a finite straight line may be produced to any length in a straight line; (iii) a circle may be described with any centre at any distance from that centre; (iv) all right angles are equal; and (v) if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which the angles are less than the two right angles. The last postulate is not as obvious as the other four and Euclid himself was reluctant to use it. Later mathematicians, finding the fifth postulate to be complicated, thought it might be possible to derive it from the other four. However, they only succeeded in replacing it with equivalent statements. The most common of these is the parallel postulate.

EUCLIDEAN: Of, derived from or relating to Euclidean geometry.

EUCLIDEAN ALGORITHM (EUCLID'S ALGORITHM): A method of finding the highest common factor of two positive integers, in which the smaller number is divided into the larger. The remainder is then divided into the smaller number, obtaining a second remainder. This second remainder is then divided into the first remainder, to give a third remainder. This is divided into the second and so on, until a zero remainder is obtained. The remainder preceding this is the highest common factor of the two numbers. For example, to find the highest common factor of 53 and 77 by using Euclidean algorithm, the solution is $77 = (1)(53) + 24$, $53 = (2)(24) + 5$, $24 = (4)(5) + 4$, $5 = (1)(4) + 1$, $4 = (1)(4)$.

EUCLIDEAN CONSTRUCTION (COMPASS-AND-STRAIGHTEDGE CONSTRUCTION): The geometric constructions of figures and lengths using only a compass and a straight edge or ruler, for example, bisecting a line using only a compass. The straight edge is used only to draw lines and not for measurement.

EUCLIDEAN DISTANCE (CARTESIAN DISTANCE): The standard distance in Euclidean space, R^n . Geometrically, it is the length of a line segment connecting two points. It is calculated as the square root of the sum of the squares of the arithmetical differences of the corresponding coordinates of the two points. The distance between points x and y in a Euclidean space R^n is given by

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}, \text{ where } x = (x_1, x_2, \dots, x_n) \text{ and } y = (y_1, y_2, \dots, y_n).$$

In two-dimensional Euclidean space, the distance between points $A = (a_1, a_2)$ and $B = (b_1, b_2)$ is given

by the Pythagorean theorem $\sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$. In n dimensions,

the Euclidean distance between two points p and q is $\sqrt{\sum_{i=1}^n (p_i - q_i)^2}$,

where p_i or q_i is the coordinate of p or q in dimension i .

EUCLIDEAN DOMAIN (EUCLIDEAN RING): An integral domain on which a Euclidean valuation can be defined. Any two elements of a Euclidean domain have a greatest common divisor, which can be computed using the Euclidean algorithm. Examples of Euclidean domains are the ring Z and the polynomial rings over any field with the gauge being the degree of the polynomial.

EUCLIDEAN GEOMETRY: A term describing an ordinary three-dimensional geometry in which only one line parallel to another given line may pass through a given point. It was named after Euclid, a Greek mathematician, based on his principles, as described in his book, "Elements".

EUCLIDEAN NORM: The norm imposed on a vector by taking the square root of the sum of the squares of its entries. The standard Euclidean norm on R^n is defined by $||x|| = \sqrt{x \cdot x}$.

EUCLIDEAN POINT SPACE: The set, R^n of n -tuples over the reals (for n a positive integer); an n -dimensional real affine manifold.

EUCLIDEAN SPACE (EUCLIDEAN PLANE; CARTESIAN PLANE; CARTESIAN SPACE): A two or more dimensional space possessing a scalar product, so that a Euclidean distance may be imposed. In other words, it is a set of points, along with a set of neighbourhoods for each point, that satisfy a set of axioms relating points and neighbourhoods. It is used to denote the position of an element in the system of two or more dimensions. The number of dimensions in Euclidean space is denoted by n . The n -dimensional space denoting real coordinates is represented by R^n or E^n . A point that is defined in a two-dimensional space, is represented by (x, y) and in three-dimensions by (x, y, z) ; where, x, y, z are the distances from x -, y - and z - axes respectively. One-dimensional Euclidean space is represented by a real number line; while Euclidean space of two dimensions is composed of Cartesian plane with two mutually perpendicular lines.

EUCLIDEAN TOPOLOGY: The topology on Euclidean n -space R^n , where n belongs to the set of natural numbers. Euclidean topology states that the subsets of R^n are open sets and is supposed to follow the following axioms: (i) An open subset is a subset containing an open circle or ball surrounding all its points; (ii) union and intersection of open sets are open sets themselves; and (iii) A metric topology in which the distance within R^n is given by the function:

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2} = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

EUCLIDEAN VECTOR SPACE: The vector space R^n of n -tuples over the reals (for n a positive integer).

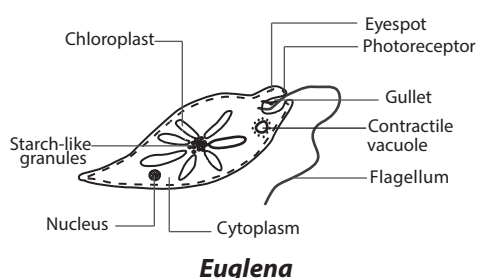
EUDICOT (TRICOLPATES): Any dicotyledonous flowering plant whose pollen grain has three apertures through which the pollen tube emerges during pollination. A large number of familiar plants are eudicots and examples include forget-me-not, cabbage, apple, dandelion, buttercup, maple and macadamia. The term literally means "true dicotyledons" and it contains most of the plants that have been regarded as dicotyledons and have real dicotyledonous features.

EUDIOMETER: A laboratory apparatus for measuring changes in the volume of gases during chemical reactions.

EUGENICS: The study of factors that influence the hereditary qualities of future generations and methods of improving the quality of human populations by the application of genetic principles. It may involve control and improved selective breeding by increasing the number of children produced by persons, who are, by some criteria considered to be superior and by reducing the number produced by people, who are physically or mentally deficient. Efforts to enhance larger families from superior parents are called **positive eugenics**, while attempts to reduce the offspring from defective parents are called **negative eugenics**.

EUGENOL: Colourless or yellowish aromatic liquid, with the chemical formula $C_{10}H_{12}O_2$. It has a spicy odour and taste and is soluble in organic solvents. It is extracted from clove oil and used in flavours, perfumes, medicines and in the manufacture of vanilla, etc.

EUGLENA: A unicellular, spindle shaped, unflagellated alga, which has both plant and animal features or any of various minute single-celled freshwater organisms of the genus *Euglena*. Their main feature is the presence of chlorophyll, a reddish eyespot and a single anterior flagellum.



EUGLENIDA: A class of mostly unicellular protists such as euglena belonging to the phylum Euglenophyta and moving by

means of undulipodia. They are mostly photosynthetic but others feed by phagocytosis or strictly by diffusion. They inhabit fresh water, especially when it is rich in organic materials, with a few marine and endosymbiotic members.

EUGLENOPHYTA: A division (phylum) of the kingdom Protista, containing mostly unicellular aquatic algae including euglena that in certain respects resemble animals more than they do plants. They lack cell walls, but the outer part of the cell consists of a firm but flexible layer called a pellicle or periplast, which cannot properly be considered a cell wall. Reproduction occurs by longitudinal cell division.

EUKARYA: A super-kingdom containing all eukaryotic organisms, embracing protists, fungi, plants and animals. It is one in the three-domain system of biological classification; the others are Archaea and Bacteria.

EUKARYOTE (EUCARYOTE): One of the two major groupings into which all organisms are divided. They are organisms made of one or more cells, each of which contains a clearly defined nucleus enclosed by a membrane, along with organelles (small, self-contained, cellular parts that perform specific functions). The organelles include mitochondria, chloroplasts, a Golgi apparatus, an endoplasmic reticulum and lysosomes. All organisms, except bacteria and archaea (blue algae) are eukaryotes. Bacteria and archaea are prokaryotes.

EUKARYOTIC CELL: A cell that contains membrane-bound compartments in which specific metabolic activities take place. Most important among these compartments is the nucleus, which houses the eukaryotic cell's DNA. Eukaryotic organisms also have other specialised, membrane-bounded structures, called organelles, which are small structures within cells that perform dedicated functions. Eukaryotic cells are typically 10 to 100 micrometres across or about 10 times the size of prokaryotic cells.

EULER CHARACTERISTIC: 1. One of the important quantities, known as a **topological invariant** that describes the shape or structure of topological space. Leonhard Euler (1707–1783), a Swiss mathematician originally formulated the characteristic for polyhedra and used it to prove various theorems about them, including the classification of the Platonic solids. For the special case of polyhedra, the Euler characteristic equals 2. If F is the number of faces of a polygon, E the number of edges and V the number of vertices, Euler's formula can be written as $F - E + V = 2$. For example, if the surface of a cube has six (square) faces, twelve edges and eight vertices, therefore, $6 - 12 + 8 = 2$.

EULER LINE: The Euler line of a triangle is the line joining the centroid and the circumcentre. It passes through the orthocentre, circumcentre, centroid and the centre of the nine-point circle of the triangle.

EULER MULTIPLIER (INTEGRATING FACTOR): A function, $m(x, y)$ by which an ordinary differential equation can be multiplied in order to make it integrable. Its form is $y'f(x, y) - g(x, y) = 0$.

EULER PATH: A path along a connected graph that connects all the vertices and traverses every edge of the graph only once. Note that a vertex with an odd degree allows one to travel through it and return by another path at least once, while a vertex with an even degree only allows a number of traversals through, but one cannot end an Euler path at a vertex with even degree. Thus, a connected graph has an Euler path which is a circuit (an Euler circuit) if all of its vertices have even degree. A connected graph has an Euler path which is non-circuitous, if it has exactly two vertices with odd degree.

EULER PHI FUNCTION (EULER'S TOTIENT FUNCTION; TOTIENT FUNCTION): The number theoretic function, written $\phi(n)$, that counts the number of distinct positive integers less than or equal to n that are relatively prime to n . ϕ is multiplicative and so determined by its value on prime powers, for which $\phi(p^{k+1}) = p^k(p - 1)$.

EULER SQUARE (GRAECO-LATIN SQUARES; GRAECO-ROMAN SQUARES; LATIN-GRAECO SQUARES): A square array made by combining n objects of two types such that the first and second

elements form a Latin square. For many years, Euler squares were known to exist for $n = 3, 4$ and for every odd n except $n = 3k$. Euler's **Graeco-Roman squares conjecture** maintained that there are no Euler squares of order $n = 4k + 2$ for $k = 1, 2, \dots$. However, such squares were found to exist in 1959 by Bose and Shrikande, refuting the conjecture.

EULER SUMMATION FORMULA (EULER-MACLAURIN SUMMATION FORMULA): A formula used in the numerical evaluation of integrals, which states that the value of an integral is equal to the sum of the value given by the trapezoidal rule and a series of terms involving the odd-numbered derivatives of the function at the end points of the interval over which the integral is evaluated.

In other words, it gives an estimation of the sum $\sum_{i=0}^n f(i)$ through the integral $\int_0^n f(t)dt$ with an error term given by an integral which involves Bernoulli numbers.

EULER-BERNOULLI LAW: Named after Swiss mathematicians, Leonhard Euler (1707-1783) and Daniel Bernoulli (1700-1782), it states that the torque on a thin elastic beam under gravity is Elk , where E is Young's modulus and I is the moment of inertia of the cross-section about an axis through the centre of mass and perpendicular to the plane of the couple.

EULER-LAGRANGE EQUATIONS: Any of the fundamental necessary conditions in the calculus of variations. For the simplest problems this requires that the extremal y_0 that $\int_b^a f(y, y', x) dx$ minimises the functional should satisfy Euler's equation at any point where y_0' is smooth.

EULER-RIEMANN ZETA FUNCTION (RIEMANN-ZETA FUNCTION; ZETA FUNCTION): In mathematics, it is the special function, denoted $\zeta(s)$ that is defined over the complex plane for one complex variable, s . The sum of the infinite series $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, is

infinite when $s = 1$; when s is greater than 1, it converges to a finite number as successive terms are added; if s is less than 1, the sum is again infinite. It is a very important function in number theory.

EULER, LEONHARD (1707-1783): Swiss mathematician; and the second most prolific mathematician in history in terms of publications, after Paul Erdős. His greatest contributions were to number theory, but Euler also did important work in calculus, geometry, algebra, probability, acoustics, optics, mechanics, astronomy, artillery, navigation and finance. Euler also worked on magic squares and the problem of the knight's tour.

EULER'S CIRCLE (NINE POINT CIRCLE; FEUERBACH'S CIRCLE; NINE-POINT CIRCLE; TERQUEM'S CIRCLE): The circle that passes through nine significant midpoints in a triangle. The points are: the midpoint of each side of the triangle, the foot of each altitude and the midpoint of the line segment from each vertex of the triangle to the orthocentre, where the three altitudes meet.

EULER'S CONSTANT (EULER-MASCHERONI CONSTANT): The constant 0.5772157 ..., usually denoted by the Greek letter γ that is the limit as n tends to infinity of the difference between the sum of n terms of the harmonic series and the natural logarithm of n . It is not known whether this number is algebraic or transcendental, nor even whether it is rational or irrational.

EULER'S EQUATION: A linear equation of the form $(ax + b)^n y'' + A_1(ax + b)^{n-1} y' + \dots + A_{n-1}(ax + b)y + A_n y = f(x)$,

where a, b, A_1 are constants. Putting $ax + b = e^t$, then $y' = ae^{-t} \frac{dy}{dt}$,

$$y'' = a^2 e^{-2t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right), \quad y''' = a^3 e^{-3t} \left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + \frac{2dy}{dt} \right), \text{ etc.}$$

Euler's equation is therefore, transformed into a linear equation with constant coefficients. For the homogeneous Euler equation, $x^n y'' + A_1 x^{(n-1)} y' + \dots + A_{n-1} x y' + y = 0$, in the form $y = x^k$, the solution may be found, if k is a real root of the characteristic equation multiplicity m , then to its corresponding m linearly independent solution. The exponent k can be found if $y, y', \dots, y^{(n)}$ put into the equation to get a characteristic equation.

EULER'S EQUATIONS OF MOTION (EULER'S EQUATIONS):

A set of three differential equations expressing relations between the force moments, angular velocities and angular accelerations of a rotating rigid body with a fixed point in terms of its principal axes. They are as follows: (i) $I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 = M_1$, (ii) $I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_3 \omega_1 = M_2$, (iii) $I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 = M_3$, where M_1, M_2 and M_3 are the components of the applied torques, I_1, I_2 and I_3 are the principal moments of inertia and ω_1, ω_2 and ω_3 are the components of the angular velocity about the principal axes.

EULER'S FORMULA FOR POLYHEDRAL: An equation in topology, which states that if F is the number of faces of a polygon, E the number of edges and V the number of vertices, Euler's formula can be written as $F - E + V = 2$. For example, a cube has six square faces, twelve edges and eight vertices, therefore, $6 - 12 + 8 = 2$. The quantity $F - E + V$ is known as the **Euler characteristic**.

EULER'S LAWS OF MOTION: Two laws of motion, which extend Newton's laws of motion for point particle to rigid body motion, which state as follows: (i) the first law states that the force acting on a body is equal to the rate of change of its linear momentum; and (ii) the second law states that the torque on any body is equal to the rate of change of its angular momentum.

EULER'S METHOD: It provides a way to approximate the solution to a differential equation. The goal is to calculate the function $y = f(x)$. The differential equation allows the calculation of $\frac{dy}{dx} = f'(x)$ at each point. If an initial condition is known, $(x_0, y_0 = f(x_0))$, then calculate $\frac{dy}{dx}$ at that point and calculate the new point $(x_0 + h, f(x_0 + h))$, $f(x_0 + h) \approx f(x_0) + f'(x_0)h$. A smaller value of h will lead to a more accurate approximation, but it will require more work to calculate the curve.

EULER'S NUMBER (e, NAPIER'S CONSTANT): The irrational number named after the Swiss mathematician Leonhard Euler and defined as the unique real number such that the value of the derivative (slope of the tangent line) of the function $f(x) = e^x$ at the point $x = 0$ is equal to 1. It has the value 2.718 28. It is used as the base of natural logarithms and occurs in the exponential function, e^x .

EULERIAN ANGLES: Three angles θ, Φ and Ψ introduced by Leonhard Euler to describe the orientation of a rigid body about a fixed point O by locating a set of Cartesian axes, OXVZ, fixed in the body in terms of Cartesian axes, Oxyz, fixed in space. The angles θ and Φ are the spherical coordinates of OZ with respect to Oxyz and Ψ is the angle between the planes OXZ and OzZ.

EULERIAN CHAIN (EULER CHAIN; EULER TRAIL): A chain in a graph that uses each edge only once.

EULERIAN CIRCUIT: A circuit in a graph that uses each edge exactly once.

EULERIAN DESCRIPTION (SPATIAL DESCRIPTION): The description of physical phenomena associated with the deformation of a body in terms of fields defined over the current rather than the reference.

EULERIAN FLUID DYNAMICS: A formulation of fluid mechanism in which the velocity of the fluid at fixed position is analysed or a

formulation that describes changes at fixed positions in the fluid, in contrast to Lagrangian fluid dynamism.

EULERIAN STRAIN RATE: The symmetric part of velocity gradient; that is, the Eulerian strain rate $\sum = \frac{1}{2}(L + L^T)$, where L is the velocity gradient.

EULERIAN WALK (TRAIL): A walk (an alternating sequence of edges and vertices) in a graph in which all edges are distinct.

EUMETAZOA: Members of the sub-kingdom that includes all animals except sponges.

EUMYCOTA: A division or phylum containing the true fungi as distinct from slime moulds. Most have a well-defined cell wall that is composed of chitin and glucans. Reproductive bodies are highly variable in morphology and size and may be asexual or sexual. They are heterotrophic, with absorptive nutrition, capable of utilising insoluble food from outside the cell by secretion of digestive enzymes and absorption.

EUPHOTIC ZONE (EPIPELAGIC ZONE; PHOTIC ZONE): The topmost layer of a lake or sea in which there is sufficient light for photosynthesis or net primary production to occur. Its thickness depends on the extent of light attenuation in the water column but typical euphotic depths vary from a few centimetres in highly turbid eutrophic lakes to about 200 metres in the open ocean. The depth may be influenced by seasonal turbidity and it is generally proportional to the level of primary productivity that occurs in that area of the ocean. About 90% of all marine life lives in this region. The transparency of the water, which determines the depth of the zone, is measured with a Secchi disk or photometer.

EUPLOID: A nucleus, cell or organism that has an exact multiple of haploid number of chromosomes for the species.

EUPHORBIACEAE: Also called the **spurge family**, it is a large family of succulent flowering plants which are herbs, shrubs or trees; occurring mostly in the tropics, especially the Indo-Malayan region and tropical America, and also in tropical Africa. Plants in the family include medicinal plants such as *Ricinus communis*, the castor bean or castor oil plant and croton oil plant (*Croton tiglium*), whose seeds are used for preparing croton oil. Examples of some other plants in the family include caoutchouc, cassava (*Manihot esculenta*), rubber tree (*Hevea brasiliensis*), and croton (*Codiaeum variegatum*). *Alchornea cordifolia* with the common names christmas bush or dovewood is another medicinal plant in the family that is effective in the treatment of diseases such as gonorrhoea, white (leucorrhoea), piles, breast cancer, weak bones, tiredness, diarrhoea and checking miscarriage. The leaves are usually prepared as an infusion or chewed fresh; but the stem and roots are also used.

EUREKA (CONSTANTAN): An alloy having an electrical resistance that varies only very slightly with temperature over a limited range around normal room temperatures. It consists of copper (50–60%) and nickel (40–50%) and is used in resistance wire, thermocouples, etc.

EUROPA: The fourth largest of the 39 known moons or natural satellites of the planet Jupiter. Data collected in 2002 by the space probe Galileo indicates a liquid-water ocean, similar to oceans found on Earth and possibly capable of supporting simple life forms beneath its frozen surface.

EUROPEAN SUBTRACTION: A subtraction algorithm in which the subtrahend is increased when regrouping is necessary. The algorithm is commonly used in Europe and in certain parts of the United States.

EUROPIUM: A silvery-white, soft, rare-earth metallic element, with the chemical symbol Eu, atomic number 63, relative atomic mass

151.96, relative density 5.259, melting point 826 °C (1099.2 K) and boiling point 1439 °C (1712 K). It occurs in monazite and bastnaesite. It is used to dope lasers and to absorb neutrons in research. It is also used as a phosphor in televisions. Its compounds are used to make control rods for nuclear reactors.

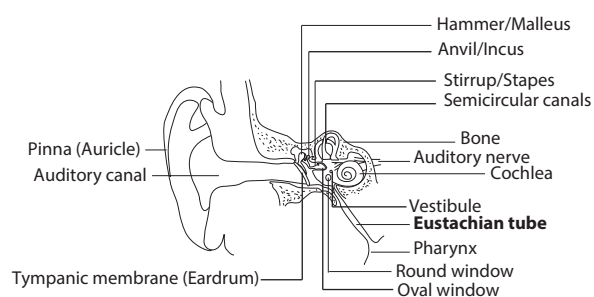
EURYPHAGIC: A term describing an animal that uses a wide range of food types. Although they have a wide food range of food types, some of them have their preferences, for example, the North African catfish is euryphagic, but it feeds predominantly on fish. An euryphage is such an animal.

EUSOCIALITY: A form of social life found in insects such as termites belonging to the order Isoptera; and honey bees, ants and wasps belonging to the order Hymenoptera. Eusocial colony is made up of special castes, such as sterile workers, drones, soldiers and reproductives, whose membership is biologically determined. Their castes can manifest in the appearance of individuals within a group, whose behaviour and sometimes anatomy is modified for the specific role they play. The only mammalian examples are the naked mole rat and the Damaraland mole rat. The analysis of eusociality is a major factor in the development of theories in sociobiology.

EUSPORANGIATE: Having the sporangium develop from a great amount of leaf tissue as opposed to only one or a few cells.

EUSPORANGIUM: A type of sporangium that is derived from several initials and typical of ferns in the *Marattiales* and *Ophioglossales*. There are several layers of cells in its wall and it usually produces a large number of spores.

EUSTACHIAN TUBE (AUDITORY TUBE; PHARYNGOTYMPANIC TUBE): A small air-filled canal connecting the middle ear to the back of the throat or the pharynx. It is found in all land vertebrates and equalises the pressure on both sides of the ear drum. It is about 35 mm long in human adults.



Eustachian tube of the human ear

EUSTATIC CHANGE (EUSTATIC SEA-LEVEL CHANGE): A global rise or fall in sea level caused by a change in the amount of water in the oceans. The sea level at any location changes constantly with changes in tides, atmospheric pressure and wind conditions, however, longer-term changes are influenced by changes in the Earth's climates. In view of this, the level is better defined as mean sea level, which is the height of the sea surface averaged over all stages of the tide over a long period of time.

EUSTELE: A dissected siphonostele with phloem only to the outside of the xylem.

EUTECTIC MIXTURE: A mixture of two or more substances that has a melting point lower than that of any of its components. Eutectic alloys are fusible, upon cooling they convert from liquids to intimately mixed solids. Examples are eutectic alloys for soldering, composed of

tin (Sn), lead (Pb) and sometimes silver (Ag) or gold (Au); and sodium chloride and water spread on roads to aid snow removal or mixed with ice to produce low temperatures, for example, in traditional ice cream making.

EUTECTIC REACTION: An isothermal, reversible reaction between two (or more) solid phases during the heating of a system, as a result of which a single liquid phase is produced.

EUTHANASIA: The act of painlessly ending the life of a person or animal who has a painful, incurable disease or incapacitating disorder in order to prevent further suffering. It was legalised in The Netherlands in 2001 and in Belgium in 2002.

EUTHERIA (PLACENTALIA): A subclass of mammals in which the embryos are retained in an uterus in the mother's body and nourished by the placenta. They are distinguished from non-eutherians by various features of the feet, ankles, jaws and teeth. One of the key variations between placental and non-placental eutherians is that the former do not have epipubic bones, which are present in all other fossil and living mammals.

EUTHERIAN MAMMALS (PLACENTAL MAMMALS): Mammals, which bear live young nourished before birth in the mother's uterus through the placenta, a specialised embryonic organ attached to the uterus wall. All mammals are placental, except monotremes and marsupials.

EUTROPHICATION: Excessive enrichment of rivers, lakes and shallow sea areas, primarily by nitrate fertilisers washed from the soil by rain, by phosphates from fertilisers and detergents in municipal sewage and by sewage itself. These encourage the growth of algae and bacteria which use up the oxygen in the water, thereby making it uninhabitable for fishes and other animal life. The process causes the death or disappearance of most species of organisms in a polluted river or lake. Other species such as the Nomura Jellyfish in Japanese waters may experience an increase in population that negatively affects other species in the direct ecosystem. Diversion of sewage, better utilisation of manure, erosion control, improved sewage treatment and harvesting of the surplus aquatic crops can reduce or stop eutrophication. **Eutrophic** describes the deterioration of life-supporting qualities of a body of water. For example, eutrophic lake is a highly productive lake supporting proliferation of plant life, especially algae, which reduces the dissolved oxygen content and often causes the extinction of other organisms.

EVAGINATION: Turning inside out or the release of the contents of membranaceous vesicles to the outside.

EVALUATE: To calculate a numerical value for a mathematical expression. For example, to evaluate $y = x^2$ at $x = 3$ is to calculate the value of 3^2 .

EVANESCENT: Fading or passing away like a vapour; remaining for a very short time only; disappearing early.

EVAPORATION: The physical process by which a liquid substance is converted to a gas or vapour. This may occur at or below the normal boiling point of the liquid (the temperature at which a liquid boils at 1 atmosphere pressure) and the process is endothermic. Evaporation, mostly from the sea and from vegetation, replenishes the humidity of the air. It is an important part of the exchange of energy in the Earth-atmosphere system that produces atmospheric motion and therefore weather and climate. The rate of evaporation depends on the temperature difference between the evaporating surface and the air, the relative humidity and wind. Generally the lower the boiling point, the faster the rate of evaporation of the liquid. **Evaporate** is to change a liquid into vapour, usually by heating to below its boiling point.

EVAPORATION RATE: The rate at which a particular substance will evaporate when compared to the rate of a known substance such as ethyl ether.

EVAPORATING DISH (EVAPORATING BASIN): It is a piece of laboratory equipment made of materials such as porcelain, ceramic or borosilicate glass that can withstand higher temperatures without breaking. It is used for heating a substance or a solution, using a Bunsen burner flame in order to remove the part that is more easily evaporated so that it is concentrated. It comes in various shapes and sizes.

EVAPORATIVE COOLING: 1. Cooling of a substance as a result of evaporation. 2. Lowering the temperature of a large mass of liquid by utilising the latent heat of vapourisation of a portion of the liquid.

EVAPORITE: The sedimentary deposit precipitated on evaporation of salt water. Examples are gypsum and rock salt. Often, evaporite deposits occur in closed sea basins, where evaporation is greater than the inflow, with the main minerals being anhydrite, halite, calcite, gypsum, polyhalite, potassium and magnesium salts such as sylvite, carnallite, kainite and kieserite.

EVAPOTRANSPIRATION: The combined processes of evaporation and transpiration by which water is transferred from the Earth's surface to the atmosphere. It is a key factor in the water cycle. Water or ice is evaporated by the Sun's rays and by the wind and plants are cooled by transpiration.

EVEN CHANCE: The same chance of an event happening as the chance of it not happening, with a probability of 0.5.

EVEN FUNCTION: A function $f(x)$ is said to be an even function, if $f(-x) = f(x)$ for all x . The value does not change if the sign of the independent variable is reversed. Examples are $f(-x) = x^2$, $f(x) = x^4$ and $f(x) = x^6$.

EVEN NUMBER: Numbers that can be divided exactly by 2 without a remainder. Examples are 2, 4, 6, 8, ..., $2n$.

EVEN PERMUTATION: A permutation obtainable from an even number of two-element swaps; a permutation with permutation symbol equal to +1. For example, (3 1 2) is derived from (1 2 3) by exchanging 3 and 1 and then 3 and 2, but (1 3 2) requires an odd number of such pairwise exchanges. For initial set {1,2,3,4}, the twelve even permutations are those with zero swaps: {1,2,3,4}; and those with two swaps: {1,3,4,2}, {1,4,2,3}, {2,1,4,3}, {2,3,1,4}, {2,4,3,1}, {3,1,2,4}, {3,2,4,1}, {3,4,1,2}, {4,1,3,2}, {4,2,1,3}, {4,3,2,1}. For a set of n elements and $n > 2$, there are $n!/2$ even permutations, which is the same as the number of odd permutations.

EVEN-ELECTRON ION: An ion containing no unpaired electrons, for example CH_3^+ in its ground state.

EVEN-EVEN NUCLEUS: An atomic nucleus containing an even number of protons and an even number of neutrons.

EVEN-ODD NUCLEUS: An atomic nucleus containing an even number of protons and an odd number of neutrons.

EVEN-PINNATE: In botany, it defines innate leaflets that are symmetrically paired along the rachis; or a pinnate leaf with a pair of leaflets at the apex.

EVENT: 1. In statistics, occurrence, final result or a possible outcome. Events can be any of the following: (i) **independent**, in which case each event is not affected by other events, for example, tossing a coin three times and coming up each time with "tails" each time; (ii) **dependent** also known as **conditional**, in which an event is affected by other events, for example, if a card is chosen at random from a standard deck of 52 playing cards and without replacing it, a second card is chosen, the outcome of choosing the first card has affected the outcome of choosing the second card; and (iii) **mutually exclusive** events that do not happen at the same time, for example,

if a 6-sided die is rolled once, rolling an odd number and rolling an even number cannot occur at the same time. 2. An occurrence defined in the theory of relativity as a single point in space-time or a phenomenon located at a single point in space-time, regarded as the fundamental observational entity in relativity theory. 3. In computer science, it is an occurrence or happening of significance to a computer program, such as user actions like the clicking of a mouse button, the completion of a write operation to a disk or pressing a key or system occurrences, such as running out of memory. Most modern applications, particularly those that run in Macintosh and Windows environments, are said to be event-driven, because they are designed to respond to events. 4. In information technology (IT), it is any detectable or discernable occurrence that has significance for the management of the IT infrastructure or the delivery of IT service and evaluation of the impact a deviation might cause to the services.

EVENT HORIZON: In general relativity, it is a boundary in spacetime beyond which events cannot affect an outside observer or the boundary of a region of space-time from which it is not possible to escape to infinity. The most common case of an event horizon is that surrounding a black hole. Light emitted from beyond the horizon can never reach the observer. Likewise, any object approaching the horizon from the observer's side appears to slow down and never quite pass through the horizon, with its image becoming more and more redshifted as time elapses. The travelling object, however, experiences no strange effects and does, in fact, pass through the horizon in a finite amount of proper time.

EVENT-DRIVEN: In computing, it is a system such as a graphical user interface that detects events as they occur, generated by the user (e.g., by clicking a mouse button or pressing keys) or the system (e.g., programs that run under graphical user interfaces such as the Macintosh operating system or Microsoft Windows).

EVERGREEN: A term describing a plant that bears leaves throughout the year, with each leaf being shed independently of the other after two or three years. Evergreen leaves are either small or waxy to cut down water loss and to withstand freezing conditions. In the tropics most trees are evergreen and nearly all have broad leaves. Examples of evergreen plants are needle-leaved trees such as pine, fir, juniper, spruce, certain broad-leaved shrubs such as rhododendron and laurel and broad-leaved trees such as magnolia.

EVIDENCE: An indication of something happening or being the case as it is. For example, the evidence that a line is straight is that the angles on the line add up to 180°.

EVISCERATION: The removal of the viscera or the internal organs or entrails of a person or an animal.

EVOCATION: The ability of experimental stimuli such as chemicals or tissue in plants to cause unspecialised embryonic tissue to develop into specialised tissue.

EVOLUTE: The locus of the centres of curvature of all the points on a given curve or the envelope of the normals to a curve. The original curve is an involute of its evolute.

EVOLUTION: 1. In mathematics, an algebraic operation in which a root of a number or expression is extracted. 2. A theory which explains the gradual process by which the present diversity of plant and animal life arose from the earliest and most primitive organisms which are believed to have been continuing for at least the past 3,000 million years. It may result in the formation of new species or extinction of others. This theory was first proposed in the nineteenth century by Charles Darwin, according to which the Earth's species have changed and diversified through time under the influence of natural selection. Life on Earth is thought to have evolved in three stages. It started with

chemical evolution, in which organic molecules were formed. This was followed by the development of single cells capable of reproducing themselves. This stage led to the development of complex organisms capable of sexual reproduction. Although debates continue over the precise mechanisms involved in the process, evolution is generally accepted as a fact by most scientists.

EVOLUTIONARY GAME THEORY (EGT): The study of the behaviour of large populations of agents who repeatedly engage in strategic interactions. In biology, EGT applies the principles of game theory to evolving populations of lifeforms using strategies that involve changes in behaviour driven either by natural selection through differences in birth and death rates or by the application of certain decision rules by individual agents. EGT is now of great interest to economists, sociologists, anthropologists and philosophers. Evolutionary game theory has been used to explain many aspects of human behaviour such as altruism, human culture, moral behaviour and social learning.

EVOLUTIONARY SPECIES CONCEPT: The idea that evolutionary lineages and ecological roles can form the basis of species identification.

EVOLUTIONARY STABLE STRATEGY (ESS): A strategy of a population, which if adopted by all members of a population in a given environment is resistant to replacement by any alternative strategy, for example, by natural selection through differences in birth and death rates. Strategy in this case consists of behavioural or physical characters of an individual.

EVOLUTIONARY SYNTHESIS (MODERN SYNTHESIS; NEO-DARWINISM; NEW SYNTHESIS): A consolidation of the results of various lines of investigation from the 1920s through the 1950s that supported and reconciled the Darwinian theory of evolution and the Mendelian laws of inheritance in terms of natural selection acting on genetic variation. It is a union of ideas from several biological specialties which provides a widely accepted account of evolution.

EVOLUTIONARY TOXICOLOGY: The study of the effects of environmental pollution on natural populations that are chronically exposed to it. It has proved effective in studies representing a diversity of organisms and ecological settings. Sources of environmental toxicity include pesticides such as DDT and sulfuric fluoride. A polluted habitat may cause behavioural, biochemical, physiological and developmental stress responses in organisms that are chronically exposed to it. An example is industrial melanism, acquired by some animals and insects such as the peppered moth that live in an industrial region, where the environment is soot-darkened evolve to be darker and therefore better camouflaged against predators such as birds, increasing the probability that they will survive and reproduce because of natural selection of the lighter, more conspicuous animals.

EVOLUTIONARY TREE (PHYLOGENETIC TREE): A diagram showing the evolutionary history of organisms based on differences in amino acid sequences or the hypothesised genealogical ties and sequence of evolutionary relationships linking individual organisms, populations or taxa. Organisms with fewer differences are placed closer together while those with more differences are further apart.

EVOLVED GAS ANALYSIS (EGA): A technique in which the nature and/or amount of volatile product(s) released by a substance subjected to a controlled temperature programme is determined. The method of analysis should always be clearly stated.

EVOLVED GAS DETECTION (EGD): A technique in which the evolution of gas from a substance is detected as a function of temperature while the substance is subjected to a controlled temperature programme.

EXA-: A prefix used in the metric system to denote 10^{18} , with symbol E. **Exabyte** is a unit of information or computer storage equal to 10^{18} bytes. The unit is so large that they are rarely used in a practical context.

EXACT: 1. Also known as **accurate**, strictly correct, precise, without any approximation. 2. Of a differential equation, obtained by putting the exact differential of a function equal to zero, so that when written in the form $y'g(x, y) - h(x, y) = 0$, the left-hand side is the derivative of some $F(x, y)$. This holds if and only if the coefficients of the differentials in each variable are continuously differentiable and satisfy the Integrability condition $g_x = h$. The possibility exists that one can usefully obtain exactness by multiplying both g and h by the Integrating factor $m(x, y)$. 3. Of a differential form, being the derivative of another form. This forces the form to be closed and if the region on which it is defined as simply connected then this form is also exact.

EXACT SEQUENCE: A sequence of mappings between objects such that the kernel of each mapping is the range of the previous one. A **short exact sequence** is a five-term exact sequence of which the initial and terminal objects are trivial. In other words one has $g: X \rightarrow Y$ and $h: Y \rightarrow Z$ such that g is a monomorphism while h is an epimorphism whose kernel is the range of g .

EXALBUMINOUS SEED: A seed that lacks an endosperm at maturity. In such seeds the cotyledons absorb the food reserves from the endosperm and act as storage organs.

EXAPTATION: A morphological or physiological character that predisposes an organism to adapt to a changed environment or lifestyle. It is common in both anatomy and behaviour. For example, bird feathers initially evolved for temperature regulation, but later were adapted for flight. Interest in this concept relates to both the process and product of evolution: the process that creates complex traits and the product that may be imperfectly designed.

EXARCH: A term used to describe xylem maturation in which the older cells (protoxylem) are further from the centre of the axis than the younger cells. Development is thus centripetal.

EXCAVATES: An assemblage of eukaryotic protists characterised by one or more feeding grooves equipped with endulipodia (flagella) that propel food to the mouth. They are a major kingdom of unicellular eukaryotes, often known as Excavata and they include the diplomads, trichomonads, jakobids and other gut parasites or symbionts. There are a variety of free-living and symbiotic forms. It also includes some important parasites of humans.

EXCEL (MICROSOFT EXCEL; MS EXCEL): A spreadsheet program for Windows computers, which is part of the Microsoft Office suite, which includes other productivity programs, such as Word and PowerPoint. Microsoft Excel is a good program for users and businesses that deal with large amounts of data. According to Microsoft, Excel can support up to 1 million rows by 16,000 columns.

EXCENTRE: The centre of an excircle or of the escribed circle of a given triangle.

EXCESS ELECTRON: An electron in a semiconductor that is not required in the bonding system of the crystal lattice and has been donated by an impurity atom.

EXCESSIVE DRAINAGE: The rapid removal of water from soils, either by percolation or by surface flow. The loss is greater than that necessary to prevent the development of an anaerobic condition for an appreciable length of time. Excessive water loss can rob soils of their nutrients by carrying them away to other locations.

EXCHANGE ACIDITY (EXCHANGEABLE ACIDITY): The titratable hydrogen and aluminum that can be replaced from the adsorption

complex by a neutral salt solution. It is usually expressed as milliequivalents per 100 g of soil (meq/g soil).

EXCHANGE CAPACITY: The total ionic charge of the adsorption complex that is active in the adsorption of ions.

EXCHANGE EXTRACTION: An extraction operation or process in which a metal from one phase is exchanged with the equivalent amount of a second metal from the other phase.

EXCHANGE FORCE: A force resulting from the continued interchange of particles in a manner that bonds their hosts together.

EXCHANGE TRANSFUSION: A blood transfusion procedure in which blood is removed through a catheter and replaced with an IV transfusion of blood obtained from a donor. This is carried out on a newly born baby or adult, who may have haemolytic disease such as sickle cell anaemia or severe jaundice.

EXCHANGE-INVERSION TRANSITION: A transition between antiferromagnetic and ferromagnetic coupling between two subarrays of magnetic atoms.

EXCHANGEABLE CATION PERCENTAGE: The extent to which the adsorption complex of a soil is occupied by a particular cation. It is expressed as:
$$\text{ECP} = \frac{\text{exchangeable cation (meq / 100g soil)}}{\text{cation exchange capacity (meq / 100g soil)}} \times 100.$$

EXCHANGEABLE PHOSPHATE: The phosphate anion reversibly attached to the surface of the solid phase of the soil in such form that it may go into solution by anionic equilibrium reactions with isotopes of phosphorus or with other anions of the liquid phase without solution of the colloid phase to which it was attached.

EXCHANGEABLE POTASSIUM: The potassium that is held by the adsorption complex of the soil and is easily exchanged with the cation of neutral non-potassium salt solutions.

EXCHANGEABLE SODIUM PERCENTAGE: The extent to which the adsorption complex of a soil is occupied by sodium. It is expressed as:
$$\text{ESP} = \frac{\text{exchangeable sodium (meq / 100g soil)}}{\text{cation exchange capacity (meq / 100g soil)}} \times 100.$$

EXCIMER: A short-lived dimeric or heterodimeric molecule formed from two species, in which one of the two bound atoms is in a higher energy state. Excimers are often diatomic and are formed between two atoms or molecules that would not bond if both were in the ground state. The lifetime of an excimer is very short, in the order of nanoseconds. Binding of a larger number of excited atoms form Rydberg matter clusters, the lifetime of which can exceed many seconds.

EXCIPIENT: A more or less inert substance added to a drug to give suitable consistency or form to the drug.

EXCIPLEX: A combination of two different atoms that exists only in an excited state. An example is a complex formed by the interaction of an excited molecular entity with a ground state counterpart of a different structure. When an exciplex emits a photon of electromagnetic radiation, it dissociates immediately into the atoms, rather than reverting to the ground state.

EXCIRCLE (ESCRIBED CIRCLE): A circle lying outside a triangle, tangent to one of its sides and tangent to the extensions of the other two. The **inscribed circle** of a triangle is the largest circle contained in the triangle, which touches or is tangent to the three sides. The centre of the incircle is called the triangle's **incentre**.

EXCISION: 1. Also known as **resection**, it is the complete removal of an organ, tissue or tumour from a body, as opposed to a biopsy. 2. The process by which a genetic element is removed from a strand of deoxyribonucleic acid.

EXCISION REPAIR: A form of DNA repair found in living cells in which damaged or mismatched bases in one of the two strands of

DNA are cut out and replaced with the correct bases using the other strand as a template. The various types include base excision repair, nucleotide excision repair and mismatch repair.

EXCITATION: 1. The process of applying current to the winding of an electromagnet as in an electric motor or the process of applying a signal to the base of a transistor or the control electrode of a thermionic valve. 2. A process in which a nucleus, electron, atom, ion or molecule acquires energy that raises it to a quantum state higher than that of its ground state or the addition of a discrete amount of energy to a system that changes it usually from a state of lowest energy (ground state) to one of higher energy (excited state). For example, in a hydrogen atom, excitation energy of 10.2 electron volts is required to move the lone electron from its ground state to its first excited state. The excitation energy stored in excited atoms and nuclei is usually emitted as ultraviolet radiation from atoms and as gamma radiation from nuclei as they return to their ground states.

EXCITATION ENERGY: The amount of energy a nucleus has above the stable ground state energy of the atom.

EXCITATION LEVEL: An excitation level (exciton) is an electrically neutral X-ray level with an expelled electron bound in the field of a core electron vacancy.

EXCITATION SPECTRUM: A plot of the spectral radiant exitance or of the spectral photon exitance against the frequency (or wavenumber or wavelength) of excitation. When corrected for wavelength dependent variations in the excitation radiant power this is called a **corrected excitation spectrum**.

EXCITATORY POSTSYNAPTIC POTENTIAL (EPSP): The electric potential that is generated in a postsynaptic neuron during the transmission of a nerve impulse or the temporary depolarisation of postsynaptic membrane potential caused by the flow of positively charged ions into the postsynaptic cell as a result of opening of ligand-sensitive channels. It can also result from a decrease in outgoing positive charges, while inhibitory postsynaptic potential (IPSPs) are sometimes caused by an increase in positive charge outflow.

EXCITED STATE: An energy-rich state of an atom or a molecule, produced when the species have absorbed energy and become elevated to a higher energy state as compared with the ground state. The excitation may be electronic, vibrational, rotational, etc. Excited states tend to have short lifetimes. They lose energy either through collisions or by emitting photons to go back to their ground states.

EXCITEMENT POTENTIAL: A term generally used in connection with electron excitation, however, it refers to the difference in potential between an excited atomic or molecular state and the ground state. It can also be applied to excited molecular vibrational and rotational state.

EXCITON: An electron-hole pair in a crystal that is bound in a manner analogous to the electron and proton of a hydrogen atom. It behaves like an atomic excitation that passes from one atom to another and may be long-lived. It can form when a photon enters a semiconductor, exciting an electron from the valence band into the conduction band, leaving a localised hole of opposite electric charge behind, to which the electron is attracted by the Coulomb force. It can be regarded as an elementary excitation of condensed matter which can transport energy without transporting net electric charge. Exciton activity is very crucial in semiconductors.

EXCLUDED MIDDLE LAW (LAW OF EXCLUDED MIDDLE; PRINCIPLE OF EXCLUDED MIDDLE): A law in (two-valued) logic which states that there is no third alternative to truth or falsehood. In other words, for any statement A, either A or not-A must be true and the other must be false. This law no longer holds in three-valued logic, in which "undecided" is a valid state, nor does it hold in fuzzy logic.

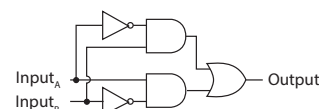
EXCLUDED VOLUME OF A MACROMOLECULE: The volume from which a macromolecule in a dilute solution of a macromolecule effectively excludes all other macromolecules. The excluded volume of a macromolecule depends on the Gibbs and Helmholtz energies of mixing of solvent and polymer, i.e. on the thermodynamic quality of the solvent and is not a measure of the geometrical volume of that macromolecule.

EXCLUDED VOLUME OF A SEGMENT: The volume from which a segment of a macromolecule in solution effectively excludes all other segments, i.e. those belonging to the same macromolecule as well as those belonging to other macromolecules. The excluded volume of a segment depends on the Gibbs and Helmholtz energies of mixing of solvent and polymer, i.e. on the thermodynamic quality of the solvent and is not a measure of the geometrical volume of that segment.

EXCLUSION CHROMATOGRAPHY: Chromatography in which molecular size in solution are separated by their size and in some cases molecular weight. The term **size-exclusion chromatography** may also be used when separation is based on molecular size. The term **ion-exclusion chromatography** is specifically used for the separation of ions in an aqueous phase.

EXCLUSION PRINCIPLE: A quantum mechanical principle in physics formulated by the Austrian physicist, Wolfgang Pauli in 1925, which states that no two electrons in a single atom may have the same set of quantum numbers. Thus, it is impossible to pack together certain elementary particles such as electrons beyond a certain critical density; otherwise they would share the same location and quantum number. In its simplest form for electrons in a single atom, it states that no two electrons can have the same four quantum numbers, that is, if n , l and m_l are the same, m_s must be different such that the electrons have opposite spins.

EXCLUSIVE OR (XOR; EXCLUSIVE DISJUNCTION): A Boolean logic operation for two inputs with the output true only if the two inputs are different; if the two inputs are the same, the output is false. The diagrams show exclusive or (XOR) equivalent circuit and logic gate truth table. In computer science, it is a search filter meaning A or B but not both.



**Exclusive Or (XOR)
Equivalent Circuit And
Logic Gate Truth Table**

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Logic Gate Truth Table

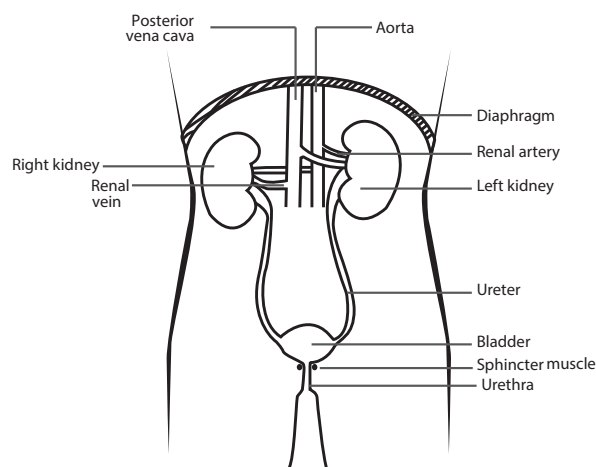
EXCRESCENCE: 1. An abnormal, distinct outgrowth on a human, animal body or a plant. 2. An abnormal projection or growth.

EXCRETA: Liquid and solid waste including faeces, urine and sweat passed from the body.

EXCRETION: The removal of poisonous or waste substances produced in metabolism from the cells of living organisms. In plants and simple animals, this is by diffusion but in higher animals this is by specialised organs. In single-celled organisms, waste products are discharged directly through the surface of the cell. Multicellular organisms utilise more complex excretory methods. Higher plants eliminate gases through the stomata or pores, on the surface of leaves. Animals have special excretory organs such as the kidneys, sweat glands, lungs and liver.

EXCRETION BY SINGLE-CELLED AQUATIC ANIMALS: Single-celled organisms such as amoeba, paramecium, bacteria, yeast and even human cells all excrete molecular waste through a process called exocytosis. In exocytosis little vesicles or membrane-bound collections are transported to the cell's outer membrane.

EXCRETORY SYSTEM: Organs which remove metabolic waste products from the body so as to help maintain homeostasis within the organism and prevent damage to the body. In mammals these organs include the kidneys, sweat gland in the skin, lungs and the liver.



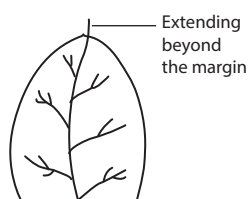
*Drawing of a mammalian(human)
excretory system*

EXCURRENT: 1. In botany, of veins, extending beyond the margin of the leaf, as shown in the diagram. 2. Flowing or running in an outward direction extending beyond what is typical, such as a leaf base that extends down the stem. 3. Having an undivided main stem or trunk, as seen in the spruce and other conifers.

EXCURVED: Curving away from the centre.

EXECUTABLE FILE: A file that is used to perform various functions or operations on a computer. Unlike a data file, an executable file cannot be read because it has been compiled. The two primary types of executable files are compiled programs and scripts. Compiled programs have an .EXE file extension and are often referred to as "EXE files" in Windows systems. On Macintosh computers, compiled programs have an .APP extension, which is short for application. Both types of executable files have been compiled from source code into binary machine code that is directly executable by the central processing unit (CPU). However, EXE files only run in Windows, while APP files only run in Mac OS X. This is because the code is executed by the operating system and therefore must be compiled in format that the operating system can understand. Uncompiled executable files are often referred to as scripts. These files are saved in a plain text format, rather than a binary format. In other words, you can open a script file and view the code in a text editor. Since scripts do not contain executable machine code, they require an interpreter to run. For example, a PHP file can execute code only when run through a PHP interpreter. If a PHP interpreter is not available, the PHP script can only be opened as a text file.

EXERCISE: Any physical or bodily exertion which contributes to good health, as for example, jogging. Evidence suggests that regular



Excurrent leaf

exercise reduces the risk of heart disease, probably as a result of improved blood circulation.

EXERGONIC REACTION: A chemical reaction that takes place with a negative change in free energy or a spontaneous biochemical reaction that releases energy.

EXFOLIATED GRAPHITE: The product of very rapid heating (or flash heating) of graphite intercalation compounds, such as graphite hydrogensulfate of relatively large particle diameter (flakes). The vapourising intercalated substances force the graphite layers apart. The exfoliated graphite assumes an accordion-like shape with an apparent volume often hundreds of times that of the original graphite flakes. Exfoliated graphite is usually prepared from well-crystallised natural flake graphite. It is used for the production of graphite foils. Exfoliated graphite is different from the deflagration product of graphite oxide (graphitic acid).

EXFOLIATION: 1. A process of weathering in which rock flakes off in small pieces. It is caused by large differences in day and night temperatures, as occurs in hot deserts. During the day the rock heats up and expands; during the night it cools and contracts. Eventually the rock is weakened by the repeated expansion and contraction and pieces flake off. The presence of dew or other water hastens the process. 2. In botany, it refers to the removal or loss of leaves from a plant as a natural part of a plant's life cycle such as that of deciduous trees or because of some trauma or outside cause such as disease or physical forces such as strong winds. It also refers to the natural process and condition of the bark peeling-away from a tree trunk, typically in large pieces that remain partially attached to the trunk until such time as they are completely detached by the elements or the eventual and subsequent exfoliation of additional layers of bark. Examples of trees with exfoliating bark are the paperbark maple and various species of Plane (Sycamore) and birch.

EXHALATION (EXPIRATION): A process by which gas is expelled from the bronchial tubes, through the airways, to the external environment during breathing. Exhaled air is rich in carbon dioxide, a waste product of cellular respiration during the production of energy, which is stored in adenosine triphosphate (ATP). In mammals, the volume of the thoracic cavity is reduced by contraction of the internal intercostals muscles while the external intercostal muscles relax. The volume of the thoracic cavity decreases and the pressure in it increases. The ribs and sternum move down and inwards; at the same time there is relaxation of the muscles of the diaphragm and the diaphragm returns to its dome-shape. During forced exhalation, expiratory muscles including the abdominal muscles and internal intercostal muscles generate abdominal and thoracic pressure, which forces air out of the lungs.

EXHAUST GAS (FLUE GAS): A mixture of gases continuously given out by an engine as it burns fuel such as natural gas, gasoline/petrol, diesel fuel, fuel oil or coal. For example, the exhaust gases of such engines contain mostly carbon dioxide, water vapour, smaller amounts of carbon monoxides, oxides of nitrogen, sulfur and unburned hydrocarbons that cause air pollution.

EXHAUST STREAM: The stream of gaseous, atomic or radiant particles emitted from the nozzle of a reaction engine.

EXHAUST VELOCITY: The average actual velocity at which exhaust material leaves the nozzle of a rocket engine. In practice, this differs from the effective exhaust velocity which takes into account other factors and is more useful for calculating quantities such as thrust.

EXHAUSTIVE: Of a set of elements of some domain, covering the whole domain, having a union equal to the whole domain. For example, the set of residue classes modulo n constitute an exhaustive covering of the integers.

EXINE: The outer covering of pollen grains, often containing sporopollenin, an acid-resistant polysaccharide that allows pollen grains to become fossils.

EXISTENTIAL GENERALISATION (EXISTENTIAL INTRODUCTION): In logic, the introduction rule for the existential quantifier that permits the inference $(\exists x)(Fx)$ from any instance Fa ; for example, from the statement that Montmorency is a unicorn, it can be validly inferred that unicorns exist.

EXISTENTIAL INSTANTIATION (EXISTENTIAL ELIMINATION): In logic, the elimination rule for the existential quantifier that licenses the inference of a conclusion from an existential premise, when the same conclusion can be derived from an instance without using any other premises about the subject of that instance.

EXISTENTIAL QUANTIFIER: The backward letter of E, $\exists$ used to represent that the open sentence that follows is true of at least one member of the relevant universe; for example, the expression "There exists at least one x such that $x^2 = x$ " can be written with the symbols: $\exists x (x^2 = x)$.

EXIT PUPIL: The minimum diameter of the light beam leaving an eyepiece through which all of the light from the eyepiece passes. It can be calculated in either of two ways: (i) by dividing the diameter of the telescope's primary objective by the magnification; or (ii) by dividing the focal length of the eyepiece by the focal ratio (f-number) of the telescope. For example, a telescope with a 20-cm primary mirror using a magnification 38x results in an exit pupil of 5.3 mm. The observer's eye should be located at the exit pupil to see the full and brightest field of view.

EXITANCE (IRRADIANCE; RADIANT EMITTANCE; RADIANT EXITANCE): Formerly called emittance, it is the radiant or luminous flux emitted per unit area of a surface. The radiant exitance (M_e) with the symbol M is measured in watt per square metre (Wm^{-2}), while the luminous exitance (M_v) is measured in lumens per square metre (lm m^{-2}).

EXO: External, outside or beyond, for example exoskeleton.

EXOBIOLOGY (ASTROBIOLOGY): The study of life forms that may possibly exist elsewhere in the universe and of the effect of extra terrestrial environments on Earth organisms.

EXOCARP (EPICARP): The outer layer of the pericarp of a fruit.

EXOCRINE GLAND: Glands that secrete their products into ducts. Examples are sweat glands, salivary glands, mammary glands and parts of the pancreas. On the other hand, endocrine or ductless glands secrete their products (hormones) directly into the bloodstream through ductless glands or release hormones such as paracrine that affect only target cells close to the release site.

EXOCYTOSIS (REVERSE PINO-CYTOSIS): The passage of material or the contents of secretory vesicles from the inside of the cell to the cell surface within membrane-bound vesicles. These membrane-bound vesicles contain soluble proteins to be secreted to the extracellular environment, as well as membrane proteins and lipids that are sent to become components of the cell membrane. The membranes of the vesicles fuse with the plasma membrane, a process involving various docking proteins in both the vesicle coat and target membrane.

EXODERMIS (HYPODERMIS): The outermost supportive and protective layer of cells in the plant cortex, lying immediately below the epidermis in many but not all plants. It develops Casparian strips, suberin deposits and cellulose deposits impregnated with phenolic or quinoidal substances and it may produce substances that act as a barrier to the entry of pathogens. In certain plants it may play a role in the absorption of water and the selection of ions that enter the plant.

EXOENZYME: An enzyme that cleaves monomers from one end of a polymer chain.

EXOERGIC: A term denoting a nuclear or chemical reaction that produces energy. Endoergic reactions, on the other hand, involve energy absorption.

EXOGENOUS: A term describing substances, stimuli, etc. originating from outside an organism. In biology, for example, an exogenous factor is any material that is present and active in an individual organism or living cell but that originated outside that organism. These

include pathogens and therapeutics, carcinogens and DNA introduced to cells via transfection or viral infection (transduction).

EXOMORPHIC: Relating to the external form.

EXON: A segment within a gene that carries part of the coding information for a protein; or a nucleotide sequence in a gene that codes for part or all of the gene product and is therefore expressed in mature messenger RNA, ribosomal RNA or transfer RNA; or a nucleic acid sequence that is represented in the mature form of an RNA molecule after either portions of a precursor RNA have been removed by cis-splicing or when two or more precursor RNA molecules have been ligated by trans-splicing. In eukaryotes, exons are separated by non-coding sequences called introns. Depending on the context, exon can refer to the sequence in the DNA or its RNA transcript.

EXONUCLEASE: An enzyme that catalyses the cleavage of nucleotides from the end of a nucleic acid molecule or breaks a phosphodiester linkage at one or the other end of a polynucleotide chain so as to release a single nucleotide or small oligonucleotides. Eukaryotes and prokaryotes have three types of exonucleases involved in the normal turnover of mRNA: 3' to 5' exonuclease, which is a dependent decapping protein; 5' to 3' exonuclease, an independent protein; and poly(A)-specific 3' to 5' exonuclease.

EXOPEPTIDASE: A protein-digesting enzyme produced in the pancreas, that catalyses the removal of an amino acid from the end of a polypeptide chain or cleaves amino acids from the ends of a polypeptide chain. An example of an exopeptidase is carboxypeptidase, which breaks down proteins in the small intestine.

EXOPLANETS (EXTRASOLAR PLANET): A planet that orbits a star other than the Sun or a planet outside the Solar System, mostly detected by the changes in radiation associated with the 'wobble' they cause in the star they orbit. More than 1000 such planets have been discovered as of February 2014. Studying extrasolar planets helps astronomers learn more about the formation of our solar system and also contributes to the study of possible life in the universe, because life is more likely to develop on planets than in the extremes of stars and empty space.

EXOPTYERYGOTE (EXOPTYERYGOTA; HEMIPTYERODEA): Any winged insect showing hemimetabolous development, the eggs hatch into young ones that resemble the adults but lack wings. With one exception (mayflies), only the mature, adult stages of insects have fully developed wings. They undergo a change between immature and adult, without going through a pupal stage and the nymphs develop gradually into adults through a process of moulting. They are a super order of insects of the subclass Pterygota in the infraclass Neoptera. There are at least 130,000 living species divided between 15 orders. They include termites, locusts, thrips, lice and stick insects.

EXOSCOPIC EMBRYOGENY: A term used to describe embryogeny in which, of the two cells resulting from the first division of the egg cell, the outer gives rise to the embryo and the inner to the foot. It is seen in the Bryophyta, Psilopsida and Sphenopsida.

EXOSKELETON (EXTERNAL SKELETON): Skeleton lying outside the body of some invertebrates or the hardened external skeleton that supports and protects the soft tissues of lower animals, derived from the ectoderm. Examples are shells of crabs, insect, spider and other arthropods. It provides attachment for muscles and protection for the internal organs as well as support.

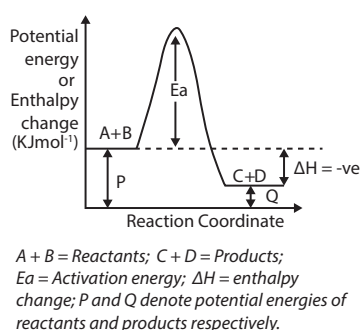
EXOSMOSIS: The passage of a fluid or solvent from a cell into a solution of lower concentration through a semipermeable membrane. This mechanism is employed in desalination. In this case, seawater of a higher concentration is converted to freshwater suitable for human consumption.

EXOSPHERE: The uppermost layer of the atmosphere. It begins at about 700 km and fades off into the vacuum of space. The gas is extremely thin with hydrogen as the main constituent.

EXTERNAL COST: It defines the costs arising from any human activity, when the agent responsible for the activity does not take full

account of the impacts on others of his or her actions. Equally, when the impacts are positive and not accounted for in the actions of the agent responsible they are referred to as external benefits. Emissions of particulate pollution from a power station affect the health of people in the vicinity, but this is not often considered, or is given inadequate weight, in private decision making and there is no market for such impacts. Such a phenomenon is referred to as an "externality," and the costs it imposes are referred to as the external costs.

EXOTHERMIC REACTION: A reaction in which heat energy is produced or released from the reactants to the surroundings. Rusting and combustion both involve combination with air to produce oxide and are both exothermic reactions. Also, if sulfuric acid is added to water, the reaction which occurs is exothermic. At the end of an exothermic reaction, the temperature of the reactants is higher than that before the reaction owing to the heat given out. It can be expressed in a chemical equation as follows: Reactants $\rightarrow$ Products + Energy. The reaction releases energy usually not only in the form of heat, but also in the form of light such as a spark, flame or explosion, electricity such as from a battery or sound such as coming from burning hydrogen. The concept is frequently applied in the physical sciences, to chemical reactions where chemical bond energy is converted to thermal energy (heat). **Endothermic reaction** is a reaction in which energy is absorbed in the form of heat.



Reaction profile diagram for exothermic reaction

EXOTIC: Not native, introduced from elsewhere, but not completely neutralised.

EXOTIC ATOM: An atom in which an electron has been replaced by another negatively charged particle, such as a meson. The negatively charged particle (meson) collides with the nucleus with the emission of x-ray photons.

EXOTIC MATTER: 1. A hypothetical kind of matter that has both a negative energy density and a negative pressure or tension that exceeds the energy density. All known forms of matter have positive energy density and pressures or tensions that are always less than the energy density in magnitude. In a stretched rubber band, for example, the energy density is 100 trillion times greater than the tension. A possible source of exotic matter lies in the behaviour of certain vacuum states in quantum field theory. If such matter exists or could be created, it might make possible schemes for faster-than-light travel, such as stable wormholes and the Alcubierre warp drive. 2. Any kind of matter that is non-baryonic or not made of baryons. Baryons are the subatomic particles, such as protons and neutrons, of which ordinary matter is composed.

EXOTIC SPECIES: Organisms introduced into new habitats, where they are not natives.

EXOTOXIN: A poison produced by microorganisms such as bacteria, fungi, algae and protozoa, which affects parts of the body away from the place of infection. They may be secreted or, like endotoxins, released during lysis of the cell. The common ones include the botulinum toxin produced by *Clostridium botulinum* and the *Corynebacterium diphtheriae* exotoxin, which is produced during

life threatening symptoms of diphtheria. They can cause damage to the host by destroying cells or disrupting normal cellular metabolism. They are susceptible to antibodies produced by the immune system, but many are so toxic that they may be fatal to the host before the immune system has a chance to mount defences against it. Most can be destroyed by heating.

EXPAND: In mathematics, it is to express in an equivalent form, for example, $(x + y)^2 = x^2 + 2xy + y^2$.

EXPANDED MEMORY (EXPANDED MEMORY SPECIFICATION, EMS): In computing, it is a technique for adding additional memory beyond 1 MB in an MS-DOS-based computer, usually installed on an expanded-memory board. The first versions developed are EMS 3.0 and 3.2, enabling programs to use an additional 8MB of memory, but for data only. The second version called Extended EMS (EEMS) enables programs to use extra memory for code as well as for data and was developed by AST, Quadram and Ashton-Tate. The most recent version is EMS 4.0 or LIM 4.0; it has a memory of 32MB and was developed in 1987 jointly by Lotus, Intel and Microsoft of EMS. Software is available under both MS-DOS and Windows to simulate expanded memory for those applications that require it.

EXPANDED NOTATION: Writing a number to show the value of each digit; shown as a sum of each digit multiplied by its matching place value (units, tens, hundreds, etc.). For example, the number 5,456 in its expanded form is $5 \times 1000 + 4 \times 100 + 5 \times 10 + 6 \times 1$.

EXPANDED PLASTIC (CELLULAR PLASTIC): A plastic in the form of rigid solid foam, with apparent density decreased substantially by the presence of numerous cells disposed throughout its mass. An example is polystyrene, used as a packing material.

EXPANSION: 1. Increase in volume or size of a constant mass of substance, caused by, for example, increasing its temperature or the internal pressure of the substance. This increase in volume is not constant for all substances for any given rise in temperature, but is a specific property of each kind of matter. For example, zinc and lead undergo greater expansion in a one-degree rise in temperature than do silver or brass. 2. The writing of a function or quantity as a series of terms. The series may be finite or infinite.

EXPANSION CARD: In computing, it is a printed circuit board that can be inserted into an expansion slot of a computer motherboard to add functionality to a computer system. For example, a user may add a new graphics card to a computer to give it more 3D graphics processing power.

EXPANSION OF THE UNIVERSE: The hypothesis based on the evidence of the red-shift which states that the distance between the galaxies is continuously increasing in the universe with time. It is an intrinsic expansion, defined by the relative separation of parts of the universe and not by motion outward into pre-existing space. It does not expand into anything outside itself.

EXPANSION RATIO: The ratio between the area of the combustion chamber exit and the area of the nozzle exit. A large expansion ratio improves the performance of an engine in a vacuum since the exhaust is expanded further, thus converting potential energy into kinetic energy. However, at sea level a high expansion ratio can result in flow separation, which can drastically reduce or eliminate the net thrust of the engine.

EXPANSION SLOT (SLOT): In computers, it is an opening in the motherboard of a computer that allows additional units to be added.

EXPANSIVITY (COEFFICIENT OF EXPANSION): The fractional increase in length, area or volume of a solid per unit rise in temperature, usually expressed as a fractional change in dimensions per unit temperature change. If the material is a fluid, the expansion is described in terms of a change in volume but if it is a solid, thermal expansion is usually described in terms of change in length, height or

thickness. Since the bonding forces among atoms and molecules vary from material to material, expansion coefficients are characteristic of elements and compounds.

EXPECTED UTILITY: The weighted average utility of the possible outcomes of a probabilistic situation; the expected value of the utility function, calculated as the sum or integral of the product of the probability distribution and the utility function. In other words, it is a predicted utility value for one of several options, calculated as the sum of the utility of every possible outcome each multiplied by the probability of its occurrence.

EXPECTED VALUE (MATHEMATICAL EXPECTATION): In statistics, the sum or integral of all possible values of a random variable or any given function of it, multiplied by the respective probabilities of the values of the variable. Its symbol is $E(X)$. $E(X)$ is the mean of the distribution; $E(X - c) = E(X) - c$, where c is a constant.

EXPECTED OUTCOME: The average outcome over a large number of repetitions of a random experiment. In other words, it is an average over infinitely many repetitions. For example, the expected outcome of rolling one die is the average number of spots landing up over a large number of rolls. Because each face of a fair die has equal probability of

landing up, the expected outcome is: $\frac{(1+2+3+4+5+6)}{6} = \frac{21}{6} = 3.5$.

This means that the average of many rolls of a fair die is expected to be about 3.5.

EXPECTORANTS: Agents or substances containing properties that evacuate mucus and cleanse the throat, the lungs, bronchi, trachea and the entire air passage. They include herbs such as eucalyptus, plantain leaf and peppermint.

EXPERIMENT: 1. A practical test in science, designed with the intention that its results will be relevant to a particular theory or set of theories or a practical test of an idea, designed to answer a question about the subject under investigation or an investigation to test a hypothesis, theory or law or to discover facts. It may be useful for gathering more information about a topic that is already well understood or for confirming a new theory or in undermining long held beliefs. First, an observation is made, then a question is asked or a problem arises. Next, a hypothesis is formed. Experiment is then used to test that hypothesis. The results are analysed, a conclusion is drawn, sometimes a theory is formulated and the results are communicated through research papers. 2. In probability, it is any procedure that yields a measurement.

EXPERIMENTAL CONDITION: 1. One of the distinct states of affairs or values of the independent variable for which the dependent variables are measured in order to carry out statistical tests or calculations. For example, a test of a new drug may involve two experimental conditions, in one test, subjects receive the new drug and in the other a placebo. 2. A condition in which there is some intervention by the experimenter, as contrasted with the control condition. Only subjects who receive the drug are subject to the experimental condition.

EXPERIMENTAL ERROR: The difference between the actual value of an experiment (a determinant) and that obtained/determined as a result of an experiment/investigation or a source of variation in experimental results due to the way that a test has been conducted. Experimental error can be minimised by following a protocol that incorporates strict standardisation.

EXPERIMENTAL INVESTIGATION: A process in which a systematic approach is used to actively manipulate, control, and measure variables in an effort to gather evidence to support or refute a causal relationship. An example is testing the height of a ramp to determine how far a marble will roll. It is one of the three main types of investigations used by scientists; the others are comparative and descriptive investigations.

EXPERIMENTAL PROBABILITY: The probability found by carrying out an event many times.

EXPERIMENTAL PROCEDURE: A detailed, step-by-step method for conducting a science experiment. It is such that duplication is allowed, as this is an important step to verify that results are consistent and not just an accident.

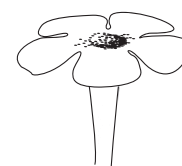
EXPERIMENTAL PSYCHOLOGY: The branch of psychology that is concerned with the application of scientific methods to the study of mental processes and behaviour.

EXPERIMENTAL UNIT: It is the smallest part of the experimental material to which treatment is applied. The experimental design specifies the number of experimental units submitted to the different treatment.

EXPERT SYSTEM: In computing, it is an artificial intelligence program that uses a knowledge base of human expertise to aid in finding solutions to problems, such as diagnosing an illness.

EXPIRATION (EXHALATION): The process by which air or water in the case of fish is expelled from the lungs (or gills). In mammals it involves a reduction in the volume of the chest cavity by relaxation of the diaphragm muscles and contraction of muscles between the ribs. This squeezes the lungs and forces gases out until the pressure in the lungs is the same as atmospheric pressure. To expire is to breathe out. Expiration and its opposite process, **inspiration**, are fundamental in aiding gas exchange, part of the complete process of respiration.

EXPLANATE: In botany, extending outwardly in a flat or horizontal form, as shown in the diagram. An example is seen in *Hypnodendron comatum* (umbrella moss).



An explanate flower

EXPLANTATION: The removal of cells, tissues or organs of animals and plants from their natural site of growth for observation of their growth and development in appropriate culture or media.

EXPLICIT FUNCTION: A function, in which the dependent variable can be expressed directly in terms of the independent variable, so that its values may be directly calculated from those of the independent variables. Examples are $y = x^2 - 3$ and $y = \log_2 x$.

EXPLORATION: The process of conscious discovery by human beings of the world around them or the act of searching or travelling a terrain for the purpose of discovery of resources or information. It occurs in all non-sessile animal species.

EXPLORATORY LEARNING (LATENT LEARNING): In animal behaviour, a type of learning in which an animal learns by exploring new surroundings without apparent reward (reinforcement) or punishment.

EXPLOSION: A rapid increase in volume and release of energy in an extreme manner, usually with the generation of high temperatures and the release of gases, which causes shock waves. An explosion is usually accompanied by the production of light, heat, sound and pressure. Low explosives create subsonic shock waves while high explosives create supersonic shock waves. Common explosives are gunpowder and dynamite. In nuclear explosions, the vast amount of energy locked within the nucleus of atoms is released and it is this energy which causes the damage in explosions.

EXPLOSIVE (EXPLOSIVE MATERIAL): Any material capable of a sudden release of energy and the rapid formation of a large volume of gas leading to the development of high pressure wave (blast) when compressed. The energy stored in an explosive material may be chemical energy such as nitroglycerine or grain dust; pressurised compressed gas such as a gas cylinder or aerosol can; or nuclear such as fissile isotopes of uranium-235 and plutonium-239. In terms of classification, explosives that explode faster than the speed of sound are said to be **high explosives** and those that deflagrate are said

to be **low explosives**. On the other hand, explosives that can be initiated by a relatively small amount of heat or pressure are called **primary explosives** while those that are relatively insensitive are called **secondary explosives**.

EXPLOSIVE ERUPTION: The most violent volcanic activity caused by a lot of gas trapped in magma, which builds up pressure until the magma eventually erupts explosively out of the volcano causing widespread damage and creating steeper cones. Explosive volcanic eruptions can be 10,000 times as powerful as the atomic bomb dropped on Nagasaki at the end of World War II in which over 40,000 people died. Explosive eruptions can also send a huge amount of lava, ash, and steam into the atmosphere forming plumes. Such eruptions can also form pyroclastic flows which sweep down valleys destroying everything in their path.

EXPLOSIVE LIMITS: The range of concentrations over which a flammable vapour mixed with proper ratios of air will ignite or explode if a source of ignition is provided or the upper and lower limits of percentage composition of a combustible gas mixed with other gases or air within which the mixture explodes when ignited.

EXPLOSIVE MECHANISM: A method of seed dispersal involving the splitting open of the fruit violently and then throwing out the seeds, as seen, for example, in flamboyant (*poinciana*) and Pride of Barbados (*Caesalpinia pulcherrima*).

EXPONENT (INDEX): A number or symbol that indicates the power to which another number or expression is raised or a number that indicates the number of times a term is multiplied by itself. For example, $x^2 = x \cdot x$ and $5^4 = 5 \times 5 \times 5 \times 5$. For example, in the expression x^n the letter n is the exponent of the base x . Fractions, zero and negative numbers are also used as exponents. For example, a negative exponent indicates the reciprocal of the quantity, for example, x^{-2} means $1/x^2$. Any nonzero quantity raised to the zero power equals one, for example, $x^0 = 5^0 = (a^2 + b^2)^0 = 1$. A fractional exponent such as $1/3$ or $1/n$ indicates the third or n^{th} root, respectively, of the base. When quantities of the same base are multiplied together, their exponents are added. For example, $x^4 \cdot x^2 = x^6$ but when they are divided, the exponents are subtracted, for example, $x^3/x^2 = x^{3-2} = x^1 = x$. When a quantity already containing an exponent is raised to a power, the exponents are multiplied; for example, $(x^2)^{3n} = x^{6n}$.

EXPONENTIAL: 1. A term describing a mathematical entity that involves the transcendental number e , the base of natural logarithms, raised to an exponent or a function in which the variable quantity is an exponent or a function that varies as the power of another quantity. For example, if $y = a^x$ then y varies exponentially with x . 2. A mathematical entity such as a curve, function, equation or series that contains, is expressed as or involves numbers or quantities raised to an exponent.

EXPONENTIAL DECAY: A function shows exponential decay if its value becomes smaller over time according to a function of the form: $x = x_0 e^{-pt}$ (assume $p > 0$).

EXPONENTIAL DISTRIBUTION: A continuous probability distribution, whose density function is given by $f(x) = ae^{-ax}$, where $a > 0$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$. The mean and standard deviation are both $1/a$.

EXPONENTIAL FUNCTION: The real or complex function $f(x) = e^x$ or a mathematical expression with the formula e^x , in which e is the base of natural logarithms. Its symbol is $\exp$.

EXPONENTIAL GROWTH: A form of population growth in which that increases at a consistent rate. If the log of population number is plotted against time a straight inclined line is produced.

EXPONENTIAL SERIES: A series derived from the expansion of an exponential expression of the form $\sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} + \dots$,

which converges for any complex number z to the exponential function $\exp z$.

EXPONENTIAL NOTATION: A way of representing repeated multiplication by the same factor. For example, 2^3 is exponential notation for $2 \times 2 \times 2$. The exponent 3 tells how many times the base 2 is used as a factor.

EXPONENTIATE: To raise a number or quantity to a power.

EXPORT: In computer science, it is a command usually found within a program's File menu and refers to saving a copy of the current open document, database, image or video into a file format required by a different application. Export command is often found in multimedia programs, like photo and video editing applications.

EXPORTATION: In logic, a rule for separating the antecedent of a conditional, deriving if P , then if Q then R from if $P \& Q$, then R .

EXPOSURE: 1. Amount or intensity of a particular physical or chemical agent or environmental agent that reaches the target population organism organ, tissue or cell, usually expressed in numerical terms of concentration, duration and frequency for chemical agents and microorganisms or intensity for physical agents. 2. A process by which a substance becomes available for absorption by the target population organism organ, tissue or cell, by any route. For X- or gamma-radiation in air, it is the sum of the electrical charges of all the ions of one sign produced when all electrons liberated by photons in a suitably small element of volume of air completely stopped, divided by the mass of the air in the volume element. 3. The amount of radiant energy needed to expose a photographic film or a photographic plate or a piece of film so exposed. 4. The act of exposing sensitised photographic film or plate.

EXPOSURE METER: A photocell that operates a meter to indicate the correct exposure for a specified film in photography or an instrument used in photography for indicating the correct exposure or the length of time the camera shutter should open under given light conditions. It enables the correct shutter speed and aperture to be chosen for any photographic circumstance.

EXPOSURE PATHWAY: A route by which a radionuclide or other toxic material can enter the body. The main exposure routes are inhalation, ingestion, absorption through the skin and entry through a cut or wound in the skin.

EXPOSURE RATE: A measure of the ionisation produced in air by x-rays or gamma rays per unit of time. It is frequently expressed in roentgens per hour.

EXPRESS: 1. To represent by a sign or a symbol; symbolise or to express a fraction as a decimal. In other words, to transform into other terms. For example, the product $(x+y)(x-y)$ can be expressed as a difference of squares as $x^2 - y^2$. 2. In genetics, it is to produce an observable inherited characteristic. Examples are to cause (itself) to produce an effect or a phenotype, for example, the gene expressed itself under specific environmental conditions; or to manifest the effects of (a gene). For example, half of the people who inherit the gene expressed it; and to manifest a genetic trait, for example, all the mice in the study expressed the defect.

EXPRESSED SEQUENCE TAG (EST): A short partial sequence, typically 200-400 bp long of a complementary DNA clone. It is produced by one-shot sequencing of a cloned mRNA (i.e. sequencing several hundred base pairs from an end of a cDNA clone taken from a cDNA library). Because cDNAs (complementary DNAs) are prepared by reverse transcription of messenger RNA molecules, ESTs act as markers for genes that are expressed in particular tissues or organs. They may be present in the database as either cDNA/mRNA sequence

or as the reverse complement of the mRNA, the template strand. They may be used to identify gene transcripts and are instrumental in gene discovery and gene sequence determination.

EXPRESSION: 1. In biology, the result where an allele causes an organism to have a particular characteristic or the process by which information from a gene is used in the synthesis of a functional gene product. The allele is expressed in the characteristic. The process is used by all known eukaryotes including multicellular organisms, prokaryotes (bacteria and archaea) and viruses, to generate the macromolecular machinery for life. It gives the cell control over structure and function and is the basis for cellular differentiation, morphogenesis and the versatility and adaptability of any organism. It may also serve as a substrate for evolutionary change, since control of the timing, location and amount of gene expression can have a profound effect on the functions of the gene in a cell or in a multicellular organism. 2. In mathematics, a sign or a group of signs that represents a quantity. An example of an expression is $x^2 + 4$. In other words, any symbol or well-formed string of symbols of a particular theory. Examples are formulae, names, variables, predicates, relations, functions and sentences.

EXPRESSION VECTOR (EXPRESSION CONSTRUCT): A vector used in genetic engineering that enables a particular gene not only to be cloned but also expressed in a host cell. The vector is constructed to contain appropriate regulatory sequences, such as a promoter and operator, so that the host-cell machinery can transcribe the gene and translate the resultant messenger RNA to synthesise the corresponding protein. Once the expression vector is inside the cell, the protein that is encoded by the gene is produced by the cellular-transcription and translation machinery ribosomal complexes. The plasmid is frequently engineered to contain regulatory sequences that act as enhancer and promoter regions and lead to efficient transcription of the gene carried on the expression vector. The aim of a well-constructed expression vector is the production of large quantities of stable messenger RNA and therefore proteins. They are primary instruments for biotechnology and the production of proteins such as insulin that are important for medical treatments of specific diseases like diabetes.

EXPRESSIVITY: In genetics, it is the degree to which a particular genotype is expressed in the phenotype of individuals with that genotype.

EXRADIUS (ERADIUS): The radius of an escribed circle of a triangle.

EXSERTED: Pushing out beyond the surrounding parts. This is seen in flowers where the stamen protrudes from a corolla.

EXTANT: Describing members of a taxon that are living in the present time.

EXTENDED: In mathematics, of a cycloid, epicycloid or hypocycloid, described by a point attached to but lying outside, rather than on or within the circumference of a circle as it rolls without slipping round some other figure.

EXTENDED ASCII (EXTENDED CHARACTER; HIGH ASCII): A set of characters with ASCII values between 128 and 255 stored in read-only memory (ROM) in personal computers (PCs). These characters may include special symbols, graphics characters and accented characters. The assignment of extended ASCII characters is not standard. It depends on the particular computer system and may also depend on the font being used. For example, in Standard ASCII, two bits are used for encoding different character sets. The Extended ASCII uses eight bits and hence is able to encode extra characters and symbols.

EXTENDED DATA OUT RAM (EDO RAM): A faster type of RAM introduced in the mid-1990s that improved the performance of fast page mode (FPM).

EXTENDED FACTS (EXTENDED MULTIPLICATION FACTS): Basic arithmetic facts with one of the numbers multiplied by a power of 10, such as 10, 100, 1,000, etc. An extended fact looks like a basic fact, except each number has one or more zeros after it. For example, for a basic addition fact of $5 + 8 = 13$, the extended facts (multiplied by powers of 10: 10, 100, 1000, etc.), are as follows:
 $(5 + 8 = 13) \times 10 = 50 + 80 = 130$;
 $(5 + 8 = 13) \times 100 = 500 + 800 = 1300$;
 $(5 + 8 = 13) \times 1000 = 5000 + 8000 = 13000$; etc.

EXTENDED PHENOTYPE: The concept that the phenotype of an organism extends beyond its body to encompass the organisms behaviour and the consequences of that behaviour or that phenotype should not be limited to biological processes such as protein biosynthesis or tissue growth, but extended to other aspects such as cognitive activities and more generally to the behaviour of the animal in its environment. Examples include birds' nests, termite mounds and spiders' webs. This concept was advanced by the British evolutionary biologist, Richard Dawkins in 1982.

EXTENDED PLANE: The set of complex numbers together with a point at infinity, which is equivalent to the Riemann sphere. It is denoted by $\mathbb{C}^*$.

EXTENDED PRECISION: A floating point number formats that provide greater precision and more exponent range than the basic floating point formats. Extended precision formats support a basic format by minimising roundoff and overflow errors in intermediate values of expressions on the base format.

EXTENDED REAL NUMBERS: The set $[-\infty, \infty]$ of real numbers, together with the positive and negative infinite cardinals. It is usually denoted by R . Often the field and order operations are partially extended to $[-\infty, \infty]$, so that, for example, $r + \infty = +\infty$ for $r \neq -\infty$; the extended real line is then order equivalent and topologically equivalent to $[0, 1]$.

EXTENDED X-RAY ABSORPTION FINE STRUCTURE (EXAFS): Oscillations of the x-ray absorption coefficient beyond an absorption edge. Its physical cause is the modification of the final state of a photoelectron wave function caused by back-scattering from atoms surrounding the excited atom.

EXTENDED BINARY-CODED DECIMAL INTERCHANGE CODE (EBCDIC): In computing, a binary (8-bit) code used for storing and communicating alphabetic and numeric characters. It was developed by IBM for its larger operating systems and is capable of holding 256 different characters, although only 85 of these are defined in the standard version. It has now been largely superseded by ASCII code, although it is still used in many mainframe computers.

EXTENDER: An inert substance added to a product like paint, rubber, washing powder, etc., to dilute it or to modify its physical properties or adulterate it. An example is a meat loaf that contains oatmeal.

EXTENSIBLE HYPERTEXT MARKUP LANGUAGE (XHTML): A family of markup languages designed for network devices and a method of displaying web pages on network and portable devices. It is the reformulation of HTML and XML based on the HTML 4.0 syntax, but has been modified to follow the guidelines of the Extensible Markup Language (XML).

EXTENSIBLE MARKUP LANGUAGE (XML): A specification emanating from the World Wide Web Consortium (W3C) that allows

Web designers to create their own language for displaying documents on the Web. XML is an extension to the standard language for creating Web pages, HTML and makes it possible to create websites containing more complex interactivity.

EXTENSION: 1. The position attained by a limb after stretching it or the stretching out of a limb after it has been bent. 2. A mathematical set that includes as subsets all the members of a given set and of another similar set. 3. In logic, a mapping, whose domain and range are supersets of those of the original mapping and such that the restriction of the larger mapping to the original domain agrees with the original mapping. For example, the principal complex square root is an extension of the positive square root defined for positive numbers. Alternatively, a function or operator defined on a superspace of the domain of a given function and coinciding for arguments for which both are defined. 4. In logic, the class of entities to which a given expression correctly applies. For example, the extension of the phrase "satellite of Mars" is "the set of which the only members are Phobos and Deimos." 5. Also called **conservative extension**, a formal theory whose primitive terms, formation rules and axioms include those of a given theory and that contains the given theory in the sense that everything true in the given theory is also true in the extended theory. For example, Zermelo-Frankel set theory is a conservative extension of Peano arithmetic and first-order predicate calculus of sentential calculus. 6. In algebra, of a ring, a ring H , of which a given ring, G , is an ideal such that the factor ring H/G is isomorphic to N , where G is extended by N . Alternatively, of a group, a group H , of which a given group, G , is a normal subgroup such that the factor group H/G is isomorphic to N , where G is extended by N .

EXTENSION FIELD: Of a smaller field, a field containing the smaller field as a sub-field.

EXTENSIONAL: In logic, explicable purely in terms of extension. In other words, ignoring differences of meaning that do not affect the extension. For example, since substitution of co-referential terms may change a truth into a contingent one, the truth-value of statements of necessity is not a function of the references of the terms of these statements and so necessary truth is not an extensional notion.

EXTENSIONALITY: The principle and axiom of set theory stating that sets are definable in terms of their elements alone, however way they may have been selected. For example, $\{a, b\} = \{b, a\} = \{x: x \text{ is one of the first two letters of the alphabet}\}$.

EXTENSIVE: In logic, of or relating to extension. Alternatively, of a definition, in terms of the objects to which the term applies rather than a property that they satisfy, for example, defining a set as $\{1, 2, 3\}$ rather than $\{x: 1 \leq x \leq 3\}$.

EXTENSIVE AGRICULTURE (EXTENSIVE FARMING): A type of farming in which a relatively small amount of produce is generated from a large area of farmland. The crop yield depends primarily on the natural fertility of the soil, the terrain, the climate and the availability of water.

EXTENSIVE PROPERTY: A property that depends on how much matter is being considered. It is directly proportional to the system size or the amount of material in the system. For example, density is an intensive quantity (it does not depend on the quantity), while mass and volume are extensive quantities. Note that the ratio of two extensive quantities that scale in the same way is scale-invariant and hence an intensive quantity.

EXTENSIVE VARIABLE: A quantity in a macroscopic system that is proportional to the size of the system. Examples of extensive variables are the volume, mass and total energy.

EXTENSOMETER: An instrument used to measure minute deformations in a test specimen of a material or any device for

measuring the extension of a specimen of a material under longitudinal stress or a device for measuring small changes of length in a specimen, especially those caused by thermal expansion in a metal or by stress.

EXTENSOR: A muscle that straightens or extends a limb or a part of the body.

EXTERIOR: The interior of the complement of a set or, equivalently, the complement of its closure.

EXTERIOR ANGLE: 1. Any of the angles formed on the outside of a polygon, between a side and an extension of an adjacent side. In the diagram, Fig. III, for example, the exterior angles are A, B, C, D and E . The interior and exterior angles add up to 180° , given a straight line. The sum of the exterior angles of any polygon is 360° . 2. Any of the four angles formed on the outside of a pair of lines that are crossed by a third line called a transversal. In the diagram, the exterior angles are angles A, B, G and H . 3. For a triangle, it is any of the angles formed by extending one side of the triangle at the vertex. In the diagram, the exterior angles are $1, 2$ and 3 . The measure of any exterior angle of a triangle equals to the sum of the measures of the two remote or opposite interior angles of the triangle.

EXTERIOR DIFFERENTIAL: The differential of a differential form, producing a $(k+1)$ -form from a k -form.

EXTERIOR POINT: A point lying simultaneously on two tangents to a given conic.

EXTERIOR PRODUCT (WEDGE PRODUCT): Of vectors, an algebraic construction used in Euclidean geometry to study areas, volumes and their higher-dimensional analogues. The exterior product of two vectors α and β is denoted by $\alpha \wedge \beta$. If α and β are differential k -forms of degrees p and q , respectively, then $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$. The magnitude of α and β can be interpreted as the area of the parallelogram with sides α and β , which in three dimensions can also be computed using the cross product of the two vectors. The **exterior algebra** or **Grassmann algebra** is the algebraic system whose product is the exterior product.

EXTERNAL DIRECT PRODUCT: 1. Also known as **external direct sum** of groups, it is the Cartesian product $M = M_1 \times M_2 \times \dots \times M_n$, of a finite or infinite set of modules over a ring, but with only finitely many non-zero entries in each sequence, where multiplication by a member of the ring, r , is defined as $r(x_1, \dots, x_n) = (rx_1, \dots, rx_n)$. 2. Also known as **Cartesian product**, it is an external direct sum over infinitely many spaces in which the sum is not required to be finite.

EXTERNAL DIVISION: In geometry, the external division of a segment is the construction of a point E , lying beyond the endpoints A and B of a given line segment AB , in such a way that the ratio of the directed lengths $|AE|$ to $|EB|$ is some given negative number λ_1 .

EXTERNAL FEMALE GENITALIA (VULVA; PUDENDUM): The accessory structures of the female reproductive system that are external to the vagina. They include the labia majora, mons pubis, labia minora, clitoris and glands within the vestibule.

EXTERNAL HARD DRIVES: A high-capacity data storage device used to back up data, especially for audio and video data. They have memory ranging from 120GB to 1TB and can be used to store extensive collection of pictures and photos, files, programs, applications, movies and MP3 files. **External memory** refers to storage in an external hard drive or on the Internet.

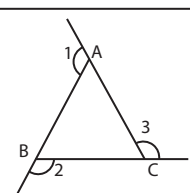


Fig I:
Exterior angles
of a triangle

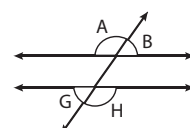


Fig II:
Exterior angles
formed by a
transversal
crossing two lines

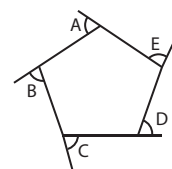


Fig III:
Exterior angles
of a polygon

EXTERNAL MODEM: A type of self-contained modem located outside a personal computer (PC) having lights indicating various modem functions and connected to the PC by a cable via a serial or USB port. This is in contrast to internal modem that is on a printed circuit board within the computer.

EXTERNAL OBLIQUE MUSCLE: A broad, thin, muscular sheet that arises from the fifth to twelfth ribs, with insertion into the anterior lateral lip of the iliac crest, the inguinal ligament and the anterior layer of the sheath of the rectus muscle of the abdomen, with nerve supply from the ventral branches of the lower thoracic nerves and whose action diminishes the capacity of the abdomen and draws the chest down.

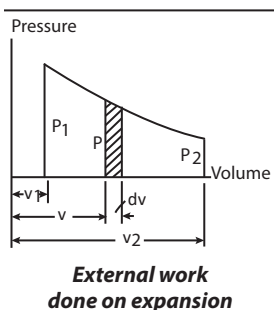
EXTERNAL RADIATION: It is the radiation (hazard) that exists where an absorber (typically a person) is being exposed to a source of ionising radiation external to the body. Examples include exposures to sealed sources, dental x-rays and cosmic rays. One feature of this hazard is that moving the absorber away from the source, usually results in a reduction in the dose of ionising radiation to that absorber, which is not the case for internal radiation hazards.

EXTERNAL STANDARD: A compound present in a standard sample of known concentration and volume which is analysed separately from the unknown sample under identical conditions. It is used to facilitate the qualitative identification and/or quantitative determination of the sample components. The volume of the external standard (standard sample) need not to be known if it is identical to that of the unknown sample.

EXTERNAL WORK: Work done, w by a substance expanding against an external resistance. It is expressed as

$$w = \int_{V_1}^{V_2} p dv, \text{ where } V_1 \text{ and } V_2 \text{ are the initial and final volumes respectively.}$$

In the diagram shown, P_1 is the applied external pressure. The work done per cycle is given by the area enclosed by the cycle or the area under the curve which goes from P_1V_1 to P_2V_2 . Therefore, there are many possible curves, which would connect the points P_1V_1 and P_2V_2 and each curve would have a different area and give a different value for the work done, w .



EXTEROCEPTOR: Any receptor that detects external stimuli or a sense organ such as the ear, that receives and responds to stimuli originating from outside the body. Other examples of exteroceptors are the thermoreceptors in the skin, which monitor the temperature of the external environment.

EXTINCT (EXTINCTION): The complete disappearance of a species or the state where the number of organisms in a species or a group of taxa falls to zero. This happens when the environment changes negatively, relative to their survival needs, so that they are unable to adapt, protect or defend themselves. Factors which lead to extinction include inability of the species to adapt quickly enough to a naturally changing environment or activities of human beings. The moment of extinction is generally considered to be the death of the last individual of that species, although the capacity to breed and recover may have been lost before this point. Since a species' potential range may be very large, determining this moment is difficult. This leads to phenomena such as the Lazarus taxa, where a species presumed extinct abruptly re-appears, typically in the fossil record after a period of apparent absence.

EXTINCTION COEFFICIENT (ABSORPTIVE POWER; ABSORBENCY INDEX; ABSORPTION CONSTANT): Any measure of the extent to which the intensity of a beam of light is reduced by passing through a distance of a solution having a molar concentration of a dissolved substance. It is also called **Molar Absorptivity** in the Beer's Law (Beer's law, Beer-Lambert law, Lambert-Beer law or the

Beer-Lambert-Bouguer law), which relates the absorption of light to the properties of the material through which the light is travelling). It is the constant, a , in the Beer's law relation $A=ebc$, where A is absorbance (no units, since $A = \log_{10} P_0 / P$), e is the molar absorptivity with units of $L \text{ mol}^{-1} \text{ cm}^{-1}$, b is the path length of the sample – i.e. the path length of the cuvette in which the sample is contained. This measurement in centimetres c is the concentration of the compound in solution, expressed in mol L^{-1} . Therefore, if all the light passes through a solution without any absorption, then absorbance is zero and percent transmittance is 100%. If all the light is absorbed, then percent transmittance is zero and absorption is infinite. The reason for stating the law with this equation is because absorbance is directly proportional to the other parameters, as long as the law is obeyed.

EXTIRPATION: A localised extinction in which a species disappears from part of its range, usually due to human activities or habitat destruction.

EXTRA-AXILLARY: Growing outside the axils.

EXTRACELLULAR (EXTRACELLULAR SPACE): Located or occurring outside the cell. The composition of the extracellular space includes metabolites, ions, proteins and many other substances that might affect cellular function. The term is often used in reference to the extracellular fluid (ECF) compartment which composes about 15 litres of an average adult (70 kg) human's body which is assumed to contain a total of about 50 litres of water, therefore about 30% of the body's water is in the ECF compartment.

EXTRACELLULAR MATRIX (BIOLOGICAL GLUE): A strong material naturally occurring in animals and plants, made up of proteins and long chain sugars in which cells are embedded to form part of connective tissues such as bone and skin. It includes the interstitial matrix and the basement membrane. Bacteria and the exoskeletons of insects and other arthropods are also formed by certain types of extracellular matrix. It usually provides structural support to the animal cells in addition to performing various other important functions. It is the defining feature of the connective tissue in animals.

EXTRACT: 1. To obtain a substance from a compound, in solid, gas or liquid form, by using an industrial or chemical process. 2. To calculate the value of the root, for example, the square root or cube root, of a number. 3. To transfer a solute from a liquid phase to another immiscible or partially miscible liquid phase in contact with it. The term is also applied to the dissolution of material from a solid phase with a liquid in which it is not wholly soluble (i.e. leaching).

EXTRACTANT (EXTRACTING AGENT): The active component(s) primarily responsible for the transfer of a solute from one phase to the other.

EXTRACTION: 1. A separation process consisting of the separation of a substance from a matrix. It may refer to *Liquid-liquid extraction* or *Solid phase extraction*. Other techniques include supercritical carbon dioxide extraction, ultrasonic extraction, heat reflux extraction, microwave-assisted extraction and DIC (instant controlled pressure drop). Extractions use two immiscible phases to separate a solute from one phase into the other. Boiling tea leaves in water extracts substances like tannins, theobromine and caffeine (the good stuff) out of the leaves and into the water. More typical laboratory extractions are of organic compounds out of an aqueous phase and into an organic phase. 2. The process of obtaining a metal from its ore. 3. To pull something out, often using force or obtain something from a source, usually by separating it out from other materials.

EXTRACTION FRACTIONATION: A process in which a polymeric material, consisting of macromolecules differing in some characteristic affecting their solubility, is separated from a polymer-rich phase into fractions by successively increasing the solution power of the solvent, resulting in the repeated formation of a two phase system in which the more soluble components concentrate in the polymer-poor phase.

EXTRACTIVE DISTILLATION: A distillation method that uses a separation solvent, which is generally nonvolatile, has a high boiling point and is miscible with the mixture, but does not form an azeotropic

mixture. An example is the separation of an azeotropic mixture of benzene and cyclohexane, where aniline is one suitable solvent. It is used for mixtures having a low value of relative volatility, nearing unity. Such mixtures cannot be separated by simple distillation, due to the fact that the volatility of the two components in the mixture is nearly the same, causing them to evaporate at nearly the same temperature at a similar rate, making normal distillation impractical. It is important to select a suitable separation solvent for this type of distillation and the solvent must alter the relative volatility by a wide enough margin for a successful result. The quantity, cost and availability of the solvent is a key consideration in using this method. It is used to separate p-xylene and m-xylene, using n-pentane as the solvent.

EXTRAEMBRYONIC MEMBRANES (EMBRYONIC MEMBRANES):

Tissues produced by an animal embryo for protection and nutrition but otherwise taking no part in its development.

EXTRAFLORAL: In botany, it is any part that is located outside the flower.

EXTRAGALACTIC: Outside of, beyond, occurring, or originating outside the Milky Way Galaxy.

EXTRANEOUS ROOT: In mathematics, a root that is introduced into an equation that is not a root of the original equation because it was excluded from the domain of the original equation.

EXTRANET: An extension of an institution's intranet, especially over the World Wide Web, enabling communication between the institution and people it deals with, often by providing limited access to its intranet. An intranet is a privately maintained computer network that can be accessed only by authorised persons, especially members or employees of the organisation that owns it.

EXTRANUCLEAR GENES: Genes included in the DNA present in organelles other than the nucleus, such as the mitochondria and chloroplasts, some of which code for the synthesis of proteins. The DNA of this organelle is inherited by the offspring via the cytoplasm of the gametes.

EXTRAORDINARY RAY: 1. Also known as **extraordinary component**, it is one of two rays into which a ray incident on an anisotropic uniaxial crystal is split. Its deviation at the crystal's surface depends on the orientation of the crystal and it is deviated even in the case of normal incidence. 2. Also known as **X wave**, it is a magneto-ionic wave component which, when viewed below the ionosphere in the direction of propagation, has clockwise or counterclockwise elliptical polarisation respectively, accordingly as the Earth's magnetic field has a positive or negative component in the same direction.

EXTRAPOLATED RANGE: The distance from a radiation source, calculated by extrapolation to zero of the flux density, of the tangent to the flux density versus distance curve, taken at the point where the flux density has decreased to one half of its initial value.

EXTRAPOLATION: An approximation technique for finding the value of a function or measurement beyond the values already known, especially by the extension of a curve. The techniques used in extrapolation are usually not as good as those used in interpolation.

EXTRASENSORY PERCEPTION (ESP): The purported sensation other than by the recognised senses of an event or object; and, by extension, those powers of the mind (such as **telekinesis**, the moving of distant objects by the exercise of willpower) that has so far eluded scientific evaluation. The best known and most researched area of ESP is **telepathy**, the ability of two or more individuals to communicate without sensory contact. Laboratory tests have been inconclusive. **Empathy** is the communication across distance of emotion. Another important area of ESP is **precognition**, the prior knowledge of an event. Also, laboratory tests have been inconclusive.

EXTRASOLAR PLANET (EXOPLANETS): A planet that orbits a star other than the Sun or a planet outside the Solar System, mostly detected by the changes in radiation associated with the 'wobble' they cause in the star they orbit. More than 1000 such planets have been

discovered as of February 2014. Studying extrasolar planets helps astronomers learn more about the formation of our solar system and also contributes to the study of possible life in the universe, because life is more likely to develop on planets than in the extremes of stars and empty space.

EXTRASTAMINAL: Located outside the stamen whorl of a flower.

EXTRATERRESTRIAL: Originating or existing beyond the Earth or a life form that originated outside our world.

EXTREMAL: 1. Of or relating to maximal or minimal values or degrees of inclusiveness. 2. In logic, the clause in a recursive definition that specifies that no items other than those generated by the stated rules fall within the definition, for example, in "1 is an integer; if n is an integer so is $n + 1$ and nothing else is", "and nothing else is" is the extremal.

EXTREME: 1. The first or last term of a series or proportion. 2. A maximum or minimum value of a given function.

EXTREME AND MEAN RATIO (GOLDEN MEAN; GOLDEN SECTION):

A special number often encountered when taking the ratios of distances in simple geometric figures such as the pentagon, pentagram, decagon and dodecahedron and found by dividing a line into two parts so that the longer part divided by the smaller part is also equal to the whole length divided by the longer part. It is often denoted by the Greek letter, ϕ .

EXTREME POINT: Any point of a convex set that cannot be expressed as the average of distinct points of the set; a corner of a polytope. Let C be a convex subset of a vector space X . A point $x \in C$ is called an extreme point if it is not an interior point of any line segment in C . In other words, x is extreme if and only if whenever $x = ty + (1-t)z$, $t \in (0,1)$, $z \neq y$, implies either $y \notin C$ or $z \notin C$. For example, the set $[0,1] \in \mathbb{R}$ is a convex set and 0 and 1 are the extreme points. Every compact convex subset of Euclidean space is the convex hull of its extreme points.

EXTREME WEATHER EVENT: A weather event such as snow, rain, drought, flood, tornado, and hurricane that is rare for the place where it occurs. The Intergovernmental Panel on Climate Change (IPCC) suggests that "rare" means in the bottom 10% or top 10% of severity for a given event type in a given location. By definition, the characteristics of what is called extreme weather may vary from place to place. For example, normal temperatures in the tropics would be felt as heat wave if they occurred in Greenland or Alaska.

EXTREMELY HIGH FREQUENCY (EHF): A radio frequency between 30,000 megahertz (30 gigahertz) and 300 gigahertz, used in various applications such as radio astronomy and remote sensing. Above this frequency, electromagnetic radiation is considered to be low or far infrared or terahertz radiation. This band has a wavelength of ten to one millimetre, giving it the name millimetre band or millimetre wave, sometimes abbreviated MMW or mmW. Compared to lower bands, terrestrial radio signals in this band are extremely prone to atmospheric attenuation, making them of very little use over long distances. In particular, signals in the 57–64 GHz region are subject to a resonance of the oxygen molecule and are severely attenuated. Even over relatively short distances, rain fade is a serious problem, caused when absorption by rain reduces signal strength.

EXTREMOPHILE: A bacterium that thrives under extreme conditions, for example, at very high or very low temperatures or in very salty or acidic environments. For example, certain archaeobacteria, termed hyperthermophiles, live in hot springs at temperatures near or even above 100 degrees. Organisms that live in more moderate environments may be termed **mesophiles** or **neutrophiles**.

EXTREMUM: 1. The point located farthest from the middle of something or the most extreme possible amount or value. 2. A point on a curve $f(x)$ at which there is an instantaneous change of sign of the derivative $f'(x)$. If $f'(x)$ is defined at this point it has a value of 0. There are two types of extrema: maximum and minimum. At a **maximum**, $f'(x)$ changes from positive to negative; at a **minimum**, $f'(x)$ changes from negative to positive. Points of inflection are not extrema.

EXTRINSIC: Of a property, relating to the space in which the bearer of the property is embedded rather than to its own internal structure. For example, left- and right-hand spirals are extrinsically but not intrinsically distinct.

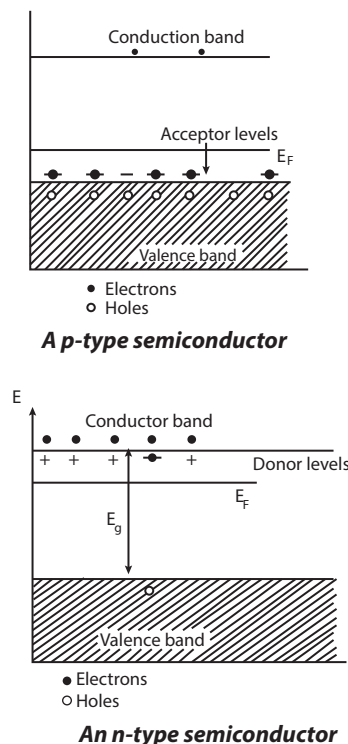
EXTRINSIC SEMICONDUCTOR:

A semiconductor that has been lightly and moderately doped giving it different electrical properties than the intrinsic or pure semiconductor. At the time of manufacture of the semiconductor crystal, this is done by incorporating chemically a trace element called a doping agent into the crystal in order to give it different electrical properties than the pure semiconductor crystal. The electrical properties of extrinsic semiconductors make them essential components of many electronic devices. A semiconductor doped to high levels that it acts more like a conductor than a semiconductor is referred to as **degenerate**. Doping involves adding dopant atoms to an intrinsic semiconductor, which changes the electron and hole carrier concentrations of the semiconductor at thermal equilibrium. Dominant carrier concentrations in an extrinsic semiconductor is classified as either an n-type or p-type semiconductor.

EXTORSE: Describing anthers that release their pollen to the outside of the flower to promote cross pollination.

EXTRUSION: 1. An object formed by forcing a semisoft material through a specially shaped nozzle or mould. It is a common method of shaping metals, plastics and other materials. The materials are usually hot and are forced through the hole in a metal to take its cross-sectional shape. The process is used to create objects of a fixed cross-sectional profile. Rods, tubes and sheets may be made in this way. The advantages of this process over other manufacturing processes are its ability to create very complex cross-sections and work materials that are brittle, because the material only encounters compressive and shear stresses. It also forms finished parts with an excellent surface finish. 2. The movement of molten material magma from a volcano or through cracks in the Earth's surface to form a solidified igneous rock of lava flow. The crystals in extrusive rocks are small, since the lava solidifies rapidly, giving little time for crystal growth

EXTRUSION TRANSFORMATION: A transformation in which an atom or group Y connected to two other atoms or groups X and Z is lost from a molecule, leading to a product in which X is bonded to Z .

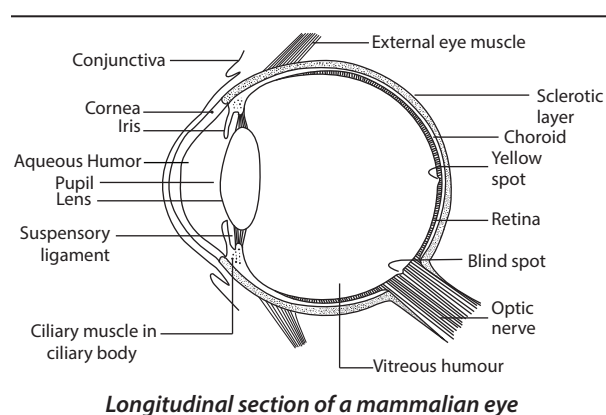


An example is $X-Y-Z \rightarrow X-Z+Y$. The reverse of an extrusion is called insertion.

EXTRUSIVE ROCK (VOLCANIC ROCK): Igneous rock formed on the surface of the Earth. It is usually distinguished from intrusive rocks on the basis of their texture and mineral composition. It is commonly glassy such as obsidian or finely crystalline such as basalts and felsites.

EXUDATE: 1. In medicine, it is a pus-like or clear fluid which occurs as a result of inflammation. 2. A substance such as gum, sap, resin and latex that oozes out of a plant. It is used as a defensive mechanism against insect attack or competitive plants. Organisms that feed on exudates are known as **exudativore**.

EYE: 1. An organ of vision that detects light and converts it into electro-chemical impulses in neurons or a sense organ, ranging from very simple structures in invertebrates to the complex organs of insects and vertebrates that responds to visible light. The components of an eye are cornea, iris, lens, ciliary muscle, retina and the optic nerve. 2. Centre of circulation in a hurricane, where the conditions are calm.



EYE MUSCLE: Any of the muscles associated with the functioning of the eye, which are divided into two groups. Four of the muscles control the movement of the eye in the four cardinal directions: up, down, left and right. The remaining two muscles control the adjustments involved in counteracting head movement.

EYE RELIEF: The distance from the observer's cornea to the nearest optical surface of the eyepiece. This is the distance at which the entire (or largest) field of view is visible.

EYE TOOTH (EYE TOOTH): The type of tooth found in the upper jaw of a canine such as a dog and also in humans. They are relatively long and pointed, however, they can appear more flattened. They evolved and are used primarily for firmly holding food in order to tear it apart and occasionally as weapons. In humans, it is also called **cuspid**, which refers to any pointed tooth located behind the incisors and in front of the premolar teeth. It is so called because of its position under the eye.

EYE WALL: Area surrounding the eye of a hurricane, where winds are most intense.

EYE WASH STATION: A standard place having acceptable equipment set apart for the purpose of emergencies in which the eyes must be quickly flushed with water to remove contaminants. Eyes should be flushed for 15 minutes and the person in charge should be notified immediately.

EYELID: A protective fold of skin and muscle that can be closed to cover the front of the eyeball. With the exception of the prepuce and the labia minora, it has the thinnest skin of the whole body. The human eyelid features a row of eyelashes which serve to heighten

the protection of the eye from dust and foreign debris, as well as from perspiration.

EYEPIECE (OCULAR): The lens in an optical instrument, such as a microscope, closest to the eye and through which an image is viewed. It offers to the eye the image originating from another system (the objective), at a suitable viewing distance. The image can be virtual. In modern instruments, most eyepieces are not independently corrected for all errors but designed to balance out certain residual aberrations of the objective or as in the microscope of a group of objectives, for example, chromatic difference of magnification.

E

EYEPOINT: The position in which the eye is placed to see the full circumference of the image.

EYESHINE: The effect that light has on the pupils of certain animals and birds when the pupils seem to glow when light is shone into their eyes at night. This effect is created by the layers of flattened cells covered by a zone of doubly refracting crystals, called *tapetum lucidum* that lie immediately behind the retina.

EYESPOT: 1. A structure found in some free swimming unicellular algae such as *chlamydomonas* and plant reproductive cells, that contain orange or red pigments and is sensitive to light. It enables the cell to move in relation to a light source. 2. A simple visual organ of certain invertebrates consisting of a sensory ending covered by light-sensitive, pigmented cells. 3. Either of two fungal diseases that affect grasses and are characterised by oval lesions. 4. A rounded eyelike marking, as on the tail of a peacock.



1-FACTOR MATCHING (COMPLETE MATCHING; PERFECT MATCHING): A matching of a graph in which all vertices are matched by an edge. In other words, every vertex of the graph is incident to exactly one edge of the matching.

6-FURFURYLAMINOPURINE (KINETIN): A synthetic cytokinin, first isolated from DNA extracts of animal origin but has now also been synthesised commercially. Cytokinins are compounds that stimulate plants growth. It is freely soluble in water (50 mg/ml). Its molecular formula is $C_{10}H_9N_5O$, molecular weight 215.21136, melting point 269-271 °C and decomposes. When used in association with auxin, it stimulates cell division in plant tissue culture. It has shown to increase callus cell growth in rice, when grown with sucrose and 2,4-D concentrations. It has also has been shown to have anti-aging effects on several different systems including plants and human cells.

f: 1. The symbol for a function, for example, $f(x) = 2x + 3$. 2. Abbreviation and symbol for femto-, which is a prefix used in the metric system to denote a factor of 10^{-15} for physical units, for example, femtosecond.

F: 1. The number 15 in the hexadecimal notation. 2. A function, as in $F(x) = \int f(x)dx$, especially in contrast to another function, f , of which it is an indefinite integral. 3. In logic, an unspecified predicate or property: Fa represents 'a is (or has) F '. The symbol may be used with subscripts to distinguish different predicates and in some circumstances with superscripts to denote the number of arguments it requires: $F_2^3 abc$ is a ternary predicate and the second such designated F . 4. In logic, symbol for falsehood. Falsehood can also be denoted by 0 or $\perp$.

F FACTOR (F AGENT; F GENOTE; F PLASMID; FERTILITY AGENT; FERTILITY FACTOR; SEX FACTOR): A fertility factor in bacteria, in which DNA segment confers the ability to form pili for conjugation and associated functions required for the transfer of DNA from donor to recipient. May exist as a plasmid or integrated into the bacterial chromosome. Hence, bacteria may be classified in relation with the F-factor as Hfr bacteria, F+ bacteria, F' bacteria and F- bacteria.

f-BLOCK: The block of elements in the Periodic Table consisting of lanthanoid series and the actinoid series. In the lanthanoid series elements the electrons are in 4f orbitals; and elements of the actinoid series have electrons in 5f orbitals series.

F-DISTRIBUTION: A continuous random variable distribution that is frequently used in statistical inference to compare different estimates of the same variance. There are many different F-distributions. Each one is identified by specifying two quantities, called the degree of freedom for the numerator and the degree of freedom for the denominator.

F-NUMBER (FOCAL RATIO; F-RATIO; F-STOP; RELATIVE APERTURE): The ratio of the focal length of a lens to the diameter of the entrance pupil. It is a dimensionless number that is a quantitative measure of lens speed, an important concept in photography. Standard f-numbers increase progressively by $\sqrt{2}$.

F-SIGMA SET: A subset of a topological space expressible as the union of countably many closed sets, usually denoted F_σ . In a metric space all open sets are of this form, as are the rationals as a subset of the reals.

F.P.S. SYSTEM (FOOT-POUND-SECOND SYSTEM; fps UNITS): A system of measurement based on the **foot** as a unit of length (equals 0.3048 m); the **pound** as a unit of mass (equals 0.453 592 kg) and the **second** as a unit of time. It is now replaced for all scientific purposes by the SI units.

F1 HYBRID (FILIAL 1 HYBRID): A crop variety that has been produced by crossing two selected parental pure lines. F1 hybrids are favoured because they can combine the qualities of the parental lines and because they usually show hybrid vigour. However, some seeds, especially those labelled on the packet as F1 or F1 hybrid, will form plants that go on to produce much more variable offspring that are often quite different from the original plant.

FAB FRAGMENT (FRAGMENT ANTIGEN-BINDING): The part of an antibody that binds to antigens. It is one of the fragments produced by partial digestion of immunoglobulin with the proteolytic enzyme, papain. It consists of an arm of the Y-shaped molecule and an entire light chain paired with part of a heavy chain, complete with antigen binding site.

FABACEAE (LEGUMINOSAE): A large family of dicotyledonous plants, commonly called the pea family, containing about 17000 species in about 700 genera. They usually have pinnately compound leaves and root nodules containing nitrogen-fixing bacteria of the genus *Rhizobium*. The inflorescence is usually a raceme and the individual flowers have five fused sepals and five petals often arranged in a shape fancifully resembling a butterfly, hence the name of the largest subfamily, Papilionoideae. The three subfamilies Mimosaceae, Caesalpinioideae and Papilionaceae are sometimes classified as families. The fruit is typically a pod or legume. Many of the Papilionoideae such as *Phaseolus* and *Vicia* (various kinds of bean), *Pisum sativum* (pea), *Lens culinaris* (lentil) and *Arachis hypogaea* (peanut) are important food crops. Others such as *Trifolium* (clovers) and *Medicago sativa* (lucerne) are used for forage. Ornamentals include *Wisteria* and *Lupinus*. One species in the family, *Abrus precatorius*, commonly known as jequirity, crab's eye, cock's eyes, or rosary pea is effective in treating diseases like cough, asthma, bronchitis, pneumonia, rheumatism, arthritis, boils, eye problems, sterility, impotence, low sperm count and cancer.

FABOIDEAE: A subfamily of flowering trees, shrubs, herbs, sometimes climbing or twining vines of the family Fabaceae or Leguminosae, with about 500 genera and 12,000 species, distributed widely in the world. The leaves are simple or compound (mostly pinnate or palmate). The flowers are classically pea shaped and root nodulation is very common. The fruit is commonly a pod or sometimes a lomentum with seeds that are usually arillate, though often inconspicuous. The seeds and pods of many of the herbaceous species have high protein content and are eaten by humans as food. Some, especially species of *Trifolium* and *Medicago*, are cultivated and used for feeding livestock or ploughed in as a fertiliser. A number of species are used for timber, while many species are widely grown as ornamentals.

FABRY LENS: A lens that forms an image of the telescope's primary mirror onto the photocathode of a photomultiplier tube and, in this way, serves as a key element of photometer. A mask, placed in front of it, isolates the object whose light intensity is to be measured. The use of such a lens was first proposed by the French physicist Charles Fabry.

FABRY-PÈROT INTERFEROMETER: An instrument with a transparent plate with two reflecting surfaces or two parallel highly reflecting mirrors, in which monochromatic light is passed through, producing circular interference fringes, used in spectroscopy. One of the glass plates is adjustable, enabling the separation of the plates to be varied. The wavelength of the light can be determined by observing the fringes while adjusting the separation.